



US012402525B2

(12) **United States Patent**  
**Um et al.**

(10) **Patent No.:** **US 12,402,525 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **HETEROCYCLIC COMPOUND,  
LIGHT-EMITTING DEVICE INCLUDING  
THE SAME, AND ELECTRONIC APPARATUS  
INCLUDING THE LIGHT-EMITTING  
DEVICE**

(71) Applicants: **Samsung Display Co., Ltd.**, Yongin-si  
(KR); **INDUSTRY-ACADEMIC  
COOPERATION FOUNDATION  
GYEONGSANG NATIONAL  
UNIVERSITY**, Jinju-si (KR)

(72) Inventors: **Hyunah Um**, Yongin-si (KR); **Yunhi  
Kim**, Yongin-si (KR); **Soonki Kwon**,  
Yongin-si (KR); **Heechoon Ahn**,  
Yongin-si (KR); **Jihyun Park**,  
Yongin-si (KR)

(73) Assignees: **Samsung Display Co., Ltd.**, Yongin-si  
(KR); **INDUSTRY-ACADEMIC  
COOPERATION FOUNDATION  
GYEONGSANG NATIONAL  
UNIVERSITY**, Jinju-si (KR)

(\*) Notice: Subject to any disclaimer, the term of this  
patent is extended or adjusted under 35  
U.S.C. 154(b) by 921 days.

(21) Appl. No.: **17/452,318**

(22) Filed: **Oct. 26, 2021**

(65) **Prior Publication Data**

US 2022/0140260 A1 May 5, 2022

(30) **Foreign Application Priority Data**

Oct. 29, 2020 (KR) ..... 10-2020-0142519

(51) **Int. Cl.**  
**H10K 85/40** (2023.01)  
**C07F 7/08** (2006.01)

(Continued)

(52) **U.S. Cl.**  
CPC ..... **H10K 85/40** (2023.02); **C07F 7/0812**  
(2013.01); **H10K 50/11** (2023.02); **H10K**  
**2101/10** (2023.02)

(58) **Field of Classification Search**  
CPC .. H10K 85/40; H10K 85/654; H10K 85/6572;  
H10K 85/6574; H10K 85/6576;  
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,409,729 B2 4/2013 Zeng et al.  
11,508,927 B2 11/2022 Duan et al.  
(Continued)

FOREIGN PATENT DOCUMENTS

CN 103122007 A 5/2013  
CN 109411634 A 3/2019  
(Continued)

OTHER PUBLICATIONS

Kang, Jae-Wook et al.; "Silane- and triazine-containing hole and  
exciton blocking material for high-efficiency phosphorescent organix  
light emitting diodes"; J. Mater. Chem.; 2007;17; pp. 3714-3719.  
(Continued)

*Primary Examiner* — Jenna N Chandhok

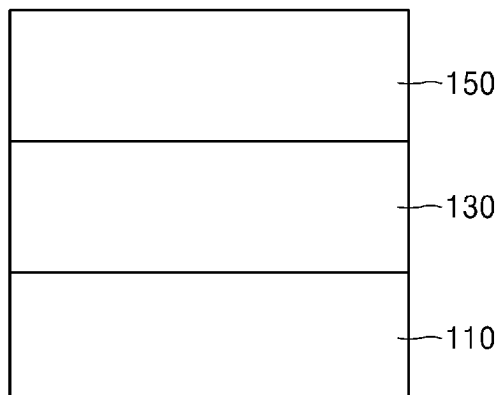
(74) *Attorney, Agent, or Firm* — Womble Bond Dickinson  
(US) LLP

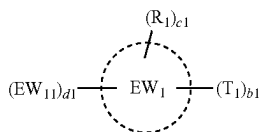
(57) **ABSTRACT**

Provided is a heterocyclic compound represented by For-  
mula 1, a light-emitting device including the same, and an  
electronic apparatus including the light-emitting device. The  
light-emitting device includes a first electrode, a second  
electrode facing the first electrode, an interlayer between the  
first electrode and the second electrode and including an  
emission layer, and the heterocyclic compound represented by  
Formula 1.

(Continued)

10



**18 Claims, 3 Drawing Sheets**(51) **Int. Cl.****H10K 50/11** (2023.01)**H10K 101/10** (2023.01)(58) **Field of Classification Search**CPC ..... C07F 7/0812; C07F 7/083; C07F 7/0805;  
C09K 2211/1018

See application file for complete search history.

(56) **References Cited**

## U.S. PATENT DOCUMENTS

2016/0028030 A1 1/2016 Jung et al.  
 2016/0141525 A1\* 5/2016 Xia ..... H10K 85/6576  
 546/14

## Formula 1

2016/0149157 A1\* 5/2016 Cho ..... H10K 85/626  
 257/40  
 2017/0162796 A1\* 6/2017 Jeong ..... H10K 85/622  
 2018/0294421 A1\* 10/2018 Wolohan ..... H10K 85/6572  
 2020/0203652 A1\* 6/2020 Duan ..... H10K 85/6574

## FOREIGN PATENT DOCUMENTS

CN 111943942 A 11/2020  
 EP 2 977 378 B1 6/2020  
 KR 10-2016-0012028 A 2/2016  
 KR 10-1919621 B1 11/2018

## OTHER PUBLICATIONS

Wang, Huan et al.; "Indole-linked triazine-dibenzothiophene/  
 dibenzofuran based host materials for high-efficiency green and red  
 phosphorescent organic light-emitting diodes"; Dyes and Pigments;  
 2018; 19pp.

Chinese Office Action for CN Application No. 202111185740.3,  
 dated Mar. 5, 2025, 8 pp.

\* cited by examiner

FIG. 1

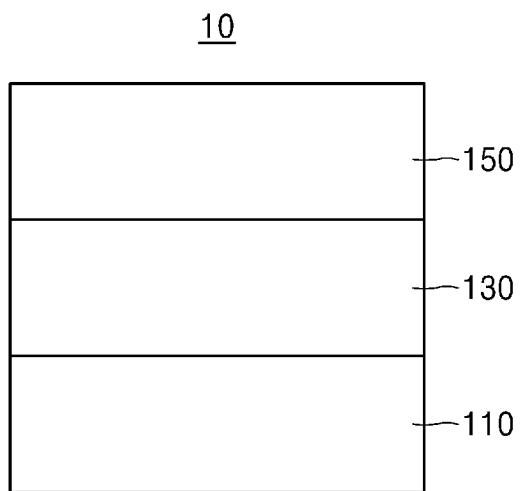


FIG. 2

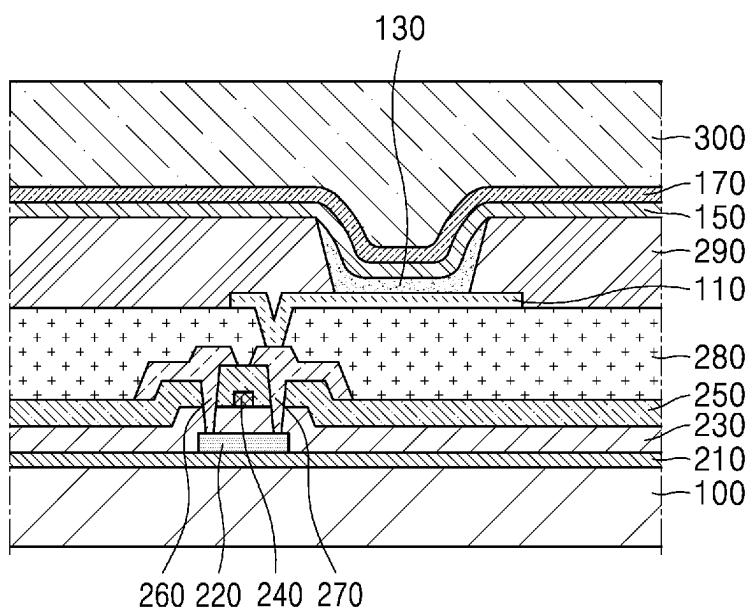
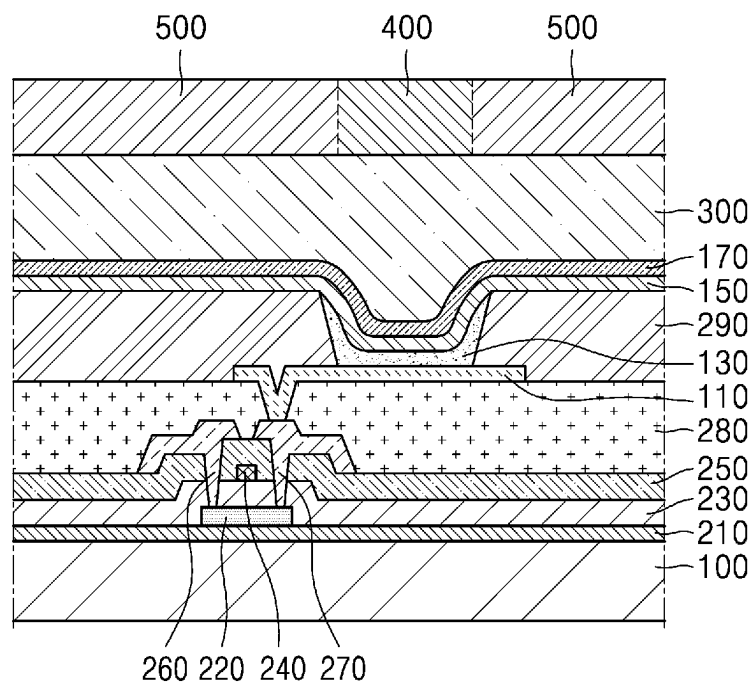


FIG. 3



# 1

## HETEROCYCLIC COMPOUND, LIGHT-EMITTING DEVICE INCLUDING THE SAME, AND ELECTRONIC APPARATUS INCLUDING THE LIGHT-EMITTING DEVICE

### CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to and the benefit of Korean Patent Application No. 10-2020-0142519, filed on Oct. 29, 2020, in the Korean Intellectual Property Office, the entire contents of which are hereby incorporated by reference.

### BACKGROUND

#### 1. Field

One or more embodiments of the present disclosure relate to a heterocyclic compound, a light-emitting device including the heterocyclic compound, and an electronic apparatus including the light-emitting device.

#### 2. Description of Related Art

Among light-emitting devices, organic light-emitting devices are self-emission devices that have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of luminance, driving voltage, and response speed, as compared to other devices in the art.

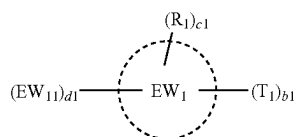
The organic light-emitting devices may include a first electrode on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode sequentially stacked on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as the holes and the electrons, recombine in the emission layer to produce excitons. These excitons transit from an excited state to a ground state to thereby generate light.

### SUMMARY

Provided is a light-emitting device including a compound having excellent luminance efficiency and an electronic apparatus including the light-emitting device.

Additional aspects of embodiments will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

According to an aspect of an embodiment, provided is a heterocyclic compound represented by Formula 1.

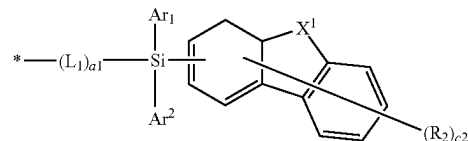


Formula 1

## 2

### -continued

Formula 2



In Formula 1,

EW<sub>1</sub> may be a C<sub>3</sub>-C<sub>60</sub> carbocyclic group or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group,

EW<sub>11</sub> may be —CN,

d1 may be 0 or 1,

wherein EW<sub>1</sub> may be a  $\pi$ -electron-deficient nitrogen-containing C<sub>1</sub>-C<sub>60</sub> cyclic group, or d1 may be 1,

T<sub>1</sub> may be a group represented by Formula 2, and

b1 may be an integer from 1 to 5,

in Formula 2,

X<sub>1</sub> may be O or S,

L<sub>1</sub> is a single bond, a C<sub>4</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

a1 may be an integer from 0 to 5, and

Ar<sub>1</sub> and Ar<sub>2</sub> may each independently be a  $\pi$  electron-rich C<sub>3</sub>-C<sub>60</sub> cyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, and

in Formulae 1 and 2,

R<sub>1</sub> and R<sub>2</sub> may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkenyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkynyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> alkoxy group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> aryloxy group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> arylthio group unsubstituted or substituted with at least one R<sub>10a</sub>, —Si(Q<sub>1</sub>)(Q<sub>2</sub>)(Q<sub>3</sub>), —B(Q<sub>1</sub>)(Q<sub>2</sub>), —P(Q<sub>1</sub>)(Q<sub>2</sub>), or —C(=O)(Q<sub>1</sub>),

c1 in Formula 1 may be an integer from 0 to 10, c2 in Formula 2 may be an integer from 0 to 7, and

R<sub>10a</sub> may be:

deuterium (—D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group,

a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, or a C<sub>1</sub>-C<sub>60</sub> alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, —Si(Q<sub>11</sub>)(Q<sub>12</sub>)(Q<sub>13</sub>), —N(Q<sub>11</sub>)(Q<sub>12</sub>), —B(Q<sub>11</sub>)(Q<sub>12</sub>), —C(=O)(Q<sub>11</sub>), —S(=O)<sub>2</sub>(Q<sub>11</sub>), —P(=O)(Q<sub>11</sub>)(Q<sub>12</sub>), or any combination thereof,

a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, or a C<sub>6</sub>-C<sub>60</sub> arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group,

3

—Si(Q<sub>21</sub>)(Q<sub>22</sub>)(Q<sub>23</sub>), —N(Q<sub>21</sub>)(Q<sub>22</sub>), —B(Q<sub>21</sub>)(Q<sub>22</sub>),  
—C(=O)(Q<sub>21</sub>), —S(=O)<sub>2</sub>(Q<sub>21</sub>), —P(=O)(Q<sub>21</sub>)  
(Q<sub>22</sub>), or any combination thereof, or  
—Si(Q<sub>31</sub>)(Q<sub>32</sub>)(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>),  
—C(=O)(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), or —P(=O)(Q<sub>31</sub>)  
(Q<sub>32</sub>),

wherein Q<sub>1</sub> to Q<sub>3</sub>, Q<sub>11</sub> to Q<sub>13</sub>, Q<sub>21</sub> to Q<sub>23</sub>, and Q<sub>31</sub> to Q<sub>33</sub>  
may each independently be: hydrogen; deuterium; —F;  
—Cl; —Br; —I; a hydroxyl group; a cyano group; a  
nitro group; a C<sub>1</sub>-C<sub>60</sub> alkyl group; a C<sub>2</sub>-C<sub>60</sub> alkenyl  
group; a C<sub>2</sub>-C<sub>60</sub> alkynyl group; a C<sub>1</sub>-C<sub>60</sub> alkoxy group;  
or a C<sub>3</sub>-C<sub>60</sub> carbocyclic group or a C<sub>1</sub>-C<sub>60</sub> heterocyclic  
group, each unsubstituted or substituted with deute-  
rium, —F, a cyano group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a  
C<sub>1</sub>-C<sub>60</sub> alkoxy group, a phenyl group, a biphenyl group,  
or any combination thereof, and

\* in Formula 2 may indicate a binding site to EW<sub>1</sub> in  
Formula 1.

According to another aspect of an embodiment, provided  
is a light-emitting device including a first electrode, a second  
electrode facing the first electrode, an interlayer between the  
first electrode and the second electrode and including an  
emission layer, and a heterocyclic compound represented by  
Formula 1.

According to another aspect of an embodiment, provided  
is an electronic apparatus including the light-emitting  
device.

## BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects and features of certain  
embodiments of the disclosure will be more apparent from  
the following description taken in conjunction with the  
accompanying drawings, in which:

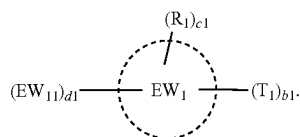
FIG. 1 is a schematic cross-sectional view of a light-  
emitting device according to an embodiment; and

FIGS. 2 and 3 are each a cross-sectional view of a  
light-emitting apparatus according to an embodiment.

## DETAILED DESCRIPTION

Reference will now be made in more detail to embodi-  
ments, examples of which are illustrated in the accompany-  
ing drawings, wherein like reference numerals refer to like  
elements throughout. In this regard, the present embodi-  
ments may have different forms and should not be construed  
as being limited to the descriptions set forth herein. Accord-  
ingly, the embodiments are merely described below, by  
referring to the figures, to explain aspects of embodiments of  
the present description. As used herein, the term “and/or”  
includes any and all combinations of one or more of the  
associated listed items. Throughout the disclosure, the  
expression “at least one of a, b and c” indicates only a, only  
b, only c, both a and b, both a and c, both b and c, all of a,  
b, and c, or variations thereof.

Provided is a heterocyclic compound represented by  
Formula 1:



Formula 1

4

EW<sub>1</sub> in Formula 1 may be a C<sub>3</sub>-C<sub>60</sub> carbocyclic group or  
a C<sub>1</sub>-C<sub>60</sub> heterocyclic group.

In an embodiment, EW<sub>1</sub> in Formula 1 may be a benzene  
group, a naphthalene group, an anthracene group, a  
phenanthrene group, a triphenylene group, a pyrene group,  
a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetra-  
hydronaphthalene group, a thiophene group, a furan group,  
an indole group, a benzoborole group, a benzophosphole  
group, an indene group, a benzosilole group, a benzoger-  
mole group, a benzothiophene group, a benzoselenophene  
group, a benzofuran group, a carbazole group, a dibenzo-  
borole group, a dibenzophosphole group, a fluorene group,  
a dibenzosilole group, a dibenzogermole group, a dibenzo-  
thiophene group, a dibenzoselenophene group, a dibenzo-  
furan group, a dibenzothiophene 5-oxide group, a 9H-fluo-  
rene-9-on group, a dibenzothiophene 5,5-dioxide group, an  
azaindole group, an azabenzoborole group, an azabenz-  
phosphole group, an azaindene group, an azabenzosilole  
group, an azabenzogermole group, an azabenzothiophene  
group, an azabenzoselenophene group, an azabenzofuran  
group, an azacarbazole group, an azadibenzoborole group,  
an azadibenzophosphole group, an azafluorene group, an  
azadibenzosilole group, an azadibenzogermole group, an  
azadibenzothiophene group, an azadibenzoselenophene  
group, an azadibenzofuran group, an azadibenzothiophene  
5-oxide group, an aza-9H-fluorene-9-on group, an azadiben-  
zothiophene 5,5-dioxide group, a pyridine group, a pyrimi-  
dine group, a pyrazine group, a pyridazine group, a triazine  
group, a quinoline group, an isoquinoline group, a quinox-  
aline group, a quinazoline group, a phenanthroline group, a  
pyrrole group, a pyrazole group, an imidazole group, a  
triazole group, an oxazole group, an isoxazole group, a  
thiazole group, an isothiazole group, an oxadiazole group, a  
thiadiazole group, a benzopyrazole group, a benzimidazole  
group, a benzoxazole group, a benzothiazole group, a ben-  
zoxadiazole group, a benzothiadiazole group, a 5,6,7,8-  
tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquino-  
line group.

EW<sub>11</sub> in Formula 1 may be —CN.

d<sub>1</sub> in Formula 1 may be 0 or 1. d<sub>1</sub> is the number of  
EW<sub>11</sub>(s).

In one or more embodiments, EW<sub>1</sub> in Formula 1 may be  
a  $\pi$ -electron-deficient nitrogen-containing C<sub>1</sub>-C<sub>60</sub> cyclic  
group, or d<sub>1</sub> may be 1. For example, when EW<sub>1</sub> is a  
 $\pi$ -electron-deficient nitrogen-containing C<sub>1</sub>-C<sub>60</sub> cyclic  
group, d<sub>1</sub> may be 0 or 1, but when EW<sub>1</sub> is not a  $\pi$ -electron-  
deficient nitrogen-containing C<sub>1</sub>-C<sub>60</sub> cyclic group, d<sub>1</sub> may  
be 1.

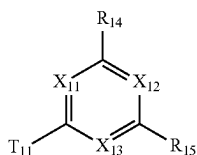
In an embodiment, EW<sub>1</sub> in Formula 1 may be an imida-  
zole group, a pyrazole group, a thiazole group, an isothiazole  
group, an oxazole group, an isoxazole group, a pyridine  
group, a pyrazine group, a pyridazine group, a pyrimidine  
group, an indazole group, a purine group, a quinoline group,  
an isoquinoline group, a benzoquinoline group, a benzoiso-  
quinoline group, a phthalazine group, a naphthyridine group,  
a quinoxaline group, a benzoquinoxaline group, a quinazo-  
line group, a cinnoline group, a phenanthridine group, an  
acridine group, a phenanthroline group, a phenazine group,  
a benzimidazole group, an benzoisothiazole group, a ben-  
zoxazole group, a benzoisoxazole group, a triazole group, a  
tetrazole group, an oxadiazole group, a triazine group, a  
thiadiazole group, an imidazopyridine group, an imidazopy-  
rimidine group, an azacarbazole group, an azadibenzofuran  
group, an azadibenzothiophene group, an azadibenzosilole  
group, an acridine group, or a pyridopyrazine group.

In one or more embodiments, EW<sub>1</sub> in Formula 1 may be  
a benzene group, a naphthalene group, an anthracene group,

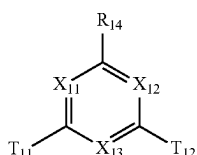
## 5

a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, or a 1,2,3,4-tetrahydronaphthalene group, and d1 may be 1.

In one or more embodiments, EW<sub>1</sub> in Formula 1 may be Formula 1(a) or 1(b):



Formula 1(a)



Formula 1(b)

In Formulae 1(a) and 1(b),

X<sub>11</sub> may be N or C(R<sub>11</sub>),

X<sub>12</sub> may be N or C(R<sub>12</sub>),

X<sub>13</sub> may be N or C(R<sub>13</sub>),

at least one selected from X<sub>11</sub> to X<sub>13</sub> may be N or C(CN), R<sub>11</sub> to R<sub>15</sub> are the same as described in connection with R<sub>1</sub>, and

T<sub>11</sub> and T<sub>12</sub> are the same as described in connection with T<sub>1</sub>.

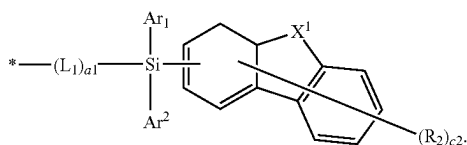
In an embodiment, at least two of X<sub>11</sub> to X<sub>13</sub> in Formulae 1(a) and 1(b) may be N.

In an embodiment, in Formulae 1(a) and 1(b), X<sub>11</sub> may be C(R<sub>11</sub>), X<sub>12</sub> may be C(R<sub>12</sub>), X<sub>13</sub> may be C(R<sub>13</sub>), and at least one selected from X<sub>11</sub> to X<sub>13</sub> may be C(CN).

b1 in Formula 1 may be an integer from 1 to 5. b1 is the number of T<sub>1</sub>(s).

In an embodiment, b1 in Formula 1 may be 1 or 2.

T<sub>1</sub> in Formula 1 may be a group represented by Formula 2:



Formula 2

X<sub>1</sub> in Formula 2 may be O or S.

In one or more embodiments, X<sub>1</sub> in Formula 2 may be O, but embodiments of the present disclosure are not limited thereto.

L<sub>1</sub> in Formula 2 may be a single bond, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>.

In one or more embodiments, L<sub>1</sub> in Formula 2 may be: a single bond; or a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, or a 1,2,3,4-tetrahydronaphthalene group, each unsubstituted or substituted with at least one R<sub>10a</sub>.

In one or more embodiments, L<sub>1</sub> in Formula 2 may be: a single bond; or a benzene group unsubstituted or substituted

## 6

with at least one R<sub>10a</sub>, but embodiments of the present disclosure are not limited thereto.

a1 in Formula 2 may be an integer from 0 to 5. a1 is the number of L<sub>1</sub>(s).

In one or more embodiments, a1 in Formula 2 may be 0, 1, or 2.

Ar<sub>1</sub> and Ar<sub>2</sub> in Formula 2 may each independently be a π electron-rich C<sub>3</sub>-C<sub>60</sub> cyclic group unsubstituted or substituted with at least one R<sub>10a</sub>.

In an embodiment, Ar<sub>1</sub> and Ar<sub>2</sub> in Formula 2 may each independently be a benzene group, a heptalene group, an indene group, a naphthalene group, an azulene group, an indacene group, an acenaphthenyl group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentacene group, a hexacene group, a pentaphene group, a rubicene group, a coronene group, an ovalene group, a pyrrole group, a furan group, a thiophene group, an isoindole group, an indole group, an indene group, a benzofuran group, a benzothiophene group, a benzosilole group, a naphtho pyrrole group, a naphthofuran group, a naphthothiophene group, a naphthosilole group, a benzocarbazole group, a dibenzocarbazole group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, a dibenzosilole group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a triindolobenzene group, a pyrrolophenanthrene group, a furanophenanthrene group, a thienophenanthrene group, a benzonaphthofuran group, a benzonaphthothiophene group, a (indolo)phenanthrene group, a (benzofurano)phenanthrene group, or a (benzothieno)phenanthrene group, each unsubstituted or substituted with at least one R<sub>10a</sub>.

In one or more embodiments, Ar<sub>1</sub> and Ar<sub>2</sub> in Formula 2 may each independently be a benzene group unsubstituted or substituted with at least one R<sub>10a</sub>, but embodiments of the present disclosure are not limited thereto.

In Formula 1 and 2,

R<sub>1</sub> and R<sub>2</sub> may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkenyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkynyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> alkoxy group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> aryloxy group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> arylthio group unsubstituted or substituted with at least one R<sub>10a</sub>, —Si(Q<sub>1</sub>)(Q<sub>2</sub>)(Q<sub>3</sub>), —B(Q<sub>1</sub>)(Q<sub>2</sub>), —P(Q<sub>1</sub>)(Q<sub>2</sub>), or —C(=O)(Q<sub>1</sub>).

c1 in Formula 1 may be an integer from 0 to 10, c2 in Formula 2 may be an integer from 0 to 7, and

R<sub>10a</sub> may be:

deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group,

a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, or a C<sub>1</sub>-C<sub>60</sub> alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, —Si(Q<sub>11</sub>)(Q<sub>12</sub>)(Q<sub>13</sub>), —N(Q<sub>11</sub>)(Q<sub>12</sub>), —B(Q<sub>11</sub>)(Q<sub>12</sub>),



7

—C(=O)(Q<sub>11</sub>), —S(=O)<sub>2</sub>(Q<sub>11</sub>), —P(=O)(Q<sub>11</sub>)  
(Q<sub>12</sub>), or any combination thereof,

a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group,  
a C<sub>6</sub>-C<sub>60</sub> aryloxy group, or a C<sub>6</sub>-C<sub>60</sub> arylthio group,  
each unsubstituted or substituted with deuterium, —F,  
—Cl, —Br, —I, a hydroxyl group, a cyano group, a  
nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl  
group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group,  
a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic  
group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group,  
—Si(Q<sub>21</sub>)(Q<sub>22</sub>)(Q<sub>23</sub>), —N(Q<sub>21</sub>)(Q<sub>22</sub>), —B(Q<sub>21</sub>)(Q<sub>22</sub>),  
—C(=O)(Q<sub>21</sub>), —S(=O)<sub>2</sub>(Q<sub>21</sub>), —P(=O)(Q<sub>21</sub>)  
(Q<sub>22</sub>), or any combination thereof, or

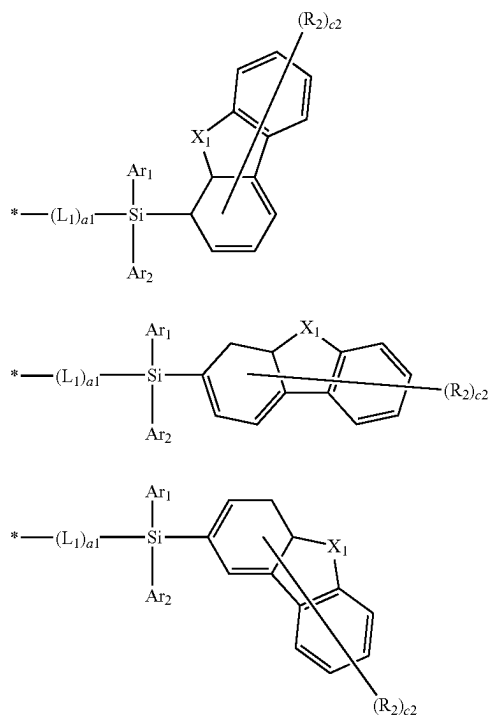
—Si(Q<sub>31</sub>)(Q<sub>32</sub>)(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>),  
—C(=O)(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), or —P(=O)(Q<sub>31</sub>)  
(Q<sub>32</sub>),

wherein Q<sub>1</sub> to Q<sub>3</sub>, Q<sub>11</sub> to Q<sub>13</sub>, Q<sub>21</sub> to Q<sub>23</sub>, and Q<sub>31</sub> to Q<sub>33</sub>  
may each independently be: hydrogen; deuterium; —F;  
—Cl; —Br; —I; a hydroxyl group; a cyano group; a  
nitro group; a C<sub>1</sub>-C<sub>60</sub> alkyl group; a C<sub>2</sub>-C<sub>60</sub> alkenyl  
group; a C<sub>2</sub>-C<sub>60</sub> alkynyl group; a C<sub>1</sub>-C<sub>60</sub> alkoxy group;  
or a C<sub>3</sub>-C<sub>60</sub> carbocyclic group or a C<sub>1</sub>-C<sub>60</sub> heterocyclic  
group, each unsubstituted or substituted with deuterium,  
—F, a cyano group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a  
C<sub>1</sub>-C<sub>60</sub> alkoxy group, a phenyl group, a biphenyl group,  
or any combination thereof, and

\* in Formula 2 may indicate a binding site to EW<sub>1</sub> in  
Formula 1.

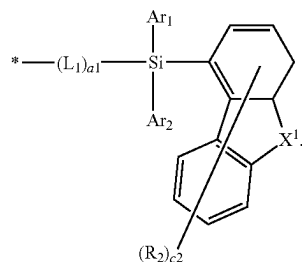
In one or more embodiments, c2 in Formula 2 may be an  
integer from 0 to 7, where c2 indicates the number of R<sub>2</sub>(s).

In an embodiment, Formula 2 may be represented by one  
selected from Formulae 2-1 to 2-4:



8

-continued



2-4

In Formulae 2-1 to 2-4,

X<sub>1</sub>, L<sub>1</sub>, a<sub>1</sub>, Ar<sub>1</sub>, Ar<sub>2</sub>, R<sub>2</sub>, and c2 are each the same as  
respectively described above, and

\* indicates a binding site to EW<sub>1</sub> in Formula 1.

R<sub>1</sub> and R<sub>2</sub> in Formulae 1 and 2 may each independently  
be hydrogen, deuterium, —F, —Cl, —Br, —I, a  
hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub>  
alkyl group unsubstituted or substituted with at least  
one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkenyl group unsubstituted or  
substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkynyl  
group unsubstituted or substituted with at least one  
R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> alkoxy group unsubstituted or substi-  
tuted with at least one R<sub>10a</sub>, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group  
unsubstituted or substituted with at least one R<sub>10a</sub>, a  
C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted  
with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> aryloxy group unsub-  
stituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub>  
arylthio group unsubstituted or substituted with at least  
one R<sub>10a</sub>, —Si(Q<sub>1</sub>)(Q<sub>2</sub>)(Q<sub>3</sub>), —B(Q<sub>1</sub>)(Q<sub>2</sub>), —P(Q<sub>1</sub>)  
(Q<sub>2</sub>), or —C(=O)(Q<sub>1</sub>).

In an embodiment, R<sub>1</sub> and R<sub>2</sub> in Formulae 1 and 2 may  
each independently be: hydrogen, deuterium, —F, —Cl,  
—Br, —I, a hydroxyl group, a cyano group, or a nitro group;  
a C<sub>1</sub>-C<sub>20</sub> alkyl group, a C<sub>2</sub>-C<sub>20</sub> alkenyl group, a C<sub>2</sub>-C<sub>20</sub>  
alkynyl group, or a C<sub>1</sub>-C<sub>20</sub> alkoxy group, each unsub-  
stituted or substituted with deuterium, —F, —Cl, —Br,  
—I, —CD<sub>3</sub>, —CD<sub>2</sub>H, —CDH<sub>2</sub>, —CF<sub>3</sub>, —CF<sub>2</sub>H,  
—CFH<sub>2</sub>, a hydroxyl group, a cyano group, a nitro  
group, a cyclopentyl group, a cyclohexyl group, a  
cycloheptyl group, a cyclooctyl group, an adamantanyl  
group, a norbornanyl group, a norbornenyl group, a  
cyclopentenyl group, a cyclohexenyl group, a cyclo-  
heptenyl group, a phenyl group, a naphthyl group, a  
pyridinyl group, a pyrimidinyl group, —Si(Q<sub>31</sub>)(Q<sub>32</sub>)  
(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>), —C(=O)  
(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), —P(=O)(Q<sub>31</sub>)(Q<sub>32</sub>), or any  
combination thereof;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl  
group, a cyclooctyl group, an adamantanyl group, a  
norbornanyl group, a norbornenyl group, a cyclopent-  
enyl group, a cyclohexenyl group, a cycloheptenyl  
group, a phenyl group, a naphthyl group, a fluorenyl  
group, a phenanthrenyl group, an anthracenyl group, a  
fluoranthenyl group, a triphenylene group, a pyrenyl  
group, a chrysenyl group, a pyrrolyl group, a thienyl  
group, a furanyl group, an imidazolyl group, a pyra-  
zoly group, a thiazolyl group, an isothiazolyl group, an  
oxazolyl group, an isoxazolyl group, a pyridinyl group,  
a pyrazinyl group, a pyrimidinyl group, a pyridazinyl  
group, an isoindolyl group, an indolyl group, an inda-  
zolyl group, a purinyl group, a quinolinyl group, an iso-  
quinolinyl group, a benzoquinolinyl group, a quino-  
xalinyl group, a quinazolinyl group, a cinnolinyl

9

group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothienyl group, a benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothienyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD<sub>3</sub>, —CD<sub>2</sub>H, —CDH<sub>2</sub>, —CF<sub>3</sub>, —CF<sub>2</sub>H, —CFH<sub>2</sub>, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>20</sub> alkyl group, a C<sub>2</sub>-C<sub>20</sub> alkenyl group, a C<sub>2</sub>-C<sub>20</sub> alkynyl group, a C<sub>1</sub>-C<sub>20</sub> alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thienyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothienyl group, a benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothienyl, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —Si(Q<sub>31</sub>)(Q<sub>32</sub>)(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>), —C(=O)(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), —P(=O)(Q<sub>31</sub>)(Q<sub>32</sub>), or any combination thereof, or —Si(Q<sub>1</sub>)(Q<sub>2</sub>)(Q<sub>3</sub>), —N(Q<sub>1</sub>)(Q<sub>2</sub>), —B(Q<sub>1</sub>)(Q<sub>2</sub>), —C(=O)(Q<sub>1</sub>), —S(=O)<sub>2</sub>(Q<sub>1</sub>), —P(=O)(Q<sub>1</sub>)(Q<sub>2</sub>), or —P(Q<sub>1</sub>)(Q<sub>2</sub>).

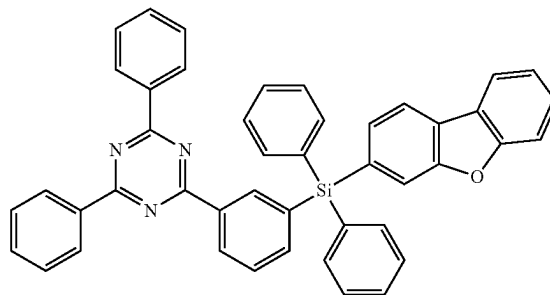
c1 in Formula 1 may be an integer from 0 to 10. c2 in Formula 2 may be an integer from 0 to 7. c1 is the number of R<sub>1</sub>(s), and c2 is the number of R<sub>2</sub>(s).

In an embodiment, the heterocyclic compound may be selected from Compounds 1 to 96 below:

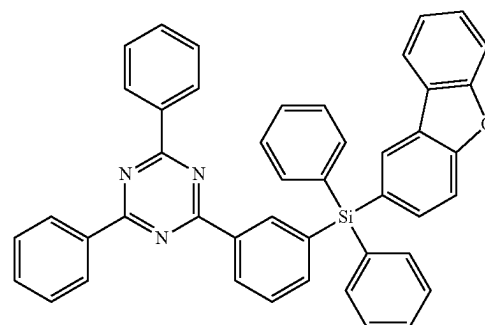
10

-continued

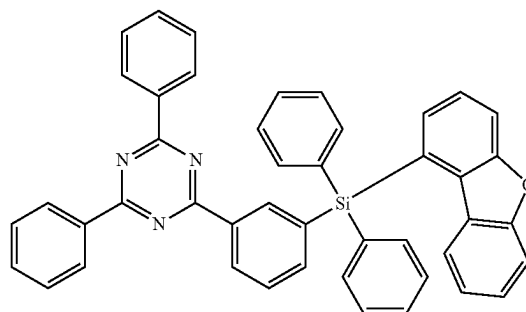
2



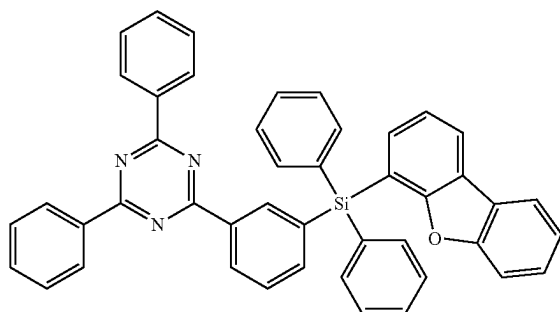
3



4



5

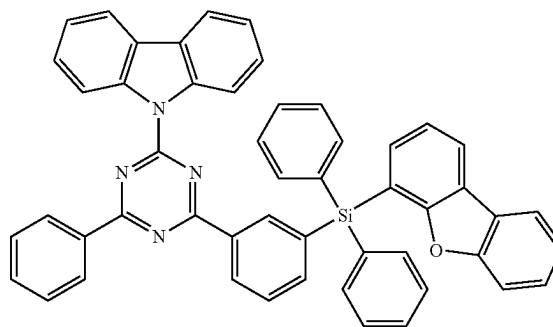


1

55

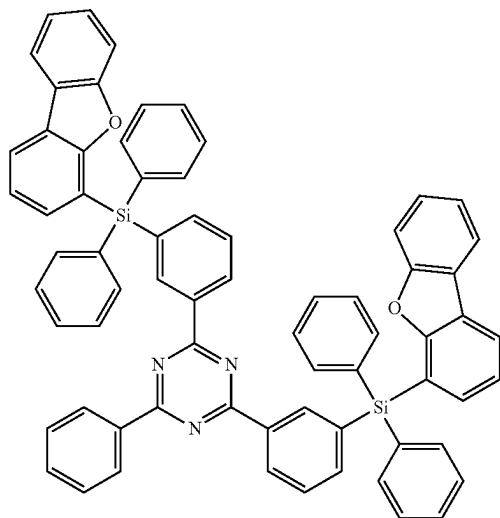
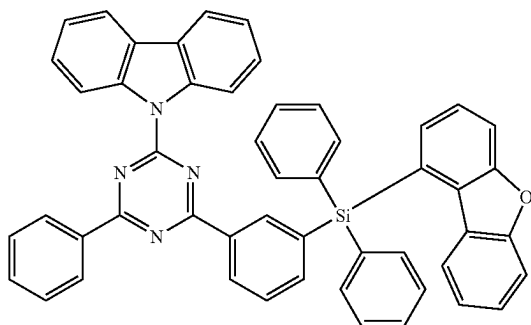
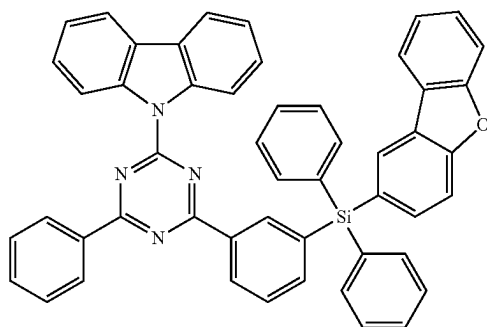
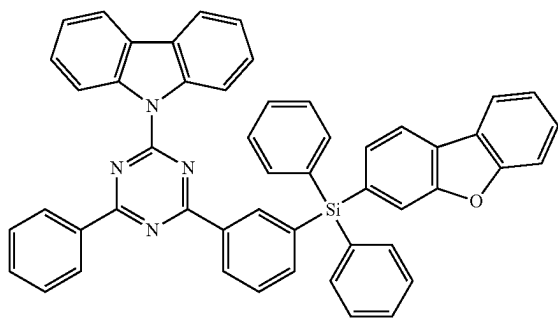
60

65

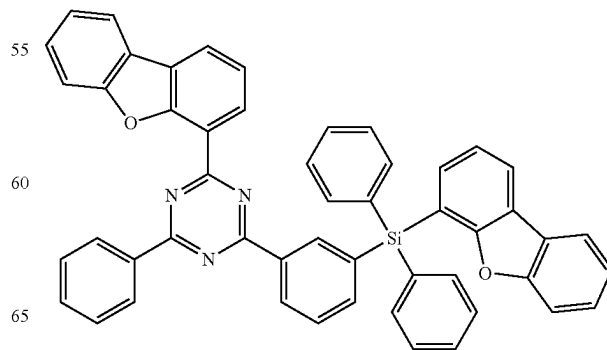
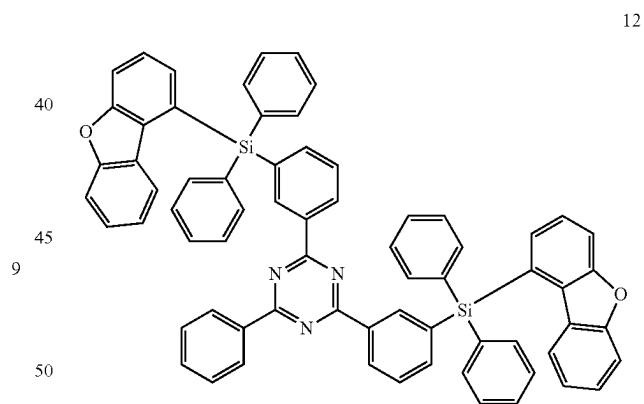
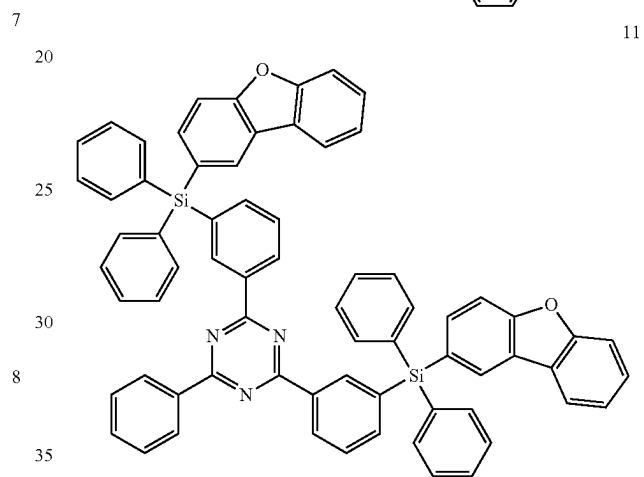
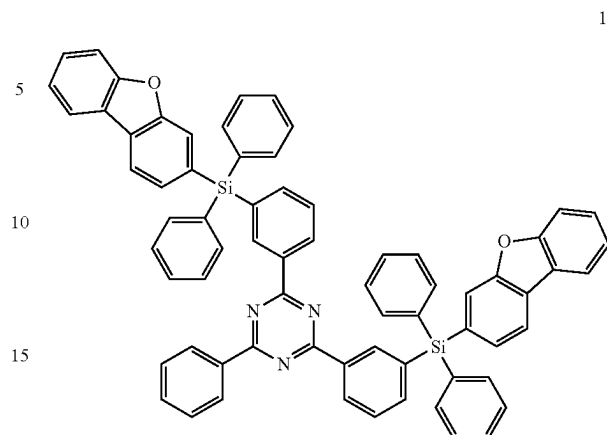


**11**

-continued

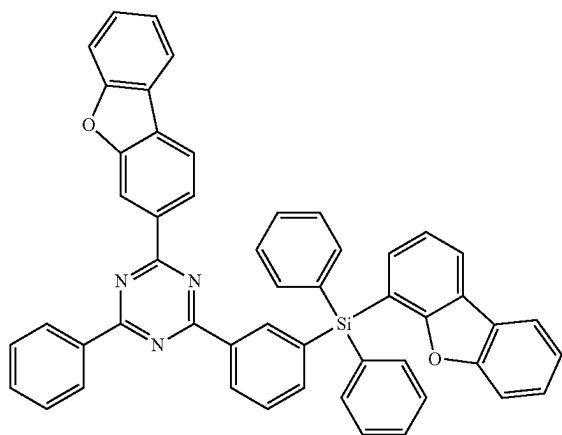
**12**

-continued



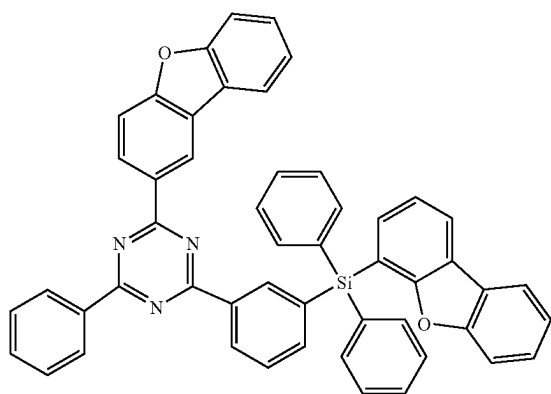
**13**

-continued



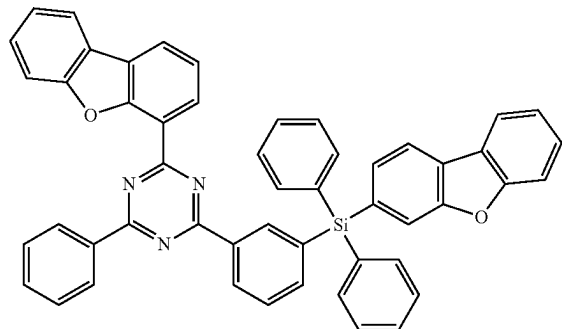
14

15

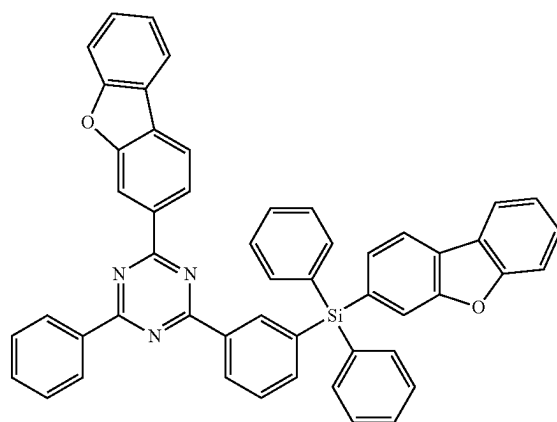


16

17

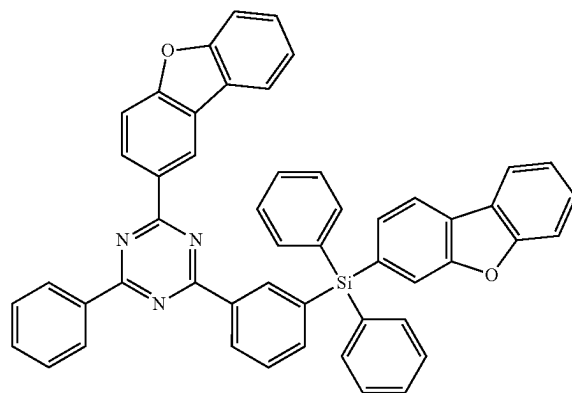
**14**

-continued



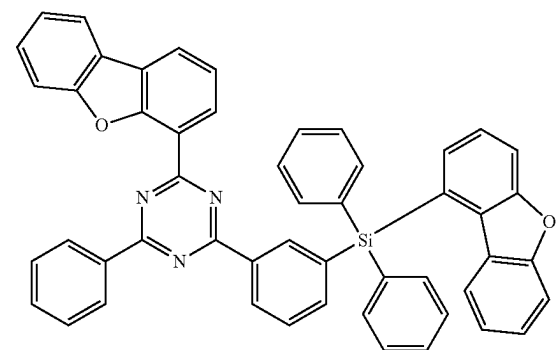
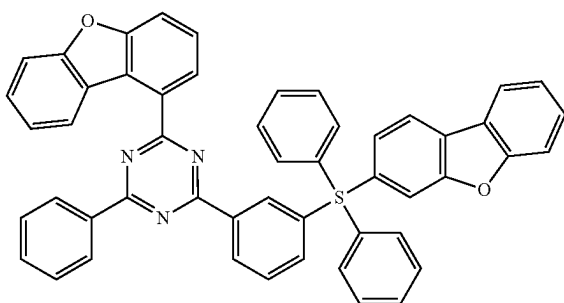
18

19



20

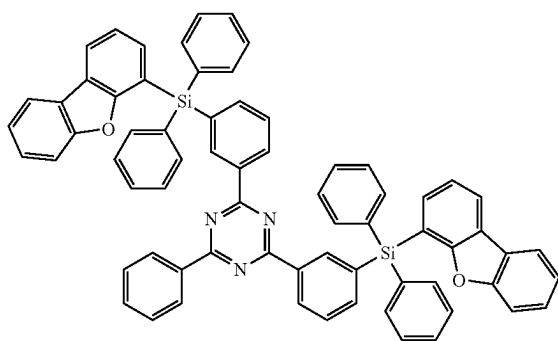
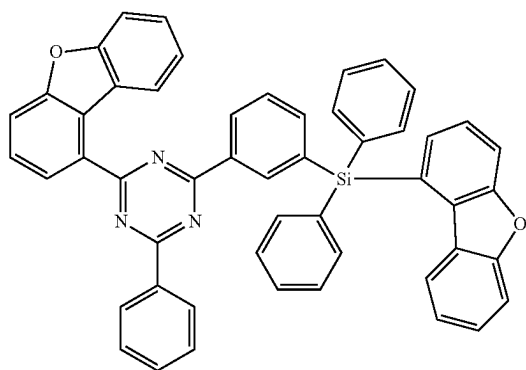
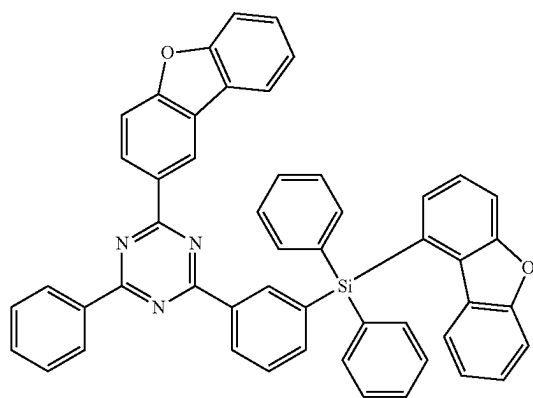
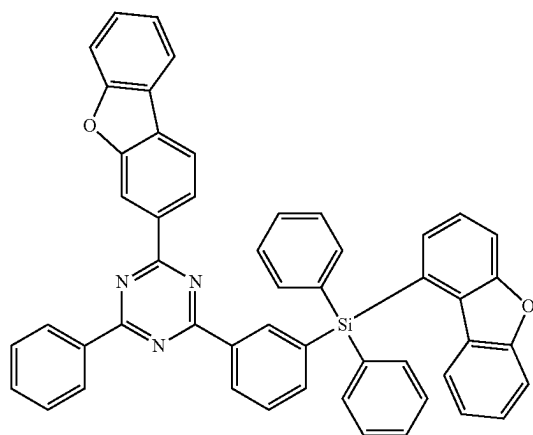
21



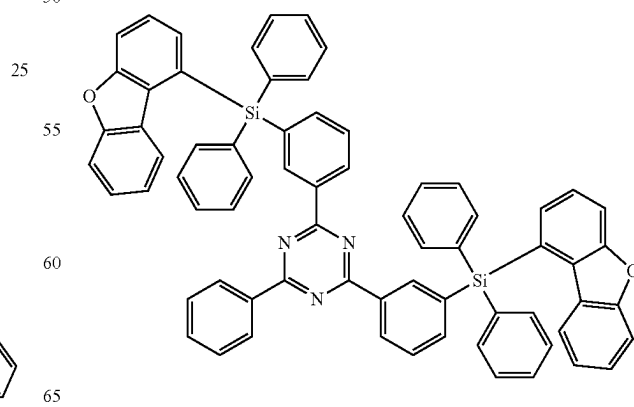
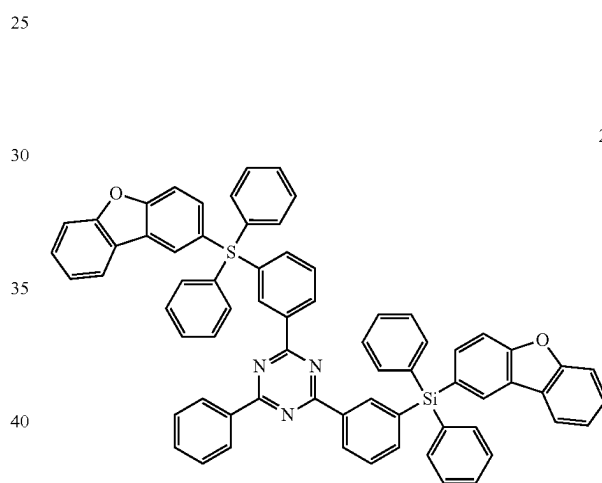
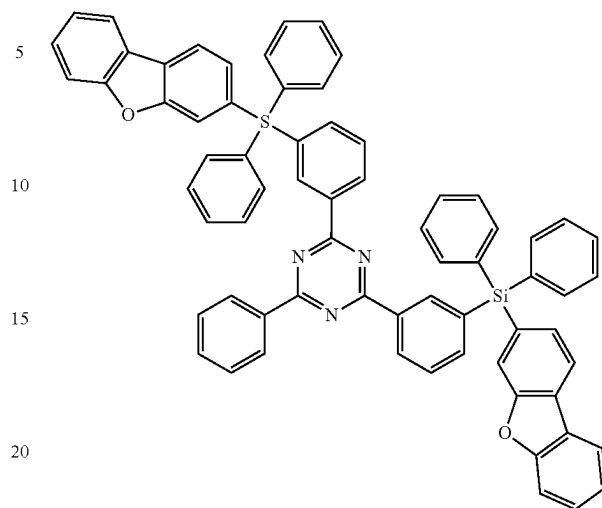
65

**15**

-continued

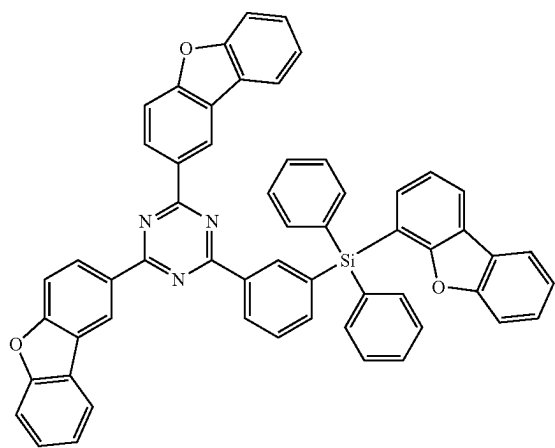
**16**

-continued



**17**

-continued



29

5

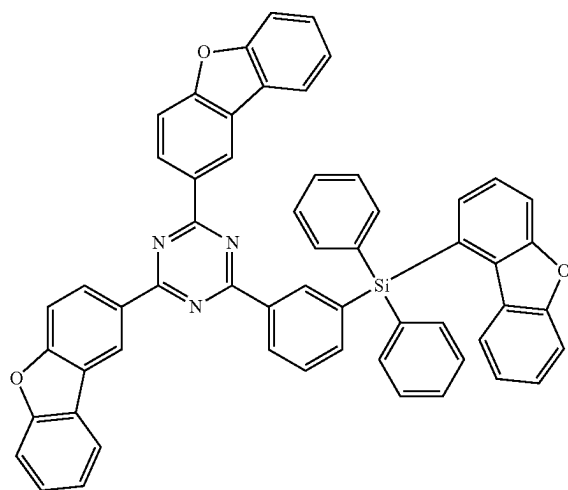
10

15

20

**18**

-continued

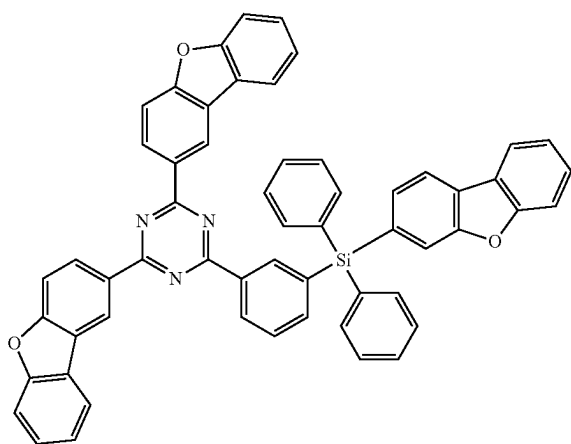


32

30

25

33

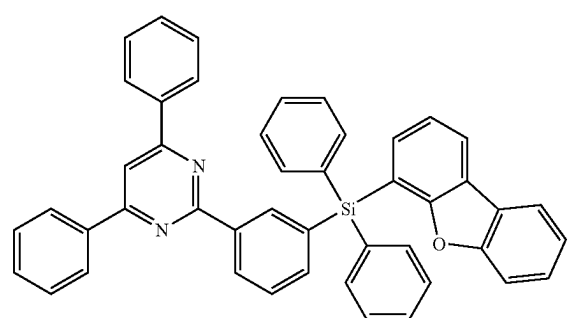


30

35

40

34



31

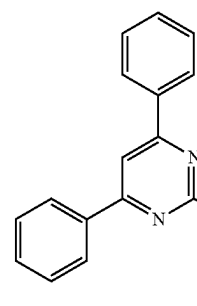
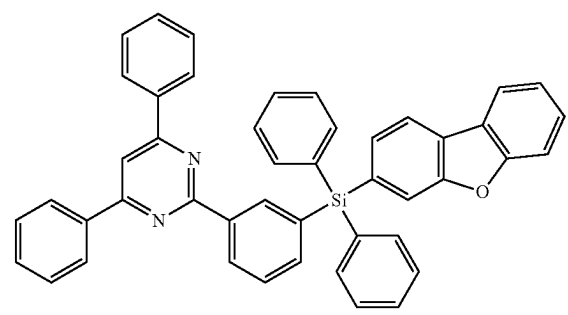
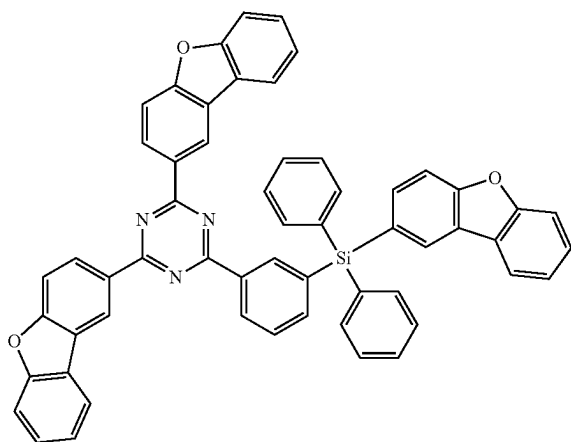
50

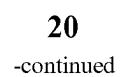
55

60

65

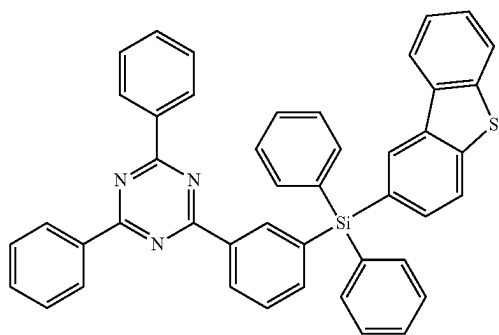
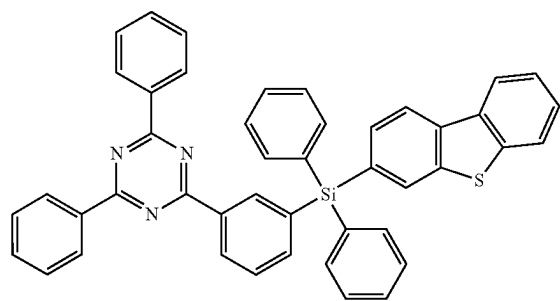
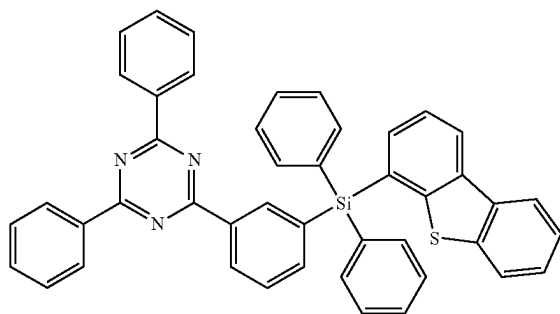
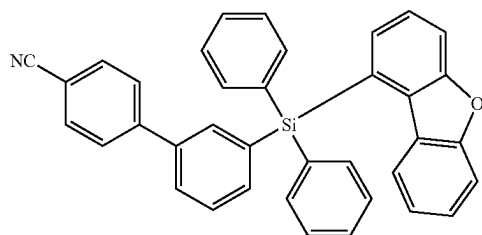
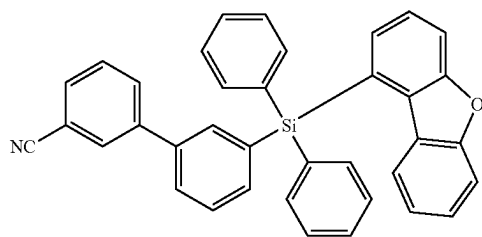
35



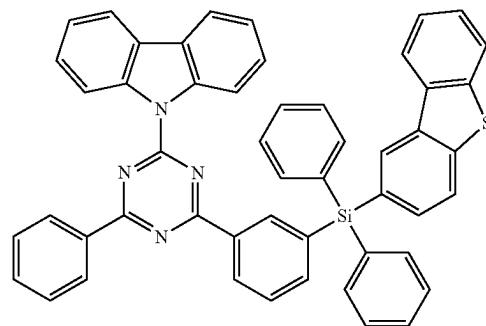
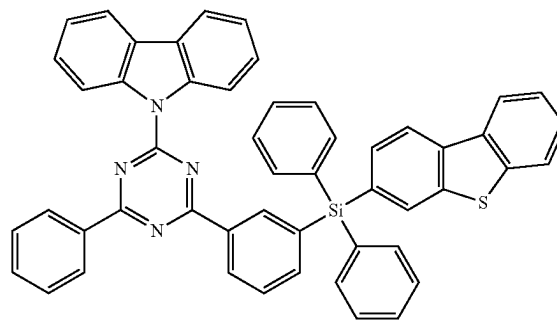
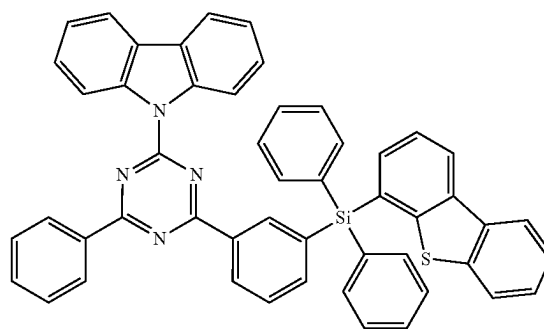
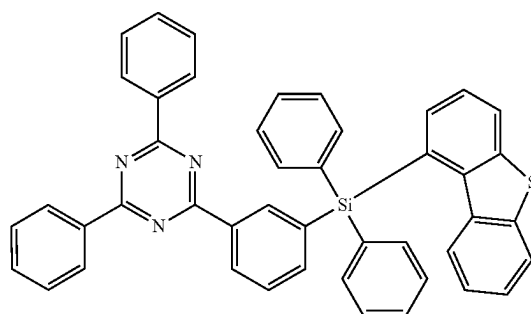


**21**

-continued

**22**

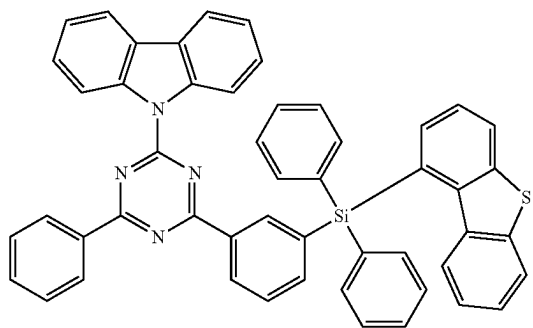
-continued





**23**

-continued



56

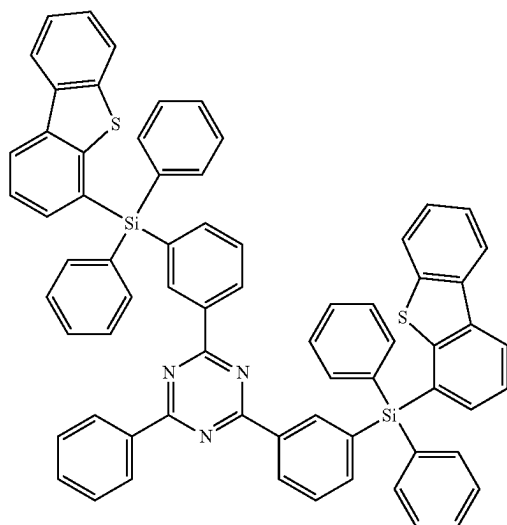
5

10

15

20

57



25

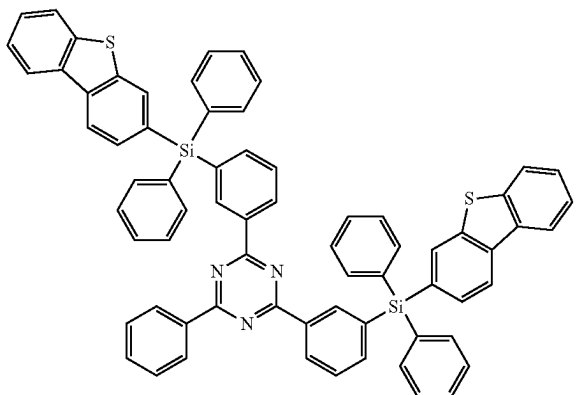
30

35

40

45

58



50

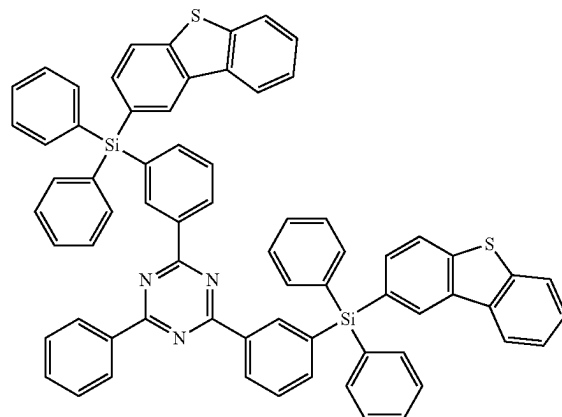
55

60

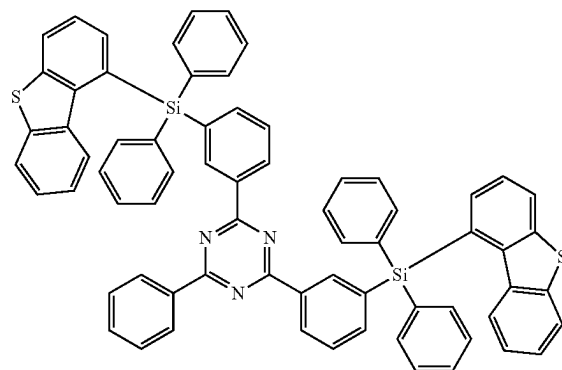
65

**24**

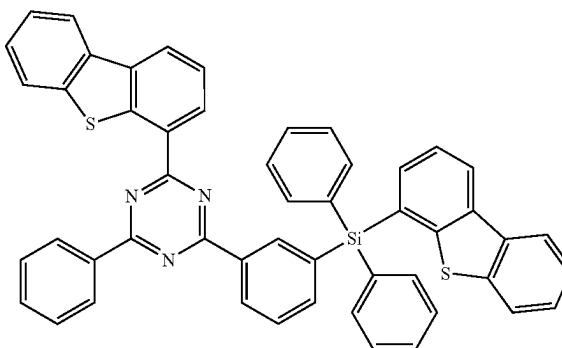
-continued



59



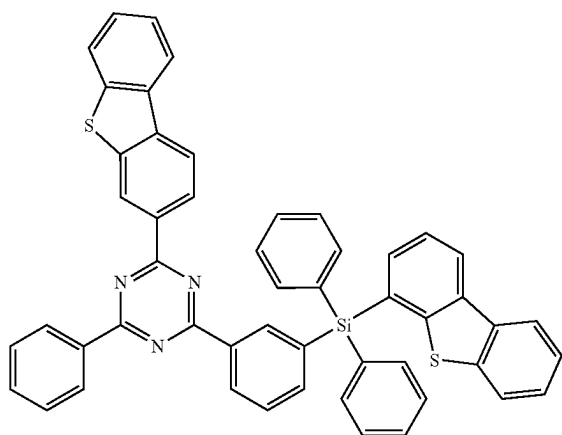
60



61

**25**

-continued



62

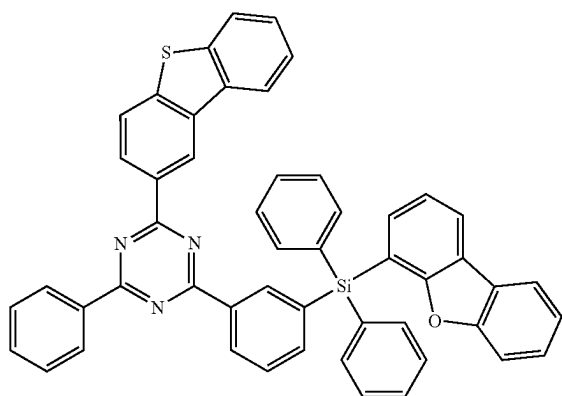
5

10

15

20

63

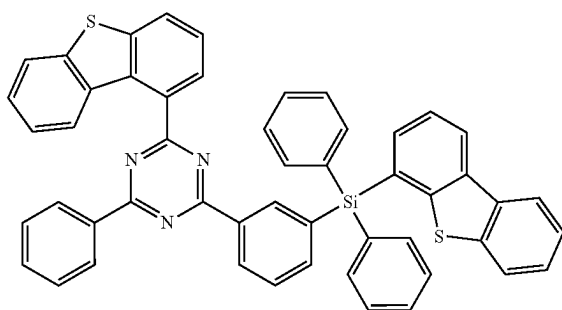


25

30

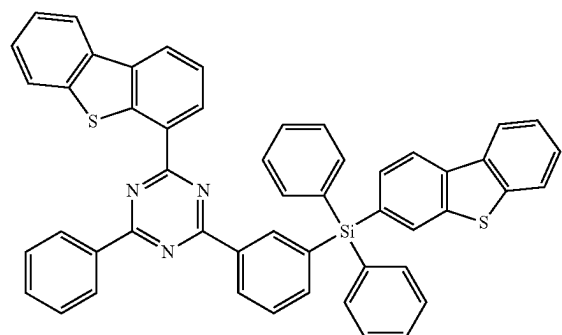
35

64



50

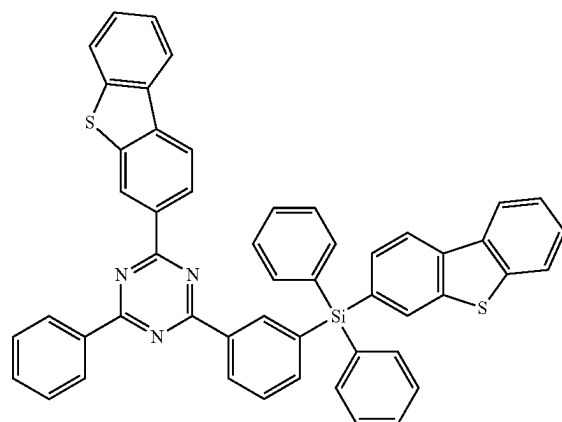
65



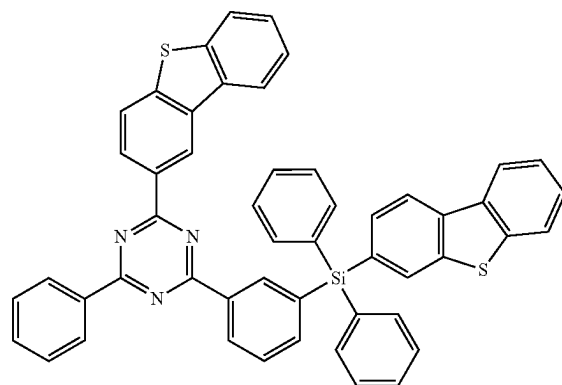
65

**26**

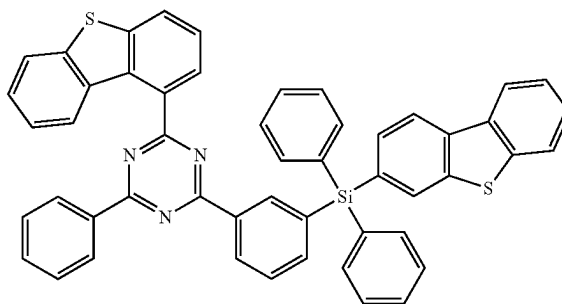
-continued



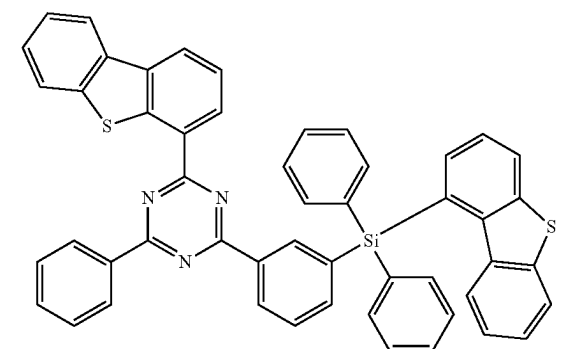
66



67



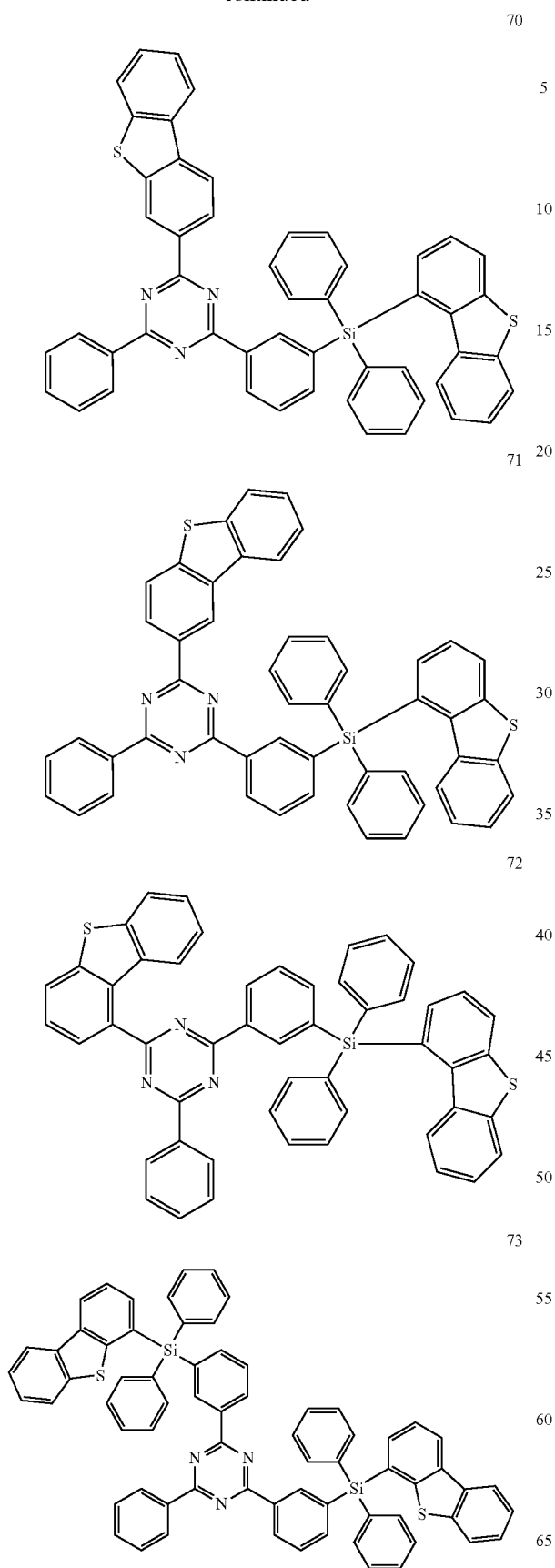
68



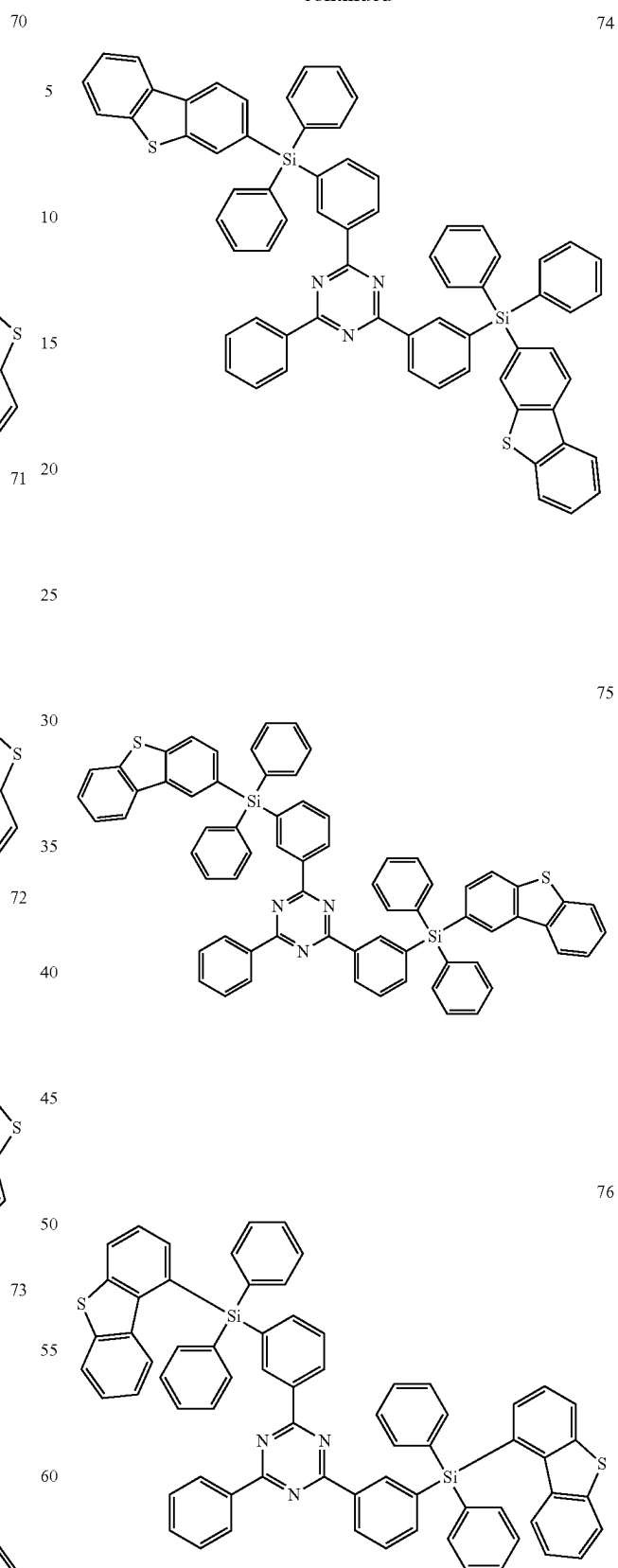
69

**27**

-continued

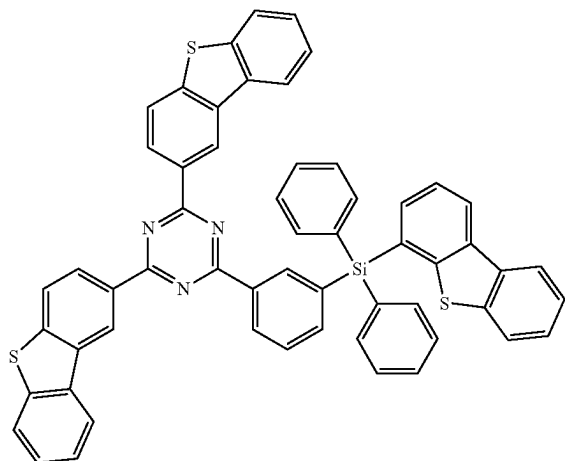
**28**

-continued



**29**

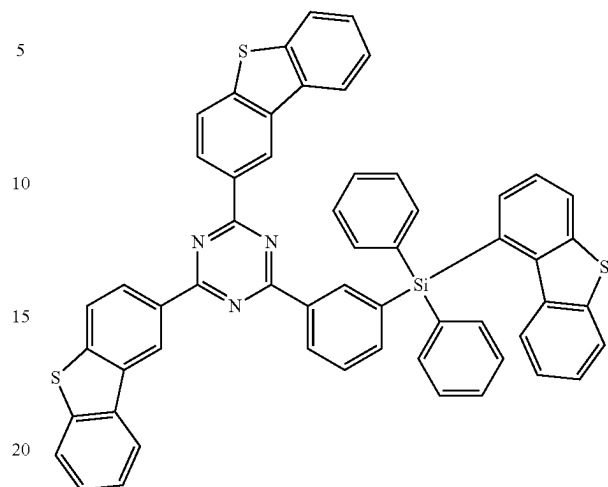
-continued



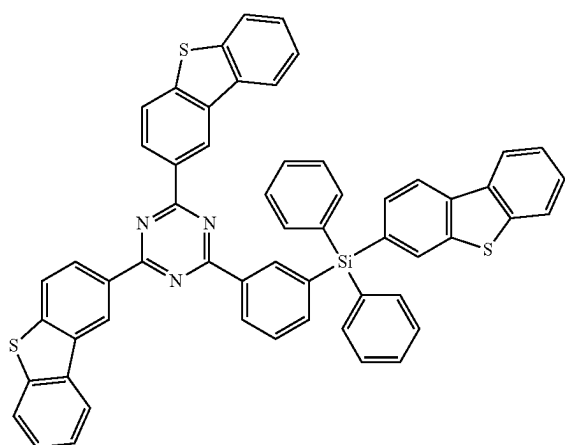
77

**30**

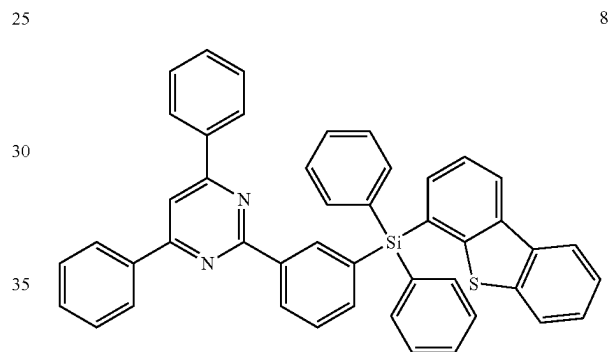
-continued



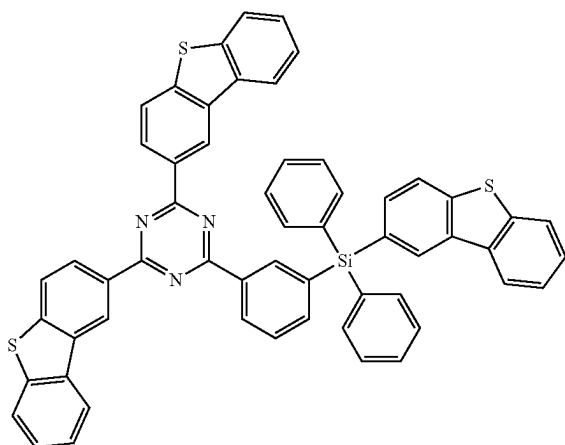
80



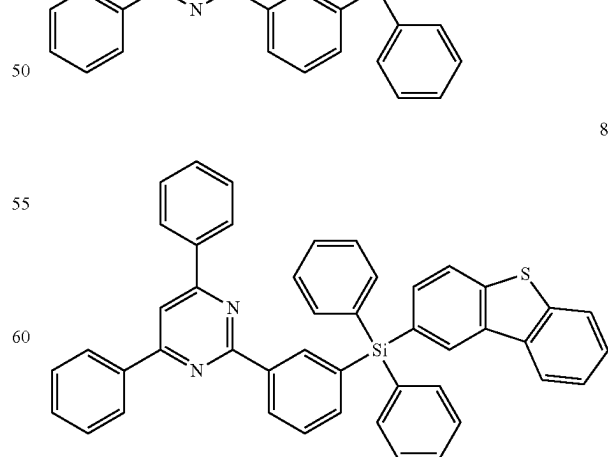
78



81



79

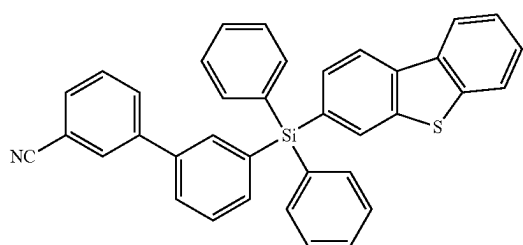
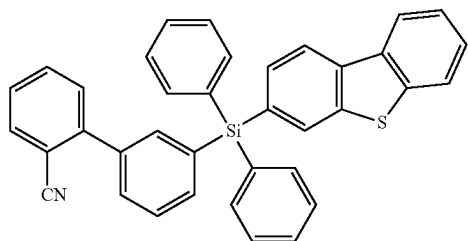
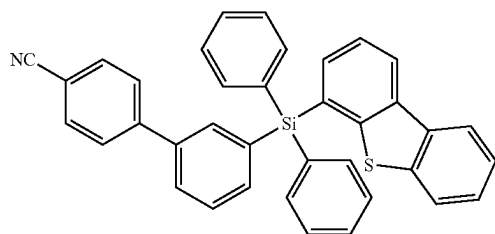
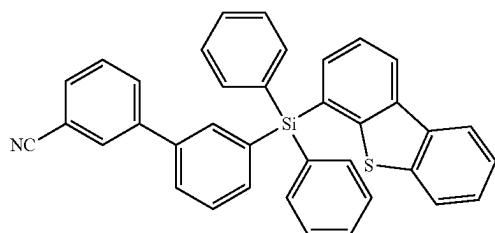
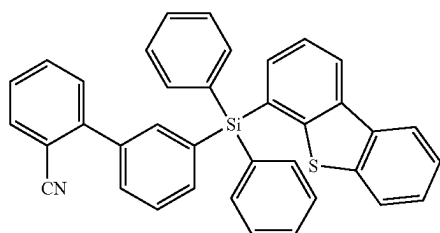
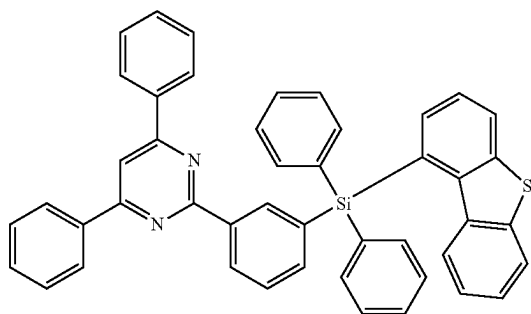


83

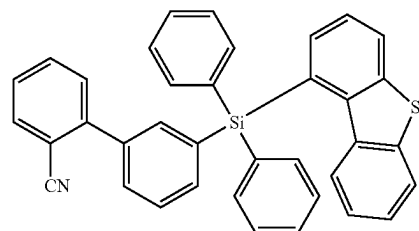
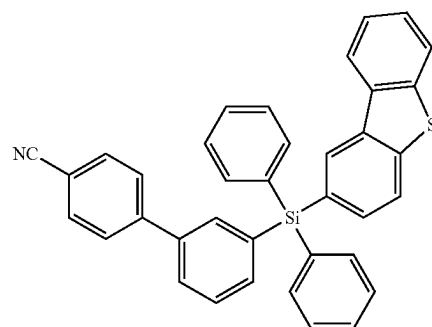
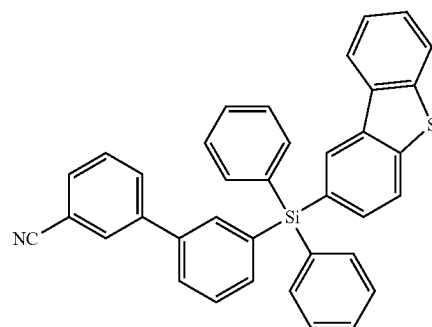
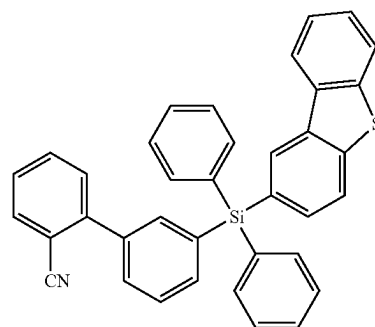
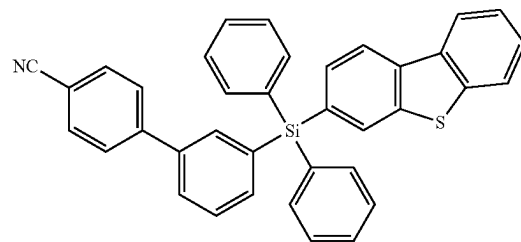
65

**31**

-continued

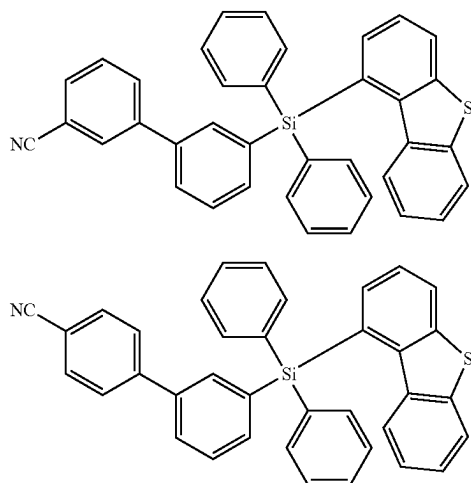
**32**

-continued



33

-continued



The heterocyclic compound may include i) a group represented by Formula 1 and 2, and  $X_1$  in Formula 2 may be O or S. When  $X_1$  is O or S, compared to a case in which  $X_1$  is N, the heterocyclic compound may have a similar lowest unoccupied molecular orbital (LUMO) energy level and a relatively low highest occupied molecular orbital (HOMO) energy level, thereby having a large energy band gap. In the case of a group represented by Formula 1 and Formula 2, a triplet ( $T_1$ ) energy level is increased due to a steric hindrance effect, and thus, a difference between a singlet ( $S_1$ ) energy level and a triplet ( $T_1$ ) energy level may be increased. Also, ii) because  $EW_1$  in Formula 1 is a  $\pi$ -electron-deficient nitrogen-containing  $C_{1-}C_{60}$  cyclic group or d1 is 1, an electron-accepting capacity is enhanced and thus, electron transport characteristics are improved. Accordingly, the heterocyclic compound may be utilized as a high-efficient delayed fluorescent luminescent material, and an electronic device, for example, an organic light-emitting device, including the heterocyclic compound may have low driving voltage, excellent luminance efficiency, and a long lifespan.

In one or more embodiments, the heterocyclic compound may emit blue light having a maximum emission wavelength of about 400 nm or more and about 465 nm or less, and may have a difference between a singlet ( $S_1$ ) energy level and a triplet ( $T_1$ ) energy level of 0.5 eV or less.

Synthesis methods of the heterocyclic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art by referring to Synthesis Examples and/or Examples provided below.

At least one heterocyclic compound represented by Formula 1 may be used in a light-emitting device (for example, an organic light-emitting device). Accordingly, provided is a light-emitting device including: a first electrode; a second electrode facing the first electrode; an interlayer between the first electrode and the second electrode and including an emission layer; and a heterocyclic compound represented by Formula 1 as described herein.

In an embodiment,

the first electrode of the light-emitting device may be an anode,

the second electrode of the light-emitting device may be a cathode,

the interlayer may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode,

34

the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and

the electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

In one or more embodiments, the heterocyclic compound may be included between the first electrode of the light-emitting device and the second electrode of the light-emitting device. In one or more embodiments, the heterocyclic compound may be included in the interlayer of the light-emitting device, for example, the emission layer or the electron transport region of the interlayer.

In one or more embodiments, the emission layer in the interlayer of the light-emitting device may include a dopant and a host, and the host may include the heterocyclic compound. In other words, the heterocyclic compound may act as a host. The emission layer may emit red light, green light, blue light, and/or white light. In one or more embodiments, the emission layer may emit blue light. The blue light may have a maximum emission wavelength of, for example, about 400 nm to about 465 nm. Additional details of the emission layer and the host are the same as described elsewhere in the present specification.

In one or more embodiments, the light-emitting device may include a capping layer outside the first electrode or outside the second electrode.

In one or more embodiments, the light-emitting device may further include at least one selected from a first capping layer outside the first electrode and a second capping layer outside the second electrode, and at least one selected from the first capping layer and the second capping layer may include the heterocyclic compound represented by Formula 1. Further detailed description of the first capping layer and the second capping layer are the same as described elsewhere in the present specification.

In an embodiment, the light-emitting device may include: a first capping layer outside the first electrode and including the heterocyclic compound represented by Formula 1; a second capping layer outside the second electrode and including the heterocyclic compound represented by Formula 1; or the first capping layer and the second capping layer.

The expression “(an interlayer and/or a capping layer) includes at least one heterocyclic compound,” as used herein, may include a case in which “(an interlayer and/or a capping layer) includes identical heterocyclic compounds represented by Formula 1” and a case in which “(an interlayer and/or a capping layer) includes two or more different heterocyclic compounds represented by Formula 1.”

In an embodiment, the interlayer and/or capping layer may include Compound 1 only as the heterocyclic compound. In this regard, Compound 1 may exist in the emission layer of the light-emitting device. In one or more embodiments, the interlayer may include, as the heterocyclic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 may all exist in an emission layer), or different layers (for example, Compound 1 may exist in an emission layer and Compound 2 may exist in an electron transport region).

The term “interlayer,” as used herein, refers to a single layer and/or all layers between the first electrode and the second electrode of the light-emitting device.

35

According to another aspect of an embodiment, provided is an electronic apparatus including the light-emitting device. The electronic apparatus may further include a thin-film transistor. In an embodiment, the electronic apparatus may further include a thin-film transistor including a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically coupled to the source electrode or the drain electrode. In one or more embodiments, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof. Further detailed description of the electronic apparatus is the same as described elsewhere in the present specification.

Description of FIG. 1

FIG. 1 is a schematic cross-sectional view of a light-emitting device 10 according to an embodiment. The light-emitting device 10 includes a first electrode 110, an interlayer 130, and a second electrode 150.

Hereinafter, a structure of the light-emitting device 10 according to an embodiment and a method of manufacturing the light-emitting device 10 will be described in connection with FIG. 1.

First Electrode 110

In FIG. 1, a substrate may be additionally under the first electrode 110 or above the second electrode 150. The substrate may be a glass substrate and/or a plastic substrate. In one or more embodiments, the substrate may be a flexible substrate. In one or more embodiments, the substrate may include plastics having excellent heat resistance and durability, such as polyimide, polyethylene terephthalate (PET), polycarbonate, polyethylene naphthalate, polyarylate (PAR), polyetherimide, or any combination thereof.

The first electrode 110 may be formed by, for example, depositing and/or sputtering a material for forming the first electrode 110 on the substrate. When the first electrode 110 is an anode, a high work function material that can easily inject holes may be used as a material for forming the first electrode 110.

The first electrode 110 may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode 110 is a transmissive electrode, a material for forming the first electrode 110 may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO<sub>2</sub>), zinc oxide (ZnO), or any combination thereof. In one or more embodiments, when the first electrode 110 is a semi-transmissive electrode or a reflective electrode, magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combination thereof may be used as a material for forming the first electrode 110.

The first electrode 110 may have a single-layered structure including (e.g., consisting of) a single layer or a multi-layered structure including a plurality of layers. In an embodiment, the first electrode 110 may have a three-layered structure of ITO/Ag/ITO.

Interlayer 130

The interlayer 130 is on the first electrode 110. The interlayer 130 includes an emission layer.

The interlayer 130 may further include a hole transport region between the first electrode 110 and the emission layer and an electron transport region between the emission layer and the second electrode 150.

The interlayer 130 may further include metal-containing compounds such as organometallic compounds, inorganic materials such as quantum dots, and/or the like, in addition to various suitable organic materials.

36

In one or more embodiments, the interlayer 130 may include, i) two or more emitting units sequentially stacked between the first electrode 110 and the second electrode 150 and ii) a charge generation layer between the two emitting units. When the interlayer 130 includes the emitting units and the charge generation layer, the light-emitting device 10 may be a tandem light-emitting device.

Hole Transport Region in Interlayer 130

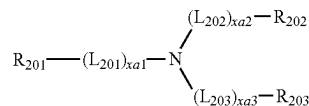
The hole transport region may have: i) a single-layered structure including (e.g., consisting of) a single layer including (e.g., consisting of) a single material, ii) a single-layered structure including (e.g., consisting of) a single layer including a plurality of different materials, or iii) a multi-layered structure including a plurality of layers including different materials.

The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof.

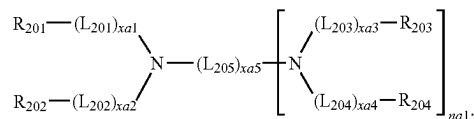
For example, the hole transport region may have a multi-layered structure including a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/emission auxiliary layer structure, wherein, in each structure, layers are stacked sequentially from the first electrode 110.

The hole transport region may include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof:

Formula 201



Formula 202



In Formulae 201 and 202,

L<sub>201</sub> to L<sub>204</sub> may each independently be a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub> or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

L<sub>205</sub> may be \*—O—\*, \*—S—\*, \*—N(Q<sub>201</sub>)—\*, a C<sub>1</sub>-C<sub>20</sub> alkylene group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>20</sub> alkenylene group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

xa1 to xa4 may each independently be an integer from 0 to 5,

xa5 may be an integer from 1 to 10,

R<sub>201</sub> to R<sub>204</sub> and Q<sub>201</sub> may each independently be a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub> or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

R<sub>201</sub> and R<sub>202</sub> may optionally be linked to each other, via a single bond, a C<sub>1</sub>-C<sub>5</sub> alkylene group unsubstituted or substituted with at least one R<sub>10a</sub>, or a C<sub>2</sub>-C<sub>5</sub> alkenylene

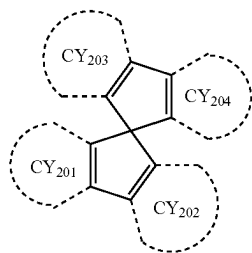
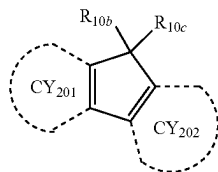
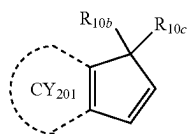
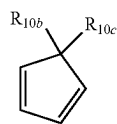
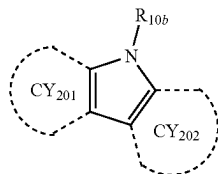
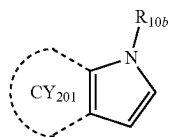
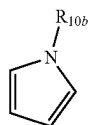
37

group unsubstituted or substituted with at least one  $R_{10a}$ , to form a  $C_8$ - $C_{60}$  polycyclic group unsubstituted or substituted with at least one  $R_{10a}$  (for example, a carbazole group or the like) (for example, refer to the following Compound HT16),

$R_{203}$  and  $R_{204}$  may optionally be linked to each other, via a single bond, a  $C_1$ - $C_5$  alkylene group unsubstituted or substituted with at least one  $R_{10a}$ , or a  $C_2$ - $C_5$  alkenylene group unsubstituted or substituted with at least one  $R_{10a}$ , to form a  $C_8$ - $C_{60}$  polycyclic group unsubstituted or substituted with at least one  $R_{10a}$ , and

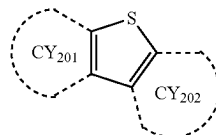
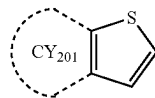
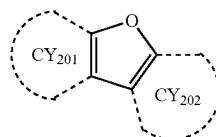
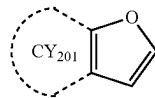
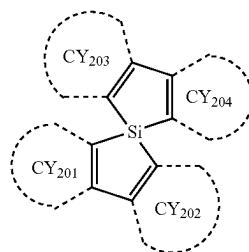
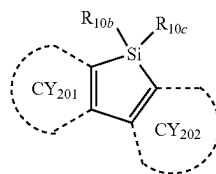
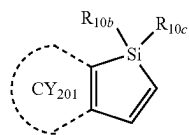
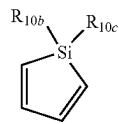
$n_1$  may be an integer from 1 to 4.

In an embodiment, Formulae 201 and 202 may each include at least one selected from groups represented by Formulae CY201 to CY217:



38

-continued



CY208

CY209

CY210

CY211

CY212

CY213

CY214

CY215

CY216

CY217



39

Regarding Formulae CY201 to CY217,  $R_{10b}$  and  $R_{10c}$  are the same as described in connection with  $R_{10a}$ , ring CY<sub>201</sub> to ring CY<sub>204</sub> may each independently be a C<sub>3</sub>-C<sub>20</sub> carbocyclic group or a C<sub>1</sub>-C<sub>20</sub> heterocyclic group, and at least one hydrogen in Formula CY201 to CY217 may be unsubstituted or substituted with at least one  $R_{10a}$  described herein.

In an embodiment, ring CY<sub>201</sub> to ring CY<sub>204</sub> in Formulae CY201 to CY217 may each independently be a benzene group, a naphthalene group, a phenanthrene group, or an anthracene group.

In an embodiment, Formulae 201 and 202 may each include at least one selected from groups represented by Formulae CY201 to CY203.

In one or more embodiments, Formula 201 may include at least one selected from groups represented by Formulae CY201 to CY203 and at least one selected from groups represented by Formulae CY204 to CY217.

In one or more embodiments, in Formula 201,  $xa_1$  may be 1,  $R_{201}$  may be a group represented by one selected from Formulae CY201 to CY203,  $xa_2$  may be 0, and  $R_{202}$  may be a group represented by one selected from Formulae CY204 to CY207.

40

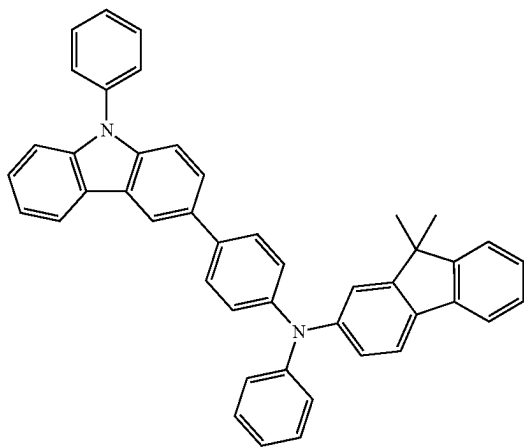
In one or more embodiments, each of Formulae 201 and 202 may not include a group represented by one selected from Formulae CY201 to CY203.

In one or more embodiments, each of Formulae 201 and 202 may not include a group represented by one selected from Formulae CY201 to CY203 and may include at least one selected from groups represented by Formulae CY204 to CY217.

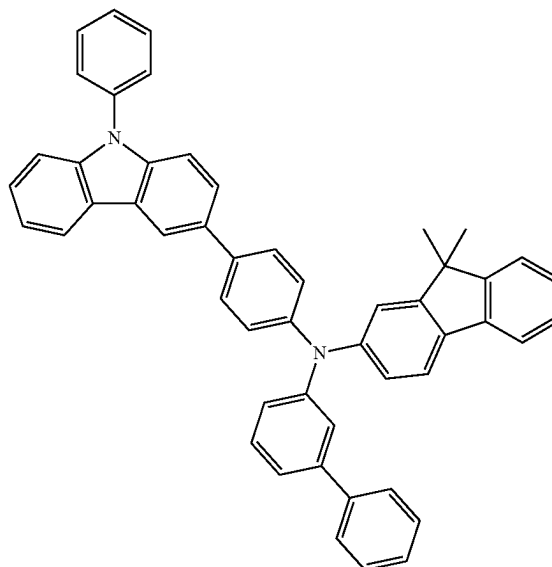
In an embodiment, each of Formulae 201 and 202 may not include a group represented by one selected from Formulae CY201 to CY217.

In an embodiment, the hole transport region may include one selected from Compounds HT1 to HT46, m-MTDATA, TDATA, 2-TNATA, NPB(NPD),  $\beta$ -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTDP, 4,4',4''-tris (N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), or any combination thereof:

HT1

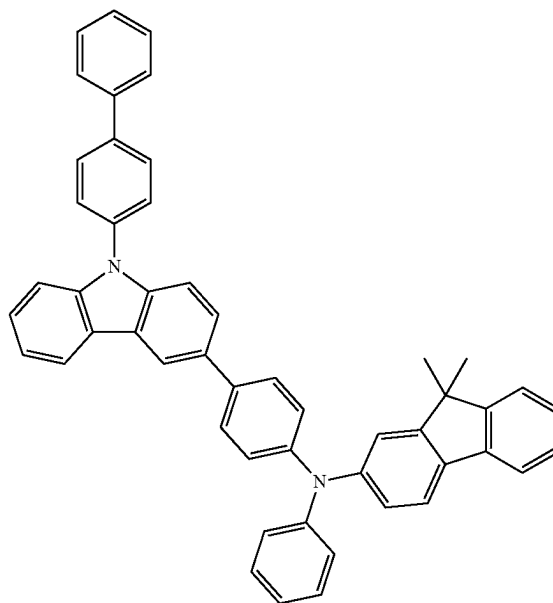
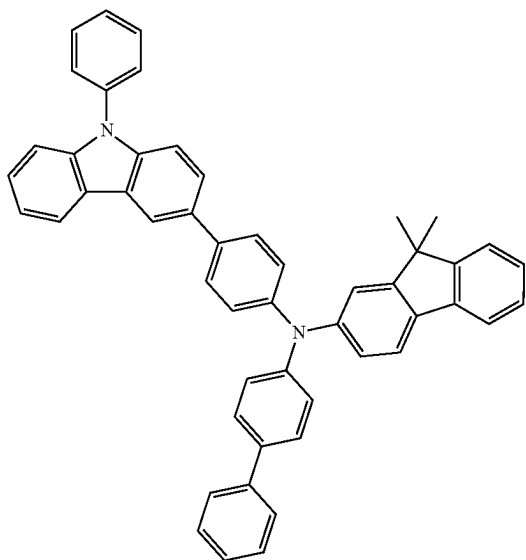


HT2



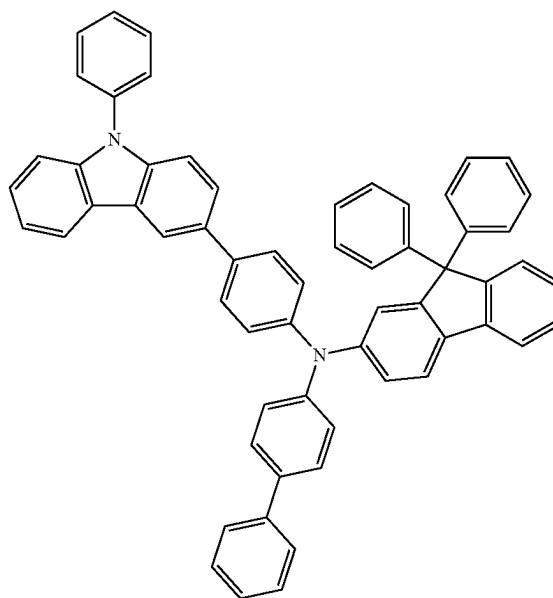
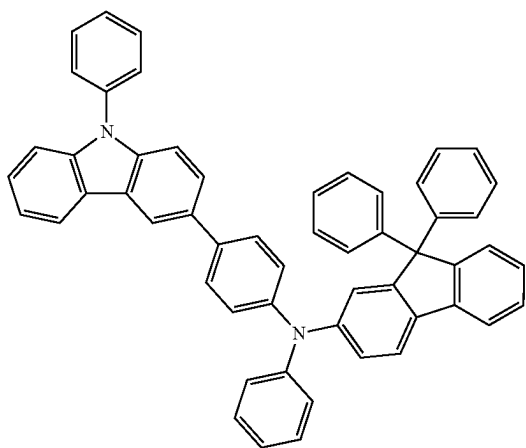
HT3

HT4

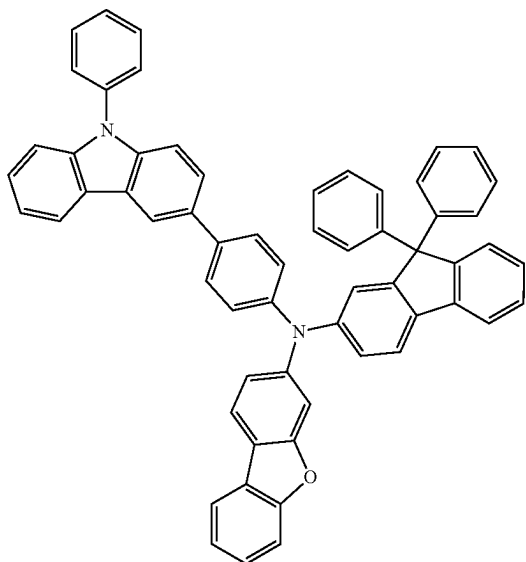


HT5

HT6

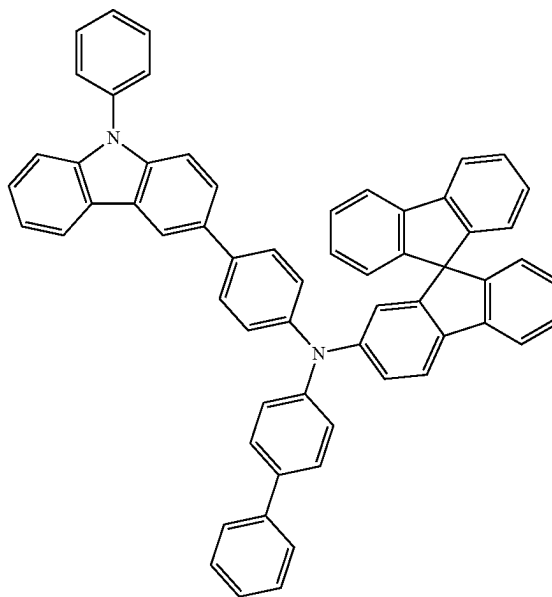


43

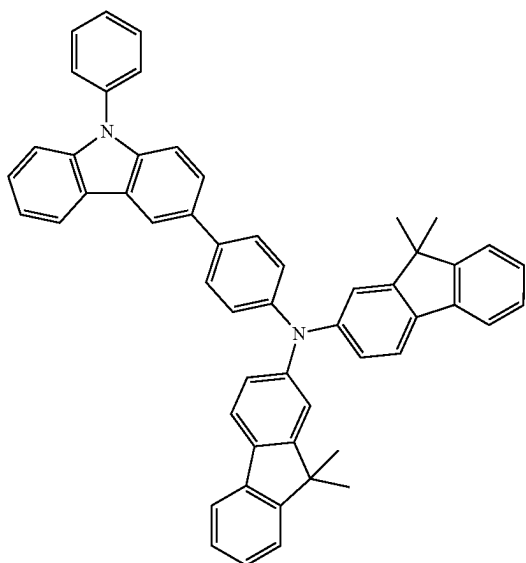
-continued  
HT7

44

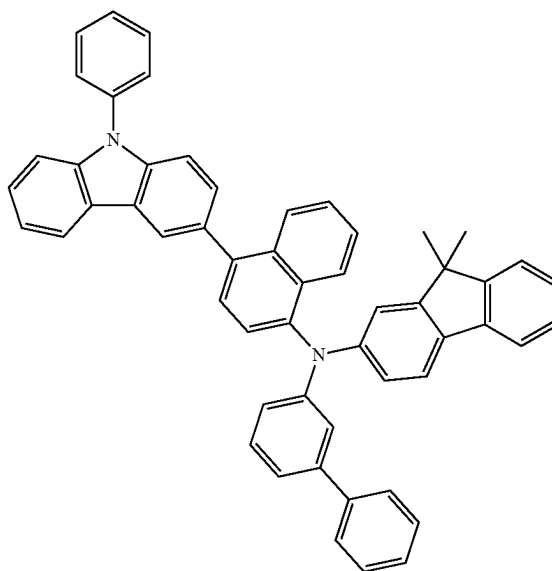
HT8



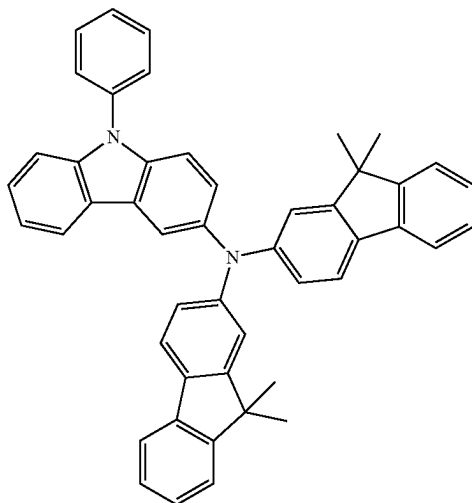
HT9



HT10

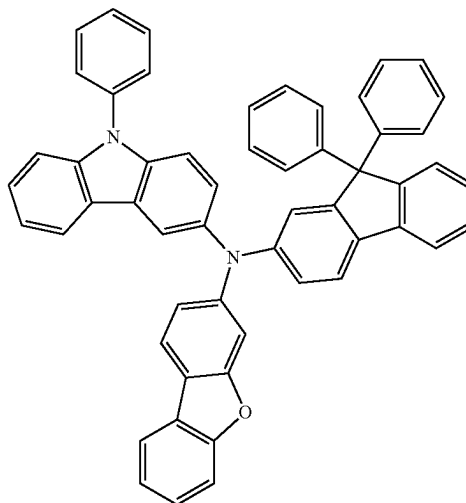


45

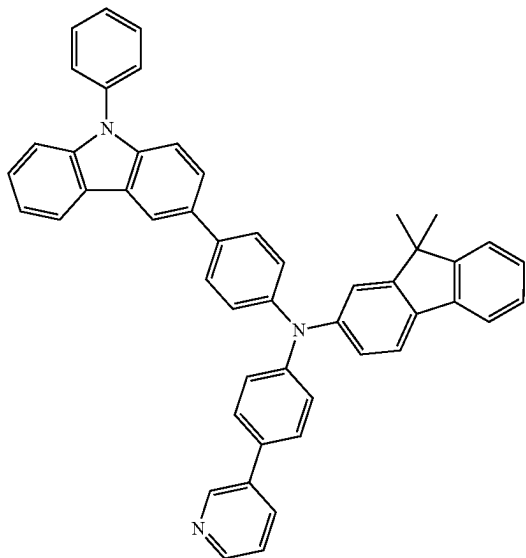
-continued  
HT11

46

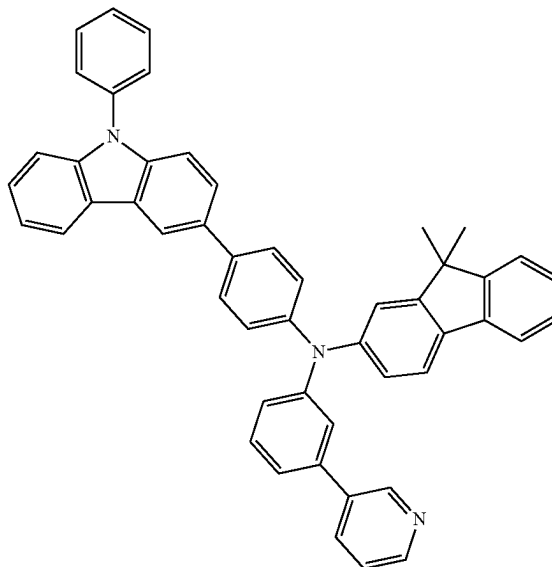
HT12



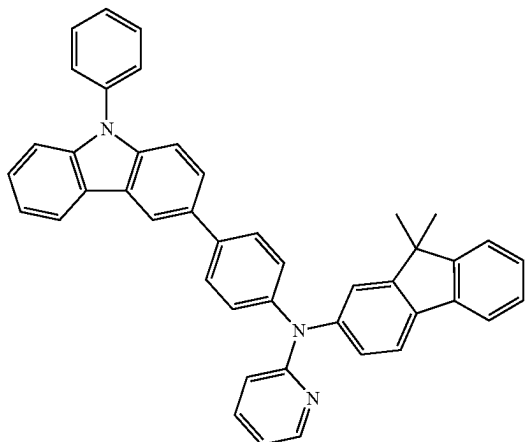
HT13



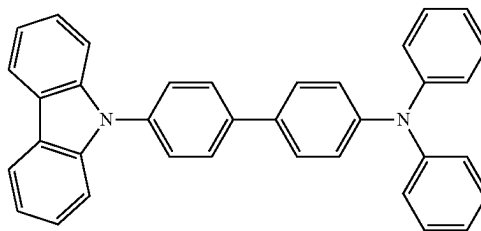
HT14



HT15



HT16

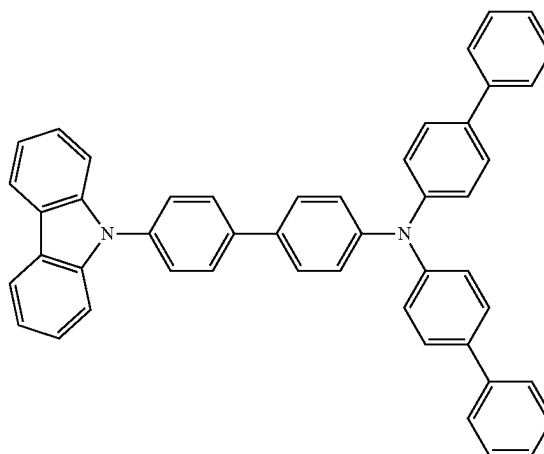
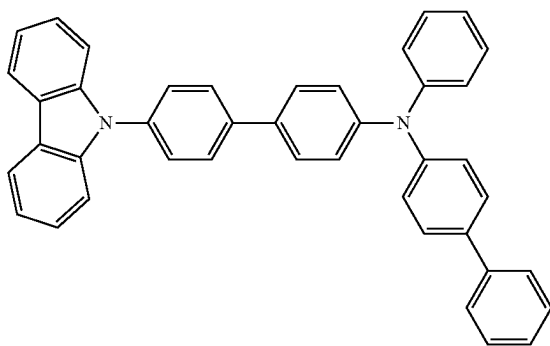


47

48

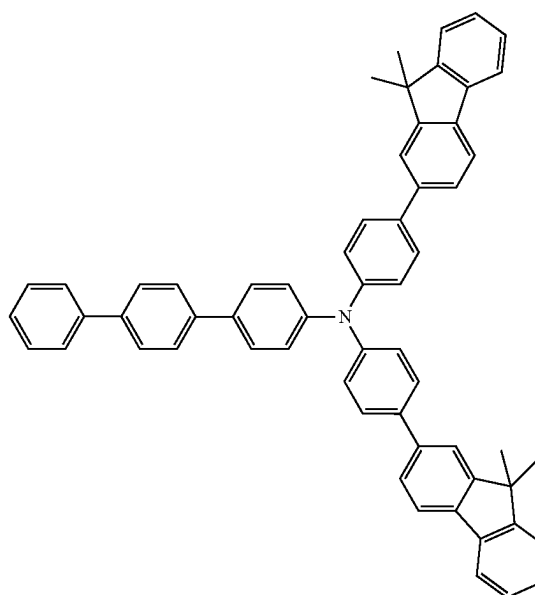
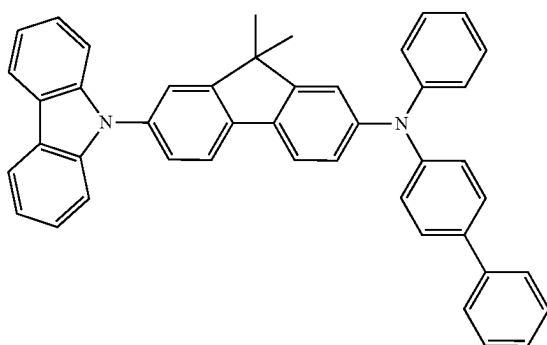
-continued  
HT17

HT18



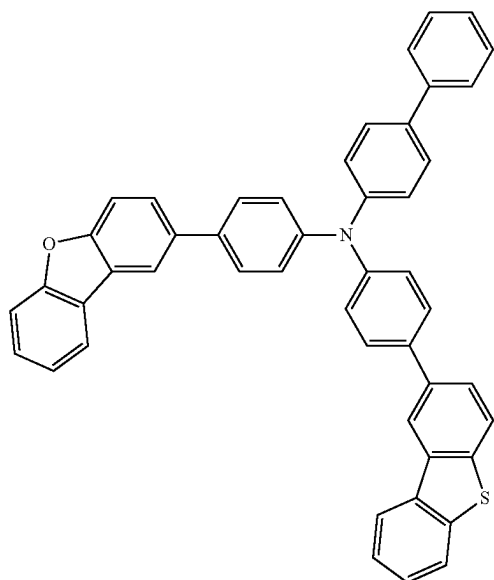
HT19

HT20



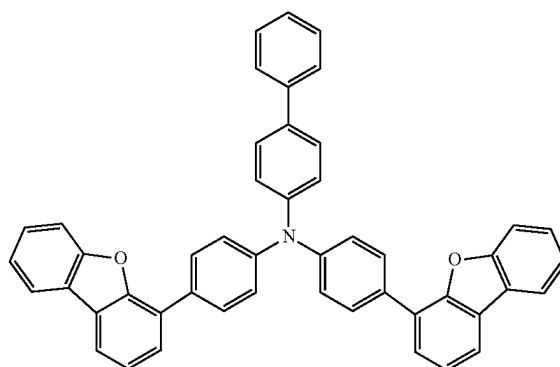
49

-continued  
HT21

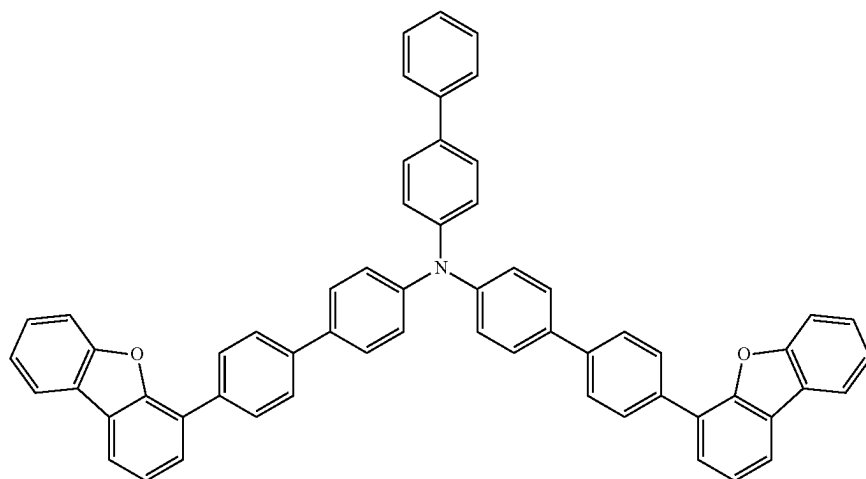


50

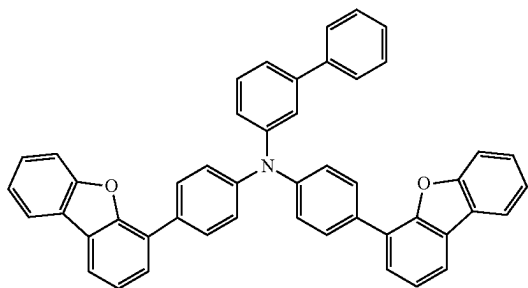
HT22



HT23

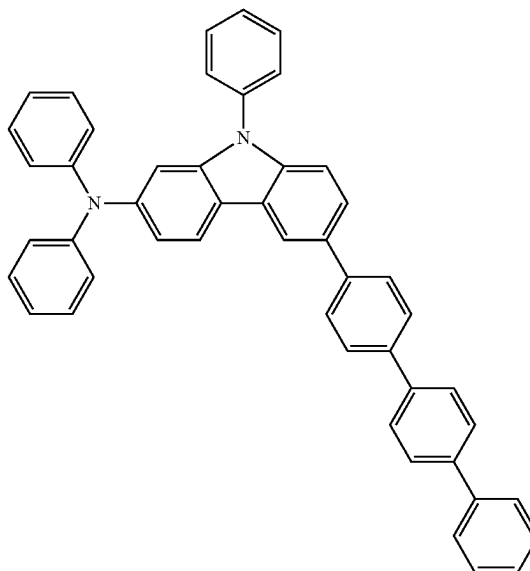


51

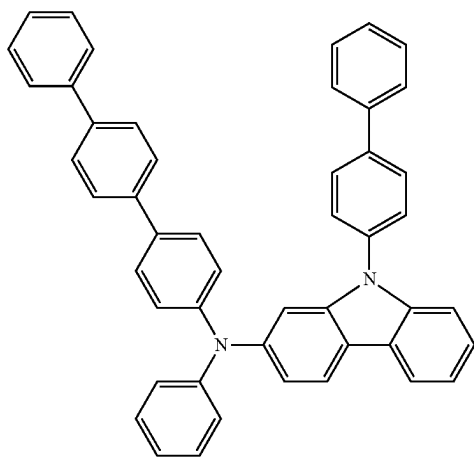
-continued  
HT24

52

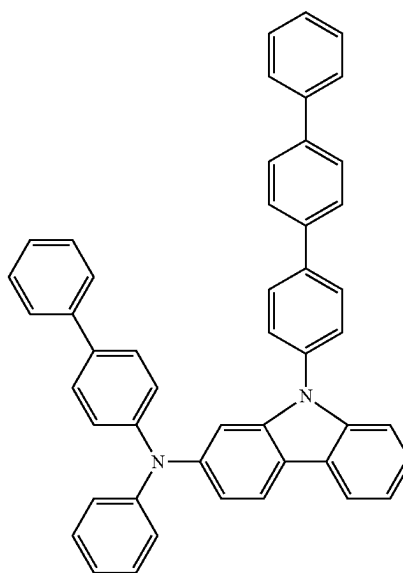
HT25



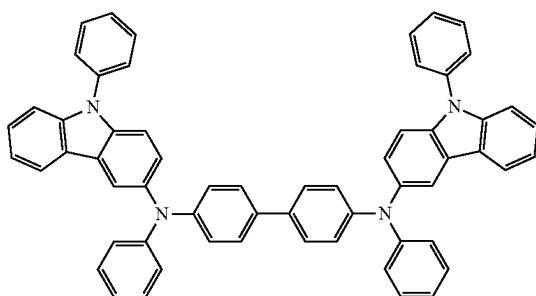
HT26



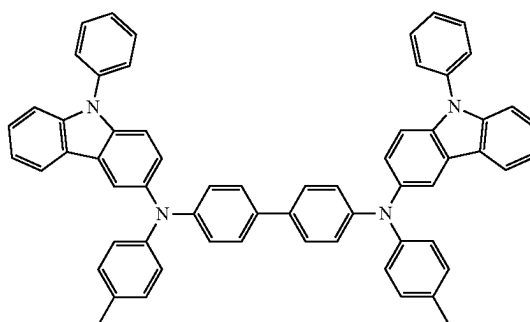
HT27



HT28



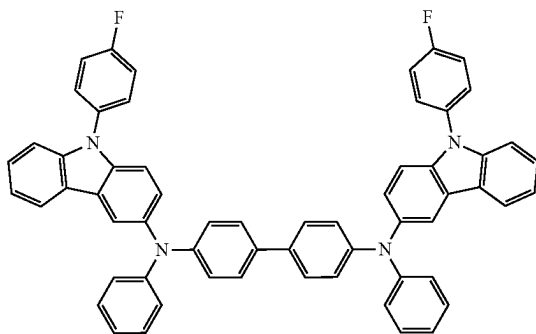
HT29



53

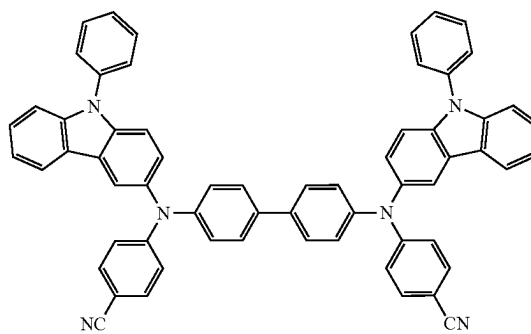
-continued

HT30

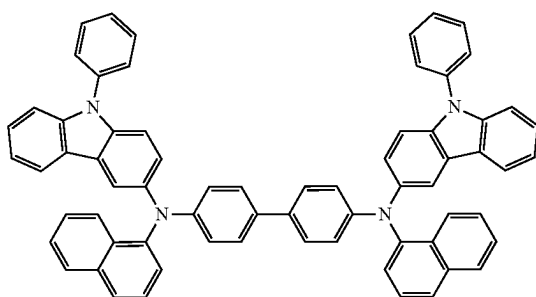


54

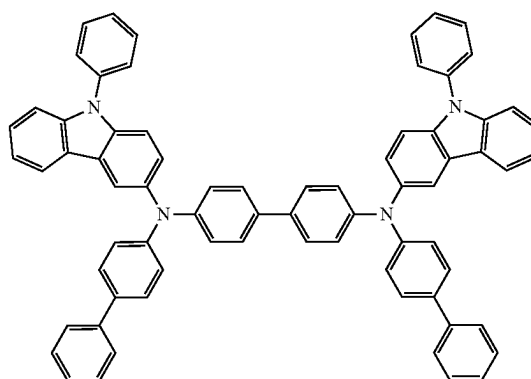
HT31



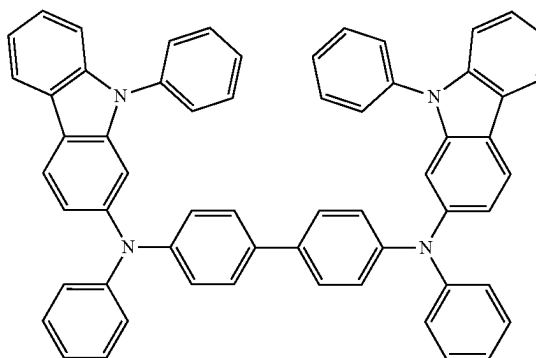
HT32



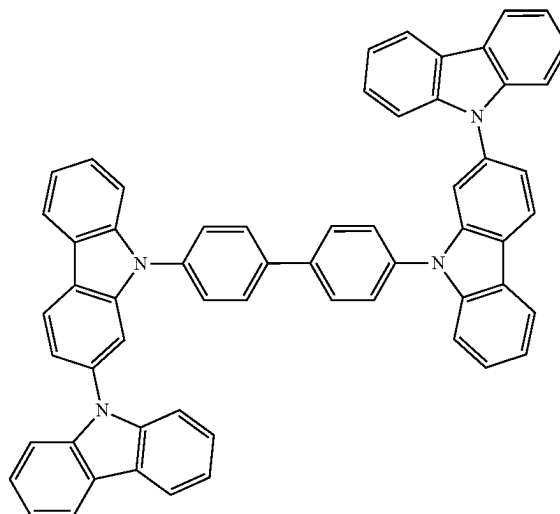
HT33



HT34



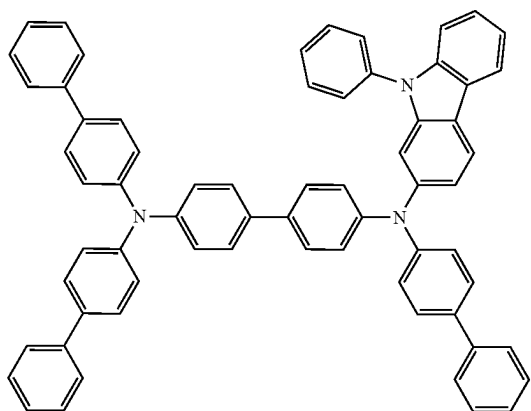
HT35





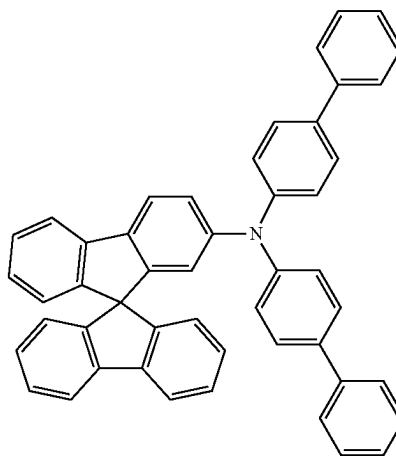
55

-continued  
HT36



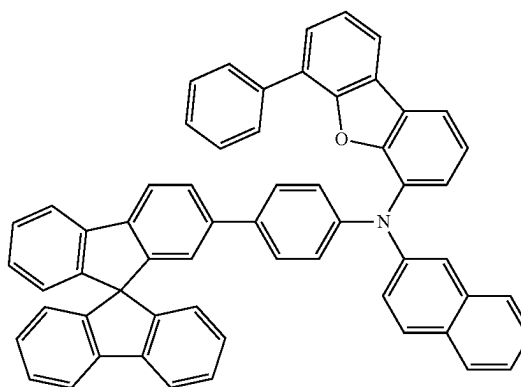
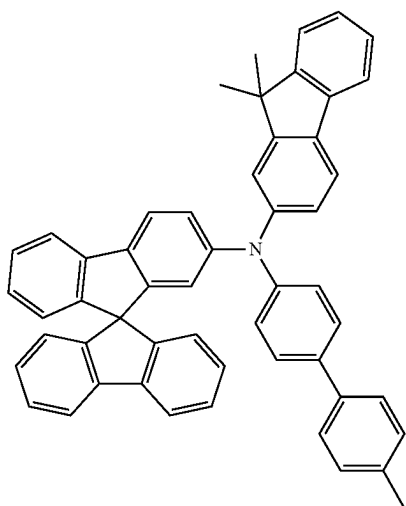
56

HT37

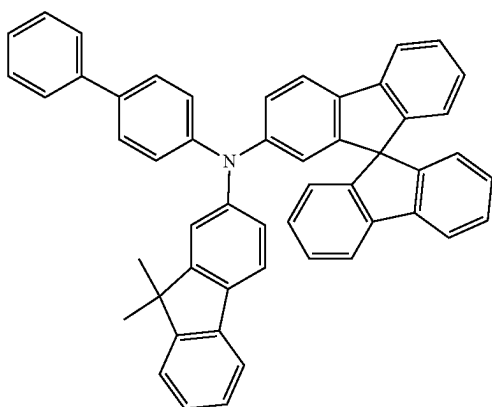


HT38

HT39



HT40

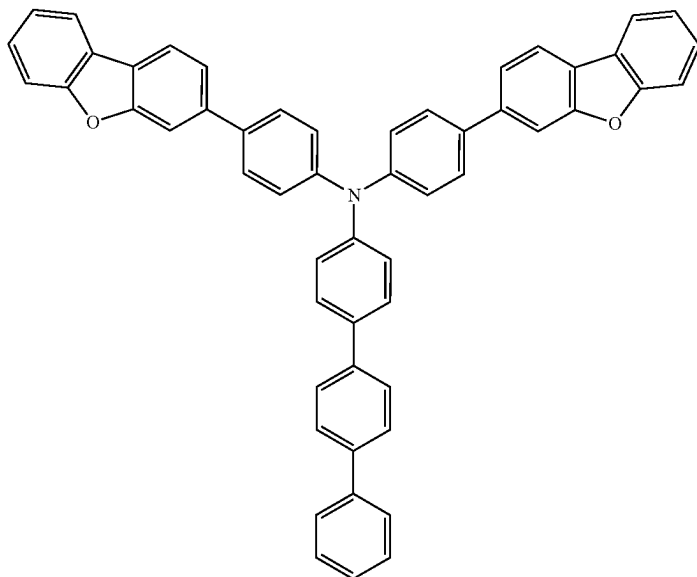


57

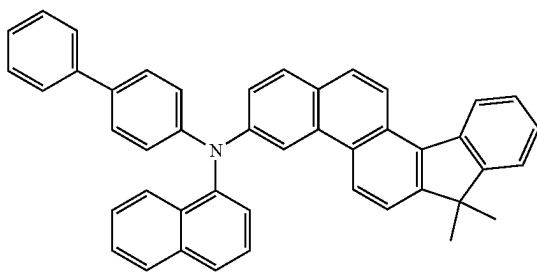
58

-continued

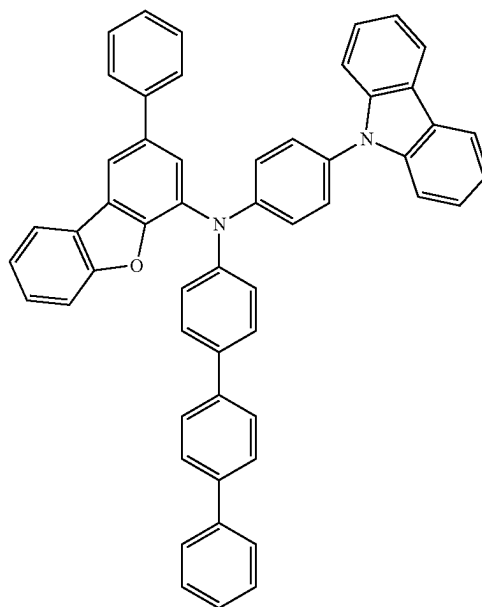
HT41



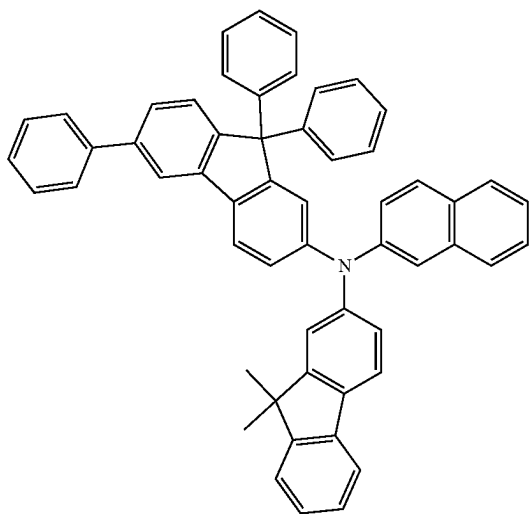
HT42



HT43

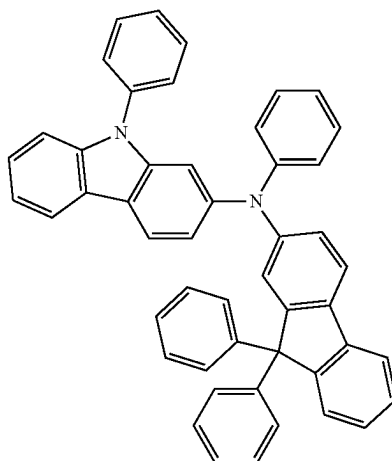


59

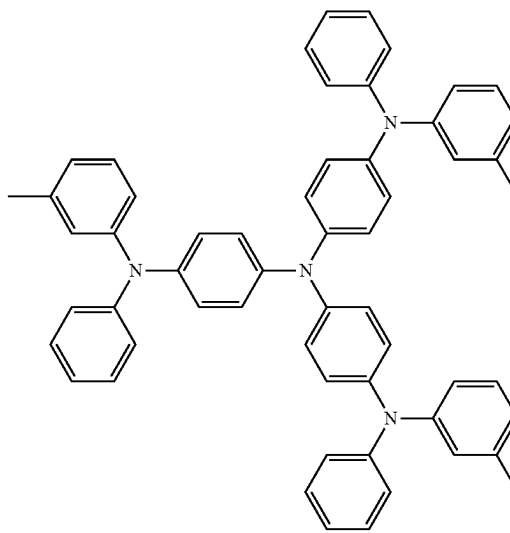
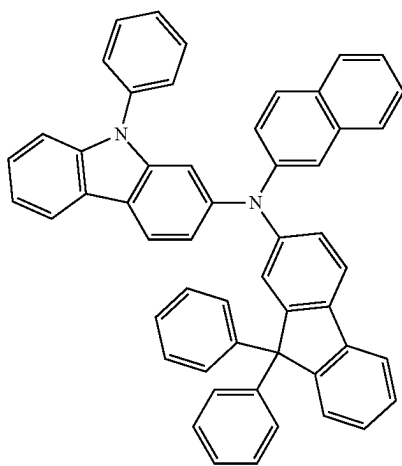
-continued  
HT44

60

HT45



HT46

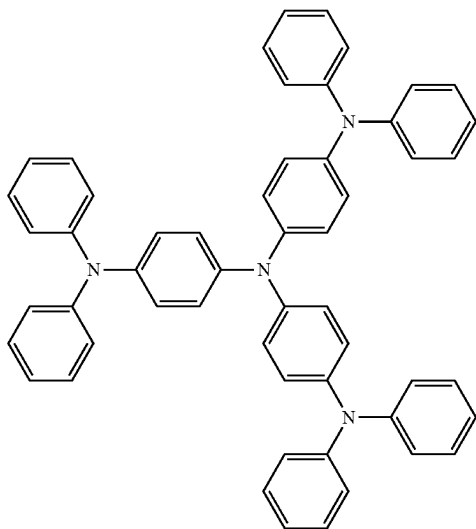


m-MTDATA

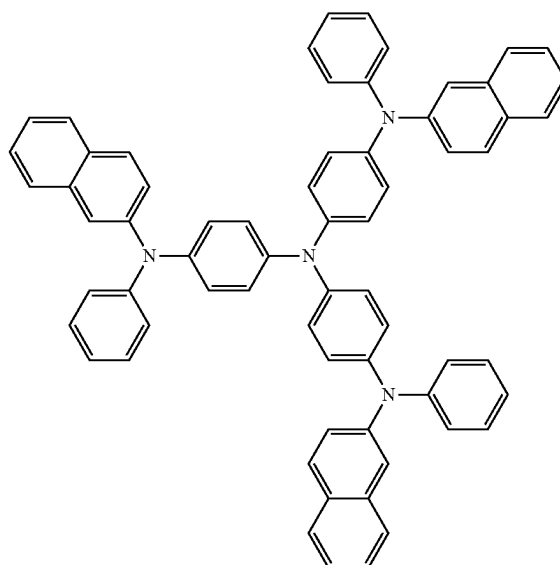
61

-continued

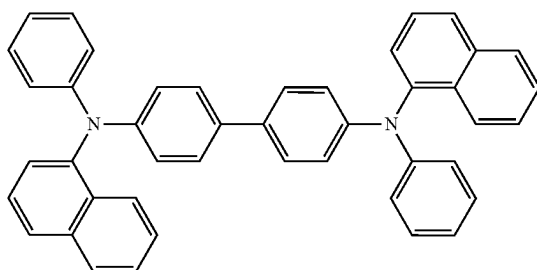
62



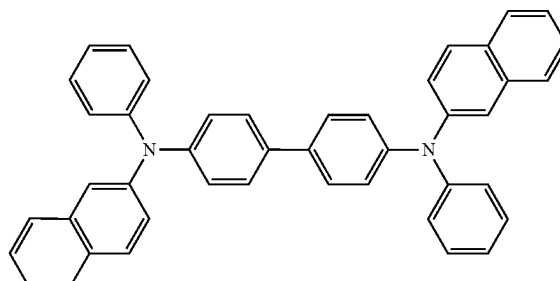
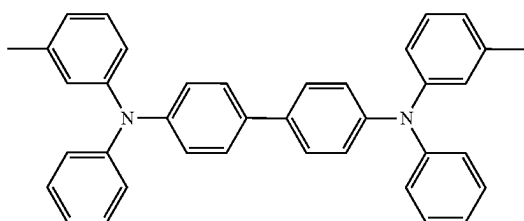
TDATA



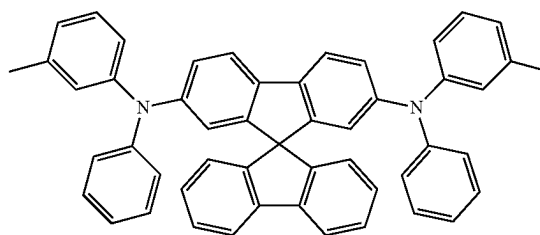
2-TNATA



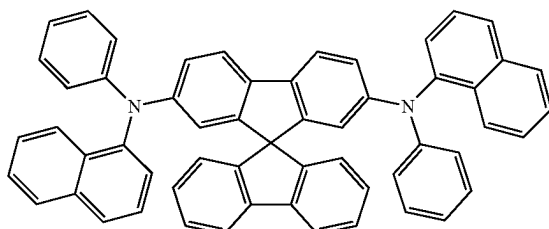
NPB

 $\beta$ -NPB

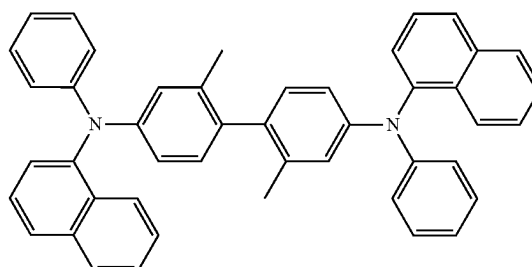
TPD



Spiro-TPD

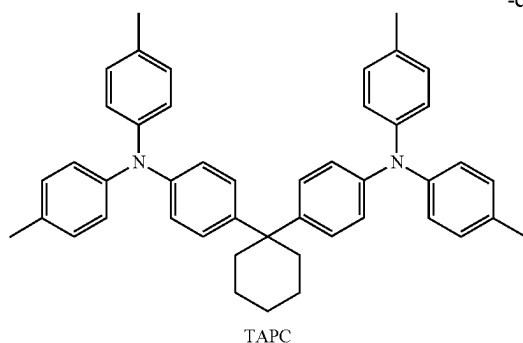


Spiro-NPD



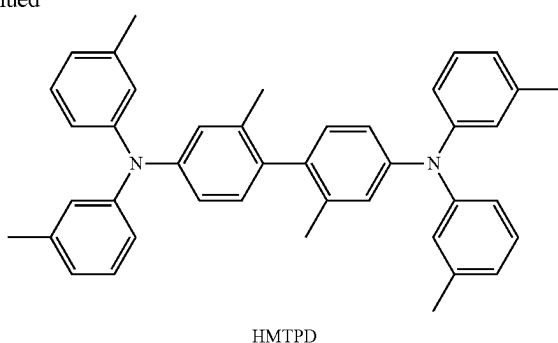
methylated-NPB

63



-continued

64



A thickness of the hole transport region may be in a range of about 50 Å to about 10,000 Å, for example, about 100 Å to about 4,000 Å. When the hole transport region includes a hole injection layer, a hole transport layer, or any combination thereof, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, for example, about 100 Å to about 1,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example, about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, suitable or satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may prevent or reduce leakage of electrons from an emission layer to a hole transport region. Materials that may be included in the hole transport region may be included in the emission auxiliary layer and the electron blocking layer.

#### p-Dopant

The hole transport region may further include, in addition to these materials, a charge-generation material for improvement of conductive properties. The charge-generation material may be uniformly or non-uniformly dispersed in the hole transport region (for example, in the form of a single layer including (e.g., consisting of) a charge-generation material).

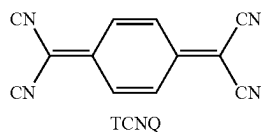
The charge-generation material may be, for example, a p-dopant.

In an embodiment, the lowest unoccupied molecular orbital (LUMO) energy level of the p-dopant may be  $-3.5$  eV or less.

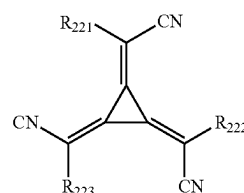
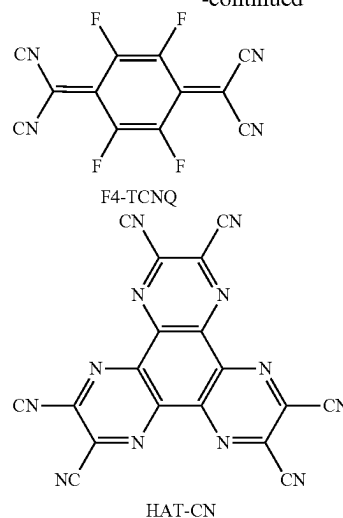
In an embodiment, the p-dopant may include a quinone derivative, a cyano group-containing compound, a compound containing element EL1 and element EL2, or any combination thereof.

Examples of the quinone derivative may include TCNQ, F4-TCNQ, etc.

Examples of the cyano group-containing compound may include HAT-CN and a compound represented by Formula 221 below.



-continued



Formula 221

In Formula 221,

$R_{221}$  to  $R_{223}$  may each independently be a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$  or a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ , and

at least one selected from  $R_{221}$  to  $R_{223}$  may each independently be a  $C_3$ - $C_{60}$  carbocyclic group or a  $C_1$ - $C_{60}$  heterocyclic group, each substituted with: a cyano group;  $-F$ ;  $-Cl$ ;  $-Br$ ;  $-I$ ; a  $C_1$ - $C_{20}$  alkyl group substituted with a cyano group,  $-F$ ,  $-Cl$ ,  $-Br$ ,  $-I$ , or any combination thereof; or any combination thereof.

Regarding the compound containing element EL1 and element EL2, element EL1 may be metal, metalloid, or a combination thereof, and element EL2 may be a non-metal, metalloid, or a combination thereof.

Examples of the metal may include: an alkali metal (for example, lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), etc.); an alkaline earth metal (for example, beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), etc.); a transition metal

(for example, titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), manganese (Mn), technetium (Tc), rhenium (Re), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), etc.); a post-transition metal (for example, zinc (Zn), indium (In), tin (Sn), etc.); and a lanthanide metal (for example, lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), etc.).

Examples of the metalloid may include silicon (Si), antimony (Sb), and tellurium (Te).

Examples of the non-metal may include oxygen (O) and halogen (for example, F, Cl, Br, I, etc.).

In an embodiment, examples of the compound containing element EL1 and element EL2 may include metal oxide, metal halide (for example, metal fluoride, metal chloride, metal bromide, and/or metal iodide), metalloid halide (for example, metalloid fluoride, metalloid chloride, metalloid bromide, and/or metalloid iodide), metal telluride, or any combination thereof.

Examples of the metal oxide may include tungsten oxide (for example, WO, W<sub>2</sub>O<sub>3</sub>, WO<sub>2</sub>, WO<sub>3</sub>, and/or W<sub>2</sub>O<sub>5</sub>), vanadium oxide (for example, VO, V<sub>2</sub>O<sub>3</sub>, VO<sub>2</sub>, and/or V<sub>2</sub>O<sub>5</sub>), molybdenum oxide (MoO, Mo<sub>2</sub>O<sub>3</sub>, MoO<sub>2</sub>, MoO<sub>3</sub>, and/or Mo<sub>2</sub>O<sub>5</sub>), and rhenium oxide (for example, ReO<sub>3</sub>).

Examples of the metal halide may include alkali metal halide, alkaline earth metal halide, transition metal halide, post-transition metal halide, and lanthanide metal halide.

Examples of the alkali metal halide may include LiF, NaF, KF, RbF, CsF, LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, RbBr, CsBr, LiI, NaI, KI, RbI, and CsI.

Examples of the alkaline earth metal halide may include BeF<sub>2</sub>, MgF<sub>2</sub>, CaF<sub>2</sub>, SrF<sub>2</sub>, BaF<sub>2</sub>, BeCl<sub>2</sub>, MgCl<sub>2</sub>, CaCl<sub>2</sub>, SrCl<sub>2</sub>, BaCl<sub>2</sub>, BeBr<sub>2</sub>, MgBr<sub>2</sub>, CaBr<sub>2</sub>, SrBr<sub>2</sub>, BaBr<sub>2</sub>, BeI<sub>2</sub>, MgI<sub>2</sub>, CaI<sub>2</sub>, SrI<sub>2</sub>, and BaI<sub>2</sub>.

Examples of the transition metal halide may include titanium halide (for example, TiF<sub>4</sub>, TiCl<sub>4</sub>, TiBr<sub>4</sub>, and/or TiI<sub>4</sub>), zirconium halide (for example, ZrF<sub>4</sub>, ZrCl<sub>4</sub>, ZrBr<sub>4</sub>, and/or ZrI<sub>4</sub>), hafnium halide (for example, HfF<sub>4</sub>, HfCl<sub>4</sub>, HfBr<sub>4</sub>, and/or HfI<sub>4</sub>), vanadium halide (for example, VF<sub>3</sub>, VCl<sub>3</sub>, VBr<sub>3</sub>, and/or VI<sub>3</sub>), niobium halide (for example, NbF<sub>3</sub>, NbCl<sub>3</sub>, NbBr<sub>3</sub>, and/or NbI<sub>3</sub>), tantalum halide (for example, TaF<sub>3</sub>, TaCl<sub>3</sub>, TaBr<sub>3</sub>, and/or TaI<sub>3</sub>), chromium halide (for example, CrF<sub>3</sub>, CrCl<sub>3</sub>, CrBr<sub>3</sub>, and/or CrI<sub>3</sub>), molybdenum halide (for example, MoF<sub>3</sub>, MoCl<sub>3</sub>, MoBr<sub>3</sub>, and/or MoI<sub>3</sub>), tungsten halide (for example, WF<sub>3</sub>, WCl<sub>3</sub>, WBr<sub>3</sub>, and/or WI<sub>3</sub>), manganese halide (for example, MnF<sub>2</sub>, MnCl<sub>2</sub>, MnBr<sub>2</sub>, and/or MnI<sub>2</sub>), technetium halide (for example, TcF<sub>2</sub>, TcCl<sub>2</sub>, TcBr<sub>2</sub>, and/or TcI<sub>2</sub>), rhenium halide (for example, ReF<sub>2</sub>, ReCl<sub>2</sub>, ReBr<sub>2</sub>, and/or ReI<sub>2</sub>), iron halide (for example, FeF<sub>2</sub>, FeCl<sub>2</sub>, FeBr<sub>2</sub>, and/or FeI<sub>2</sub>), ruthenium halide (for example, RuF<sub>2</sub>, RuCl<sub>2</sub>, RuBr<sub>2</sub>, and/or RuI<sub>2</sub>), osmium halide (for example, OsF<sub>2</sub>, OsCl<sub>2</sub>, OsBr<sub>2</sub>, and/or OsI<sub>2</sub>), cobalt halide (for example, CoF<sub>2</sub>, CoCl<sub>2</sub>, CoBr<sub>2</sub>, and/or CoI<sub>2</sub>), rhodium halide (for example, RhF<sub>2</sub>, RhCl<sub>2</sub>, RhBr<sub>2</sub>, and/or RhI<sub>2</sub>), iridium halide (for example, IrF<sub>2</sub>, IrCl<sub>2</sub>, IrBr<sub>2</sub>, and/or IrI<sub>2</sub>), nickel halide (for example, NiF<sub>2</sub>, NiCl<sub>2</sub>, NiBr<sub>2</sub>, and/or NiI<sub>2</sub>), palladium halide (for example, PdF<sub>2</sub>, PdCl<sub>2</sub>, PdBr<sub>2</sub>, and/or PdI<sub>2</sub>), platinum halide (for example, PtF<sub>2</sub>, PtCl<sub>2</sub>, PtBr<sub>2</sub>, and/or PtI<sub>2</sub>), copper halide (for example, CuF, CuCl, CuBr, and/or CuI), silver halide (for example, AgF, AgCl, AgBr, and/or AgI), and gold halide (for example, AuF, AuCl, AuBr, and/or AuI).

Examples of the post-transition metal halide may include zinc halide (for example, ZnF<sub>2</sub>, ZnCl<sub>2</sub>, ZnBr<sub>2</sub>, and/or ZnI<sub>2</sub>), indium halide (for example, InI<sub>3</sub>), and tin halide (for example, SnI<sub>2</sub>).

Examples of the lanthanide metal halide may include YbF, YbF<sub>2</sub>, YbF<sub>3</sub>, SmF<sub>3</sub>, YbCl, YbCl<sub>2</sub>, YbCl<sub>3</sub>, SmCl<sub>3</sub>, YbBr, YbBr<sub>2</sub>, YbBr<sub>3</sub>, SmBr<sub>3</sub>, YbI, YbI<sub>2</sub>, YbI<sub>3</sub>, and SmI<sub>3</sub>.

Examples of the metalloid halide may include antimony halide (for example, SbCl<sub>5</sub>).

Examples of the metal telluride may include alkali metal telluride (for example, Li<sub>2</sub>Te, Na<sub>2</sub>Te, K<sub>2</sub>Te, Rb<sub>2</sub>Te, and/or Cs<sub>2</sub>Te), alkaline earth metal telluride (for example, BeTe, MgTe, CaTe, SrTe, and/or BaTe), transition metal telluride (for example, TiTe<sub>2</sub>, ZrTe<sub>2</sub>, HfTe<sub>2</sub>, V<sub>2</sub>Te<sub>3</sub>, Nb<sub>2</sub>Te<sub>3</sub>, Ta<sub>2</sub>Te<sub>3</sub>, Cr<sub>2</sub>Te<sub>3</sub>, Mo<sub>2</sub>Te<sub>3</sub>, W<sub>2</sub>Te<sub>3</sub>, MnTe, TcTe, ReTe, FeTe, RuTe, OsTe, CoTe, RhTe, IrTe, NiTe, PdTe, PtTe, Cu<sub>2</sub>Te, CuTe, Ag<sub>2</sub>Te, AgTe, and/or Au<sub>2</sub>Te), post-transition metal telluride (for example, ZnTe), and lanthanide metal telluride (for example, LaTe, CeTe, PrTe, NdTe, PmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, and/or LuTe).

#### Emission Layer in Interlayer 130

When the light-emitting device 10 is a full-color light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer, according to a sub-pixel. In one or more embodiments, the emission layer may have a stacked structure of two or more layers of a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other to emit white light. In one or more embodiments, the emission layer may include two or more materials of a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

The emission layer may include a host and a dopant. The dopant may include a phosphorescent dopant, a fluorescent dopant, or any combination thereof.

The amount of the dopant in the emission layer may be from about 0.01 to about 15 parts by weight based on 100 parts by weight of the host.

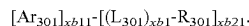
In one or more embodiments, the emission layer may include a quantum dot.

In one or more embodiments, the emission layer may include a delayed fluorescent material. The delayed fluorescent material may act as a host or a dopant in the emission layer.

A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

#### Host

The host may include the compound represented by Formula 1, a compound represented by Formula 301, or any combination thereof:



Formula 301

In Formula 301,

Ar<sub>301</sub> and L<sub>301</sub> may each independently be a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub> or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

xb11 may be 1, 2, or 3,

xb1 may be an integer from 0 to 5,

67

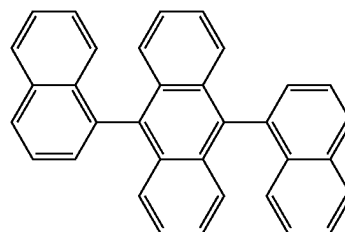
$R_{301}$  may be hydrogen, deuterium, —F, —Cl, —Br, —I, hydroxyl group, a cyano group, a nitro group, a  $C_1$ - $C_{60}$  alkyl group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_2$ - $C_{60}$  alkenyl group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_2$ - $C_{60}$  alkynyl group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_1$ - $C_{60}$  alkoxy group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ , —Si( $Q_{301}$ )( $Q_{302}$ )( $Q_{303}$ ), —N( $Q_{301}$ )( $Q_{302}$ ), —B( $Q_{301}$ )( $Q_{302}$ ), —C(=O)( $Q_{301}$ ), —S(=O)<sub>2</sub>( $Q_{301}$ ), or —P(=O)( $Q_{301}$ )( $Q_{302}$ ),  
 xb21 may be an integer from 1 to 5, and  
 $Q_{301}$  to  $Q_{303}$  are the same as described in connection with  $Q_1$ .

In one or more embodiments, when xb11 in Formula 301 is 2 or more, two or more of  $Ar_{301}(s)$  may be linked to each other via a single bond.

In an embodiment, the host may include a compound represented by Formula 301-1, a compound represented by Formula 301-2, or any combination thereof:

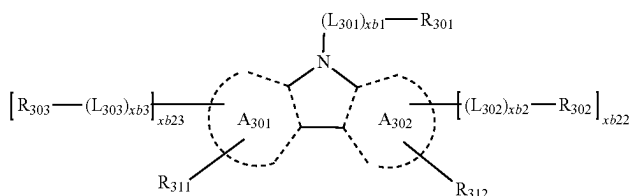
68

In an embodiment, the host may include one selected from Compounds H1 to H124, 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di-(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), or any combination thereof, but embodiments of the present disclosure are not limited thereto:

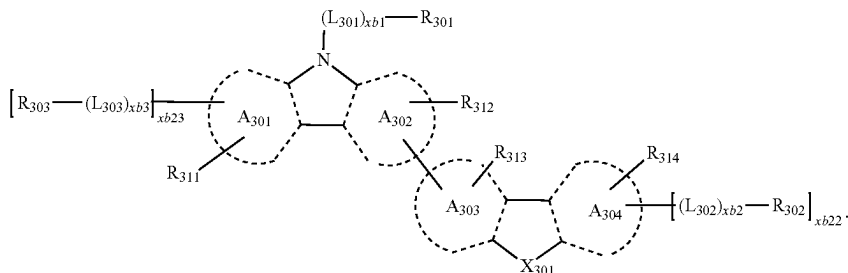


H1

Formula 301-1



Formula 301-2



45

In Formulae 301-1 and 301-2,

ring  $A_{301}$  to ring  $A_{304}$  may each independently be a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$  or a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ ,

$X_{301}$  may be O, S, N-[ $(L_{304})_{xb4}$ - $R_{304}$ ], C( $R_{304}$ )( $R_{305}$ ), or Si( $R_{304}$ )( $R_{305}$ ),

xb22 and xb23 may each independently be 0, 1, or 2,

$L_{301}$ , xb1, and  $R_{301}$  may each be the same as respectively described elsewhere in the present specification,

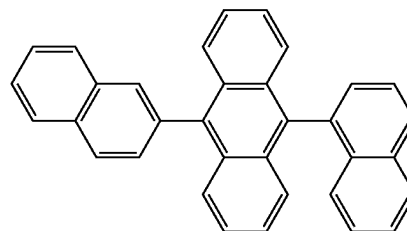
$L_{302}$  to  $L_{304}$  may each independently be the same as respectively described in connection with  $L_{301}$ ,

xb2 to xb4 may each independently be the same as respectively described in connection with xb1, and

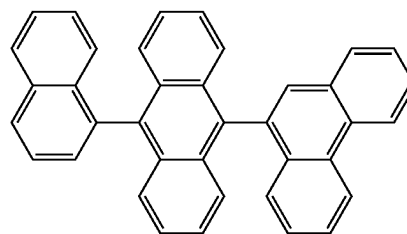
$R_{302}$  to  $R_{305}$  and  $R_{311}$  to  $R_{314}$  may each be the same as respectively described in connection with  $R_{301}$ .

In one or more embodiments, the host may include an alkaline earth-metal complex, a post-transition metal complex, or any combination thereof. In an embodiment, the host may include a Be complex (for example, Compound H55), a Mg complex, a Zn complex, or any combination thereof.

-continued



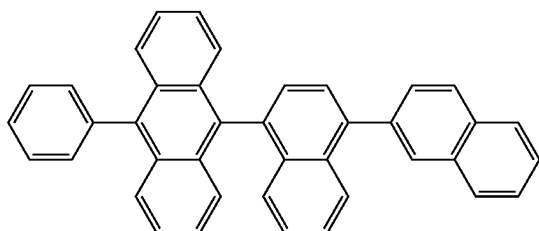
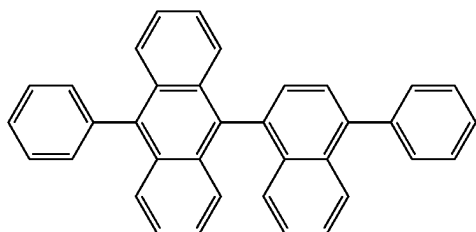
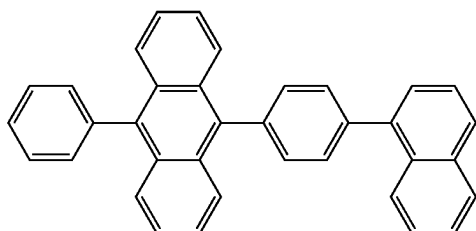
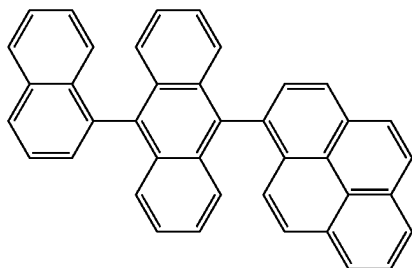
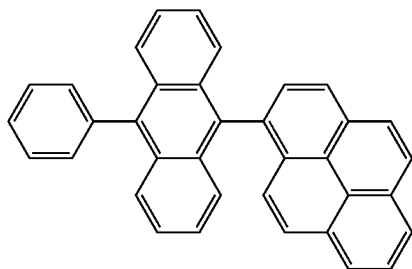
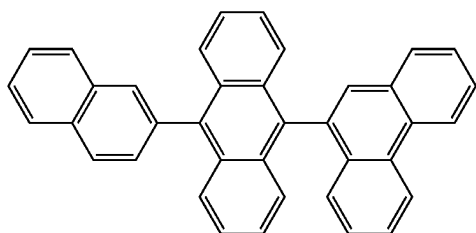
H2



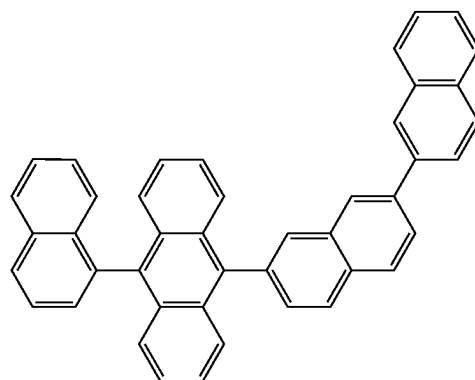
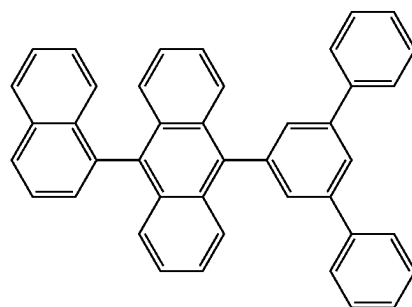
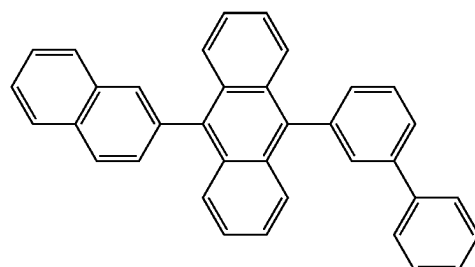
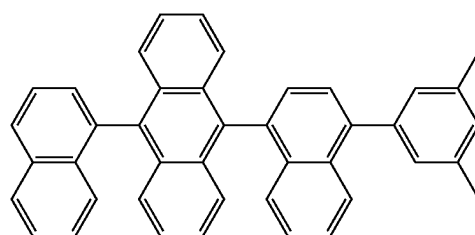
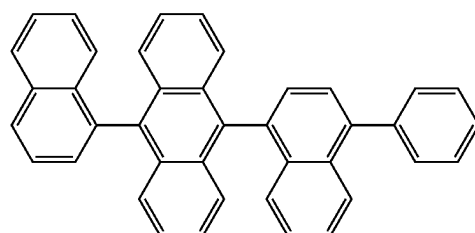
H3

**69**

-continued

**70**

-continued



H4

H10

5

10

H5

15

H11

20

H6

25

30

H12

H7

35

40

H13

H8

45

50

H14

H9

55

60

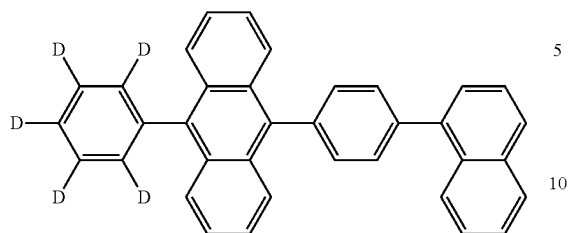
65



71

-continued

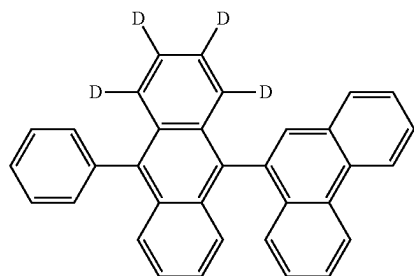
H15



5

10

H16 15

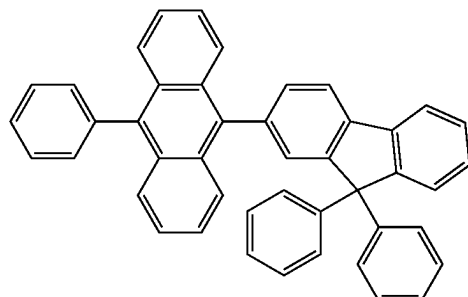


20

25

H17

30

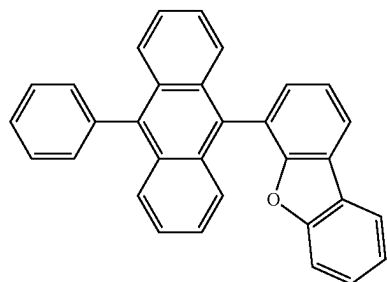


35

40

H18

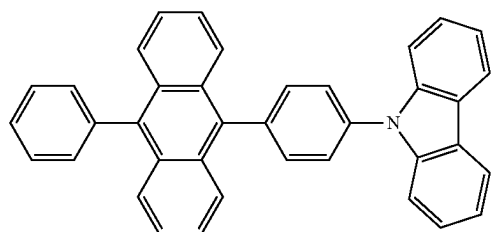
45



50

55

H19



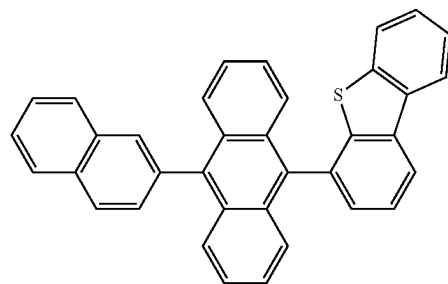
60

65

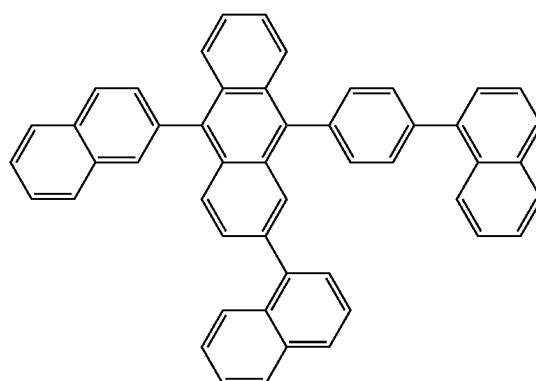
72

-continued

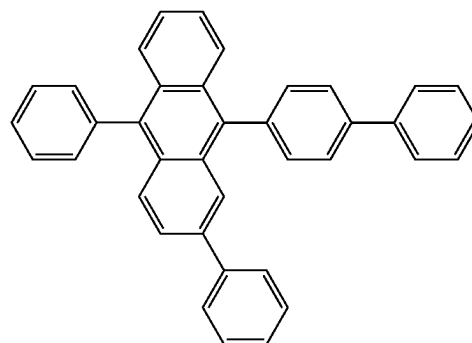
H20



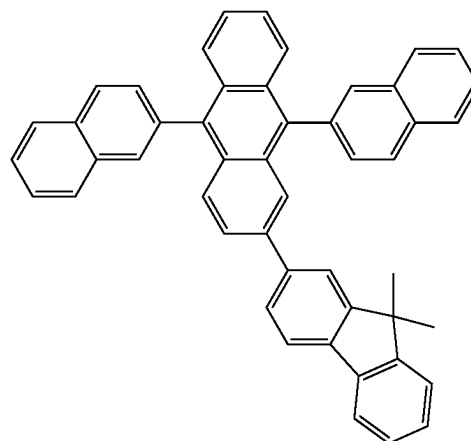
H21



H22

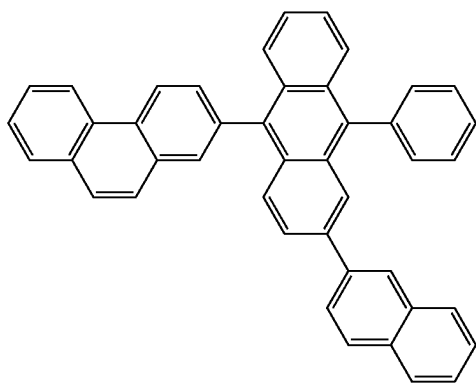


H23



**73**

-continued



H24

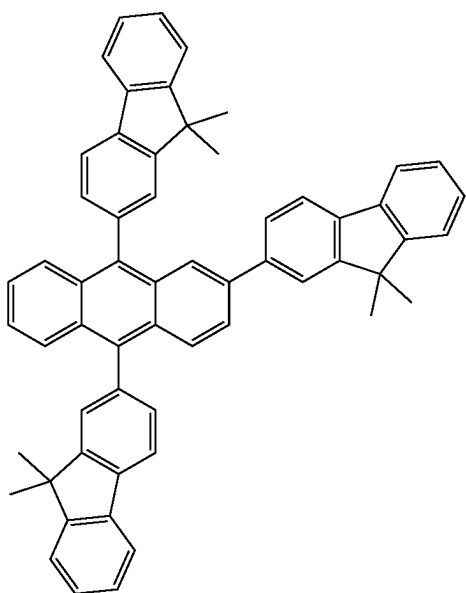
5

10

15

H25

20



25

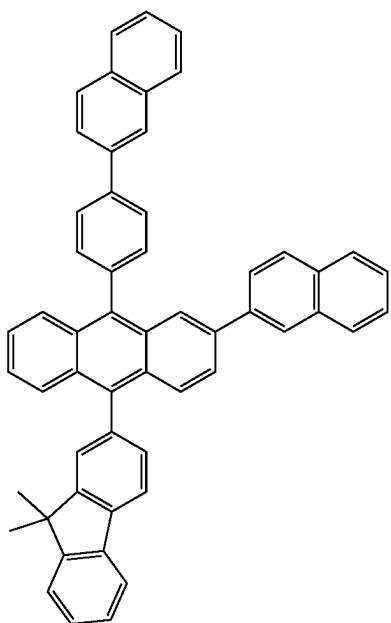
30

35

40

H26

45



50

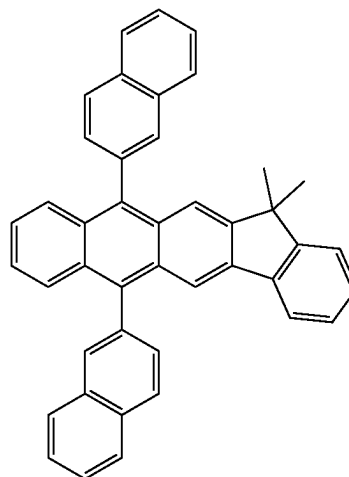
55

60

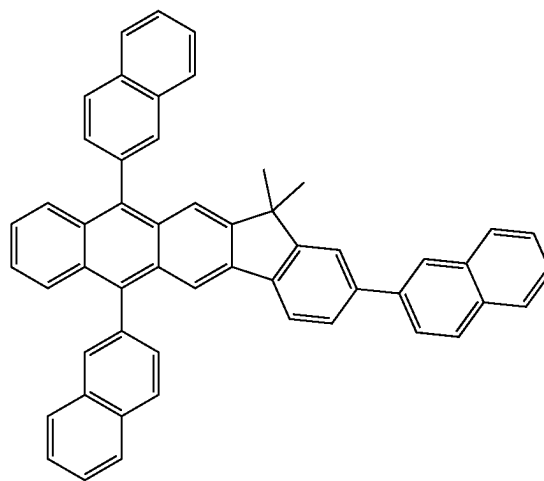
65

**74**

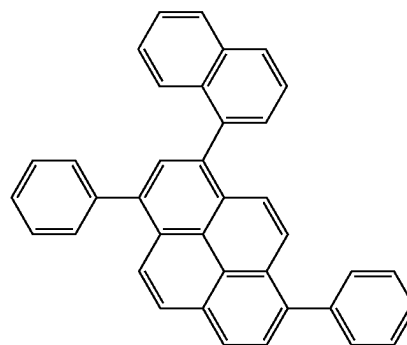
-continued



H27



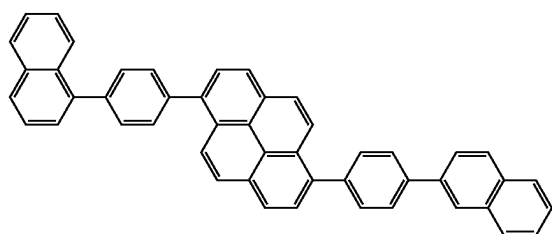
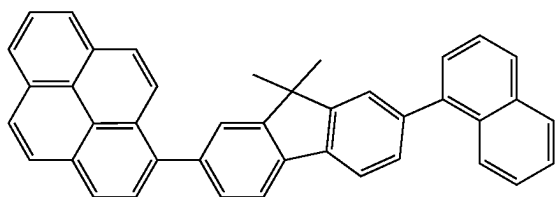
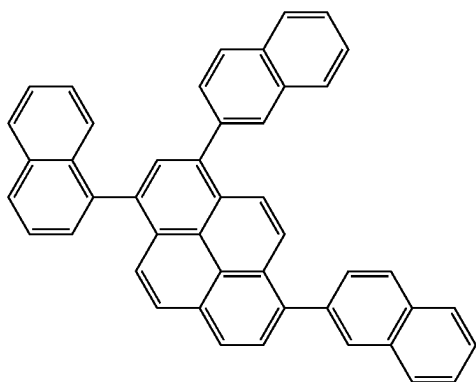
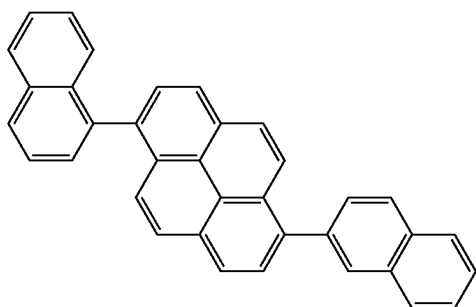
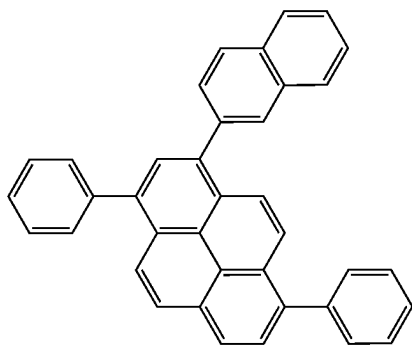
H28



H29

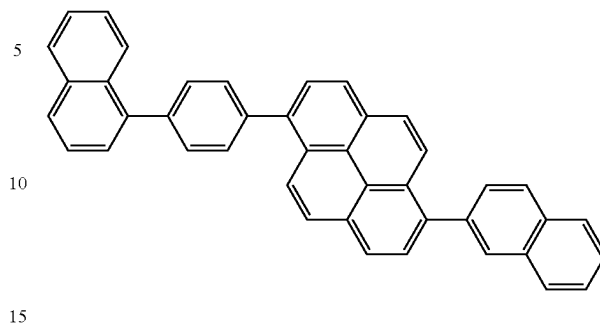
**75**

-continued

**76**

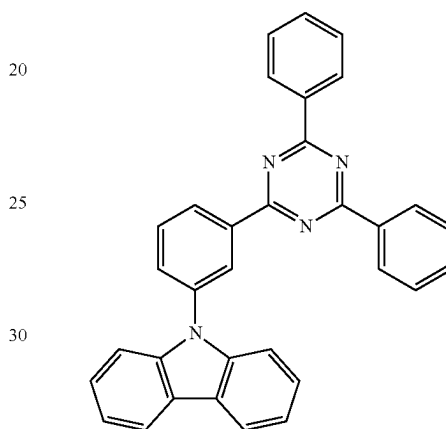
-continued

H30



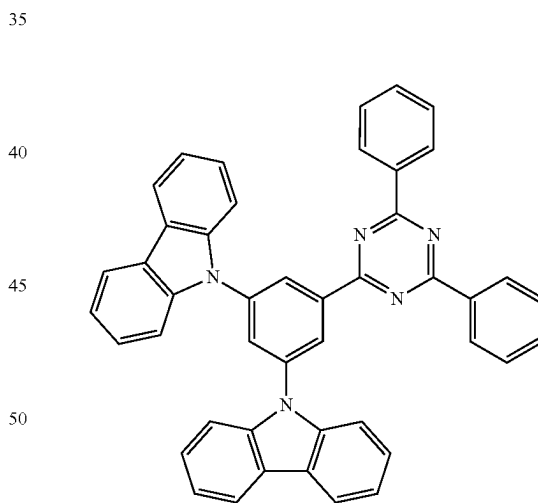
H35

H31



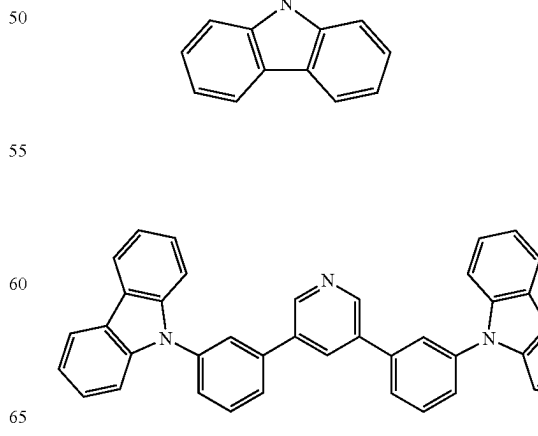
H36

H32



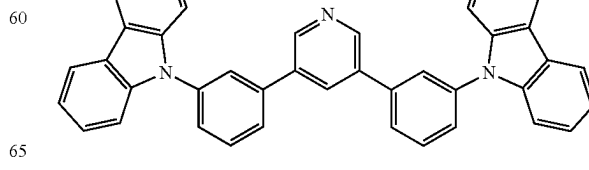
H37

H33



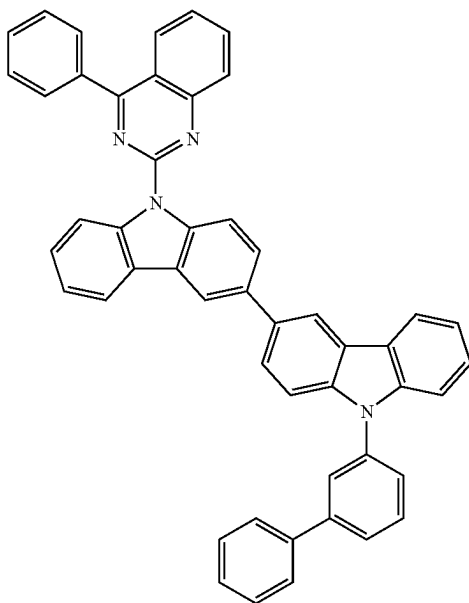
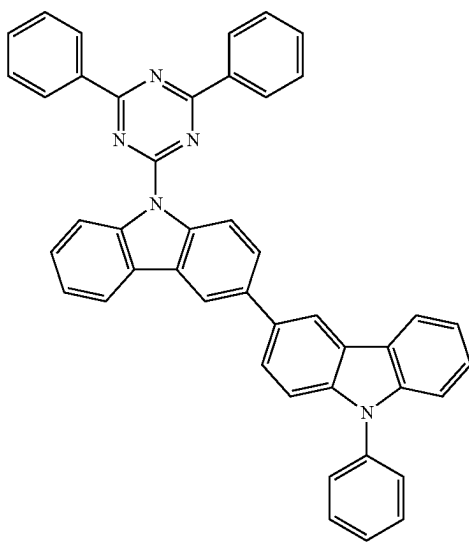
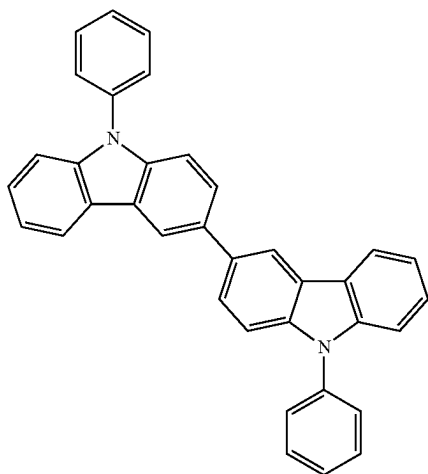
H38

H34



**77**

-continued

**78**

-continued

H39

5

10

15

20

H40

25

30

35

40

H41

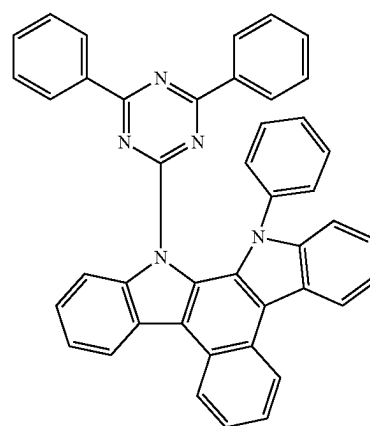
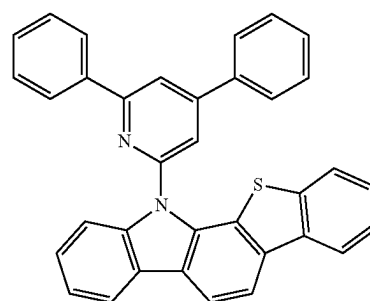
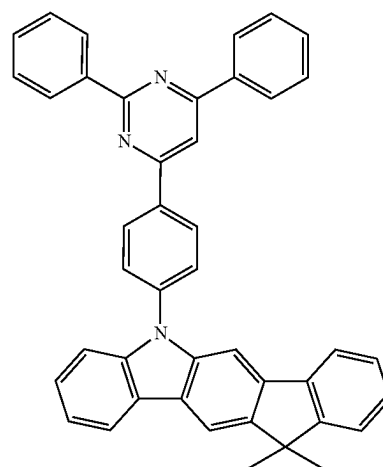
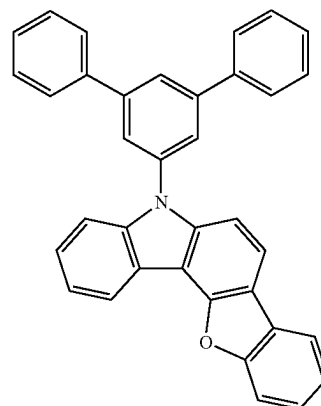
45

50

55

60

65



H42

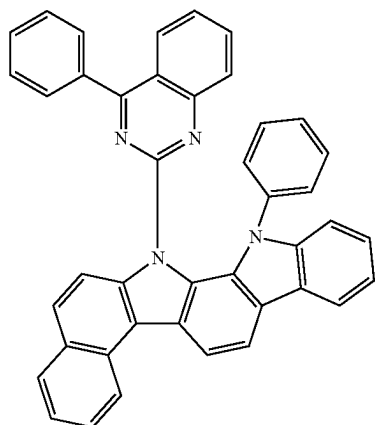
H43

H44

H45

**79**

-continued



H46

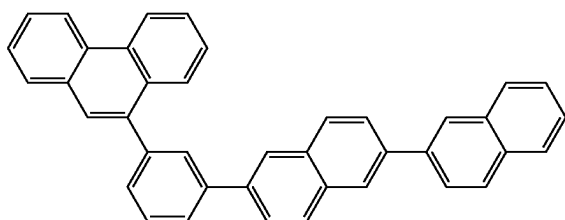
5

10

15

H47

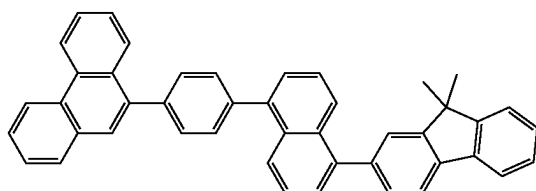
20



25

H48

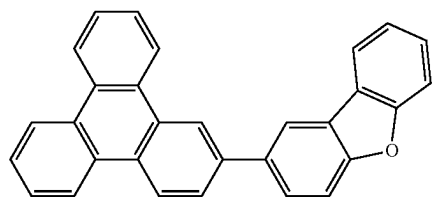
30



35

H49

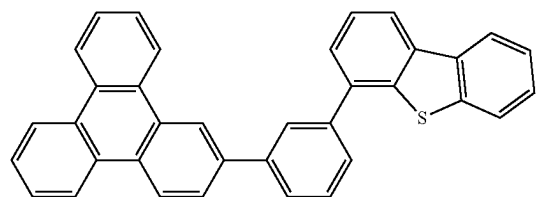
40



45

H50

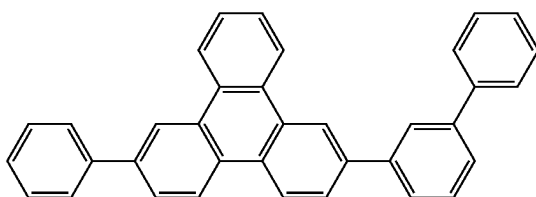
50



55

H51

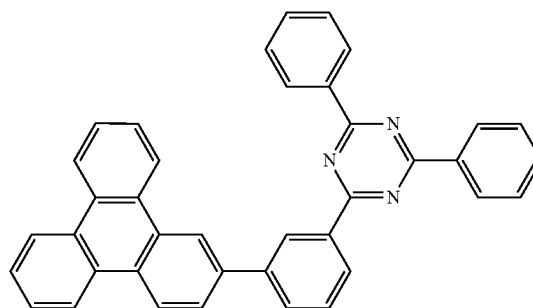
60



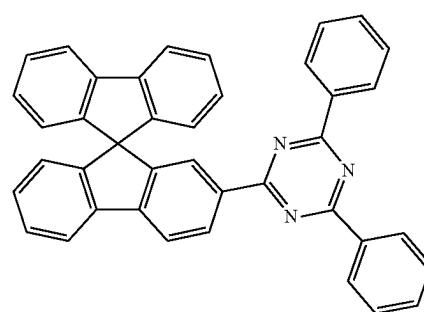
65

**80**

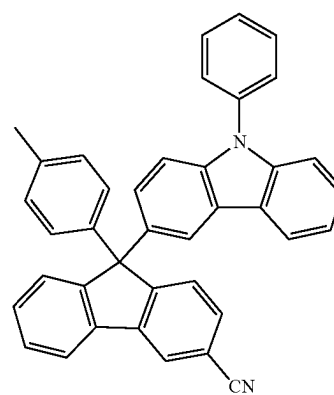
-continued



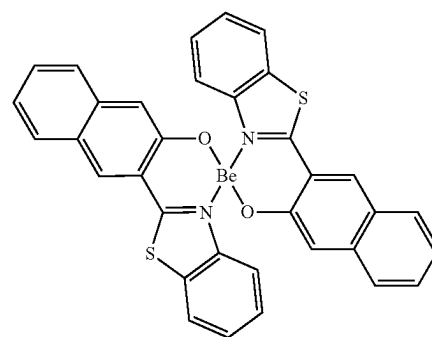
H52



H53



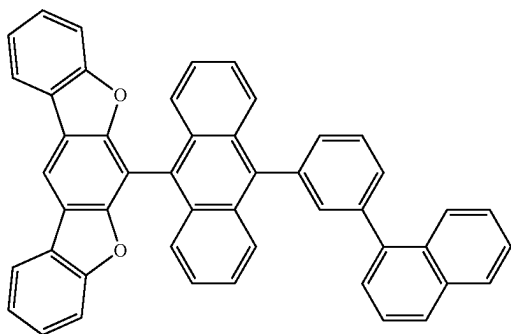
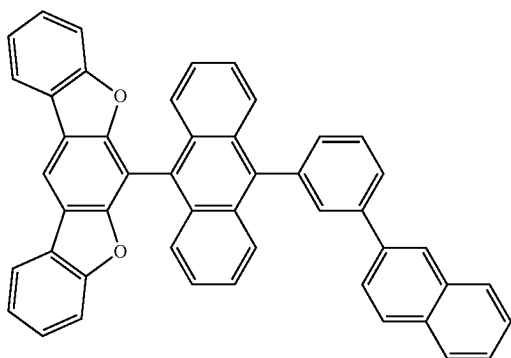
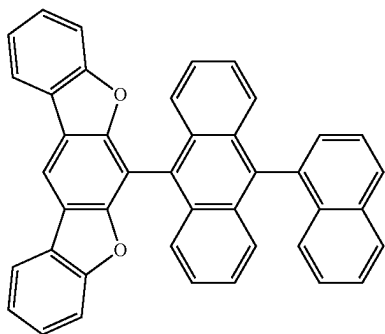
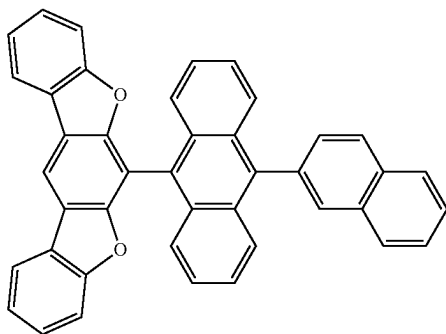
H54



H55

**81**

-continued

**82**

-continued

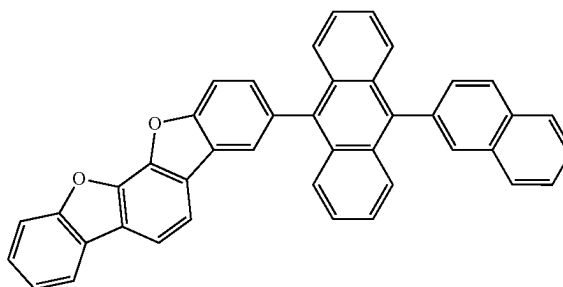
H56

H60

5

10

15



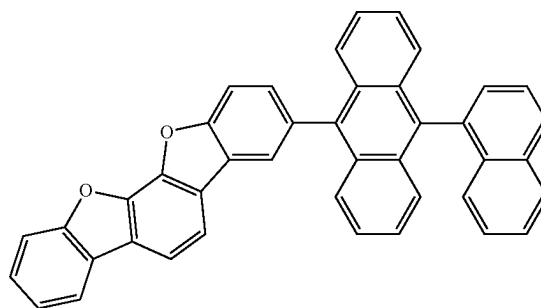
H57

H61

20

25

30



H62

H58

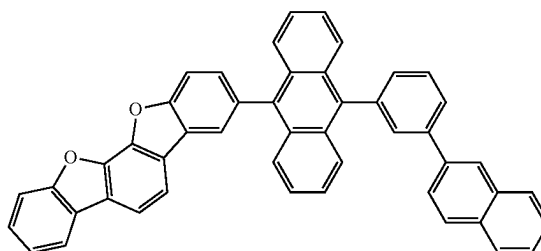
35

40

45

50

H59



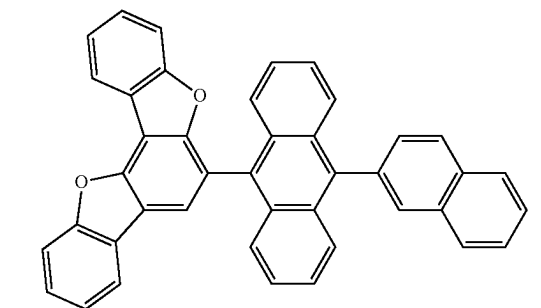
H63

55

60

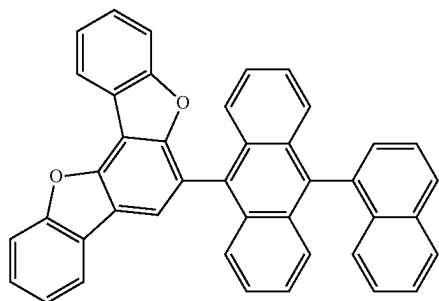
65

H64



**83**

-continued

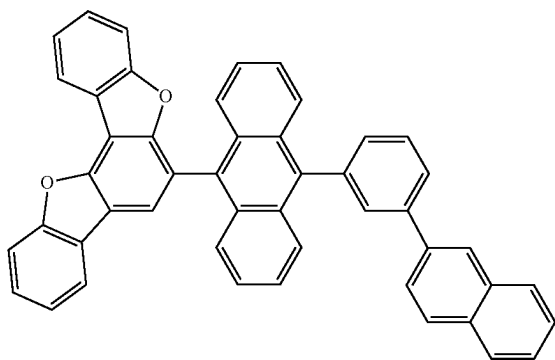


H65

5

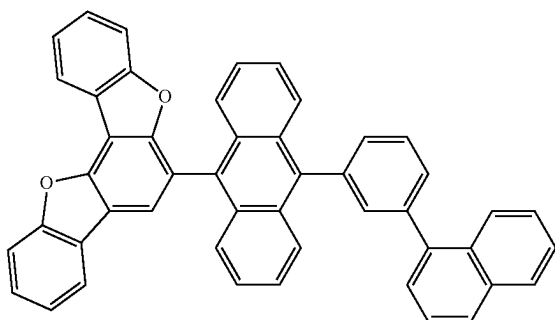
10

H66 15



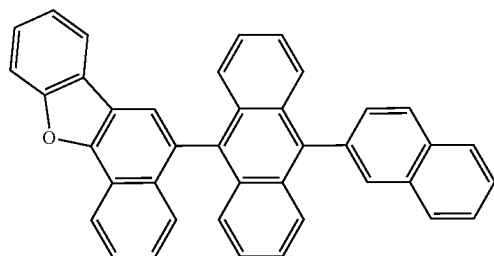
H67

30

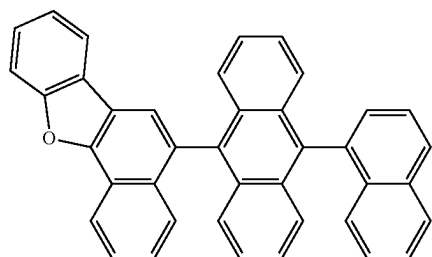


H68

45



50



H69

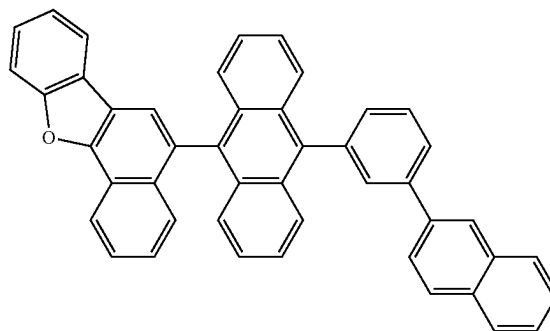
55

60

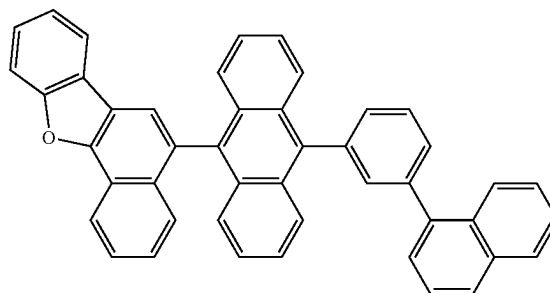
65

**84**

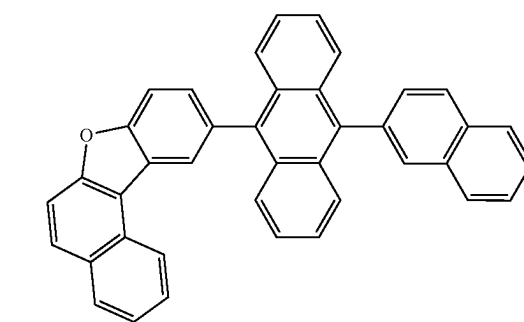
-continued



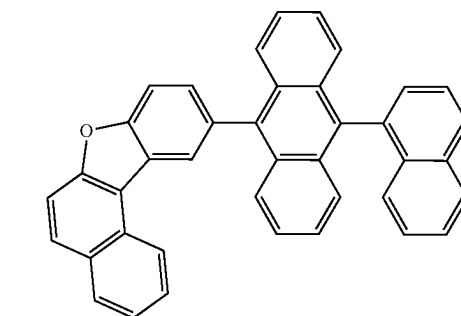
H70



H71



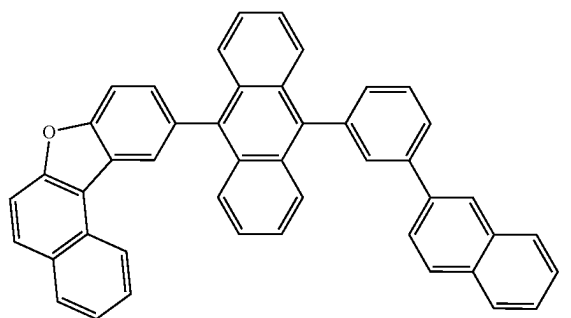
H72



H73

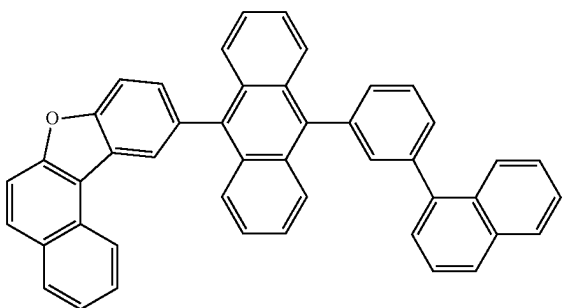
**85**

-continued

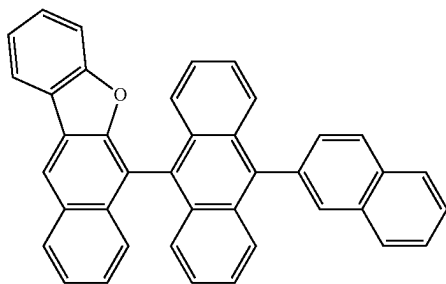


H74

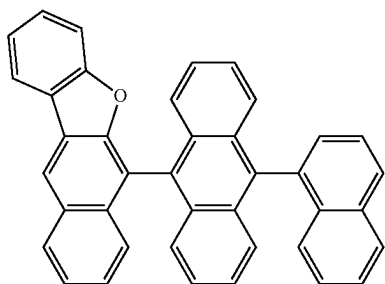
H75



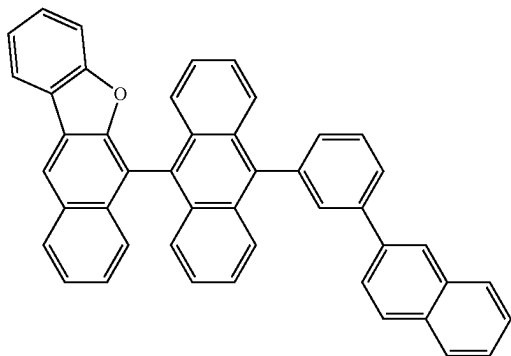
H76



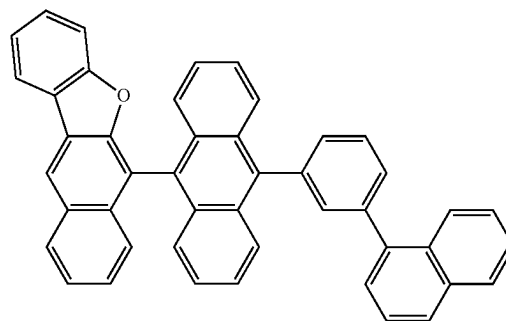
H77



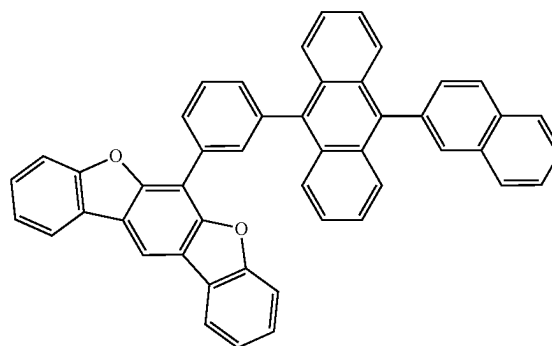
H78

**86**

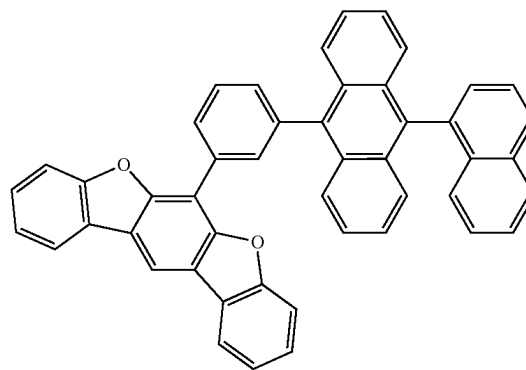
-continued



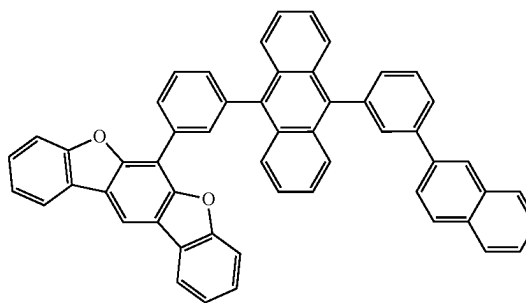
H79



H80



H81



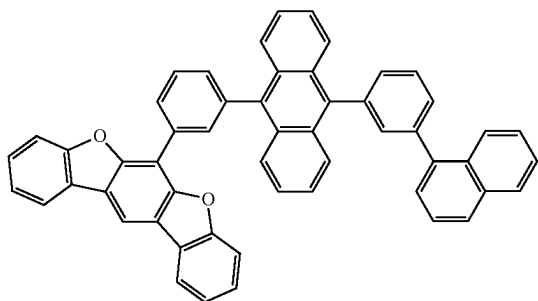
H82



**87**

-continued

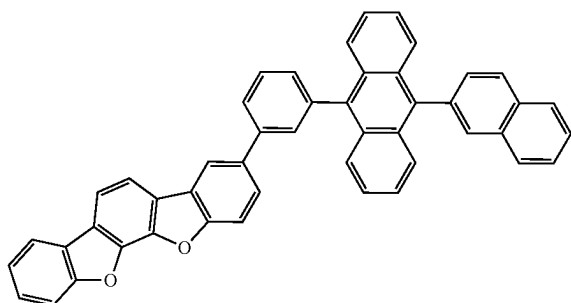
H83



5

10

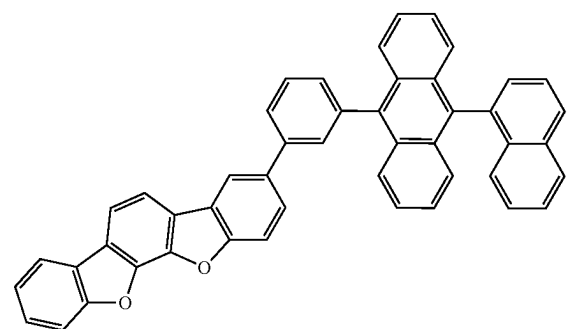
H84 15



20

25

H85

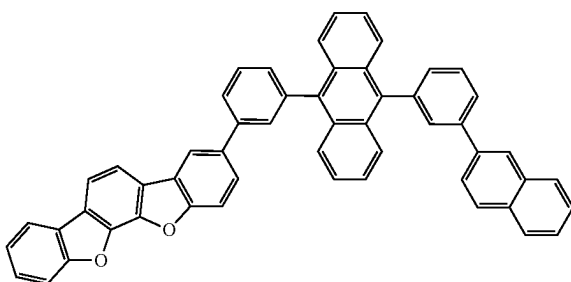


30

35

40

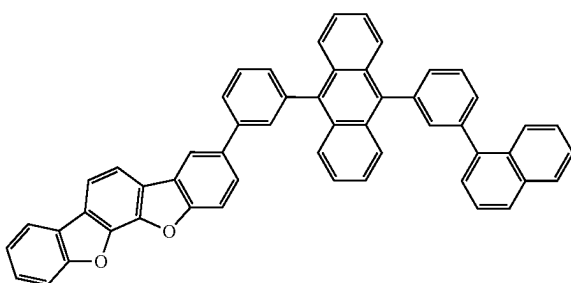
H86



45

50

H87 55



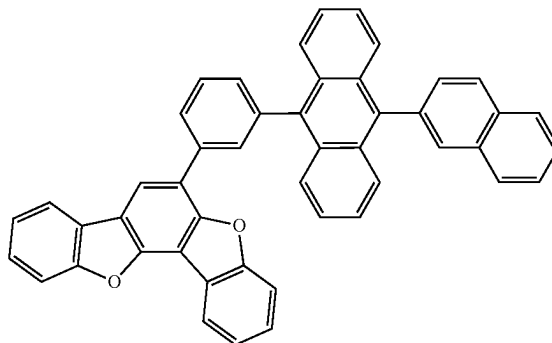
60

65

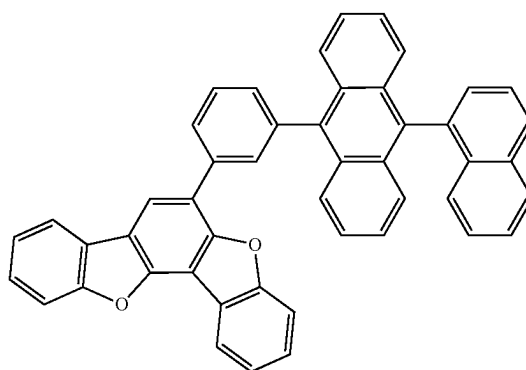
**88**

-continued

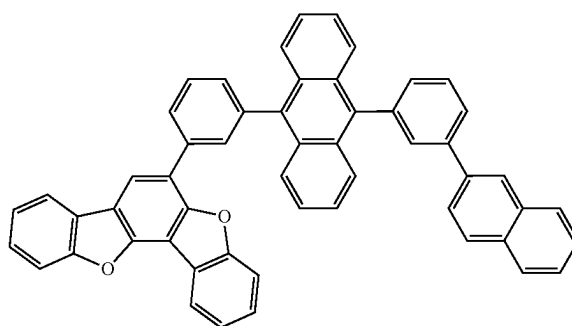
H88



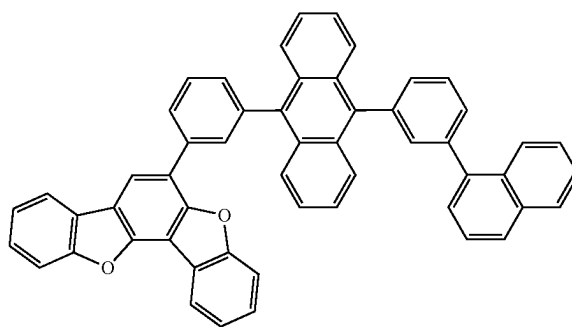
H89



H90

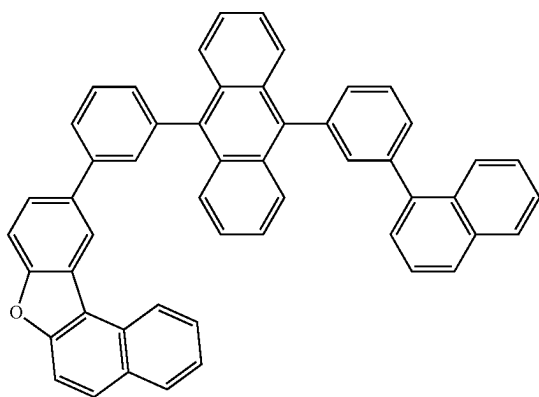
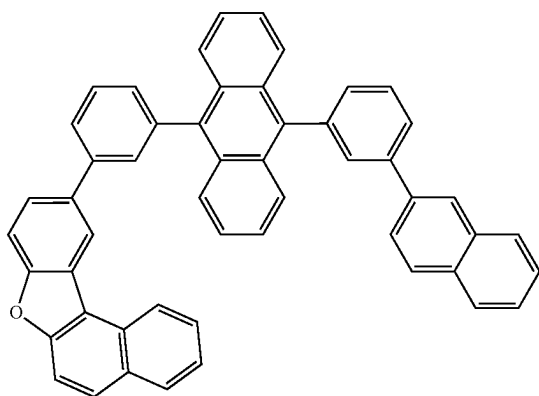
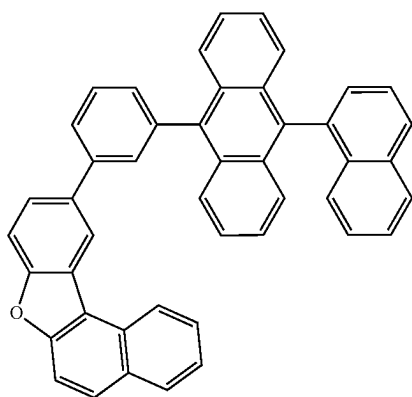
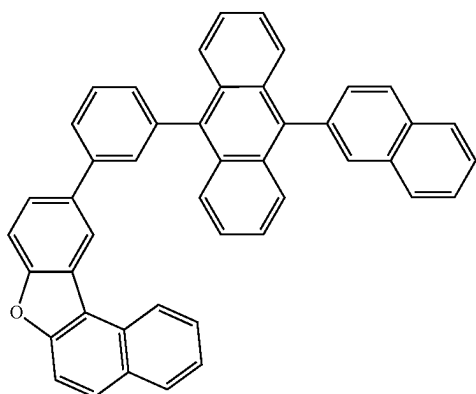


H91



**89**

-continued

**90**

-continued

H92

H96

5

10

15

H93

20

25

30

H94

35

40

45

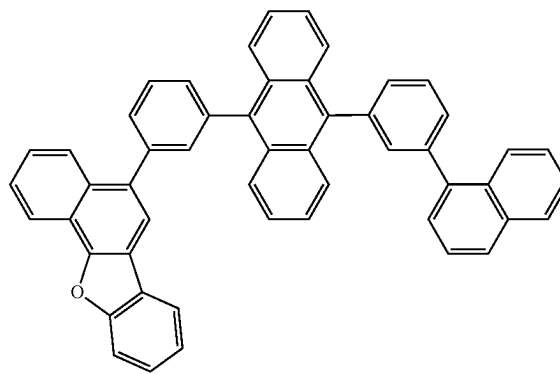
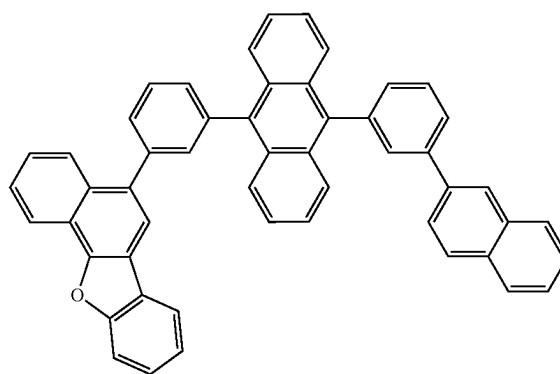
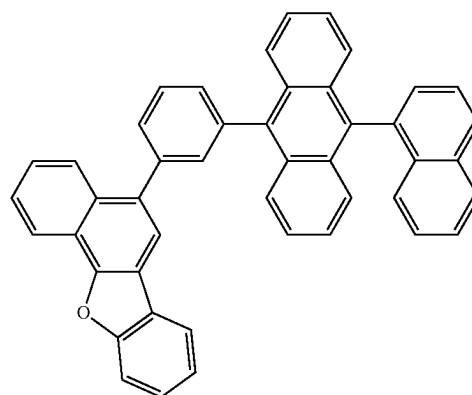
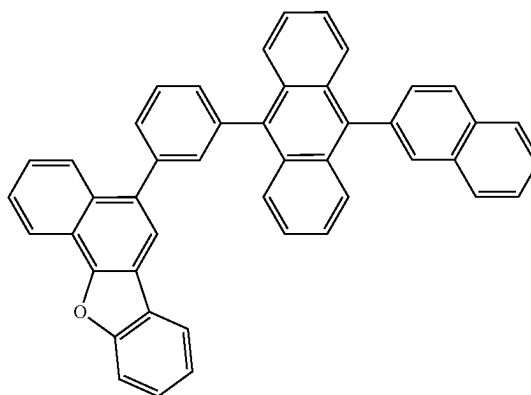
H95

50

55

60

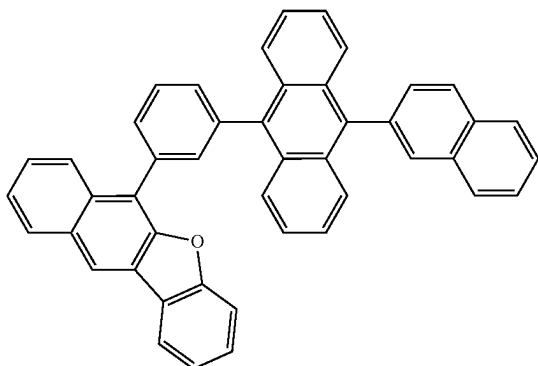
65



**91**

-continued

H100



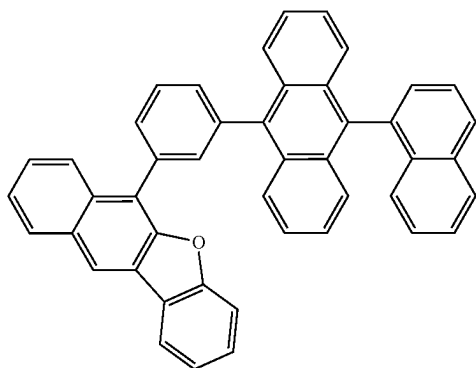
5

10

15

20

H101

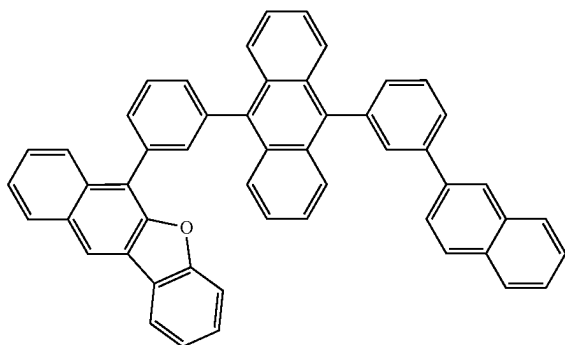


25

30

35

H102

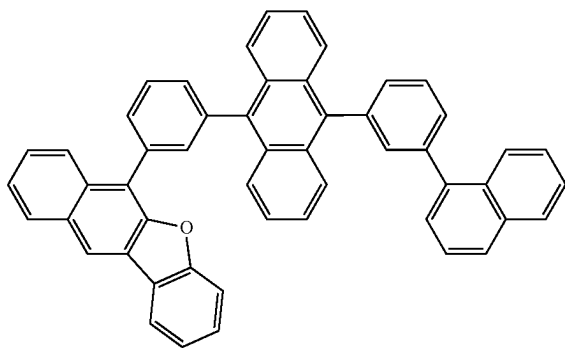


40

45

50

H103



55

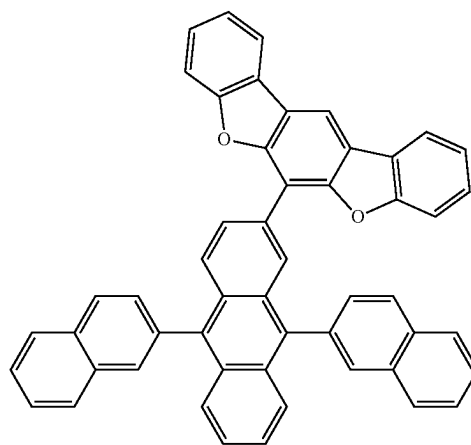
60

65

**92**

-continued

H104



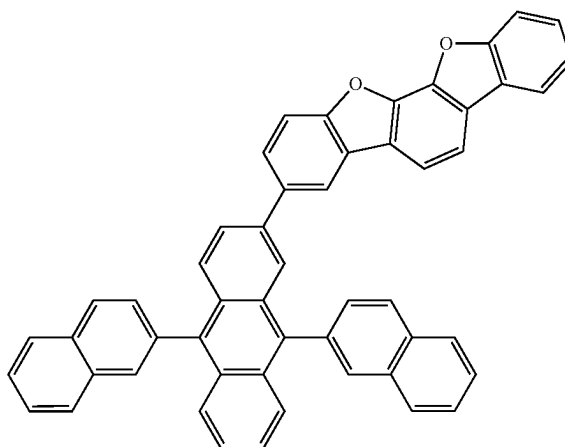
5

10

15

20

H105



25

30

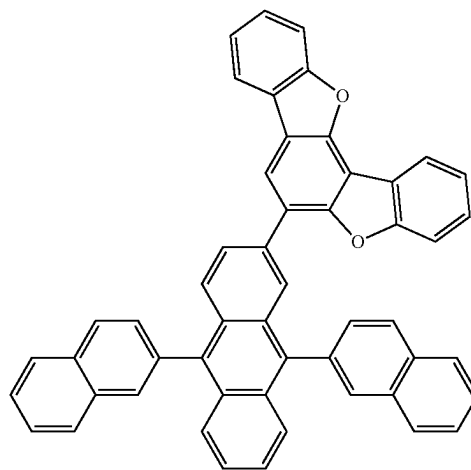
35

40

45

50

H106



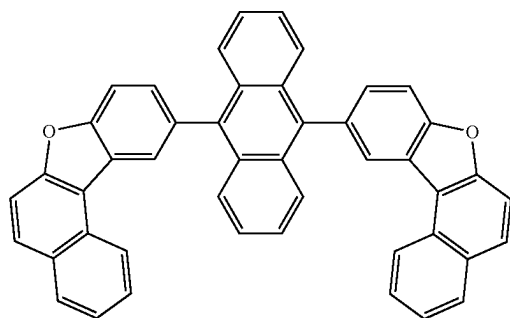
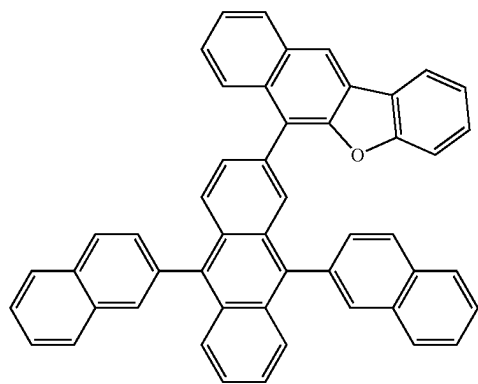
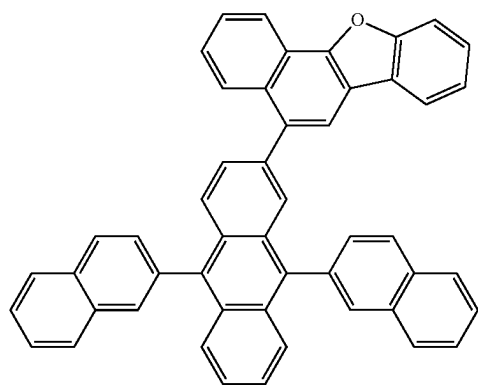
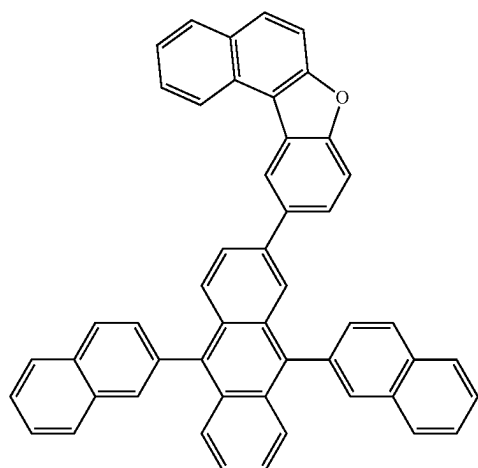
55

60

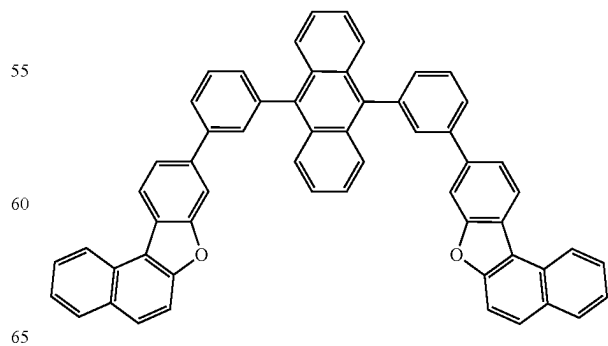
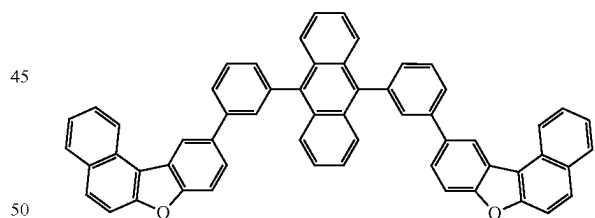
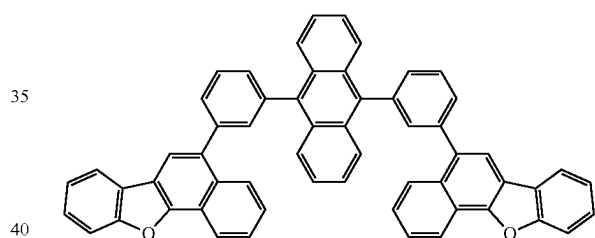
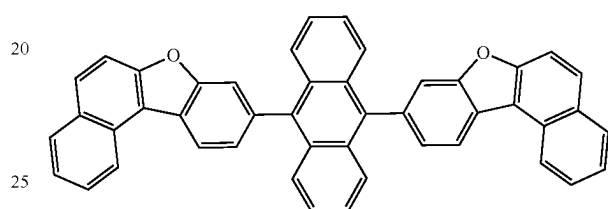
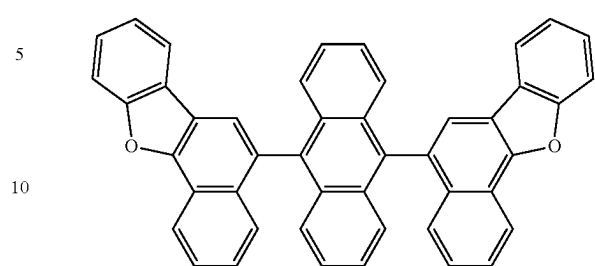
65

**93**

-continued

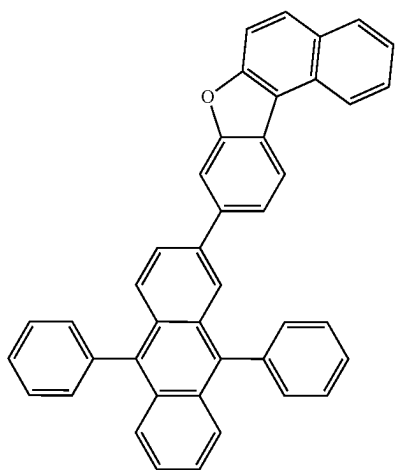
**94**

-continued



**95**

-continued



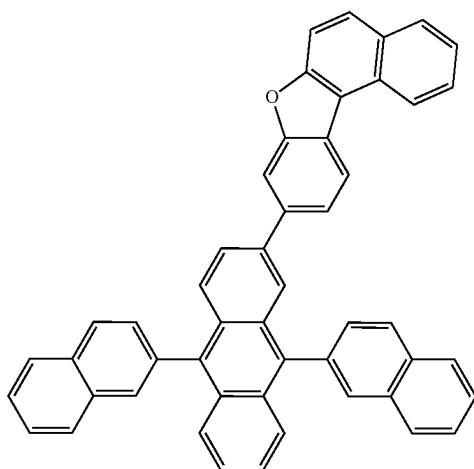
H116

5

10

15

20



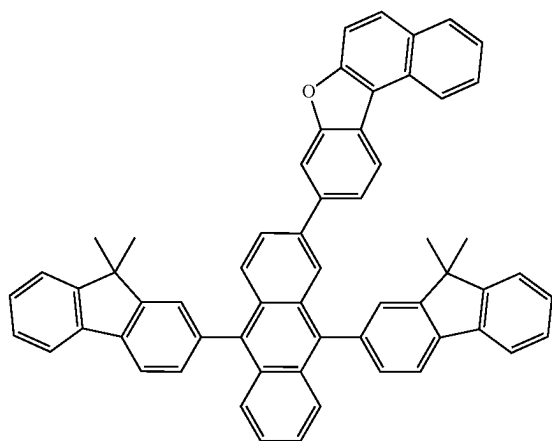
H117

25

30

35

40



H118

50

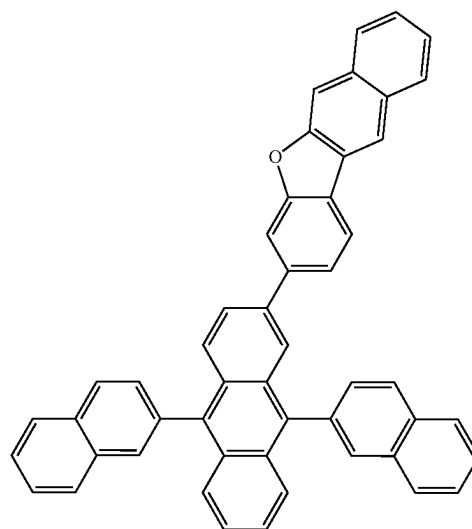
55

60

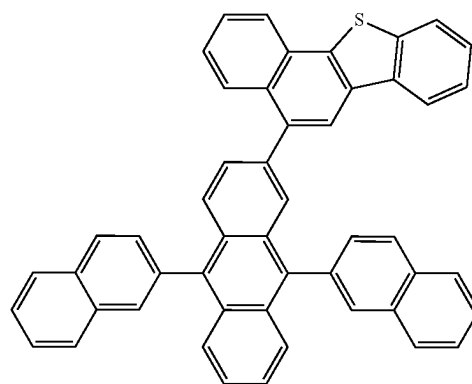
65

**96**

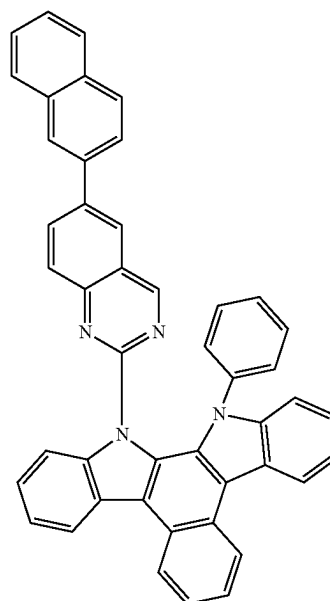
-continued



H119



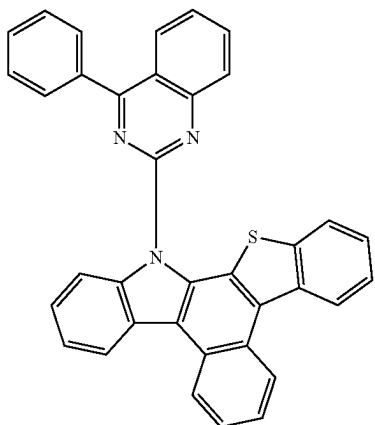
H120



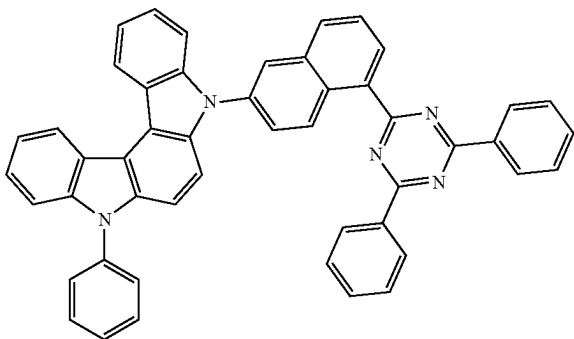
H121

97

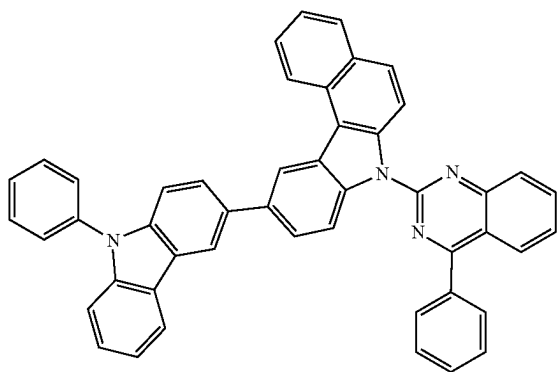
-continued



H122



H123



H124

## Phosphorescent Dopant

The phosphorescent dopant may include at least one transition metal as a central metal.

The phosphorescent dopant may include a monodentate ligand, a bidentate ligand, a tridentate ligand, a tetradentate ligand, a pentadentate ligand, a hexadentate ligand, or any combination thereof.

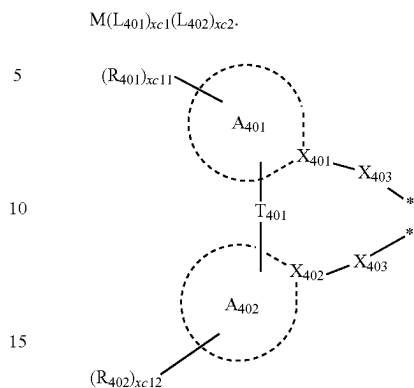
The phosphorescent dopant may be electrically neutral.

In an embodiment, the phosphorescent dopant may include an organometallic compound represented by Formula 401:

98

Formula 401

Formula 402



In Formulae 401 and 402,

M may be transition metal (for example, iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), gold (Au), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), rhenium (Re), or thulium (Tm)),

$L_{401}$  may be a ligand represented by Formula 402, and  $xc1$  may be 1, 2, or 3, wherein when  $xc1$  is 2 or more, two or more of  $L_{401}(s)$  may be identical to or different from each other,

$L_{402}$  may be an organic ligand, and  $xc2$  may be 0, 1, 2, 3, or 4, and when  $xc2$  is 2 or more, two or more of  $L_{402}(s)$  may be identical to or different from each other,

$X_{401}$  and  $X_{402}$  may each independently be nitrogen or carbon,

ring  $A_{401}$  and ring  $A_{402}$  may each independently be a  $C_3$ - $C_{60}$  carbocyclic group or a  $C_1$ - $C_{60}$  heterocyclic group,

$T_{401}$  may be a single bond,  $*-O-*$ ,  $*-S-*$ ,  $*-C(=O)-*$ ,  $*-N(Q_{411})-*$ ,  $*-C(Q_{411})(Q_{412})-*$ ,  $*-C(Q_{411})=C(Q_{412})-*$ ,  $*-C(Q_{411})=*$ , or  $*=C=*$ ,

$X_{403}$  and  $X_{404}$  may each independently be a chemical bond (for example, a covalent bond or a coordination bond), O, S, N( $Q_{413}$ ), B( $Q_{413}$ ), P( $Q_{413}$ ), C( $Q_{413}$ )( $Q_{414}$ ), or Si( $Q_{413}$ )( $Q_{414}$ ),

$Q_{411}$  to  $Q_{414}$  may each be the same as respectively described in connection with  $Q_1$ ,

$R_{401}$  and  $R_{402}$  may each independently be hydrogen, deuterium,  $-F$ ,  $-Cl$ ,  $-Br$ ,  $-I$ , a hydroxyl group, a cyano group, a nitro group, a  $C_1$ - $C_{20}$  alkyl group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_1$ - $C_{20}$  alkoxy group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ ,  $-Si(Q_{401})(Q_{402})(Q_{403})$ ,  $-N(Q_{401})(Q_{402})$ ,  $-B(Q_{401})(Q_{402})$ ,  $-C(=O)(Q_{401})$ ,  $-S(=O)_2(Q_{401})$ , or  $-P(=O)(Q_{401})(Q_{402})$ ,

$Q_{401}$  to  $Q_{403}$  may each be the same as respectively described in connection with  $Q_1$ ,

$xc11$  and  $xc12$  may each independently be an integer from 0 to 10, and

$*$  and  $*$ ' in Formula 402 each indicate a binding site to M in Formula 401.

In an embodiment, in Formula 402, i)  $X_{401}$  may be nitrogen, and  $X_{402}$  may be carbon, or ii) each of  $X_{401}$  and  $X_{402}$  may be nitrogen.

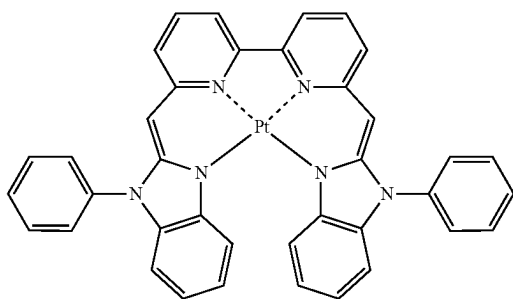
In an embodiment, when  $xc1$  in Formula 401 is 2 or more, two ring  $A_{401}$  in two or more of  $L_{401}(s)$  may be optionally linked to each other via  $T_{402}$ , which is a linking group, and

**99**

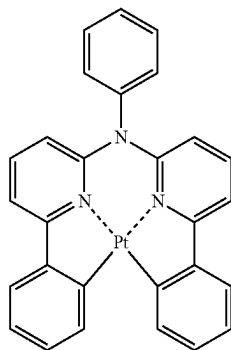
two ring A<sub>402</sub> may optionally be linked to each other via T<sub>403</sub>, which is a linking group (see Compounds PD1 to PD4 and PD7). T<sub>402</sub> and T<sub>403</sub> may each be the same as respectively described in connection with T<sub>401</sub>.

L<sub>402</sub> in Formula 401 may be an organic ligand. In an embodiment, L<sub>402</sub> may include a halogen group, a diketone group (for example, an acetylacetonate group), a carboxylic acid group (for example, a picolinate group), —C(=O), an isonitrile group, a —CN group, a phosphorus group (for example, a phosphine group, a phosphite group, etc.), or any combination thereof.

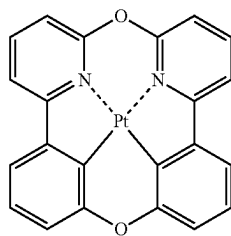
The phosphorescent dopant may include, for example, one selected from compounds PD1 to PD25 or any combination thereof:



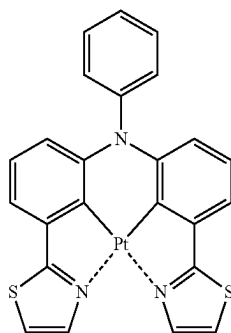
PD1



PD2



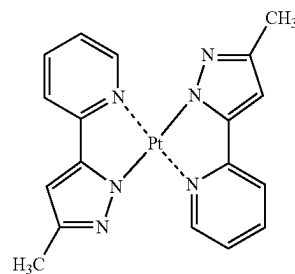
PD3



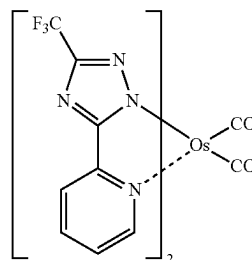
PD4

**100**

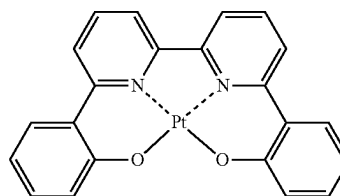
-continued



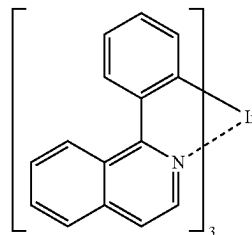
PD5



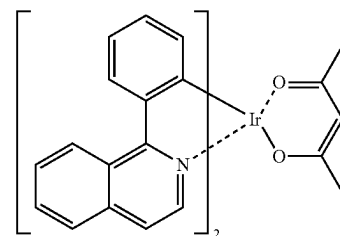
PD6



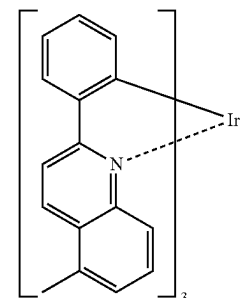
PD7



PD8



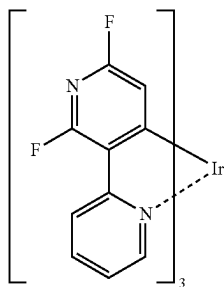
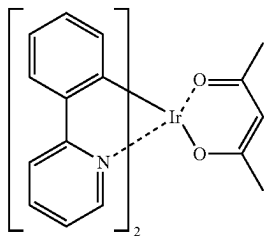
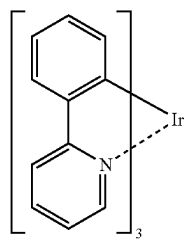
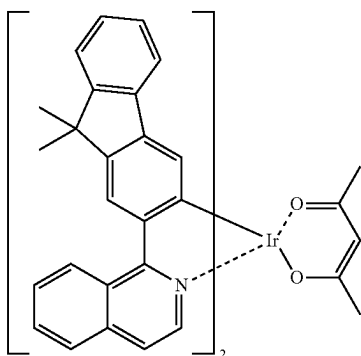
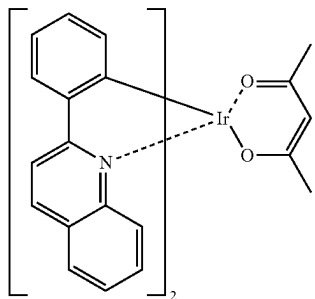
PD9



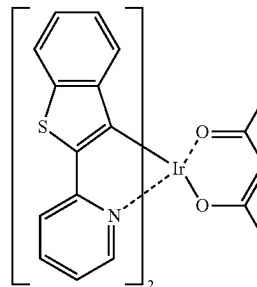
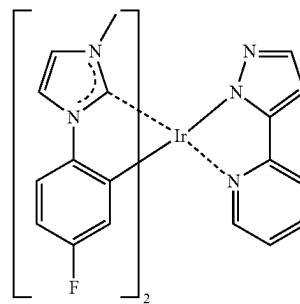
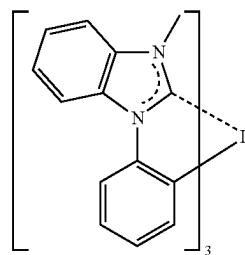
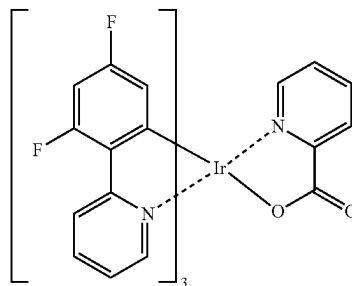
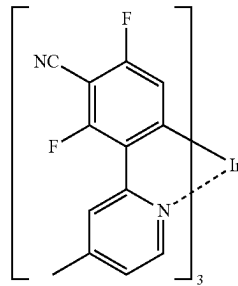
PD10

**101**

-continued

**102**

-continued



PD11

5

10

15

PD12

20

25

30

PD13

35

40

PD14

45

50

PD15

55

60

65

PD16

PD17

PD18

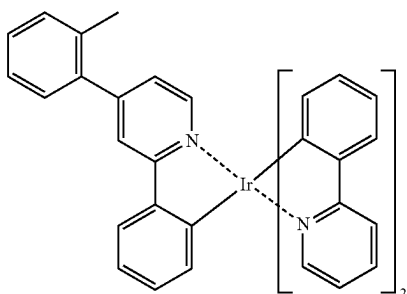
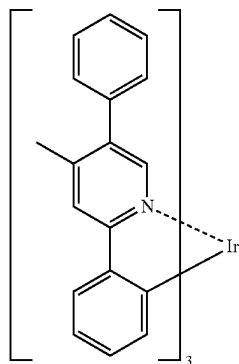
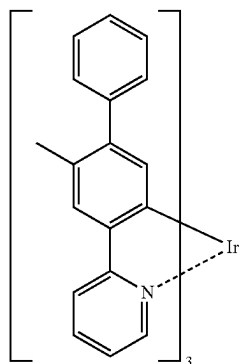
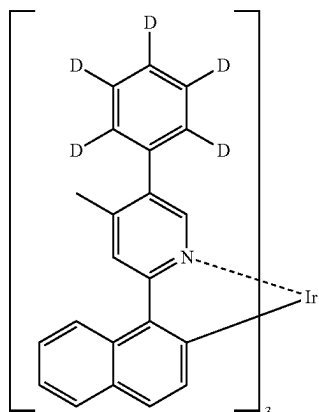
PD19

PD20



**103**

-continued

**104**

-continued

PD21

5

10

15

20

PD22

25

30

35

PD23

40

45

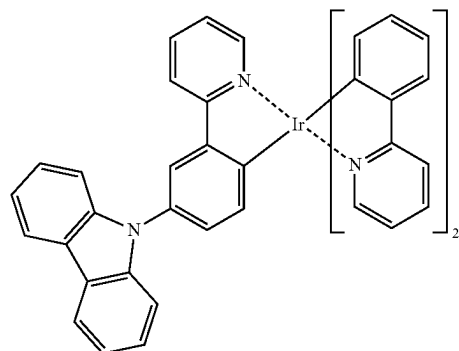
PD24

55

60

65

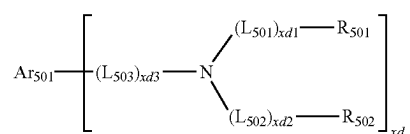
PD25



## Fluorescent Dopant

The fluorescent dopant may include an amine group-containing compound, a styryl group-containing compound, or any combination thereof.

In an embodiment, the fluorescent dopant may include a compound represented by Formula 501:



Formula 501

In Formula 501,

$Ar_{501}$ ,  $L_{501}$  to  $L_{503}$ ,  $R_{501}$  and  $R_{502}$  may each independently be a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$  or a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ ,

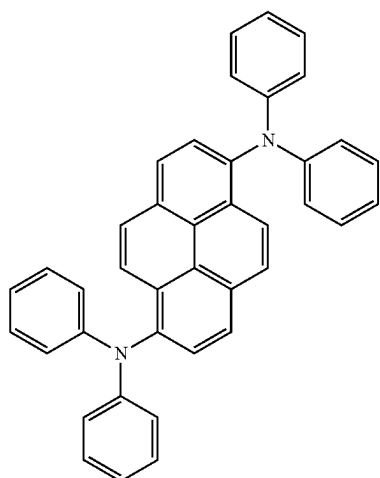
$xd1$  to  $xd3$  may each independently be 0, 1, 2, or 3, and  $xd4$  may be 1, 2, 3, 4, 5, or 6.

In an embodiment,  $Ar_{501}$  in Formula 501 may include a condensed cyclic group (for example, an anthracene group, a chrysene group, or a pyrene group) in which three or more monocyclic groups are condensed together (e.g., combined together).

In one or more embodiments,  $xd4$  in Formula 501 may be 2.

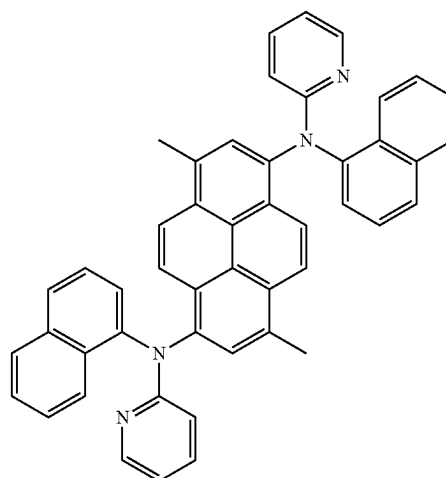
In an embodiment, the fluorescent dopant may include: one selected from Compounds FD1 to FD36; DPVBi; DPAVBi; or any combination thereof:

105



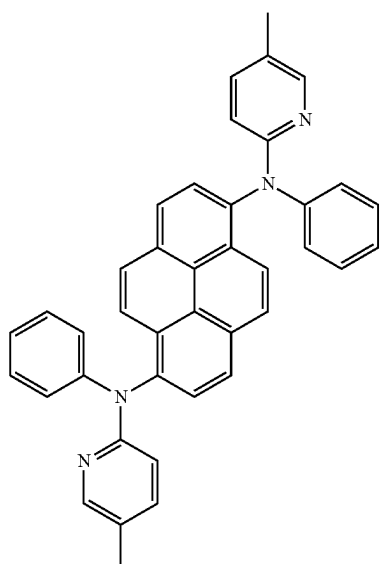
FD1

106

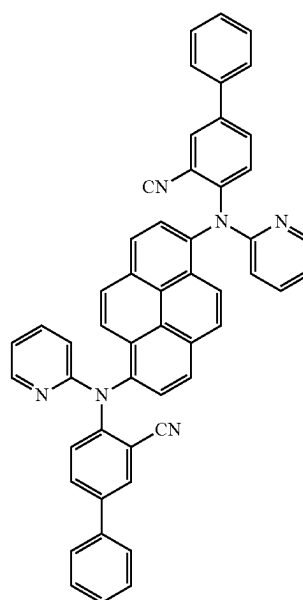


FD2

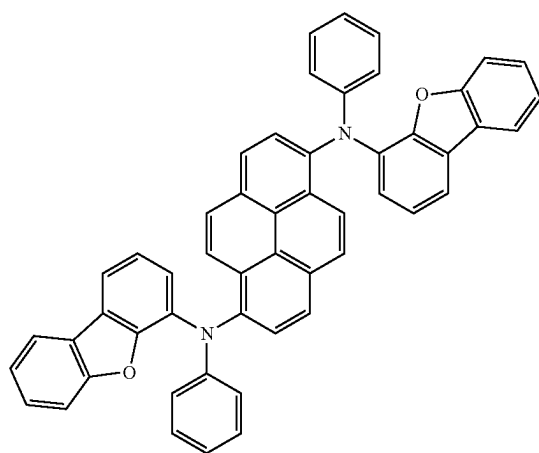
FD3



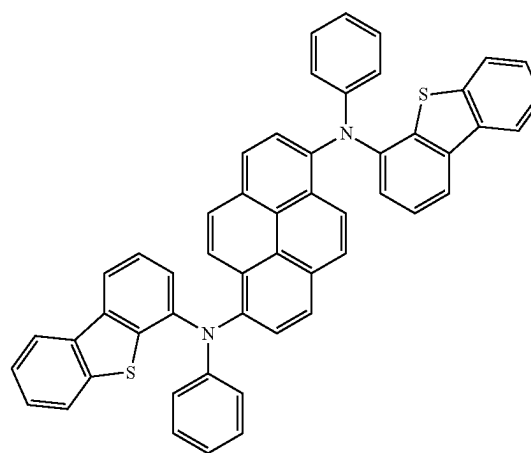
FD4



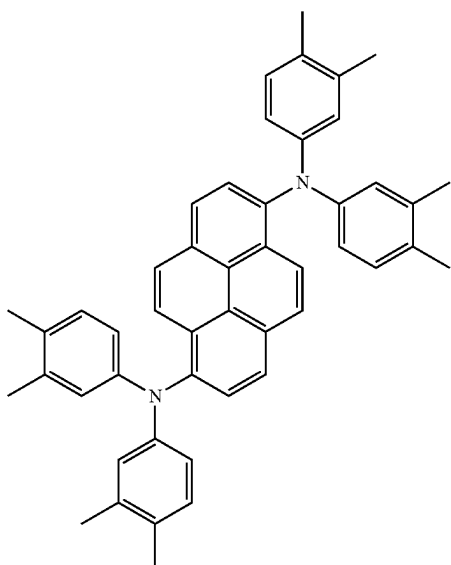
FD5



FD6

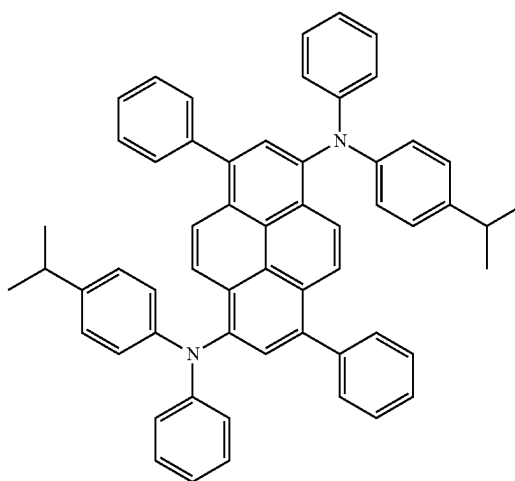


107

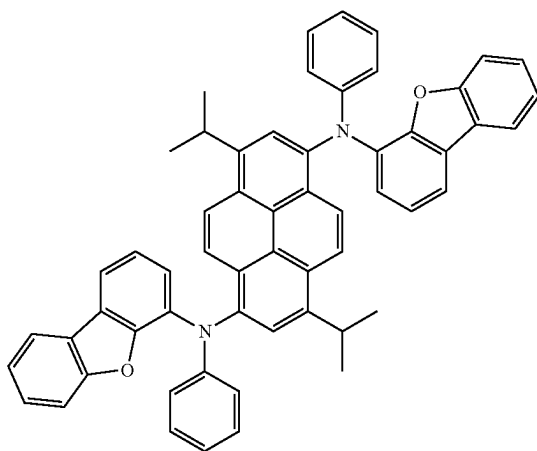
-continued  
FD7

108

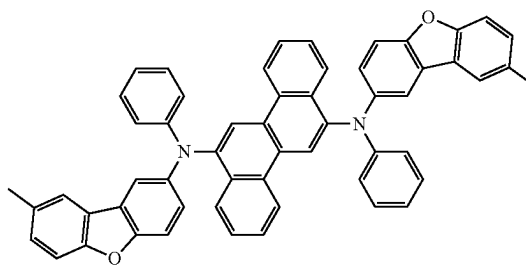
FD8



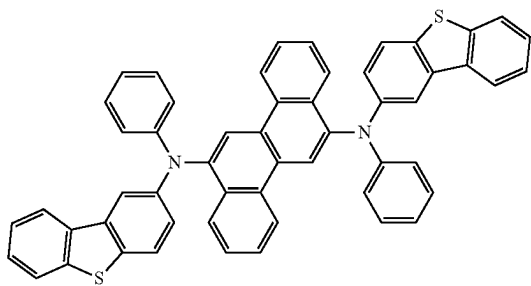
FD9



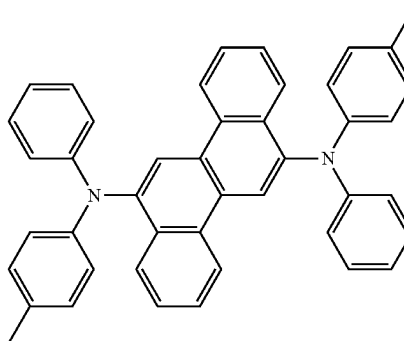
FD10



FD11



FD12



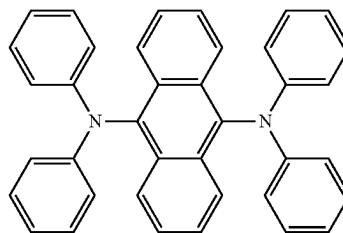
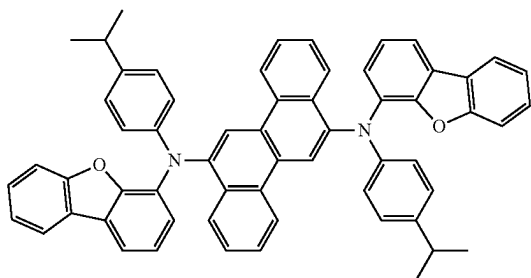
109

110

-continued

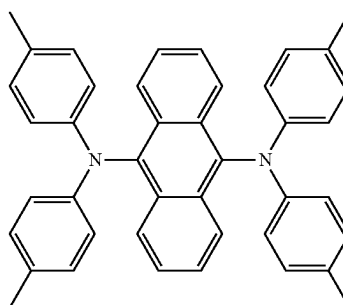
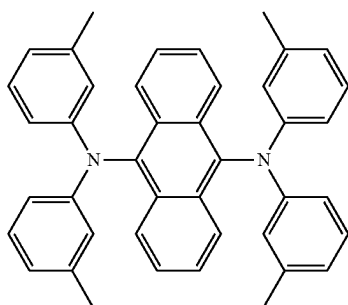
FD13

FD14



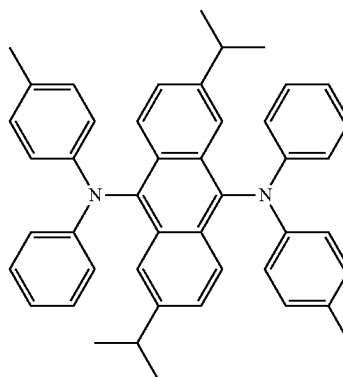
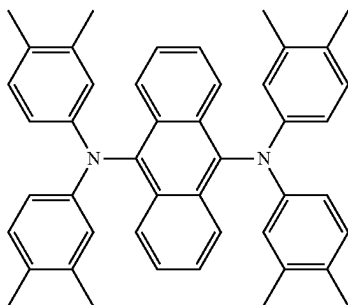
FD15

FD16



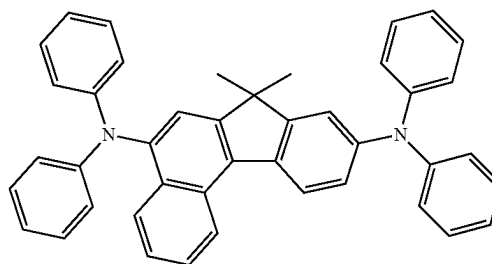
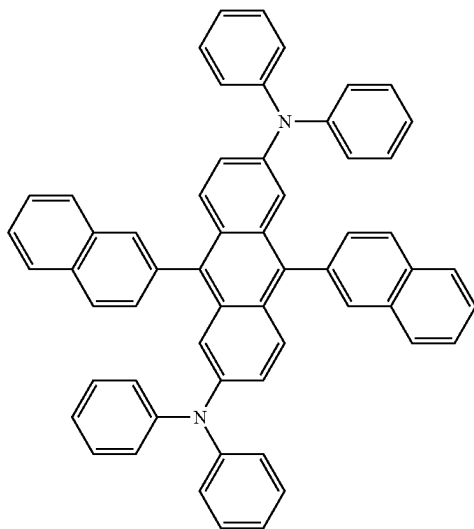
FD17

FD18



FD19

FD20



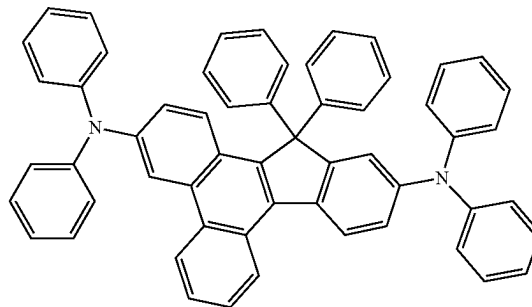
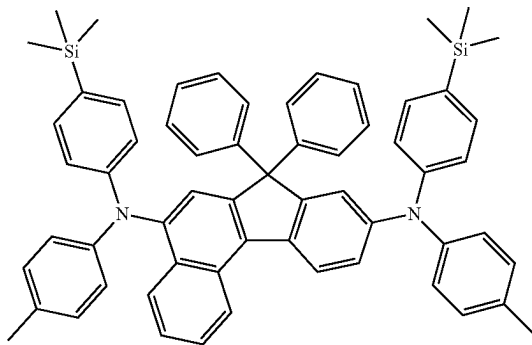
111

112

-continued

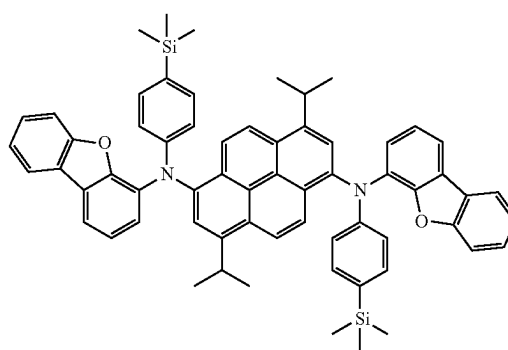
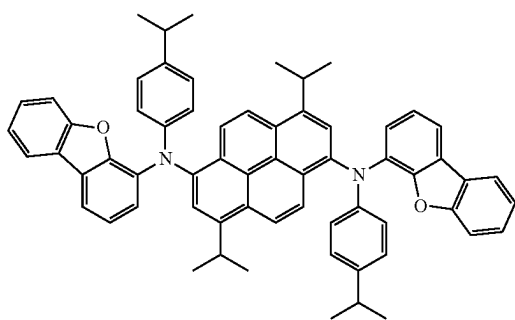
FD21

FD22



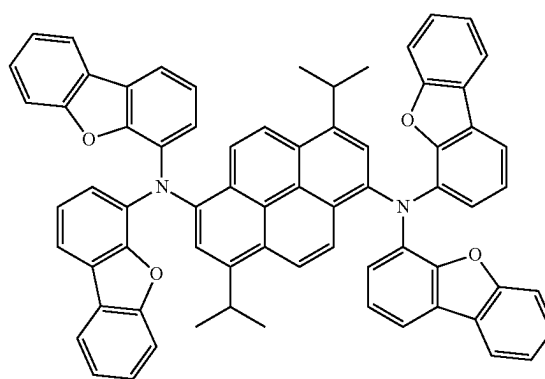
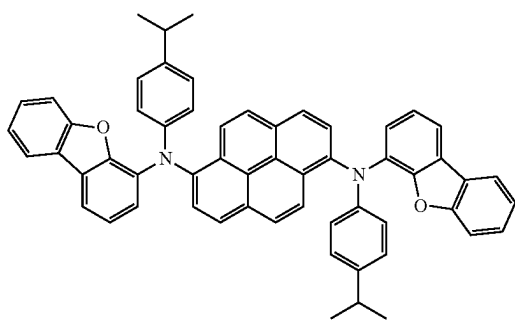
FD23

FD24



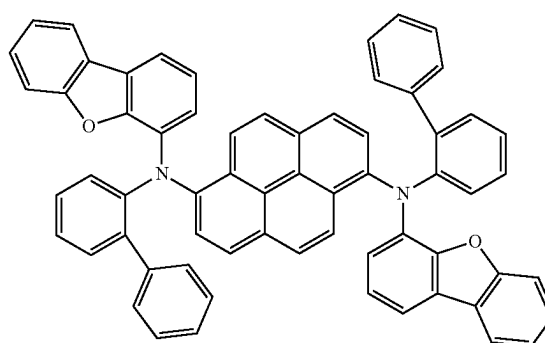
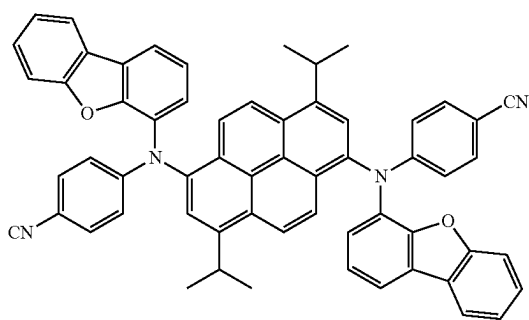
FD25

FD26



FD27

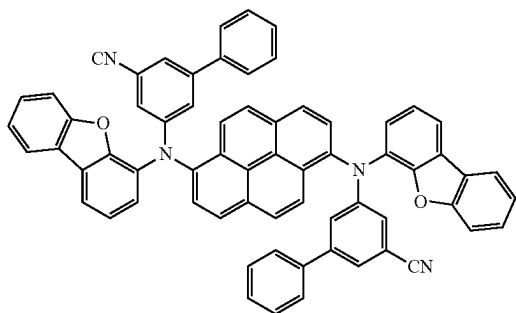
FD28



113

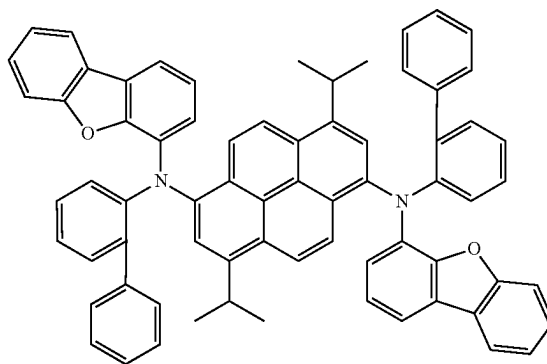
-continued

FD29



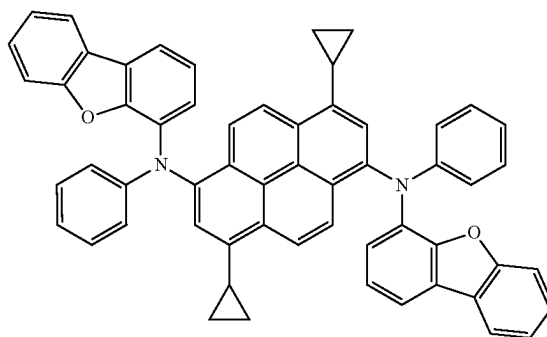
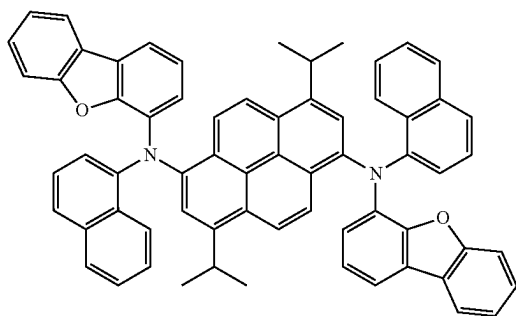
114

FD30



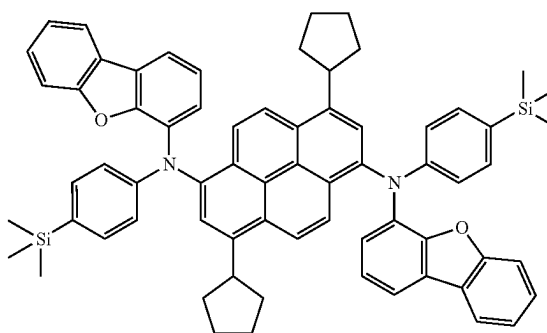
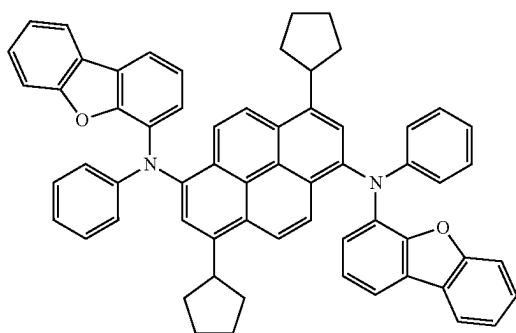
FD31

FD32



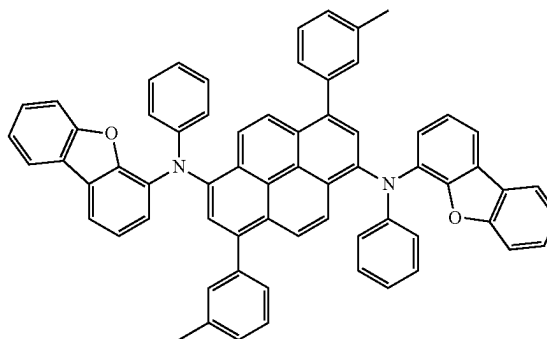
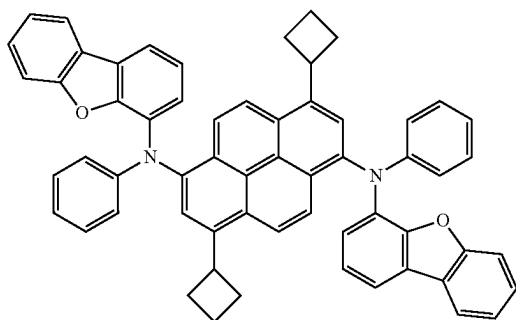
FD33

FD34



FD35

FD36

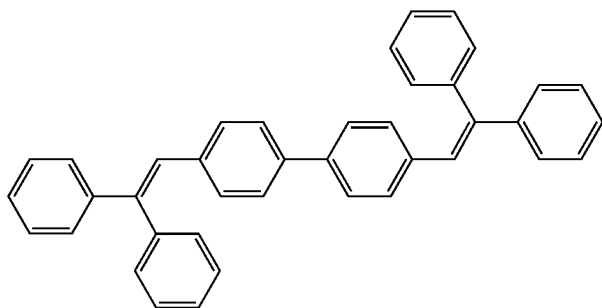


115

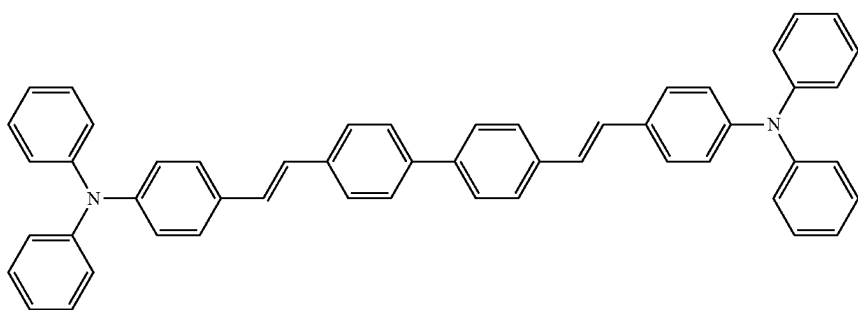
116

-continued

DPVBi



DPAVB



### Delayed Fluorescent Material

The emission layer may include a delayed fluorescent material.

The delayed fluorescence material used herein may be selected from the heterocyclic compound represented by Formula 1 or any suitable compound that may emit delayed fluorescence by a delayed fluorescence emission mechanism.

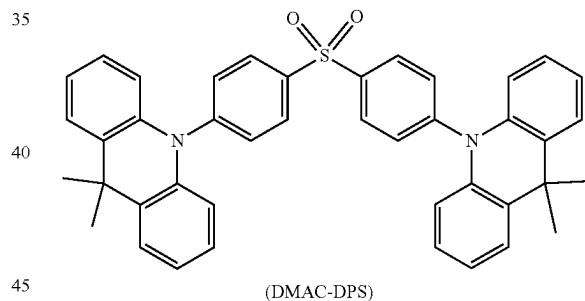
The delayed fluorescent material included in the emission layer may act as a host or a dopant depending on the type or kind of other materials included in the emission layer.

In an embodiment, a difference between a triplet energy level (eV) of the delayed fluorescent material and a singlet energy level (eV) of the delayed fluorescent material may be 0 eV or more and 0.5 eV or less. When the difference between the triplet energy level (eV) of the delayed fluorescent material and the singlet energy level (eV) of the delayed fluorescent material satisfies the above-described range, up-conversion from a triplet state to a singlet state of the delayed fluorescent materials may effectively occur, and thus, the luminescence efficiency of the light-emitting device 10 may be improved.

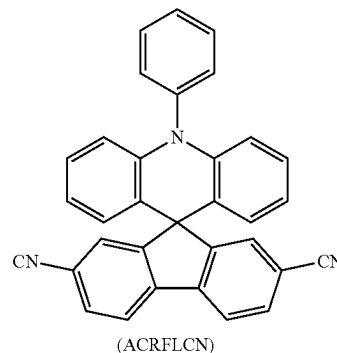
In an embodiment, the delayed fluorescent material may include i) a material that includes at least one electron donor (for example, a  $\pi$  electron-rich  $C_3$ - $C_{60}$  cyclic group, such as a carbazole group) and at least one electron acceptor (for example, a sulfoxide group, a cyano group, or a  $\pi$ -electron-deficient nitrogen-containing  $C_1$ - $C_{60}$  cyclic group), ii) a material including a  $C_8$ - $C_{60}$  polycyclic group in which two or more cyclic groups share boron (B) and are condensed with (e.g., combined together with) each other.

The delayed fluorescent material may include at least one selected from Compounds DF1 to DF9:

DF1

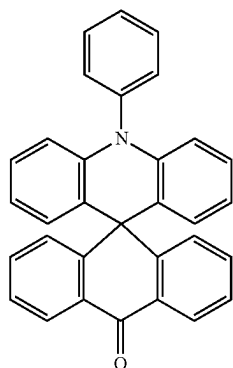


DF2

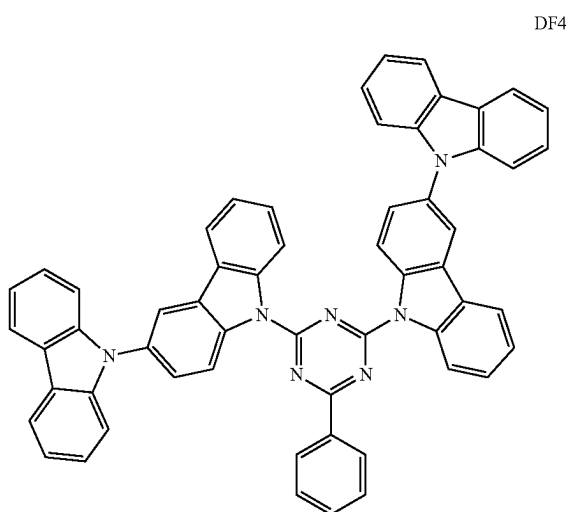


**117**

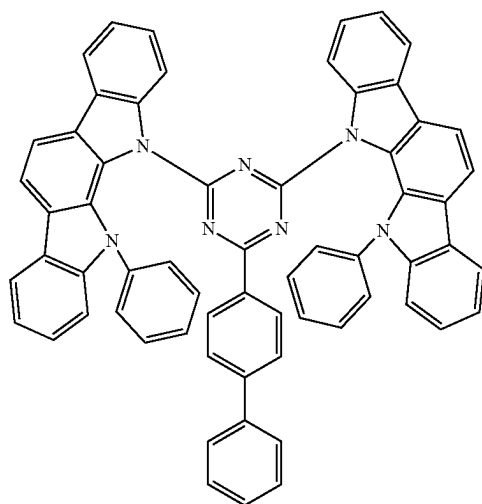
-continued



(ACRSA)



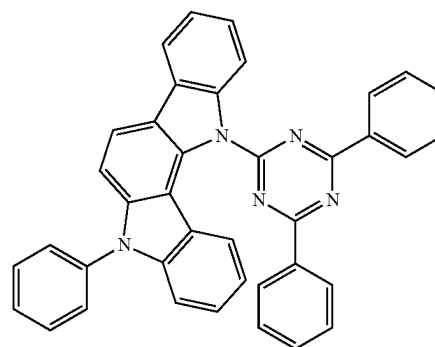
(CC2TA)



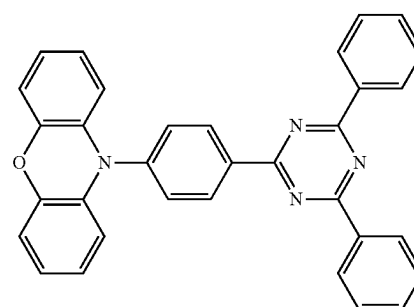
(PIC-TRZ)

**118**

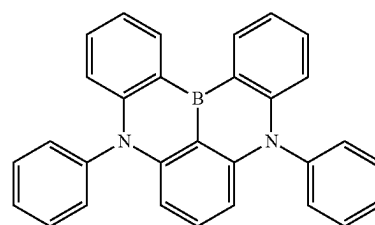
-continued



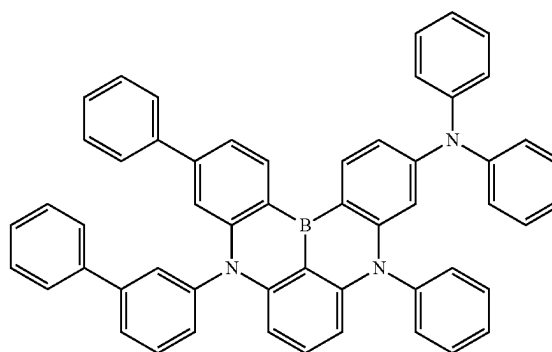
(PIC-TRZ2)



(PXZ-TRZ)



(DABNA-1)



(DABNA-2)

**Quantum Dot**

The emission layer may include a quantum dot.

The quantum dot used herein refers to a crystal of a semiconductor compound, and may include any suitable material that is capable of emitting light of various suitable emission wavelengths depending on a size of the crystal.

A diameter of the quantum dot may be, for example, in a range of about 1 nm to about 10 nm.



The quantum dot may be synthesized by a wet chemical process, a metal organic chemical vapor deposition process, a molecular beam epitaxy process, or a process that is similar to these processes.

The wet chemical process refers to a method in which a solvent and a precursor material are mixed, and then, a quantum dot particle crystal is grown. When the crystal grows, the organic solvent naturally acts as a dispersant coordinated on the surface of the quantum dot crystal and controls the growth of the crystal so that the growth of quantum dot particles can be controlled through a process which is more easily performed than vapor deposition methods, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE), and which requires low costs.

The quantum dot may include Groups II-VI semiconductor compound, Groups III-V semiconductor compound, Groups III-VI semiconductor compound, Groups I-III-VI semiconductor compound, Groups IV-VI semiconductor compound, a Group IV element or compound, or any combination thereof.

Examples of the Groups II-VI semiconductor compound may include: a binary compound, such as CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnO, HgS, HgSe, HgTe, MgSe, and/or MgS; a ternary compound, such as CdSeS, CdSeTe, CdSTe, ZnSeS, ZnSeTe, ZnSTe, HgSeS, HgSeTe, HgSTe, CdZnS, CdZnSe, CdZnTe, CdHgS, CdHgSe, CdHgTe, HgZnS, HgZnSe, HgZnTe, MgZnSe, and/or MgZnS; a quaternary compound, such as CdZnSeS, CdZnSeTe, CdZnSTe, CdHgSeS, CdHgSeTe, CdHgSTe, HgZnSeS, HgZnSeTe, and/or HgZnSTe; or any combination thereof.

Examples of the Groups III-V semiconductor compound may include: a binary compound, such as GaN, GaP, GaAs, GaSb, AlN, AlP, AlAs, AlSb, InN, InP, InAs, and/or InSb; a ternary compound, such as GaNP, GaNAs, GaNSb, GaPAs, GaPSb, AlNP, AlNAs, AlNSb, AlPAs, AlPSb, InGaP, InNP, InAlP, InNAs, InNSb, InPAs, and/or InPSb; a quaternary compound, such as GaAlNAs, GaAlNSb, GaAlPAs, GaAlPSb, GaInNP, GaInNAs, GaInNSb, GaInPAs, GaInPSb, InAlNP, InAlNAs, InAlNSb, InAlPAs, InAlPSb, and/or GaAlNP; or any combination thereof. In one or more embodiments, the Groups III-V semiconductor compound may further include a Group II element. Examples of the Groups III-V semiconductor compound further including a Group II element may include InZnP, InGaZnP, and/or InAlZnP.

Examples of the Groups III-VI semiconductor compound may include: a binary compound, such as GaS, GaSe, Ga<sub>2</sub>Se<sub>3</sub>, GaTe, InS, In<sub>2</sub>S<sub>3</sub>, InSe, In<sub>2</sub>Se<sub>3</sub>, and/or InTe; a ternary compound, such as InGaS<sub>3</sub>, and/or InGaSe<sub>3</sub>; or any combination thereof.

Examples of the Group I-III-VI semiconductor compound may include: a ternary compound, such as AgInS, AgInS<sub>2</sub>, CuInS, CuInS<sub>2</sub>, CuGaO<sub>2</sub>, AgGaO<sub>2</sub>, and/or AgAlO<sub>2</sub>.

Examples of the Group IV-VI semiconductor compound may include: a binary compound, such as SnS, SnSe, SnTe, PbS, PbSe, and/or PbTe; a ternary compound, such as SnSeS, SnSeTe, SnSTe, PbSeS, PbSeTe, PbSTe, SnPbS, SnPbSe, and/or SnPbTe; a quaternary compound, such as SnPbSSe, SnPbSeTe, and/or SnPbSTe; or any combination thereof.

In an embodiment, the Group IV element or compound may include: a single element, such as Si and/or Ge; a binary compound, such as SiC and/or SiGe; or any combination thereof.

Each element included in the multi-element compound such as the binary compound, ternary compound, and qua-

ternary compound may be present in a particle at a uniform concentration or a non-uniform concentration.

In an embodiment, the quantum dot may have a single structure having a uniform (e.g., substantially uniform) concentration of each element included in the corresponding quantum dot or a dual structure of a core-shell. In an embodiment, the material included in the core may be different from the material included in the shell.

The shell of the quantum dot may function as a protective layer for maintaining semiconductor characteristics by preventing or reducing chemical degeneration of the core and/or may function as a charging layer for imparting electrophoretic characteristics to the quantum dot. The shell may be a single layer or a multilayer. An interface between the core and the shell may have a concentration gradient in which the concentration of elements existing in the shell decreases along a direction toward the center.

Examples of the shell of the quantum dot may include an oxide of metal, metalloid, and/or non-metal, a semiconductor compound, or any combination thereof. Examples of the oxide of metal, metalloid, and/or non-metal may include a binary compound, such as SiO<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub>, TiO<sub>2</sub>, ZnO, MnO, Mn<sub>2</sub>O<sub>3</sub>, Mn<sub>3</sub>O<sub>4</sub>, CuO, FeO, Fe<sub>2</sub>O<sub>3</sub>, Fe<sub>3</sub>O<sub>4</sub>, CoO, Co<sub>3</sub>O<sub>4</sub>, and/or NiO; a ternary compound, such as MgAl<sub>2</sub>O<sub>4</sub>, CoFe<sub>2</sub>O<sub>4</sub>, NiFe<sub>2</sub>O<sub>4</sub>, and/or CoMn<sub>2</sub>O<sub>4</sub>; and any combination thereof. Examples of the semiconductor compound may include, as described herein, Groups II-VI semiconductor compound, Groups III-V semiconductor compound, Groups III-VI semiconductor compound, Groups I-III-VI semiconductor compound, Groups IV-VI semiconductor compound, or any combination thereof. In an embodiment, the semiconductor compound may include CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnSeS, ZnTeS, GaAs, GaP, GaSb, HgS, HgSe, HgTe, InAs, InP, InGaP, InSb, AlAs, AlP, AlSb, or any combination thereof.

A full width at half maximum (FWHM) of an emission wavelength spectrum of the quantum dot may be about 45 nm or less, for example, about 40 nm or less, for example, about 30 nm or less. When the FWHM of the emission wavelength spectrum of the quantum dot is within this range, color purity or color reproduction may be improved. In addition, light emitted through such quantum dots is irradiated in omnidirection (e.g., substantially every direction). Accordingly, a wide viewing angle may be increased. In addition, the quantum dot may be, for example, a spherical nanoparticle, a pyramidal nanoparticle, a multi-arm nanoparticle, a cubic nanoparticle, a nanotube particle, a nanowire particle, a nanofiber particle, and/or a nanoplate particle.

Because the energy band gap can be adjusted by controlling the size of the quantum dot, light having various suitable wavelength bands can be obtained from the quantum dot emission layer. Therefore, by using quantum dots of different sizes, a light-emitting device that emits light of various suitable wavelengths may be implemented. In more detail, the size of the quantum dot may be selected to emit red, green and/or blue light. In addition, the size of the quantum dot may be configured to emit white light by combining light of various suitable colors.

#### Electron Transport Region in Interlayer 130

The electron transport region may have: i) a single-layered structure including (e.g., consisting of) a single layer including (e.g., consisting of) a single material, ii) a single-layered structure including (e.g., consisting of) a single layer including a plurality of different materials, or iii) a multi-layered structure including a plurality of layers including different materials.

## 121

The electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

In an embodiment, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein, for each structure, constituting layers are sequentially stacked from an emission layer.

The electron transport region (for example, the buffer layer, the hole blocking layer, the electron control layer, or the electron transport layer in the electron transport region) may include a metal-free compound including at least one  $\pi$ -electron-deficient nitrogen-containing  $C_1$ - $C_{60}$  cyclic group.

In an embodiment, the electron transport region may include the heterocyclic compound represented by Formula 1, a compound represented by Formula 601, or any combination thereof.



Formula 601

In Formula 601,

$Ar_{601}$  and  $L_{601}$  may each independently be a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$  or a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ ,

$xe11$  may be 1, 2, or 3,

$xe1$  may be 0, 1, 2, 3, 4, or 5,

$R_{601}$  may be a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$ , a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ ,  $-Si(Q_{601})(Q_{602})(Q_{603})$ ,  $-C(=O)(Q_{601})$ ,  $-S(=O)_2(Q_{601})$ , or  $-P(=O)(Q_{601})(Q_{602})$ ,

$Q_{601}$  to  $Q_{603}$  are the same as described in connection with  $Q_1$ ,

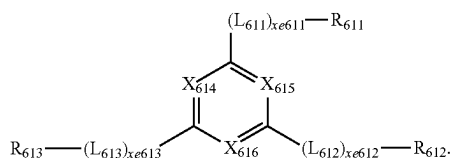
$xe21$  may be 1, 2, 3, 4, or 5, and

at least one selected from  $Ar_{601}$ ,  $L_{601}$ , and  $R_{601}$  may each independently be a  $\pi$ -electron-deficient nitrogen-containing  $C_1$ - $C_{60}$  cyclic group unsubstituted or substituted with at least one  $R_{10a}$ .

In one or more embodiments, when  $xe11$  in Formula 601 is 2 or more, two or more of  $Ar_{601}(s)$  may be linked to each other via a single bond.

In an embodiment,  $Ar_{601}$  in Formula 601 may be a substituted or unsubstituted anthracene group.

In an embodiment, the electron transport region may include a compound represented by Formula 601-1:



Formula 601-1

In Formula 601-1,

$X_{614}$  may be N or C( $R_{614}$ ),  $X_{615}$  may be N or C( $R_{615}$ ),  $X_{616}$  may be N or C( $R_{616}$ ), and at least one selected from  $X_{614}$  to  $X_{616}$  may be N,

$L_{611}$  to  $L_{613}$  may be understood by referring to the description presented in connection with  $L_{601}$ ,

## 122

$xe611$  to  $xe613$  may be understood by referring to the description presented in connection with  $xe1$ ,

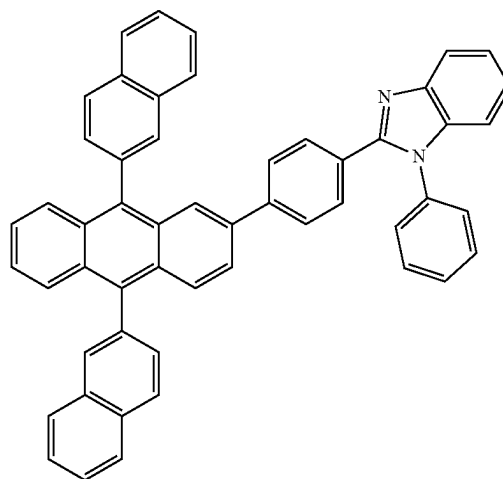
$R_{611}$  to  $R_{613}$  may be understood by referring to the description presented in connection with  $R_{601}$ , and

$R_{614}$  to  $R_{616}$  may each independently be hydrogen, deuterium,  $-F$ ,  $-Cl$ ,  $-Br$ ,  $-I$ , a hydroxyl group, a cyano group, a nitro group, a  $C_1$ - $C_{20}$  alkyl group, a  $C_1$ - $C_{20}$  alkoxy group, a  $C_3$ - $C_{60}$  carbocyclic group unsubstituted or substituted with at least one  $R_{10a}$ , or a  $C_1$ - $C_{60}$  heterocyclic group unsubstituted or substituted with at least one  $R_{10a}$ .

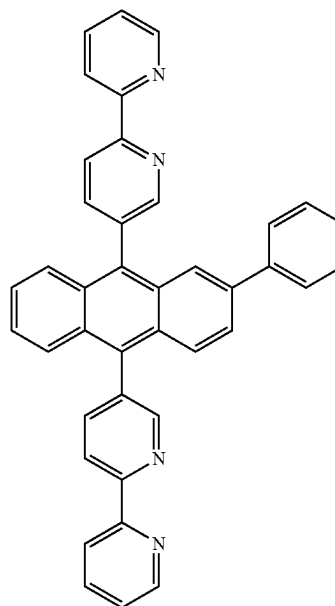
In an embodiment,  $xe1$  and  $xe611$  to  $xe613$  in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

The electron transport region may include one selected from Compounds ET1 to ET45, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq<sub>3</sub>, BALq, TAZ, NTAZ, or any combination thereof:

ET1

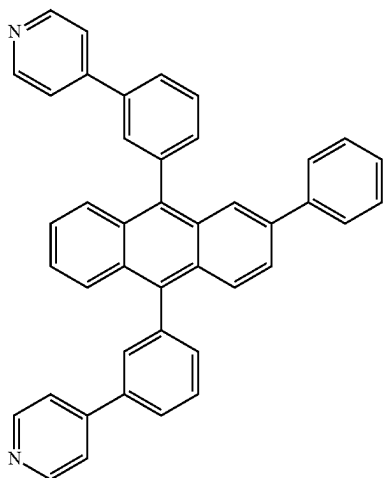


ET2



**123**

-continued



ET3

5

10

15

20

**124**

-continued

5

10

15

20

ET6

5

10

15

20

ET4

25

30

35

40

45

ET5

50

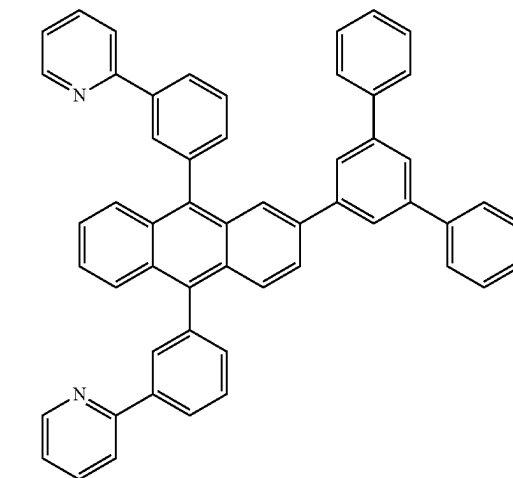
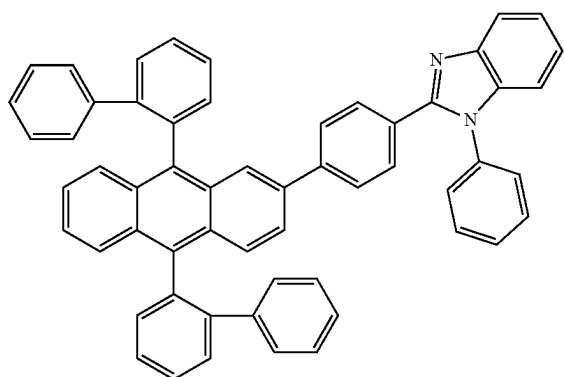
55

60

65

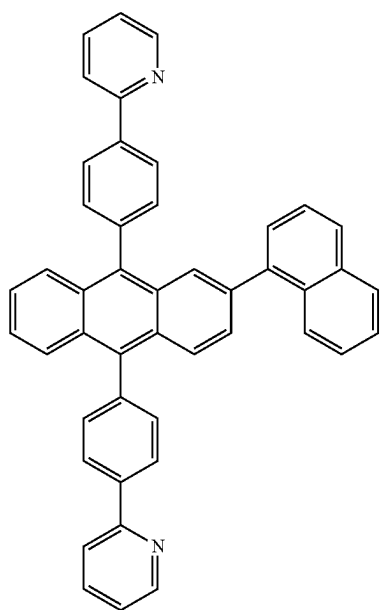
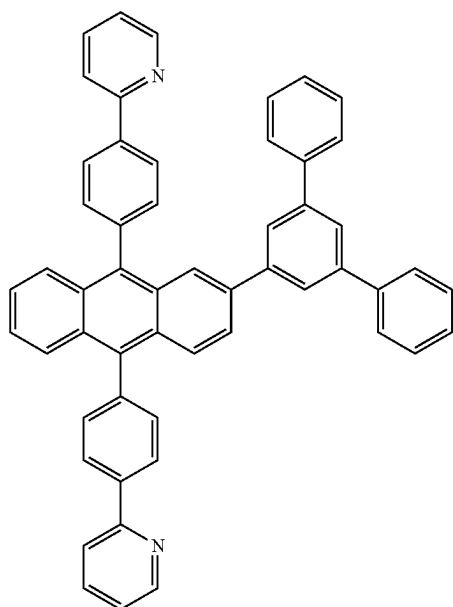
ET7

ET8



**125**

-continued

**126**

-continued

ET9

ET11

5

10

15

20

25

30

35

40

ET10

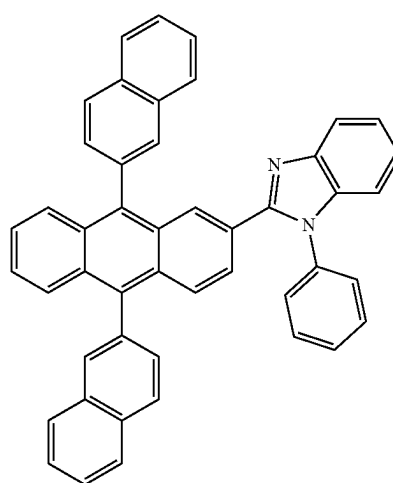
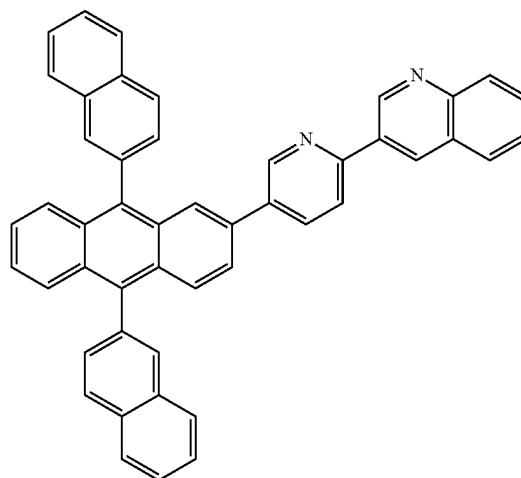
45

50

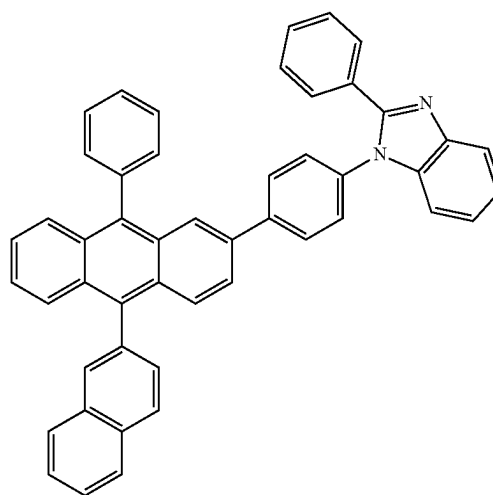
55

60

65



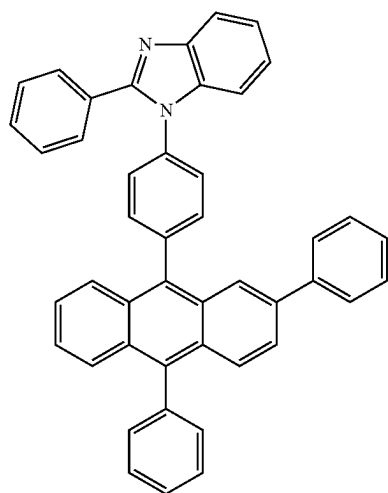
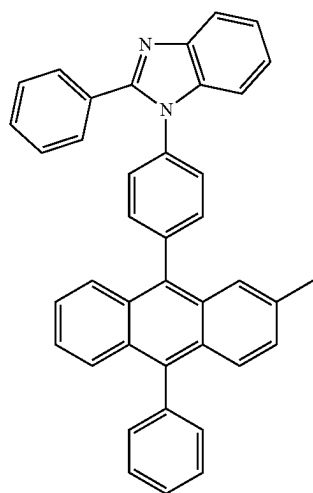
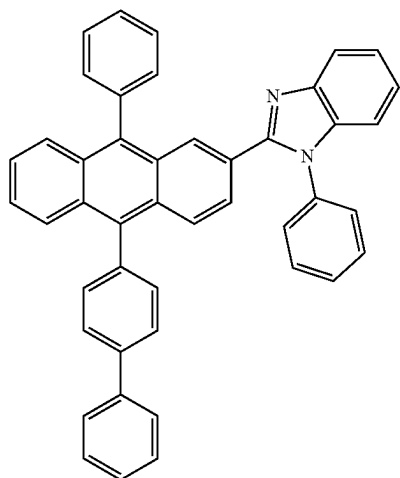
ET12



ET13

**127**

-continued

**128**

-continued

ET14

5

10

15

20

ET15

25

30

35

40

ET16

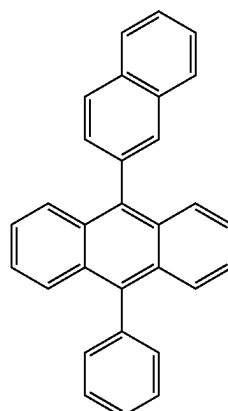
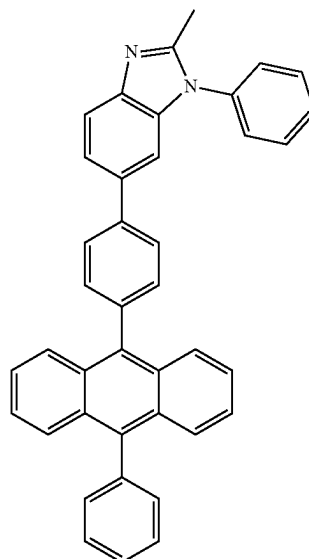
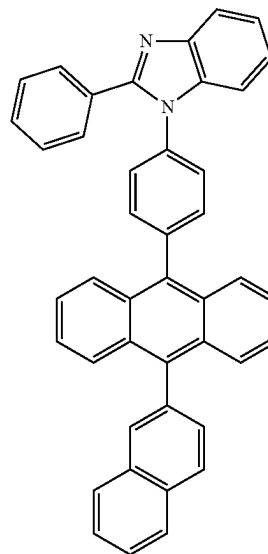
45

50

55

60

65



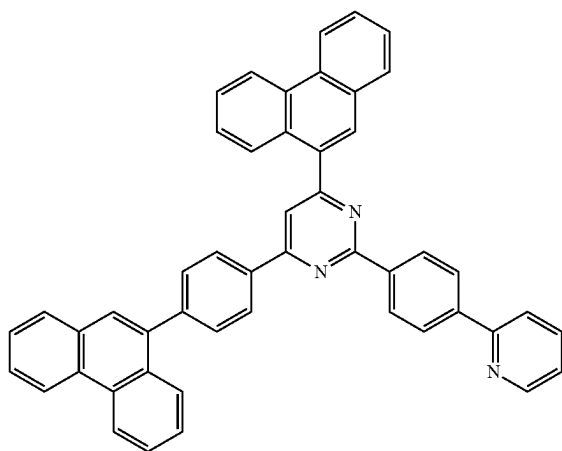
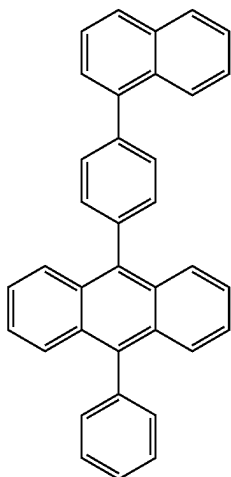
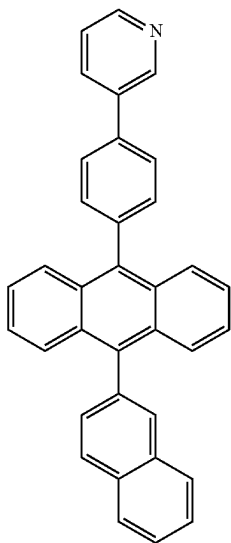
ET17

ET18

ET19

**129**

-continued

**130**

-continued

ET20

ET23

5

10

15

20

25

ET21

ET24

30

35

40

45

ET22

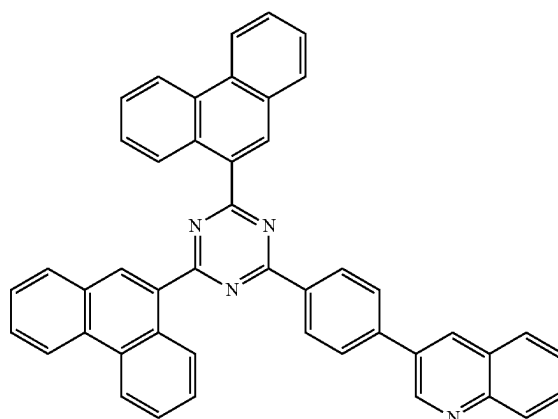
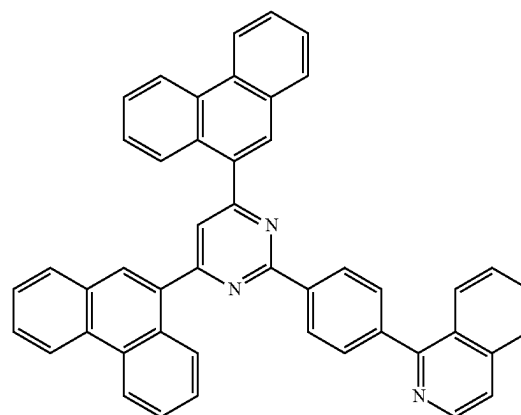
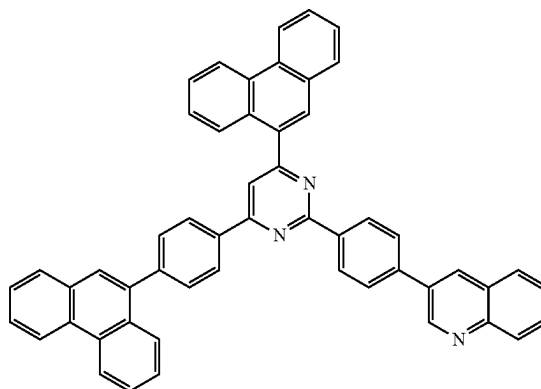
ET25

50

55

60

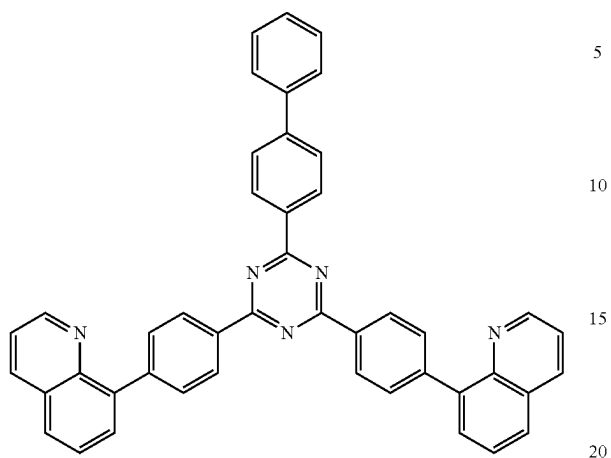
65



**131**

-continued

ET26



5

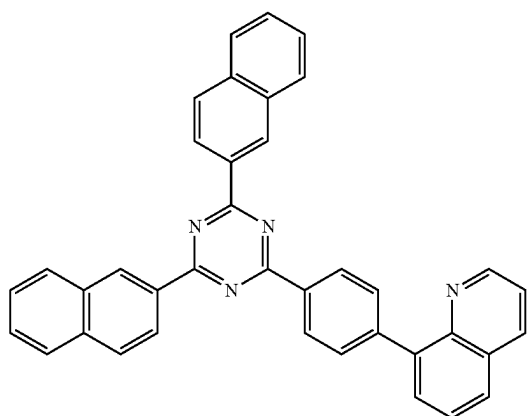
10

15

20

25

ET27



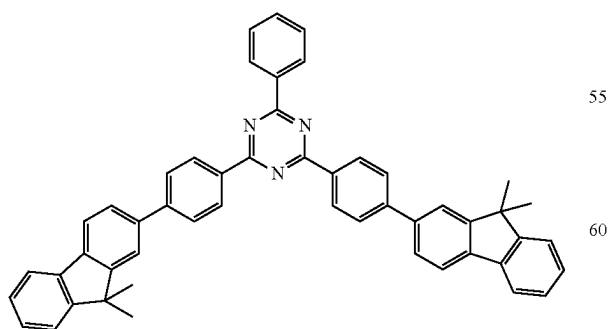
30

35

40

45

ET28



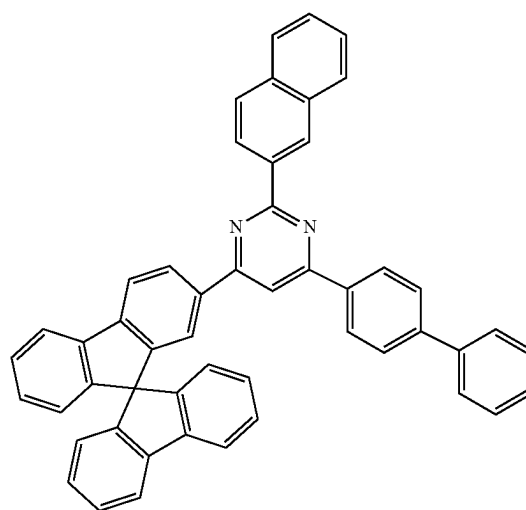
60

65

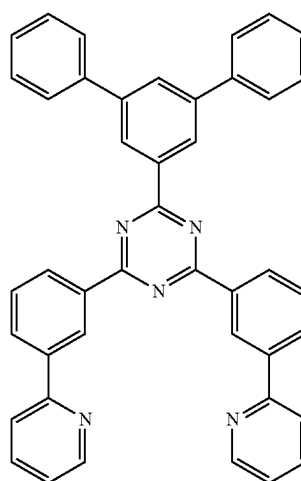
**132**

-continued

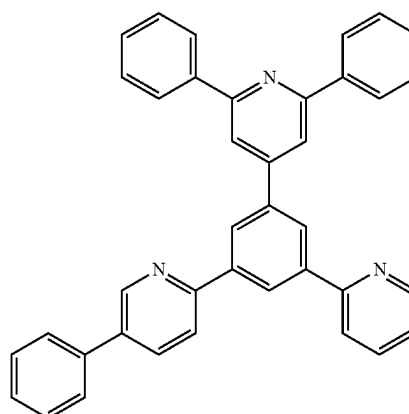
ET29



ET30

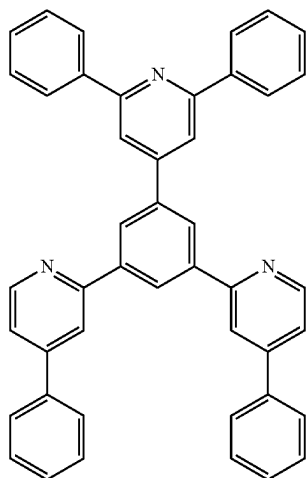


ET31



**133**

-continued



ET32

5

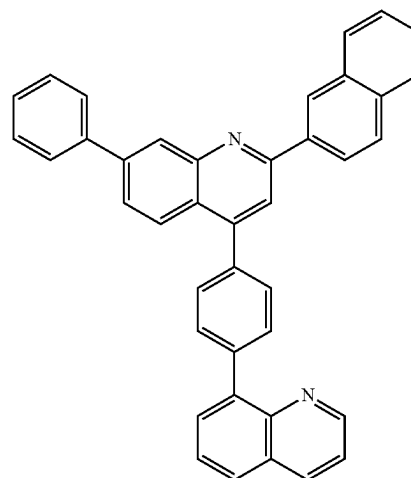
10

15

20

**134**

-continued



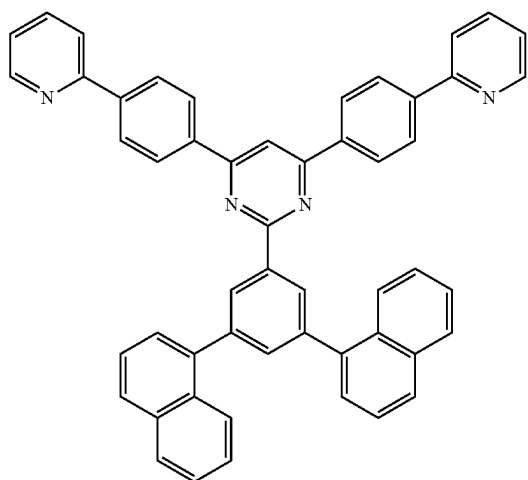
ET35

ET33

25

30

35



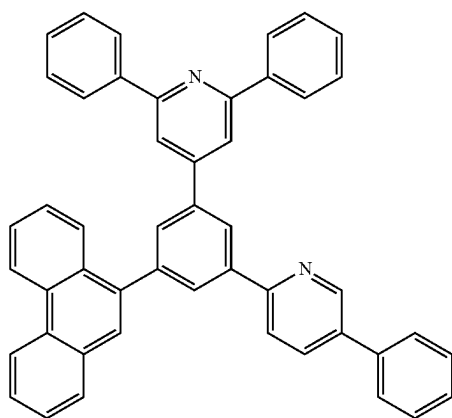
ET36

40

45

ET34

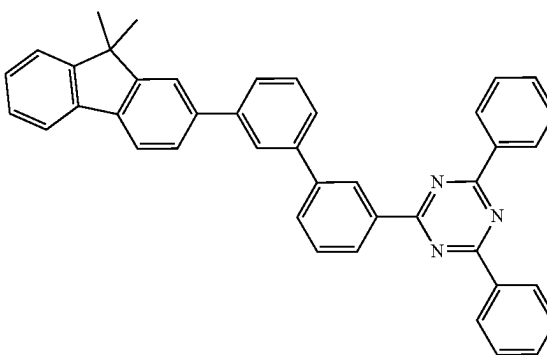
50



55

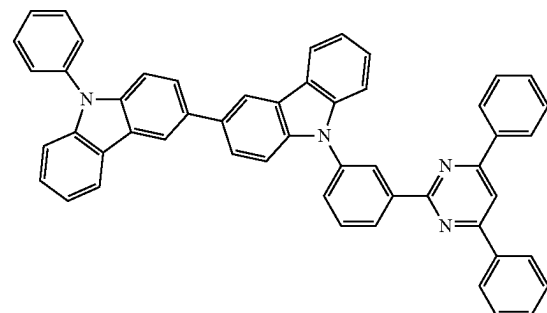
60

65



ET37

ET38

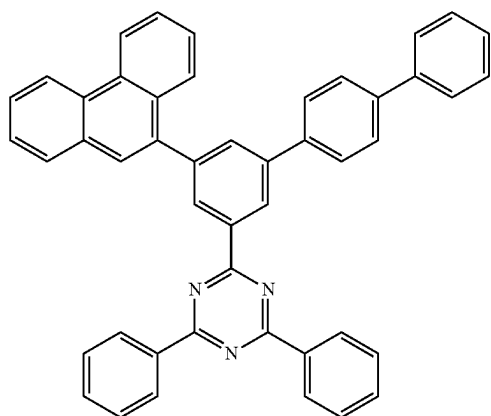




**135**

-continued

ET39

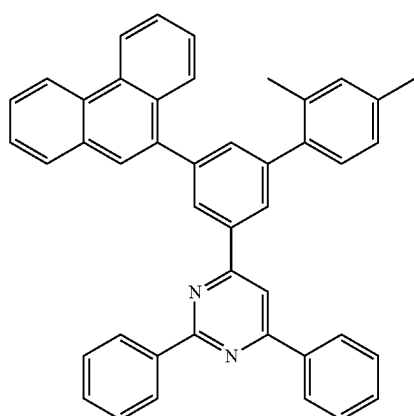


5

10

15

20



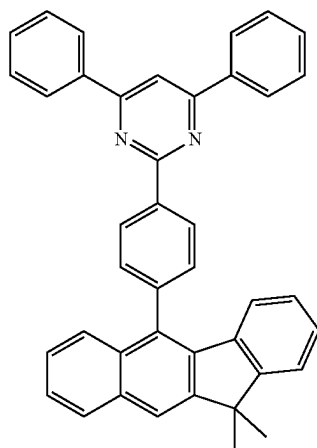
ET40

25

30

35

40



ET41

50

55

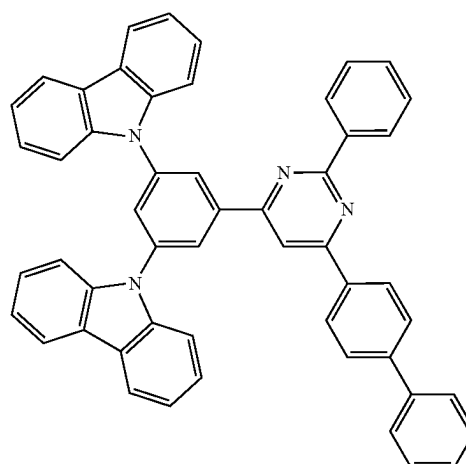
60

65

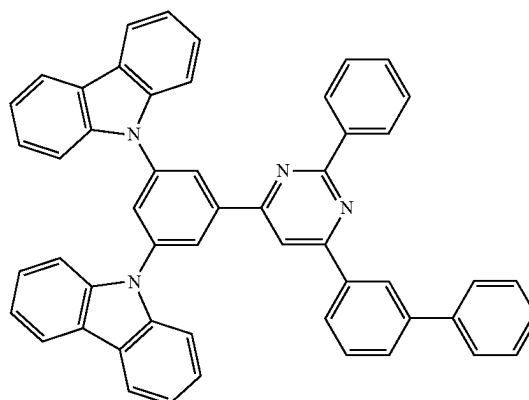
**136**

-continued

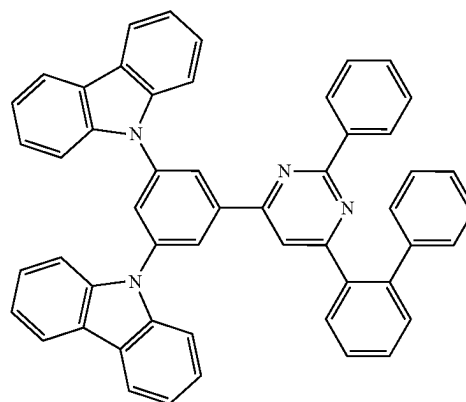
ET42



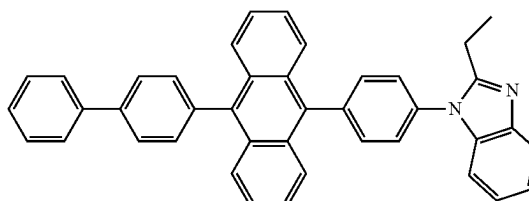
ET43



ET44

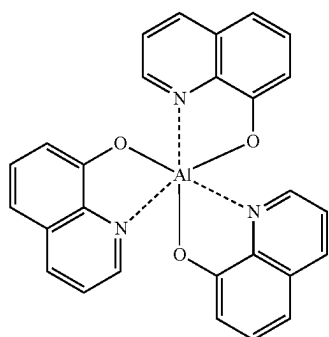
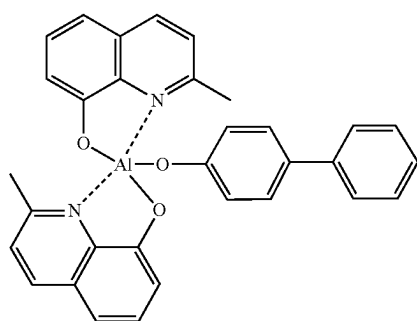


ET45

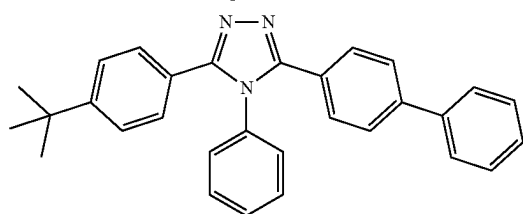


137

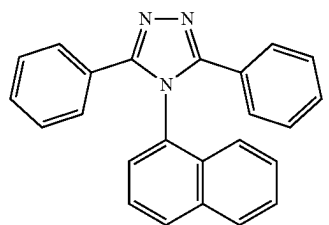
-continued

Alq<sub>3</sub>

BALq



TAZ



NTAZ

A thickness of the electron transport region may be from about 100 Å to about 5,000 Å, for example, about 160 Å to about 4,000 Å. When the electron transport region includes a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, or any combination thereof, a thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be from about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å, and a thickness of the electron transport layer may be from about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thicknesses of the buffer layer, the hole blocking layer, the electron control layer, the electron transport layer, and/or the electron transport region are within these ranges, suitable or satisfactory electron transport characteristics may be obtained without a substantial increase in driving voltage.

138

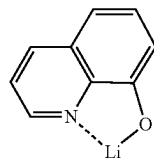
The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

- 5 The metal-containing material may include an alkali metal complex, an alkaline earth-metal complex, or any combination thereof. A metal ion of the alkali metal complex may be a Li ion, a Na ion, a K ion, a Rb ion, or a Cs ion, and a metal ion of the alkaline earth-metal complex may be a Be ion, a Mg ion, a Ca ion, a Sr ion, or a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may each independently be a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenoxazole, a hydroxy phenylthiazole, a hydroxy phenoxadiazole, a hydroxy phenylthiadiazole, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.

20 In an embodiment, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (Liq) or ET-D2:

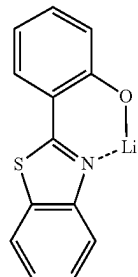
25

ET-D1



30

ET-D2



35

40

- 45 The electron injection layer may include an electron injection layer that facilitates the injection of electrons from the second electrode 150. The electron injection layer may be in direct contact (e.g., physical contact) with the second electrode 150.

- 50 The electron injection layer may have: i) a single-layered structure including (e.g., consisting of) a single layer including (e.g., consisting of) a single material, ii) a single-layered structure including (e.g., consisting of) a single layer including a plurality of different materials, or iii) a multi-layered structure including a plurality of layers including different materials.

The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

60 The alkali metal may include Li, Na, K, Rb, Cs, or any combination thereof. The alkaline earth metal may include Mg, Ca, Sr, Ba, or any combination thereof. The rare earth metal may include Sc, Y, Ce, Tb, Yb, Gd, or any combination thereof.

The alkali metal-containing compound, the alkaline earth metal-containing compound, and the rare earth metal-containing compound may be oxides and/or halides (for example, fluorides, chlorides, bromides, and/or iodides) of the alkali metal, the alkaline earth metal, and the rare earth metal, telluride, or any combination thereof.

The alkali metal-containing compound may be alkali metal oxides, such as  $\text{Li}_2\text{O}$ ,  $\text{Cs}_2\text{O}$ , and/or  $\text{K}_2\text{O}$ , alkali metal halides, such as  $\text{LiF}$ ,  $\text{NaF}$ ,  $\text{CsF}$ ,  $\text{KF}$ ,  $\text{LiI}$ ,  $\text{NaI}$ ,  $\text{CsI}$ , and/or  $\text{KI}$ , or any combination thereof. The alkaline earth metal-containing compound may include an alkaline earth metal oxide, such as  $\text{BaO}$ ,  $\text{SrO}$ ,  $\text{CaO}$ ,  $\text{Ba}_x\text{Sr}_{1-x}\text{O}$  ( $x$  is a real number that satisfies the condition of  $0 < x < 1$ ), and/or  $\text{Ba}_x\text{Ca}_{1-x}\text{O}$  ( $x$  is a real number that satisfies the condition of  $0 < x < 1$ ). The rare earth metal-containing compound may include  $\text{YbF}_3$ ,  $\text{ScF}_3$ ,  $\text{Sc}_2\text{O}_3$ ,  $\text{Y}_2\text{O}_3$ ,  $\text{Ce}_2\text{O}_3$ ,  $\text{GdF}_3$ ,  $\text{TbF}_3$ ,  $\text{YbI}_3$ ,  $\text{ScI}_3$ ,  $\text{TbI}_3$ , or any combination thereof. In an embodiment, the rare earth metal-containing compound may include lanthanide metal telluride. Examples of the lanthanide metal telluride may include  $\text{LaTe}$ ,  $\text{CeTe}$ ,  $\text{PrTe}$ ,  $\text{NdTe}$ ,  $\text{PmTe}$ ,  $\text{SmTe}$ ,  $\text{EuTe}$ ,  $\text{GdTe}$ ,  $\text{TbTe}$ ,  $\text{DyTe}$ ,  $\text{HoTe}$ ,  $\text{ErTe}$ ,  $\text{TmTe}$ ,  $\text{YbTe}$ ,  $\text{LuTe}$ ,  $\text{La}_2\text{Te}_3$ ,  $\text{Ce}_2\text{Te}_3$ ,  $\text{Pr}_2\text{Te}_3$ ,  $\text{Nd}_2\text{Te}_3$ ,  $\text{Pm}_2\text{Te}_3$ ,  $\text{Sm}_2\text{Te}_3$ ,  $\text{Eu}_2\text{Te}_3$ ,  $\text{Gd}_2\text{Te}_3$ ,  $\text{Tb}_2\text{Te}_3$ ,  $\text{Dy}_2\text{Te}_3$ ,  $\text{Ho}_2\text{Te}_3$ ,  $\text{Er}_2\text{Te}_3$ ,  $\text{Tm}_2\text{Te}_3$ ,  $\text{Yb}_2\text{Te}_3$ , and  $\text{Lu}_2\text{Te}_3$ .

The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include i) one selected from metal ions of the alkali metal, the alkaline earth metal, and the rare earth metal and ii) as a ligand linked to the metal ion, for example, a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenylloxazole, a hydroxy phenylthiazole, a hydroxy phenylloxadiazole, a hydroxy phenylthiadiazole, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.

The electron injection layer may include (e.g., consist of) an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof, or may further include an organic material (for example, a compound represented by Formula 601).

In an embodiment, the electron injection layer may include (e.g., consist of) i) an alkali metal-containing compound (for example, an alkali metal halide), or ii) a) an alkali metal-containing compound (for example, an alkali metal halide); and b) alkali metal, alkaline earth metal, rare earth metal, or any combination thereof. In an embodiment, the electron injection layer may be a  $\text{KI}:\text{Yb}$  co-deposited layer and/or a  $\text{RbI}:\text{Yb}$  co-deposited layer.

When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described above, the electron

injection layer may have suitable or satisfactory electron injection characteristics without a substantial increase in driving voltage.

#### Second Electrode 150

The second electrode 150 may be on the interlayer 130 having such a structure. The second electrode 150 may be a cathode, which is an electron injection electrode, and as a material for forming the second electrode 150, a metal, an alloy, an electrically conductive compound, or any combination thereof, each having a low work function, may be utilized.

The second electrode 150 may include lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ytterbium (Yb), silver-ytterbium (Ag—Yb), ITO, IZO, or any combination thereof. The second electrode 150 may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

The second electrode 150 may have a single-layered structure or a multi-layered structure including two or more layers.

#### Capping Layer

A first capping layer may be outside the first electrode 110, and/or a second capping layer may be outside the second electrode 150. In more detail, the light-emitting device 10 may have a structure in which the first capping layer, the first electrode 110, the interlayer 130, and the second electrode 150 are sequentially stacked in this stated order, a structure in which the first electrode 110, the interlayer 130, the second electrode 150, and the second capping layer are sequentially stacked in this stated order, or a structure in which the first capping layer, the first electrode 110, the interlayer 130, the second electrode 150, and the second capping layer are sequentially stacked in this stated order.

Light generated in an emission layer of the interlayer 130 of the light-emitting device 10 may be extracted toward the outside through the first electrode 110, which is a semi-transmissive electrode or a transmissive electrode, and the first capping layer, and light generated in an emission layer of the interlayer 130 of the light-emitting device 10 may be extracted toward the outside through the second electrode 150, which is a semi-transmissive electrode or a transmissive electrode, and the second capping layer.

The first capping layer and the second capping layer may increase external luminescence efficiency according to the principle of constructive interference. Accordingly, the optical extraction efficiency of the light-emitting device 10 is increased, so that the luminescence efficiency of the light-emitting device 10 may be improved.

Each of the first capping layer and the second capping layer may include a material having a refractive index of 1.6 or more (at a wavelength of 589 nm).

The first capping layer and the second capping layer may each independently be an organic capping layer including an organic material, an inorganic material including an inorganic material, or an organic-inorganic composite capping layer including an organic material and an inorganic material.

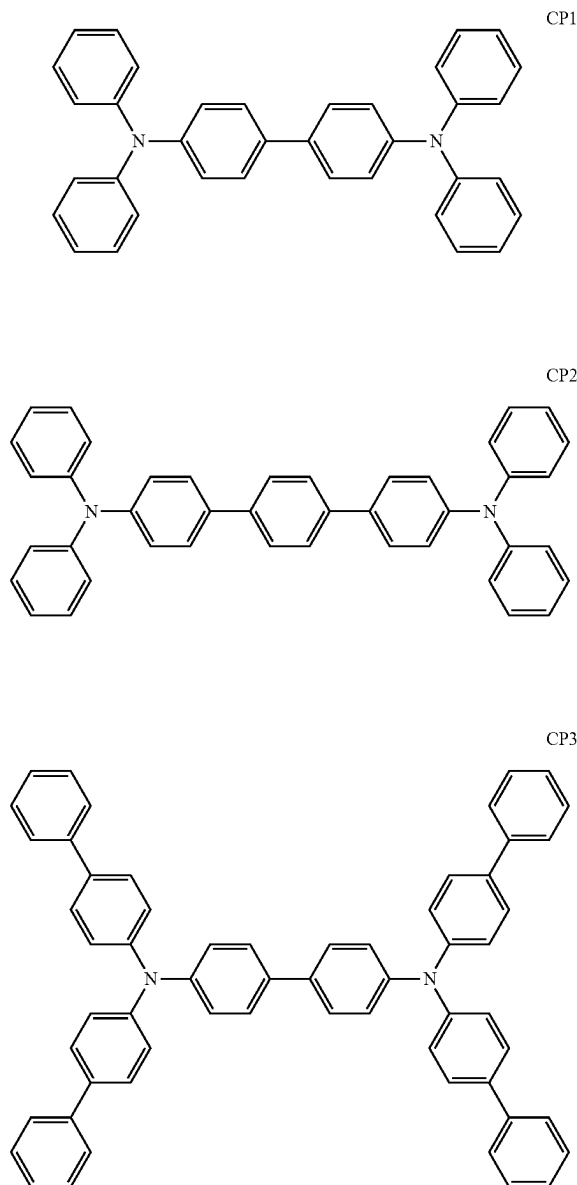
At least one selected from the first capping layer and the second capping layer may each independently include a carbocyclic compound, a heterocyclic compound, an amine group-containing compound, a porphyrine derivative, a phthalocyanine derivative, a naphthalocyanine derivative, an alkali metal complex, an alkaline earth-metal complex, or any combination thereof. The carbocyclic compound, the heterocyclic compound, and the amine group-containing

## 141

compound may be optionally substituted with a substituent containing O, N, S, Se, Si, F, Cl, Br, I, or any combination thereof. In an embodiment, at least one selected from the first capping layer and the second capping layer may each independently include an amine group-containing compound.

In an embodiment, at least one selected from the first capping layer and second capping layer may each independently include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof.

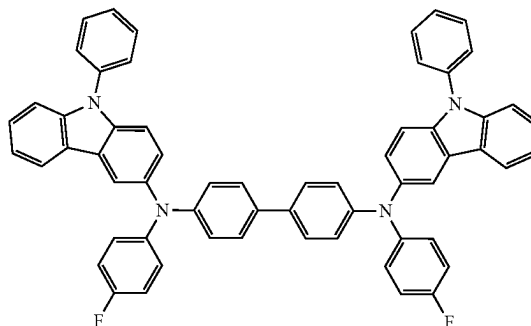
In one or more embodiments, at least one selected from the first capping layer and the second capping layer may each independently include one selected from Compounds HT28 to HT33, one selected from Compounds CP1 to CP6,  $\beta$ -NPB, or any combination thereof:



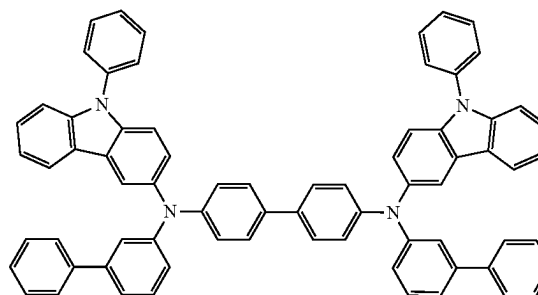
## 142

-continued

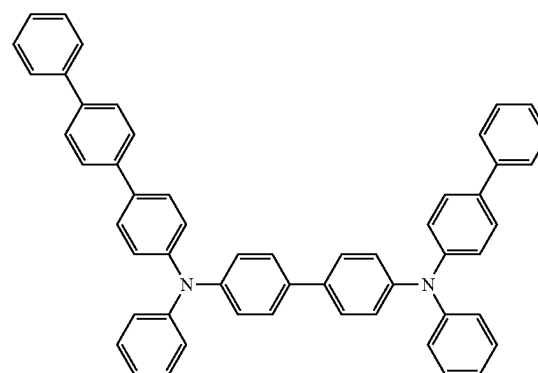
CP4



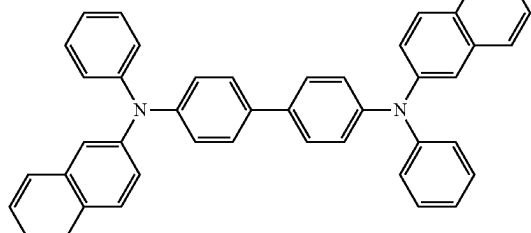
CP5



CP6



$\beta$ -NPB



#### Electronic Apparatus

The light-emitting device may be included in various suitable electronic apparatuses. In an embodiment, the electronic apparatus including the light-emitting device may be a light-emitting apparatus, an authentication apparatus, and/or the like.

The electronic apparatus (for example, light-emitting apparatus) may further include, in addition to the light-

emitting device, i) a color filter, ii) a color conversion layer, or iii) a color filter and a color conversion layer. The color filter and/or the color conversion layer may be in at least one traveling direction of light emitted from the light-emitting device. In an embodiment, light emitted from the light-emitting device may be blue light and/or white light. The light-emitting device may be the same as described above. In an embodiment, the color conversion layer may include a quantum dot. The quantum dot may be, for example, a quantum dot as described elsewhere herein.

The electronic apparatus may include a first substrate. The first substrate includes a plurality of subpixel areas, the color filter includes a plurality of color filter areas corresponding to the plurality of subpixel areas, respectively, and the color conversion layer may include a plurality of color conversion areas corresponding to the plurality of subpixel areas, respectively.

A pixel-defining layer may be between the plurality of subpixel areas to define each of the subpixel areas.

The color filter may further include a plurality of color filter areas and a light-blocking pattern between the plurality of color filter areas, and the color conversion layer may further include a plurality of color conversion areas and a light-blocking pattern between the plurality of color conversion areas.

The plurality of color filter areas (or the plurality of color conversion areas) may include a first area emitting first color light, a second area emitting second color light, and/or a third area emitting third color light, and the first color light, the second color light, and/or the third color light may have different maximum emission wavelengths from one another. In an embodiment, the first color light may be red light, the second color light may be green light, and the third color light may be blue light. In an embodiment, the plurality of color filter areas (or the plurality of color conversion areas) may include a quantum dot. In more detail, the first area may include a red quantum dot, the second area may include a green quantum dot, and the third area may not include a quantum dot. The quantum dot is the same as described elsewhere in the present specification. The first area, the second area and/or the third area may further include a scatterer.

In an embodiment, the light-emitting device may emit a first light, the first area may absorb the first light to emit a first first-color light, the second area may absorb the first light to emit a second first-color light, and the third area may absorb the first light to emit a third first-color light. In this regard, the first first-color light, the second first-color light, and the third first-color light may have different maximum emission wavelengths from one another. In more detail, the first light may be blue light, the first first-color light may be red light, the second first-color light may be green light, and the third first-color light may be blue light.

The electronic apparatus may further include a thin-film transistor in addition to the light-emitting device as described above. The thin-film transistor may include a source electrode, a drain electrode, and an active layer, wherein any one selected from the source electrode and the drain electrode may be electrically coupled to any one selected from the first electrode and the second electrode of the light-emitting device.

The thin-film transistor may further include a gate electrode, a gate insulating film, and/or the like.

The active layer may include crystalline silicon, amorphous silicon, organic semiconductor, oxide semiconductor, and/or the like.

The electronic apparatus may further include a sealing portion for sealing the light-emitting device. The sealing portion may be between the color filter and/or the color conversion layer and the light-emitting device. The sealing portion allows light from the light-emitting device to be extracted to the outside, while concurrently (e.g., simultaneously) preventing or reducing penetration of ambient air and/or moisture into the light-emitting device. The sealing portion may be a sealing substrate including a transparent glass substrate and/or a plastic substrate. The sealing portion may be a thin film encapsulation layer including at least one layer of an organic layer and an inorganic layer. When the sealing portion is a thin film encapsulation layer, the electronic apparatus may be flexible.

On the sealing portion, in addition to the color filter and/or the color conversion layer, various suitable functional layers may be further included according to the use of the electronic apparatus. The functional layers may include a touch screen layer, a polarizing layer, and/or the like. The touch screen layer may be a pressure-sensitive touch screen layer, a capacitive touch screen layer, and/or an infrared touch screen layer. The authentication apparatus may be, for example, a biometric authentication apparatus for authenticating an individual by using biometric information of a biometric body (for example, a finger tip, a pupil, and/or the like).

The authentication apparatus may further include, in addition to the light-emitting device, a biometric information collector.

The electronic apparatus may be applied to various suitable displays, light sources, lighting, personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram displays, ultrasonic diagnostic devices, and/or endoscope displays), fish finders, various suitable measuring instruments, meters (for example, meters for a vehicle, an aircraft, and/or a vessel), projectors, and/or the like.

#### Description of FIGS. 2 and 3

FIG. 2 is a cross-sectional view of a light-emitting apparatus according to an embodiment.

The light-emitting apparatus of FIG. 2 includes a substrate **100**, a thin-film transistor (TFT), a light-emitting device, and an encapsulation portion **300** that seals the light-emitting device.

The substrate **100** may be a flexible substrate, a glass substrate, and/or a metal substrate. A buffer layer **210** may be on the substrate **100**. The buffer layer **210** prevents or reduces the penetration of impurities through the substrate **100** and may provide a flat surface on the substrate **100**.

A TFT may be on the buffer layer **210**. The TFT may include an active layer **220**, a gate electrode **240**, a source electrode **260**, and a drain electrode **270**.

The active layer **220** may include an inorganic semiconductor such as silicon and/or polysilicon, an organic semiconductor, and/or an oxide semiconductor, and may include a source region, a drain region, and a channel region.

A gate insulating film **230** for insulating the active layer **220** from the gate electrode **240** may be on the active layer **220**, and the gate electrode **240** may be on the gate insulating film **230**.

An interlayer insulating film **250** may be on the gate electrode **240**. The interlayer insulating film **250** is between the gate electrode **240** and the source electrode **260** to

145

insulate the gate electrode **240** from the source electrode **260** and between the gate electrode **240** and the drain electrode **270** to insulate the gate electrode **240** from the drain electrode **270**.

The source electrode **260** and the drain electrode **270** may be on the interlayer insulating film **250**. The interlayer insulating film **250** and the gate insulating film **230** may be formed to expose the source region and the drain region of the active layer **220**, and the source electrode **260** and the drain electrode **270** may be in contact with the exposed portions of the source region and the drain region of the active layer **220**.

The TFT may be electrically coupled to a light-emitting device to drive the light-emitting device, and is covered by a passivation layer **280**. The passivation layer **280** may include an inorganic insulating film, an organic insulating film, or a combination thereof. A light-emitting device is provided on the passivation layer **280**. The light-emitting device includes the first electrode **110**, the interlayer **130**, and the second electrode **150**.

The first electrode **110** may be on the passivation layer **280**. The passivation layer **280** may not completely cover the drain electrode **270** and expose a portion of the drain electrode **270**, and the first electrode **110** may be coupled to the exposed portion of the drain electrode **270**.

A pixel-defining layer **290** including an insulating material may be on the first electrode **110**. The pixel-defining layer **290** may expose a portion of the first electrode **110**, and the interlayer **130** may be formed in the exposed portion of the first electrode **110**. The pixel-defining layer **290** may be a polyimide and/or polyacrylic organic film. In one or more embodiments, at least some layers of the interlayer **130** may extend beyond the upper portion of the pixel-defining layer **290** and may thus be in the form of a common layer.

The second electrode **150** may be on the interlayer **130**, and a capping layer **170** may be additionally formed on the second electrode **150**. The capping layer **170** may cover the second electrode **150**.

The encapsulation portion **300** may be on the capping layer **170**. The encapsulation portion **300** may be on a light-emitting device and protects the light-emitting device from moisture and/or oxygen. The encapsulation portion **300** may include: an inorganic film including silicon nitride ( $\text{SiN}_x$ ), silicon oxide ( $\text{SiO}_x$ ), indium tin oxide, indium zinc oxide, or a combination thereof; an organic film including polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyimide, polyethylene sulfonate, polyoxymethylene, polyarylate, hexamethyldisiloxane, an acrylic resin (for example, polymethyl methacrylate and/or polyacrylic acid), an epoxy-based resin (for example, aliphatic glycidyl ether (AGE)), or a combination thereof; or a combination of an inorganic film and an organic film.

FIG. **3** is a cross-sectional view showing a light-emitting apparatus according to an embodiment of the present disclosure.

The light-emitting apparatus of FIG. **3** is the same as the light-emitting apparatus of FIG. **2**, except that a light-blocking pattern **500** and a functional region **400** are additionally on the encapsulation portion **300**. The functional region **400** may be i) a color filter area, ii) a color conversion area, or iii) a combination of the color filter area and the color conversion area. In an embodiment, the light-emitting device included in the light-emitting apparatus of FIG. **3** may be a tandem light-emitting device.

#### Preparation Method

Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region

146

may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about  $100^\circ\text{C}$ . to about  $500^\circ\text{C}$ ., a vacuum degree of about  $10^{-8}$  torr to about  $10^{-3}$  torr, and a deposition speed of about  $0.01\text{ \AA/sec}$  to about  $100\text{ \AA/sec}$  by taking into account a material to be included in a layer to be formed and the structure of a layer to be formed.

#### Definition of at Least Some of the Terms

The term " $\text{C}_3\text{-C}_{60}$  carbocyclic group," as used herein, refers to a cyclic group that includes (e.g. consists of) only carbon as a ring-forming atom and has three to sixty carbon atoms, and the term " $\text{C}_1\text{-C}_{60}$  heterocyclic group," as used herein, refers to a cyclic group that has one to sixty carbon atoms and further has, in addition to carbon, a heteroatom as a ring-forming atom. The  $\text{C}_3\text{-C}_{60}$  carbocyclic group and the  $\text{C}_1\text{-C}_{60}$  heterocyclic group may each be a monocyclic group that includes (e.g. consists of) one ring or a polycyclic group in which two or more rings are condensed with (e.g., combined together with) each other. In an embodiment, the number of ring-forming atoms of the  $\text{C}_1\text{-C}_{60}$  heterocyclic group may be from 3 to 61.

The term "cyclic group," as used herein, includes the  $\text{C}_3\text{-C}_{60}$  carbocyclic group and the  $\text{C}_1\text{-C}_{60}$  heterocyclic group.

The term " $\pi$  electron-rich  $\text{C}_3\text{-C}_{60}$  cyclic group," as used herein, refers to a cyclic group that has three to sixty carbon atoms and does not include  $\text{*—N=*}$  as a ring-forming moiety, and the term " $\pi$ -electron-deficient nitrogen-containing  $\text{C}_1\text{-C}_{60}$  cyclic group," as used herein, refers to a heterocyclic group that has one to sixty carbon atoms and includes  $\text{*—N=*}$  as a ring-forming moiety.

In an embodiment,

the  $\text{C}_3\text{-C}_{60}$  carbocyclic group may be i) a group T1 or ii) a condensed cyclic group in which two or more groups T1 are condensed with (e.g., combined together with) each other (for example, a cyclopentadiene group, an adamantane group, a norbornane group, a benzene group, a pentalene group, a naphthalene group, an azulene group, an indacene group, acenaphthenyl group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a perylene group, a pentaphene group, a heptalene group, a naphthacene group, a picene group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, an indenophenanthrene group, or an indenoanthracene group),

the  $\text{C}_1\text{-C}_{60}$  heterocyclic group may be i) a group T2, ii) a condensed cyclic group in which two or more groups T2 are condensed with (e.g., combined together with) each other, or iii) a condensed cyclic group in which at least one groups T2 and at least one group T1 are condensed with (e.g., combined together with) each other (for example, a pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a

147

carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothieno dibenzothiophene group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, or an azadibenzofuran group),

the  $\pi$  electron-rich  $C_3$ - $C_{60}$  cyclic group may be i) a group T1, ii) a condensed cyclic group in which two or more groups T1 are condensed with (e.g., combined together with) each other, iii) a group T3, iv) a condensed cyclic group in which two or more groups T3 are condensed with (e.g., combined together with) each other, or v) a condensed cyclic group in which at least one group T3 and at least one group T1 are condensed with (e.g., combined together with) each other (for example, the  $C_3$ - $C_{60}$  carbocyclic group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, etc.),

the  $\pi$ -electron-deficient nitrogen-containing  $C_1$ - $C_{60}$  cyclic group may be i) a group T4, ii) a condensed cyclic group in which two or more groups T4 are condensed with (e.g., combined together with) each other, iii) a condensed cyclic group in which at least one group T4 and at least one group T1 are condensed with (e.g., combined together with) each other, iv) a condensed cyclic group in which at least one group T4 and at least one group T3 are condensed with (e.g., combined together with) each other, or v) a condensed cyclic group in which at least one group T4, at least one group T1, and at least one group T3 are condensed with (e.g., combined together with) each other (for example, a

148

pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, or an azadibenzofuran group),

the group T1 may be a cyclopropane group, a cyclobutane group, a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclobutene group, a cyclopentene group, a cyclopentadiene group, a cyclohexene group, a cyclohexadiene group, a cycloheptene group, an adamantane group, a norbornane group (or, a bicyclo[2.2.1]heptane group), a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, or a benzene group,

the group T2 may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a tetrazine group, a pyrrolidine group, an imidazolidine group, a dihydropyrrole group, a piperidine group, a tetrahydropyridine group, a dihydropyridine group, a hexahydropyrimidine group, a tetrahydro pyrimidine group, a dihydropyrimidine group, a piperazine group, a tetrahydropyrazine group, a dihydropyrazine group, a tetrahydropyridazine group, or a dihydropyridazine group,

the group T3 may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, or a borole group, and

the group T4 may be a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a tetrazine group.

The terms "the cyclic group," "the  $C_3$ - $C_{60}$  carbocyclic group," "the  $C_1$ - $C_{60}$  heterocyclic group," "the  $\pi$  electron-rich  $C_3$ - $C_{60}$  cyclic group," or "the  $\pi$  electron-deficient nitrogen-containing  $C_1$ - $C_{60}$  cyclic group," as used herein, refer to a monovalent or polyvalent group (for example, a divalent group, a trivalent group, a tetravalent group, or the like) that is condensed with (e.g., combined together with) a cyclic group, according to the structure of a formula described with corresponding terms. In an embodiment, "a benzene group" may be a benzo group, a phenyl group, a phenylene group, or the like, which may be easily understood by one of

ordinary skill in the art according to the structure of a formula including the "benzene group."

In an embodiment, examples of the monovalent  $C_3$ - $C_{60}$  carbocyclic group and the monovalent  $C_1$ - $C_{60}$  heterocyclic group may include a  $C_3$ - $C_{10}$  cycloalkyl group, a  $C_1$ - $C_{10}$  heterocycloalkyl group, a  $C_3$ - $C_{10}$  cycloalkenyl group, a  $C_1$ - $C_{10}$  heterocycloalkenyl group, a  $C_6$ - $C_{60}$  aryl group, a  $C_1$ - $C_{60}$  heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, and examples of the divalent  $C_3$ - $C_{60}$  carbocyclic group and the divalent  $C_1$ - $C_{60}$  heterocyclic group may include a  $C_3$ - $C_{10}$  cycloalkylene group, a  $C_1$ - $C_{10}$  heterocycloalkylene group, a  $C_3$ - $C_{10}$  cycloalkenylene group, a  $C_1$ - $C_{10}$  heterocycloalkenylene group, a  $C_6$ - $C_{60}$  arylene group, a  $C_1$ - $C_{60}$  heteroarylene group, a divalent non-aromatic condensed polycyclic group, and divalent non-aromatic condensed heteropolycyclic group.

The term " $C_1$ - $C_{60}$  alkyl group," as used herein, refers to a linear or branched aliphatic hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, and a tert-decyl group. The term " $C_1$ - $C_{60}$  alkylene group," as used herein, refers to a divalent group having substantially the same structure as the  $C_1$ - $C_{60}$  alkyl group.

The term " $C_2$ - $C_{60}$  alkenyl group," as used herein, refers to a monovalent hydrocarbon group having at least one carbon-carbon double bond at a main chain (e.g., in the middle) or at a terminal end (e.g., the terminus) of a  $C_2$ - $C_{60}$  alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term " $C_2$ - $C_{60}$  alkenylene group," as used herein, refers to a divalent group having substantially the same structure as the  $C_2$ - $C_{60}$  alkenyl group.

The term " $C_2$ - $C_{60}$  alkynyl group," as used herein, refers to a monovalent hydrocarbon group having at least one carbon-carbon triple bond at a main chain (e.g., in the middle) or at a terminal end (e.g., the terminus) of a  $C_2$ - $C_{60}$  alkyl group, and examples thereof include an ethynyl group and a propynyl group. The term " $C_2$ - $C_{60}$  alkynylene group," as used herein, refers to a divalent group having substantially the same structure as the  $C_2$ - $C_{60}$  alkynyl group.

The term " $C_1$ - $C_{60}$  alkoxy group," as used herein, refers to a monovalent group represented by  $-OA_{101}$  (wherein  $A_{101}$  is the  $C_1$ - $C_{60}$  alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

The term " $C_3$ - $C_{10}$  cycloalkyl group," as used herein, refers to a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or a bicyclo[2.2.1]heptyl group), a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, and a bicyclo[2.2.2]octyl group. The term " $C_3$ - $C_{10}$  cycloalkylene group," as used herein,

refers to a divalent group having substantially the same structure as the  $C_3$ - $C_{10}$  cycloalkyl group.

The term " $C_1$ - $C_{10}$  heterocycloalkyl group," as used herein, refers to a monovalent cyclic group that further includes, in addition to a carbon atom, at least one heteroatom as a ring-forming atom and has 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranlyl group, and a tetrahydrothienyl group. The term " $C_1$ - $C_{10}$  heterocycloalkylene group," as used herein, refers to a divalent group having substantially the same structure as the  $C_1$ - $C_{10}$  heterocycloalkyl group.

The term " $C_3$ - $C_{10}$  cycloalkenyl group," as used herein, refers to a monovalent cyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity (e.g., is not aromatic), and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term " $C_3$ - $C_{10}$  cycloalkenylene group," as used herein, refers to a divalent group having substantially the same structure as the  $C_3$ - $C_{10}$  cycloalkenyl group.

The term " $C_1$ - $C_{10}$  heterocycloalkenyl group," as used herein, refers to a monovalent cyclic group that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, 1 to 10 carbon atoms, and at least one double bond in the cyclic structure thereof. Examples of the  $C_1$ - $C_{10}$  heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranlyl group, and a 2,3-dihydrothienyl group. The term " $C_1$ - $C_{10}$  heterocycloalkenylene group," as used herein, refers to a divalent group having substantially the same structure as the  $C_1$ - $C_{10}$  heterocycloalkenyl group.

The term " $C_6$ - $C_{60}$  aryl group," as used herein, refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term " $C_6$ - $C_{60}$  arylene group," as used herein, refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Examples of the  $C_6$ - $C_{60}$  aryl group include a fluorenyl group, a phenyl group, a pentalenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a heptalenyl group, a naphthacenyl group, a picenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, and an ovalenyl group. When the  $C_6$ - $C_{60}$  aryl group and the  $C_6$ - $C_{60}$  arylene group each include two or more rings, the two or more rings may be fused to each other (e.g., combined together with each other).

The term " $C_1$ - $C_{60}$  heteroaryl group," as used herein, refers to a monovalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. The term " $C_1$ - $C_{60}$  heteroarylene group," as used herein, refers to a divalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. Examples of the  $C_1$ - $C_{60}$  heteroaryl group include a carbazolyl group, a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, a benzoquinolinyl group, an isoquinolinyl group, a benzoisoquinolinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthrolinyl group, a phthalazinyl group, and a naphthyridinyl group. When the  $C_1$ - $C_{60}$  heteroaryl group and the  $C_1$ - $C_{60}$  heteroarylene group each include two or more rings,



## 151

the two or more rings may be condensed with (e.g., combined together with) each other.

The term “monovalent non-aromatic condensed polycyclic group,” as used herein, refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with (e.g., combined together with) each other, only carbon atoms as ring-forming atoms, and non-aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed polycyclic group include an indenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, an indenophenanthrenyl group, and an indenanthracenyl group. The term “divalent non-aromatic condensed polycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed polycyclic group.

The term “monovalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other (e.g., combined together with each other), at least one heteroatom other than carbon atoms, as a ring-forming atom, and non-aromaticity in its entire molecular structure (e.g., is not aromatic when considered as a whole). Examples of the monovalent non-aromatic condensed heteropolycyclic group include a pyrrolyl group, a thienyl group, a furanyl group, an indolyl group, a benzindolyl group, a naphthoindolyl group, an isoindolyl group, a benzoisoindolyl group, a naphthoisoindolyl group, a benzosilolyl group, a benzothienyl group, a benzofuranyl group, a carbazolyl group, a dibenzosilolyl group, a dibenzothienyl group, a dibenzofuranyl group, an azacarbazolyl group, an azafluorenyl group, an azadibenzosilolyl group, an azadibenzothienyl group, an azadibenzofuranyl group, a pyrazolyl group, an imidazolyl group, a triazolyl group, a tetrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, an oxadiazolyl group, a thiadiazolyl group, a benzopyrazolyl group, a benzimidazolyl group, a benzoxazolyl group, a benzothiazolyl group, a benzoxadiazolyl group, a benzothiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an imidazotriazinyl group, an imidazopyrazinyl group, an imidazopyridazinyl group, an indenocarbazolyl group, an indolocarbazolyl group, a benzofurocarbazolyl group, a benzothienocarbazolyl group, a benzosilolocarbazolyl group, a benzoinolocarbazolyl group, a benzocarbazolyl group, a benzonaphthofuranyl group, a benzonaphthothienyl group, a benzonaphthosilolyl group, a benzofurodibenzofuranyl group, a benzofurodibenzothienyl group, and a benzothienodibenzothienyl group. The term “divalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

The term “C<sub>6</sub>-C<sub>60</sub> aryloxy group,” as used herein, indicates —OA<sub>102</sub> (wherein A<sub>102</sub> is the C<sub>6</sub>-C<sub>60</sub> aryl group), and the term “C<sub>6</sub>-C<sub>60</sub> arylthio group,” as used herein, indicates —SA<sub>103</sub> (wherein A<sub>103</sub> is the C<sub>6</sub>-C<sub>60</sub> aryl group).

The term “C<sub>7</sub>-C<sub>60</sub> arylalkyl group,” as used herein, indicates —A<sub>104</sub>A<sub>105</sub> (wherein A<sub>104</sub> is a C<sub>1</sub>-C<sub>54</sub> alkylene group, and A<sub>105</sub> is a C<sub>6</sub>-C<sub>59</sub> aryl group), and the term “C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group,” as used herein, indicates —A<sub>106</sub>A<sub>107</sub> (wherein A<sub>106</sub> is a C<sub>1</sub>-C<sub>59</sub> alkylene group, and A<sub>107</sub> is a C<sub>1</sub>-C<sub>59</sub> heteroaryl group). The term “R<sub>10a</sub>,” as used herein, refers to:

deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

## 152

a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, or a C<sub>1</sub>-C<sub>60</sub> alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, —Si(Q<sub>11</sub>)(Q<sub>12</sub>)(Q<sub>13</sub>), —N(Q<sub>11</sub>)(Q<sub>12</sub>), —B(Q<sub>11</sub>)(Q<sub>12</sub>), —C(=O)(Q<sub>11</sub>), —S(=O)<sub>2</sub>(Q<sub>11</sub>), —P(=O)(Q<sub>11</sub>)(Q<sub>12</sub>), or any combination thereof;

a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, or a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, —Si(Q<sub>21</sub>)(Q<sub>22</sub>)(Q<sub>23</sub>), —N(Q<sub>21</sub>)(Q<sub>22</sub>), —B(Q<sub>21</sub>)(Q<sub>22</sub>), —C(=O)(Q<sub>21</sub>), —S(=O)<sub>2</sub>(Q<sub>21</sub>), —P(=O)(Q<sub>21</sub>)(Q<sub>22</sub>), or any combination thereof; or —Si(Q<sub>31</sub>)(Q<sub>32</sub>)(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>), —C(=O)(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), or —P(=O)(Q<sub>31</sub>)(Q<sub>32</sub>).

Q<sub>1</sub> to Q<sub>3</sub>, Q<sub>11</sub> to Q<sub>13</sub>, Q<sub>21</sub> to Q<sub>23</sub> and Q<sub>31</sub> to Q<sub>33</sub>, as described herein, may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; C<sub>1</sub>-C<sub>60</sub> alkyl group; C<sub>2</sub>-C<sub>60</sub> alkenyl group; C<sub>2</sub>-C<sub>60</sub> alkynyl group; C<sub>1</sub>-C<sub>60</sub> alkoxy group; or a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, or a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

The term “heteroatom,” as used herein, refers to any atom other than a carbon atom. Examples of the heteroatom include O, S, N, P, Si, B, Ge, Se, or any combination thereof.

A third-row transition metal in the present specification includes hafnium (Hf), tantalum (Ta), tungsten (W), rhenium (Re), osmium (Os), iridium (Ir), platinum (Pt), and/or gold (Au).

The term “Ph,” as used herein, refers to a phenyl group, the term “Me,” as used herein, refers to a methyl group, the term “Et,” as used herein, refers to an ethyl group, the term “tert-Bu” or “Bu<sup>t</sup>,” as used herein, refers to a tert-butyl group, and the term “OMe,” as used herein, refers to a methoxy group.

The term “biphenyl group,” as used herein, refers to “a phenyl group substituted with a phenyl group.” In other words, the “biphenyl group” is a substituted phenyl group having a C<sub>6</sub>-C<sub>60</sub> aryl group as a substituent.

The term “terphenyl group,” as used herein, refers to “a phenyl group substituted with a biphenyl group.” In other words, the “terphenyl group” is a substituted phenyl group having, as a substituent, a C<sub>6</sub>-C<sub>60</sub> aryl group substituted with a C<sub>6</sub>-C<sub>60</sub> aryl group.

\* and \*, as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula or a moiety.

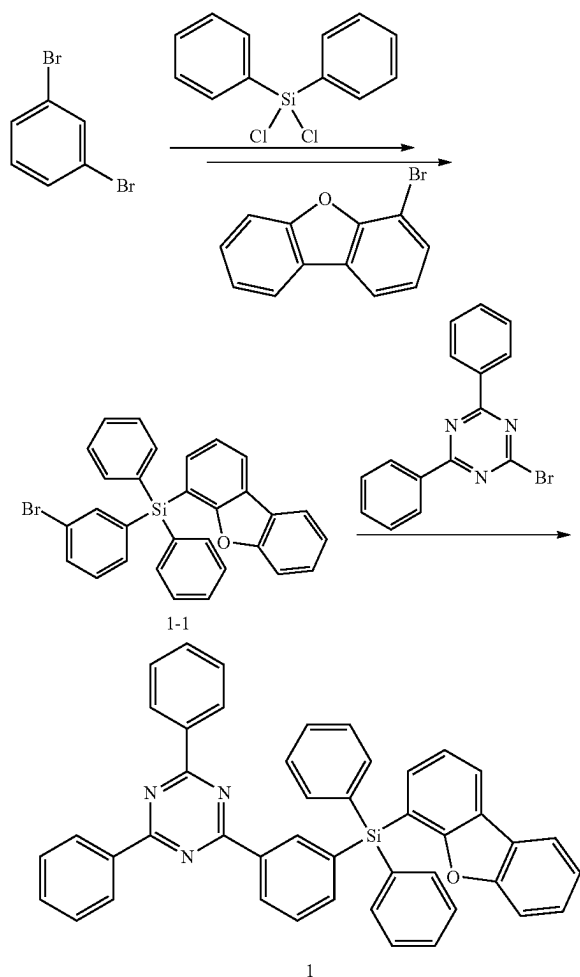
Hereinafter, a compound according to embodiments and a light-emitting device according to embodiments will be described in more detail with reference to Synthesis Examples and Examples. The wording “B was used instead

## 153

of A" used in describing Synthesis Examples refers to that an identical molar equivalent of B was used in place of A.

## EXAMPLES

## Synthesis Example 1: Synthesis of Compound 1



## 1) Synthesis of Intermediate 1-1

In order to consecutively proceed the following reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

[Reactor 1-1] While 20 g of 1,3-dibromobenzene was stirred in 40 ml of a tetrahydrofuran (THF) solvent at  $-78^{\circ}\text{C}$ ., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was further stirred at  $-78^{\circ}\text{C}$ . for an hour to thereby obtain Reaction solution 1-1.

[Reactor 1-2] 21.5 g of dichlorodiphenylsilane was dissolved in 80 ml of THF and then stirred at  $-78^{\circ}\text{C}$ . for 30 minutes, to thereby obtain Reaction solution 1-2. Reaction solution 1-1 was added dropwise to Reaction solution 1-2 at  $-78^{\circ}\text{C}$ . using a cannula. Reaction solution 1-1 and Reaction solution 1-2 were stirred at  $-78^{\circ}\text{C}$ . until the following Reaction solution 1-3 was prepared.

[Reactor 1-3] While 20.9 g of 4-bromodibenzo[b,d]furan was dissolved in 40 ml of THF and then stirred at  $-78^{\circ}\text{C}$ .,

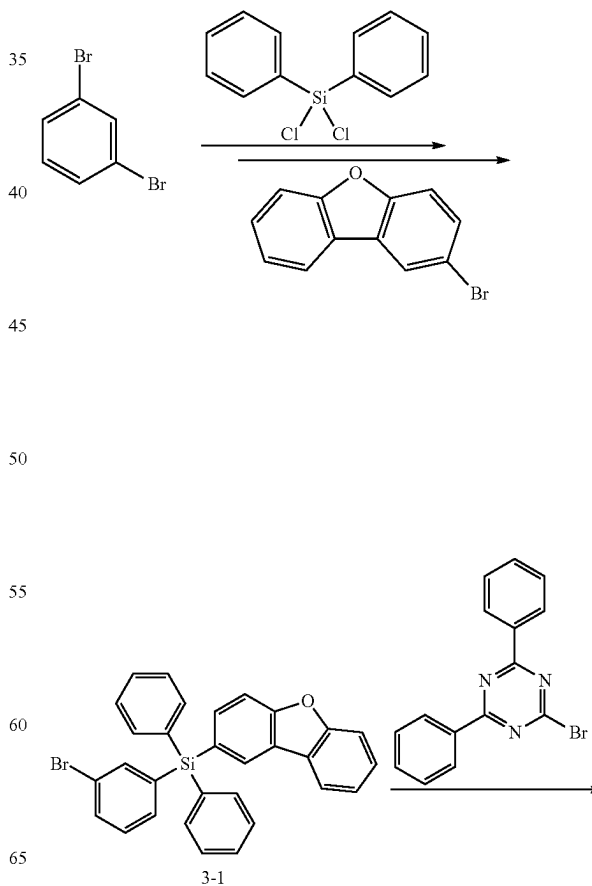
## 154

33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at  $-78^{\circ}\text{C}$ . for an hour to thereby obtain Reaction solution 1-3. Reaction solution 1-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at  $-78^{\circ}\text{C}$ . using a cannula. Afterwards, a mixture of Reaction solution 1-1 to Reaction solution 1-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 30 g (yield of 70%) of Intermediate 1-1. Intermediate 1-1 was confirmed by LC-MS. ( $\text{C}_{30}\text{H}_{21}\text{BrOSi}$ :  $\text{M}+1$  505.49)

## 2) Synthesis of Compound 1

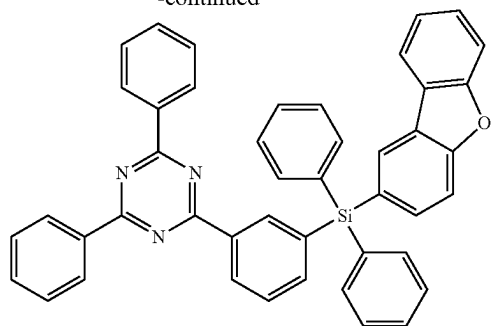
2 g of Intermediate 1-1, 1.06 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 4 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent, and then stirred at  $90^{\circ}\text{C}$ . for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.2 g (yield of 85%) of high-purity Compound 1. Compound 1 was confirmed by LC-MS and  $^1\text{H-NMR}$ .

## Synthesis Example 2: Synthesis of Compound 3



155

-continued



3

## 1) Synthesis of Intermediate 3-1

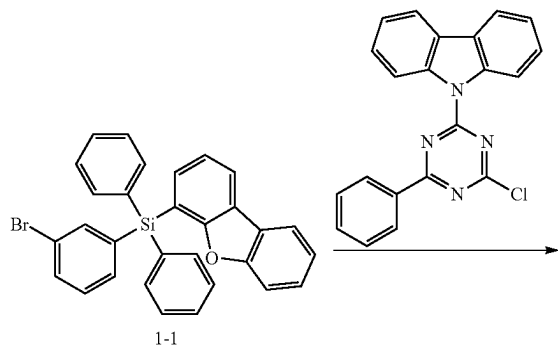
In order to consecutively proceed the reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

[Reactor 2-3] While 20.9 g of 2-bromodibenzo[b,d]furan was dissolved in 40 ml of THF and then stirred at  $-78^{\circ}\text{C}$ ., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at  $-78^{\circ}\text{C}$ . for an hour to thereby obtain Reaction solution 2-3. Reaction solution 2-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at  $-78^{\circ}\text{C}$ . using a cannula. Afterwards, a mixture of Reaction solution 1-1, Reaction solution 1-2, and Reaction solution 2-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 30 g (yield of 70%) of Intermediate 3-1. Intermediate 3-1 was confirmed by LC-MS. ( $\text{C}_{30}\text{H}_{21}\text{BrOSi}$ : M+1 505.49)

## 2) Synthesis of Compound 3

2 g of Intermediate 3-1, 1.06 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 4 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent, and then stirred at  $90^{\circ}\text{C}$ . for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.4 g (yield of 92%) of high-purity Compound 3. Compound 3 was confirmed by LC-MS and  $^1\text{H-NMR}$ .

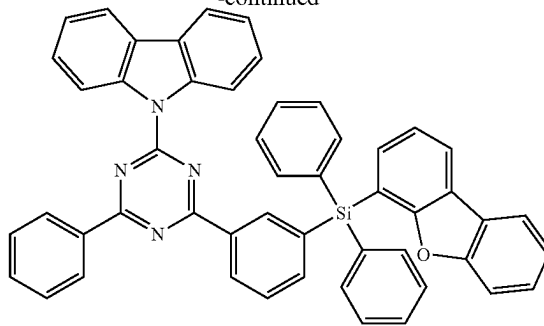
## Synthesis Example 3: Synthesis of Compound 5



1-1

156

-continued

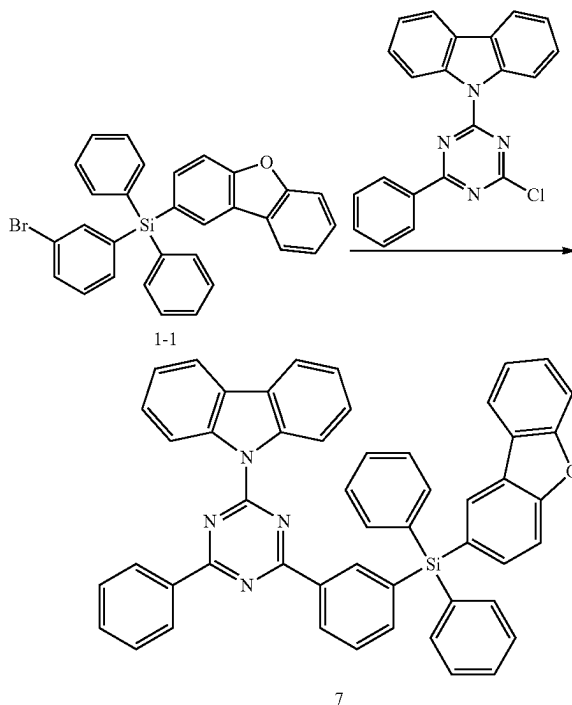


5

## 1) Synthesis of Compound 5

2 g of Intermediate 1-1, 1.41 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 4 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent and then stirred at  $90^{\circ}\text{C}$ . for 12 hours, to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.5 g (yield of 85%) of high-purity Compound 5. Compound 5 was confirmed by LC-MS and  $^1\text{H-NMR}$ .

## Synthesis Example 4: Synthesis of Compound 7



7

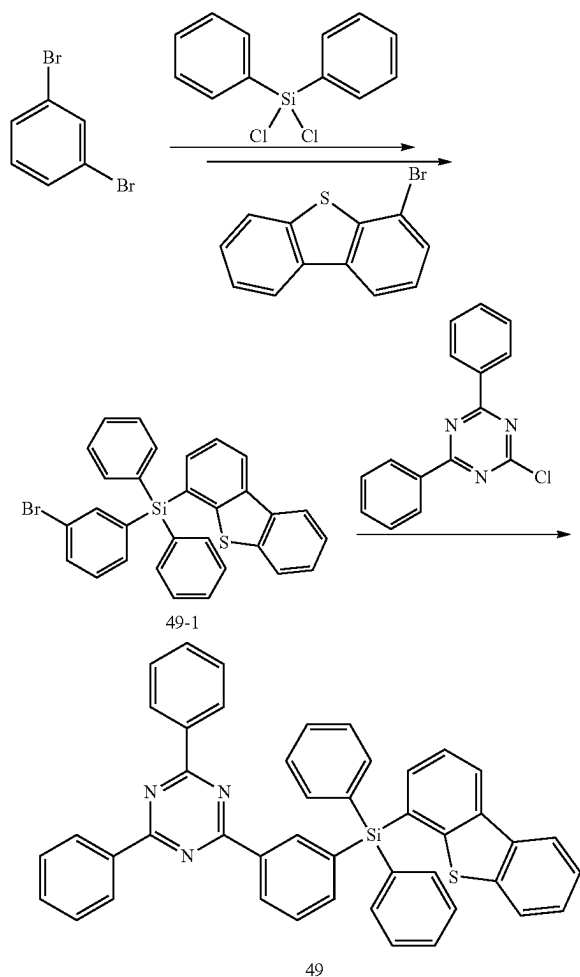
## 1) Synthesis of Compound 7

2 g of Intermediate 3-1, 1.41 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 4 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent and then stirred at  $90^{\circ}\text{C}$ . for 12 hours, to thereby obtain a reactant. An

## 157

organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.4 g (yield of 81%) of high-purity Compound 7. Compound 7 was confirmed by LC-MS and  $^1\text{H-NMR}$ .

## Synthesis Example 5: Synthesis of Compound 49



## 1) Synthesis of Intermediate 49-1

In order to consecutively proceed the reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

[Reactor 49-3] While 22.3 g of 4-bromodibenzo[b,d]thiophene was dissolved in 100 ml of THF and then stirred at  $-60^\circ\text{C}$ ., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at  $-60^\circ\text{C}$ . for an hour to thereby obtain Reaction solution 49-3. Reaction solution 49-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at  $-60^\circ\text{C}$ . using a cannula. Afterwards, a mixture of Reaction solution 1-1, Reaction solution 1-2, and Reaction solution 49-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to

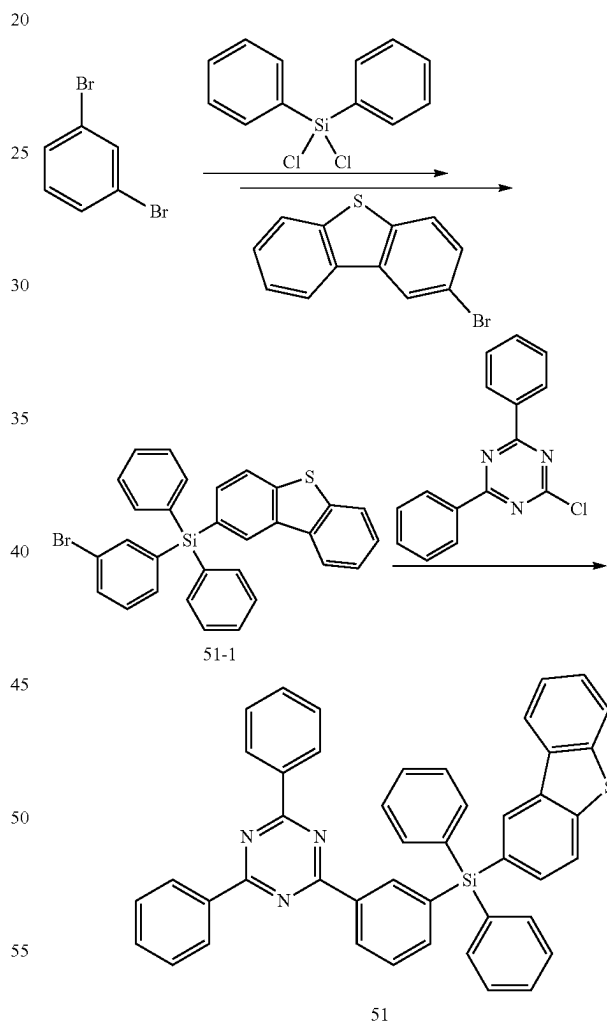
## 158

thereby obtain 24 g (yield of 54%) of Intermediate 49-1. Intermediate 49-1 was confirmed by LC-MS. ( $\text{C}_{30}\text{H}_{21}\text{BrSSi}$ : M+1 521.55)

## 2) Synthesis of Compound 49

2 g of Intermediate 49-1, 1.02 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 3.8 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent, and then stirred at  $90^\circ\text{C}$ . for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.2 g (yield of 85%) of high-purity Compound 49. Compound 49 was confirmed by LC-MS and  $^1\text{H-NMR}$ .

## Synthesis Example 6: Synthesis of Compound 51



## 1) Synthesis of Intermediate 51-1

In order to consecutively proceed the reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

[Reactor 51-3] While 22.3 g of 2-bromodibenzo[b,d]thiophene was dissolved in 100 ml of THF and then stirred at  $-60^\circ\text{C}$ ., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture

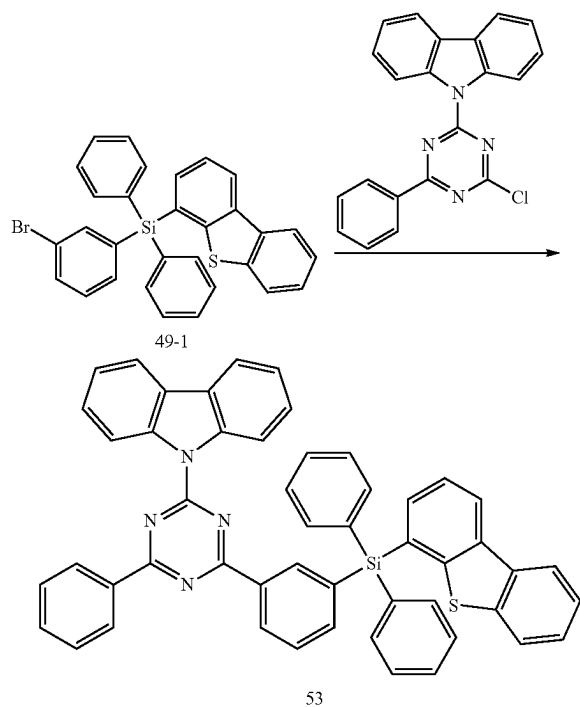
## 159

was stirred at  $-60^{\circ}\text{C}$ . for an hour to thereby obtain Reaction solution 51-3. Reaction solution 51-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at  $-60^{\circ}\text{C}$ . using a cannula. Afterwards, a mixture of Reaction solution 1-1, Reaction solution 1-2, and Reaction solution 51-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 25 g (yield of 57%) of Intermediate 51-1. Intermediate 51-1 was confirmed by LC-MS. ( $\text{C}_{30}\text{H}_{21}\text{BrSSi}$ :  $\text{M}+1$  521.55)

## 2) Synthesis of Compound 51

2 g of Intermediate 51-1, 1.02 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 3.8 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and then stirred at  $90^{\circ}\text{C}$ . for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.1 g (yield of 81%) of high-purity Compound 51. Compound 51 was confirmed by LC-MS and  $^1\text{H}$ -NMR.

## Synthesis Example 7: Synthesis of Compound 53



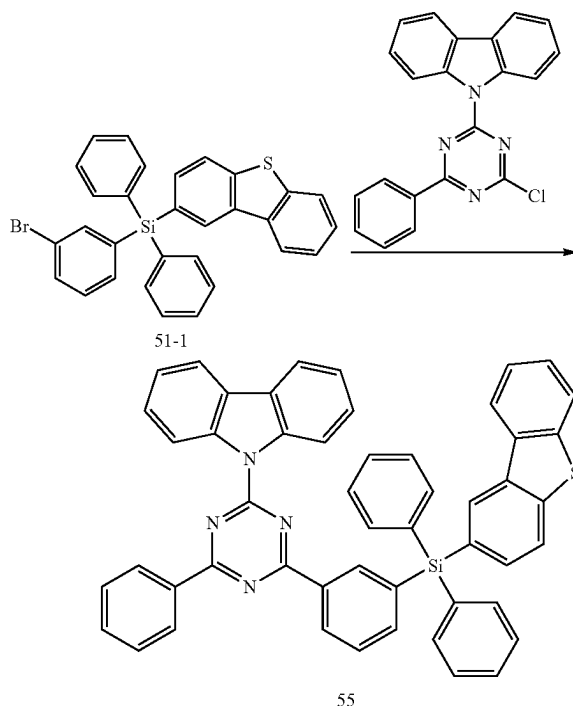
## 1) Synthesis of Compound 53

2 g of Intermediate 49-1, 1.37 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 3.8 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent and then stirred at  $90^{\circ}\text{C}$ . for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain

## 160

2.5 g (yield of 85%) of high-purity Compound 53. Compound 53 was confirmed by LC-MS and  $^1\text{H}$ -NMR.

## Synthesis Example 8: Synthesis of Compound 55



## 1) Synthesis of Compound 55

2 g of Intermediate 51-1, 1.37 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of  $\text{Pd}(\text{PPh}_3)_4$  were dissolved in 3.8 ml of a 2 M  $\text{K}_2\text{CO}_3$  aqueous solution and 20 ml of a THF solvent and then stirred at  $90^{\circ}\text{C}$ . for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.7 g (yield of 92%) of high-purity Compound 55. Compound 55 was confirmed by LC-MS and  $^1\text{H}$ -NMR.

$^1\text{H}$  NMR and MS/FAB results for the compounds synthesized according to Synthesis Examples 1 to 8 are shown in Table 1 below. Even compounds other than the compounds shown in Table 1 may be easily recognized by those skilled in the art by referring to the above synthesis routes and source materials.

TABLE 1

| Compound | $^1\text{H}$ NMR ( $\delta$ )                                                                | MS/FAB |        |
|----------|----------------------------------------------------------------------------------------------|--------|--------|
|          |                                                                                              | Calc   | found  |
| 1        | 8.36(m, 5H), 8.08(d, 1H), 7.98(d, 1H), 7.88(s, 1H), 7.31-7.50(m, 23H)                        | 657.22 | 657.85 |
| 3        | 8.36(m, 5H), 7.98(d, 1H), 7.88(s, 1H), 7.46(s, 2H), 7.31-7.70(m, 22H)                        | 657.22 | 657.85 |
| 5        | 8.55(d, 1H), 8.36-8.38(m, 3H), 8.19(d, 1H), 8.08(d, 1H), 7.88-7.98(m, 3H), 7.31-7.50(m, 25H) | 746.25 | 746.95 |

161

TABLE 1-continued

| Compound | <sup>1</sup> H NMR (δ)                                                                                  | MS/EAB |        |
|----------|---------------------------------------------------------------------------------------------------------|--------|--------|
|          |                                                                                                         | Calc   | found  |
| 7        | 8.55(d, 1H), 8.36-8.38(m, 3H),<br>8.19(d, 1H), 7.88-7.98(m, 3H),<br>7.77(s, 1H), 7.16-7.64(m, 25H)      | 746.25 | 746.95 |
| 49       | 8.55(d, 1H), 8.36-8.45(d, 6H),<br>7.88-7.93(m, 2H), 7.38-7.64(m, 22H)                                   | 673.20 | 673.91 |
| 51       | 8.36-8.45(d, 6H), 8.06(d, 2H),<br>7.88-7.93(m, 2H), 7.38-7.64(m, 21H)                                   | 673.20 | 673.91 |
| 53       | 8.55(d, 2H), 8.36-8.45(m, 4H),<br>8.19(d, 1H), 7.88-7.94(m, 3H),<br>7.16-7.64(m, 24H)                   | 762.23 | 763.01 |
| 55       | 8.55(d, 1H), 8.36-8.45(m, 4H),<br>8.19(d, 1H), 7.46-8.06(m, 2H),<br>7.88-7.94(m, 3H), 7.16-7.64(m, 23H) | 762.23 | 763.01 |

## Example 1

As an anode, a 15 Ω/cm<sup>2</sup> (1,200 Å) ITO glass substrate available from Corning Inc. was cut to a size of 50 mm×50 mm×0.7 mm, sonicated using isopropyl alcohol and pure water each for 5 minutes, and then cleaned by irradiation of ultraviolet rays and exposure of ozone thereto for 30 minutes. The ITO glass substrate was provided to a vacuum deposition apparatus.

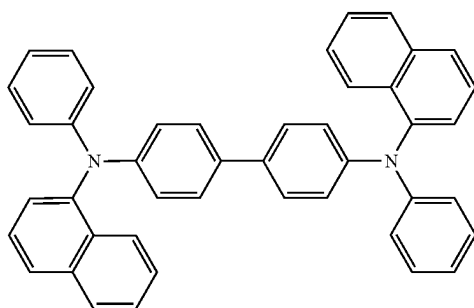
N,N'-di(1-naphthyl)-N,N'-diphenylbenzidine (NPD) was vacuum-deposited on the glass substrate to form a hole injection layer having a thickness of 300 Å.

TCTA was vacuum-deposited on the hole injection layer to form a first hole transport layer having a thickness of 200 Å, and then CzSi was vacuum-deposited on the first hole transport layer to form a second hole transport layer having a thickness of 100 Å.

Compound 1 (host) and Ir(pmp)<sub>3</sub> (dopant) were co-deposited at a weight ratio of 92:8 on the second hole transport layer to form an emission layer having a thickness of 200 Å.

Then, TSPO1 was deposited on the emission layer to form an electron transport layer having a thickness of 200 Å, and then, TPBi as a buffer electron transport compound was deposited on the electron transport layer to form a buffer layer having a thickness of 300 Å.

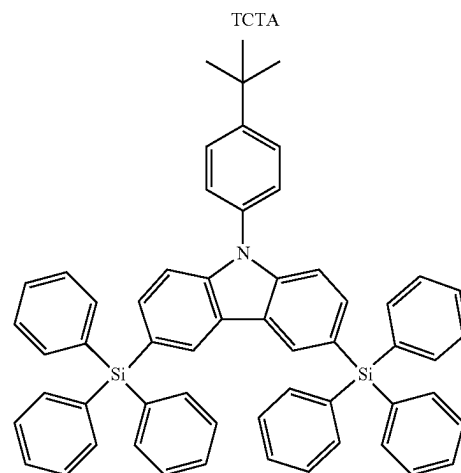
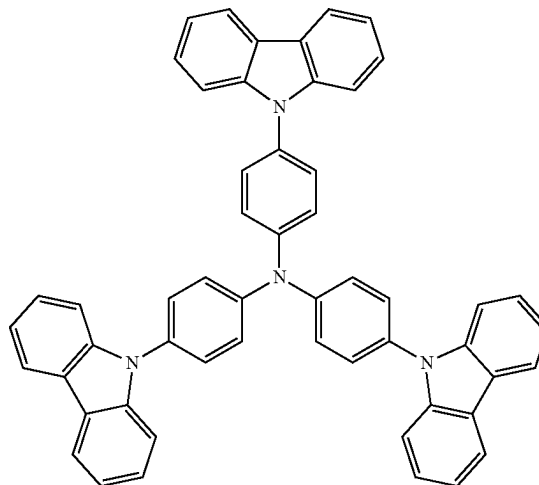
LiF, which is an alkali metal halide, was deposited on the buffer layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited thereon to form an LiF/Al electrode having a thickness of 3,000 Å, thereby completing manufacture of a light-emitting device.



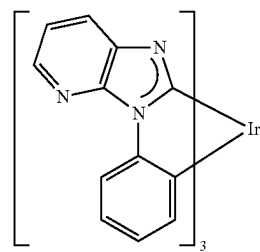
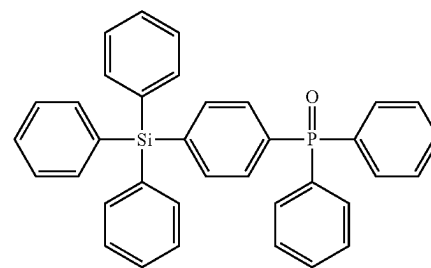
NPD

162

-continued



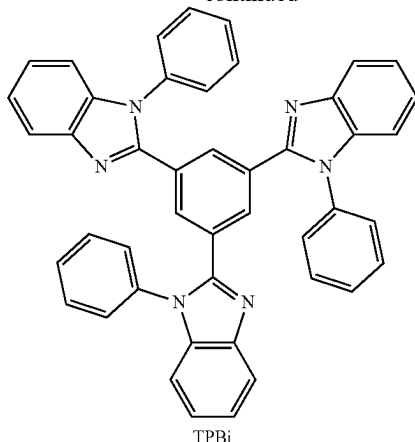
CzSi

Ir(pmp)<sub>3</sub>

TPSP01

163

-continued



Examples 2 to 8 and Comparative Examples 1 to 3

Light-emitting devices were manufactured in substantially the same manner as in Example 1, except that Com-

164

pounds shown in Table 2 were each used instead of Compound 1 in forming an emission layer.

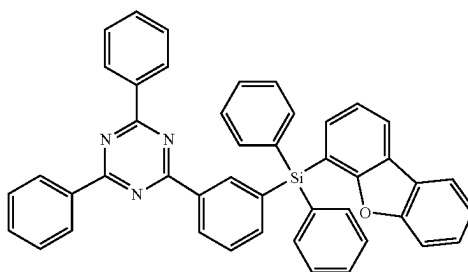
## Evaluation Example 1

5

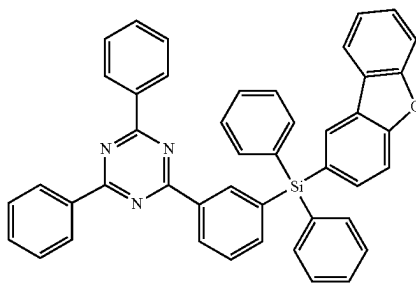
To evaluate characteristics of the light-emitting devices manufactured according to Examples 1 to 8 and Comparative Examples 1 to 3, the driving voltage at a current density of 10 mA/cm<sup>2</sup>, luminescence efficiency, and maximum external quantum efficiency (EQE) thereof were measured. The driving voltage of a light-emitting device were measured using a source meter (Keithley Instrument Inc., 2400 series), and the maximum external quantum efficiency was measured using the external quantum efficiency measurement apparatus C9920-2-12 of Hamamatsu Photonics Inc. In evaluating the maximum external quantum efficiency, the luminance/current density was measured using a luminance meter that was calibrated for wavelength sensitivity, and the maximum external quantum efficiency was converted by assuming an angular luminance distribution (Lambertian) which introduced a perfect reflecting diffuser. Table 2 below shows the evaluation results of the characteristics of the organic light-emitting devices.

TABLE 2

|                          | Host in<br>emission<br>layer | Driving<br>voltage<br>(V) | Luminescence<br>efficiency<br>(Cd/A) | Maximum<br>external<br>quantum<br>efficiency (%) | Emission<br>color |
|--------------------------|------------------------------|---------------------------|--------------------------------------|--------------------------------------------------|-------------------|
| Example 1                | Compound 1                   | 4.2                       | 22.5                                 | 22.1                                             | Blue              |
| Example 2                | Compound 3                   | 4.1                       | 22.4                                 | 21.6                                             | Blue              |
| Example 3                | Compound 5                   | 4.2                       | 21.1                                 | 20.8                                             | Blue              |
| Example 4                | Compound 7                   | 4.3                       | 22.7                                 | 21.7                                             | Blue              |
| Example 5                | Compound 49                  | 4.1                       | 22.4                                 | 21.8                                             | Blue              |
| Example 6                | Compound 51                  | 4.0                       | 22.3                                 | 21.3                                             | Blue              |
| Example 7                | Compound 53                  | 4.2                       | 22.0                                 | 21.8                                             | Blue              |
| Example 8                | Compound 55                  | 4.1                       | 22.1                                 | 22.0                                             | Blue              |
| Comparative<br>Example 1 | mCP                          | 4.8                       | 16.3                                 | 15.2                                             | Blue              |
| Comparative<br>Example 2 | Compound<br>A                | 4.9                       | 17.8                                 | 16.9                                             | Blue              |
| Comparative<br>Example 3 | Compound<br>B                | 4.8                       | 17.5                                 | 17.1                                             | Blue              |



1



3

TABLE 2-continued

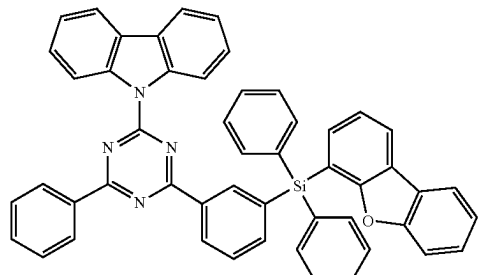
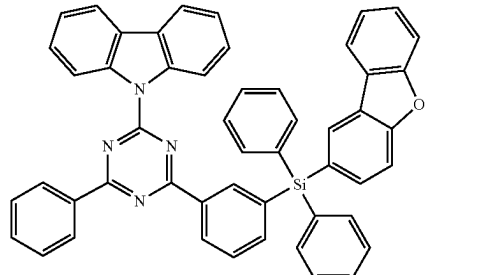
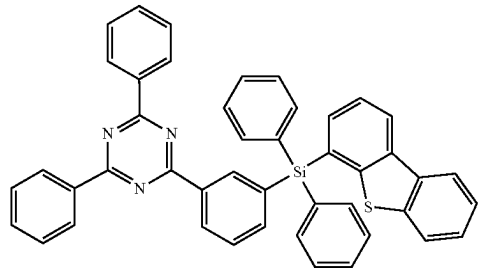
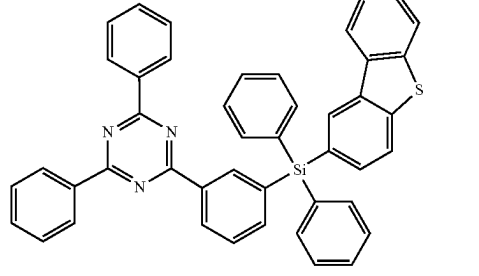
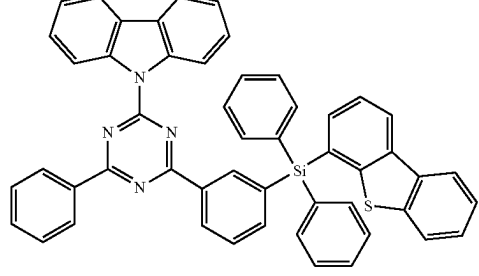
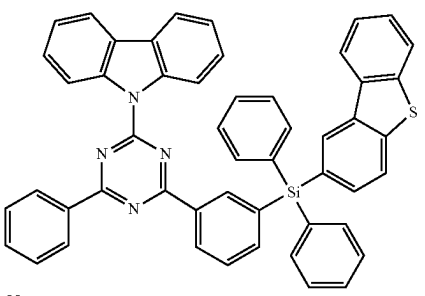
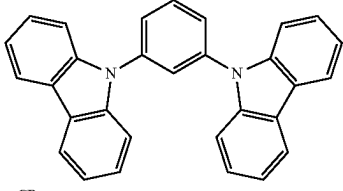
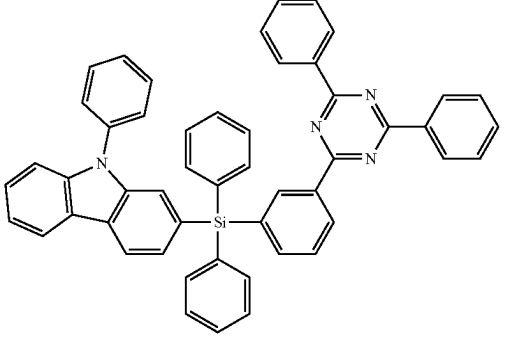
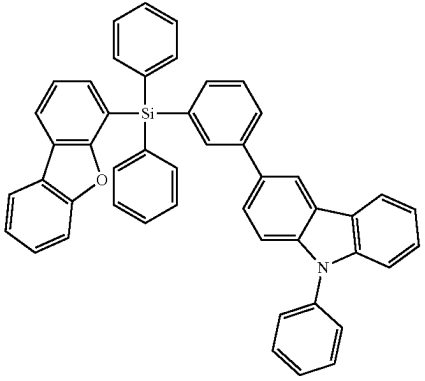
|    | Host in<br>emission<br>layer                                                        | Driving<br>voltage<br>(V) | Luminescence<br>efficiency<br>(Cd/A) | Maximum<br>external<br>quantum<br>efficiency (%) | Emission<br>color |
|----|-------------------------------------------------------------------------------------|---------------------------|--------------------------------------|--------------------------------------------------|-------------------|
|    |    |                           |                                      |                                                  |                   |
| 5  |    |                           |                                      |                                                  |                   |
| 7  |  |                           |                                      |                                                  |                   |
| 49 |  |                           |                                      |                                                  |                   |
| 51 |  |                           |                                      |                                                  |                   |
| 53 |                                                                                     |                           |                                      |                                                  |                   |



TABLE 2-continued

|     | Host in<br>emission<br>layer                                                        | Driving<br>voltage<br>(V) | Luminescence<br>efficiency<br>(Cd/A) | Maximum<br>external<br>quantum<br>efficiency (%) | Emission<br>color |
|-----|-------------------------------------------------------------------------------------|---------------------------|--------------------------------------|--------------------------------------------------|-------------------|
|     |    |                           |                                      |                                                  |                   |
| 55  |    |                           |                                      |                                                  |                   |
| mCP |                                                                                     |                           |                                      |                                                  |                   |
|     |   |                           |                                      |                                                  |                   |
| A   |                                                                                     |                           |                                      |                                                  |                   |
|     |  |                           |                                      |                                                  |                   |
| B   |                                                                                     |                           |                                      |                                                  |                   |

From Table 2, it can be seen that the light-emitting devices of Examples 1 to 8 have reduced driving voltage, increased luminescence efficiency, and increased maximum quantum efficiency, compared to the light-emitting devices of Comparative Examples 1 and 3.

The heterocyclic compound may be utilized in manufacturing a light-emitting device having high efficiency and long lifespan, and the light-emitting device may be utilized in manufacturing a high-quality electronic apparatus having excellent luminance efficiency and long lifespan.

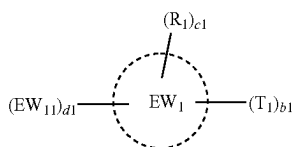
It should be understood that embodiments described herein should be considered in a descriptive sense only and

not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the drawings, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims, and equivalents thereof.

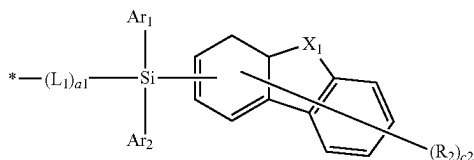
169

What is claimed is:

1. A heterocyclic compound represented by Formula 1:



Formula 1 5



Formula 2 10

wherein, in Formula 1,

EW<sub>11</sub> is —CN,

d1 is 0 or 1,

wherein EW<sub>1</sub> in Formula 1 is an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyridazine group, a pyrimidine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, a benzoisothiazole group, a benzoxazole group, a benzoisoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, an acridine group, or a pyridopyrazine group, or

T<sub>1</sub> is a group represented by Formula 2, and

b1 is an integer from 1 to 5,

wherein, in Formula 2,

X<sub>1</sub> is O or S,

L<sub>1</sub> is a single bond, a C<sub>4</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

a1 is an integer from 0 to 5, and

Ar<sub>1</sub> and Ar<sub>2</sub> are each independently a π electron-rich C<sub>3</sub>-C<sub>60</sub> cyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, and

wherein, in Formulae 1 and 2,

R<sub>1</sub> and R<sub>2</sub> are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkenyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkynyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> alkoxy group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> aryloxy group unsubstituted or substituted with

170

at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> arylthio group unsubstituted or substituted with at least one R<sub>10a</sub>, —Si(Q<sub>1</sub>)(Q<sub>2</sub>)(Q<sub>3</sub>), —B(Q<sub>1</sub>)(Q<sub>2</sub>), —P(Q<sub>1</sub>)(Q<sub>2</sub>), or —C(=O)(Q<sub>1</sub>),

c1 in Formula 1 is an integer from 0 to 10, c2 in Formula 2 is an integer from 0 to 7, and

R<sub>10a</sub> is:

deuterium (—D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, or a C<sub>1</sub>-C<sub>60</sub> alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, —Si(Q<sub>11</sub>)(Q<sub>12</sub>)(Q<sub>13</sub>), —N(Q<sub>11</sub>)(Q<sub>12</sub>), —B(Q<sub>11</sub>)(Q<sub>12</sub>), —C(=O)(Q<sub>11</sub>), —S(=O)<sub>2</sub>(Q<sub>11</sub>), —P(=O)(Q<sub>11</sub>)(Q<sub>12</sub>), or any combination thereof;

a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, or a C<sub>6</sub>-C<sub>60</sub> arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, —Si(Q<sub>21</sub>)(Q<sub>22</sub>)(Q<sub>23</sub>), —N(Q<sub>21</sub>)(Q<sub>22</sub>), —B(Q<sub>21</sub>)(Q<sub>22</sub>), —C(=O)(Q<sub>21</sub>), —S(=O)<sub>2</sub>(Q<sub>21</sub>), —P(=O)(Q<sub>21</sub>)(Q<sub>22</sub>), or any combination thereof; or —Si(Q<sub>31</sub>)(Q<sub>32</sub>)(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>), —C(=O)(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), or —P(=O)(Q<sub>31</sub>)(Q<sub>32</sub>),

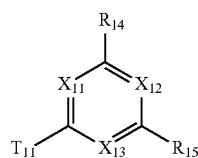
wherein Q<sub>1</sub> to Q<sub>3</sub>, Q<sub>11</sub> to Q<sub>13</sub>, Q<sub>21</sub> to Q<sub>23</sub>, and Q<sub>31</sub> to Q<sub>33</sub> are each independently:

hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C<sub>1</sub>-C<sub>60</sub> alkyl group; a C<sub>2</sub>-C<sub>60</sub> alkenyl group; a C<sub>2</sub>-C<sub>60</sub> alkynyl group; a C<sub>1</sub>-C<sub>60</sub> alkoxy group; or a C<sub>3</sub>-C<sub>60</sub> carbocyclic group or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a phenyl group, a biphenyl group, or any combination thereof,

\* in Formula 2 indicates a binding site to EW<sub>1</sub> in Formula 1, and

wherein when d1 is 0, either (i) b1 is an integer from 2 to 5, or (ii) c1 in Formula 1 is an integer from 2 to 10 and at least one R<sub>1</sub> is a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub> and at least one other R<sub>1</sub> is a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>.

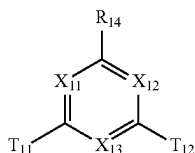
2. The heterocyclic compound of claim 1, wherein the heterocyclic compound is represented by Formula 1(a) or 1(b):



Formula 1(a) 65

171

-continued



wherein, in Formulae 1(a) and 1(b),

X<sub>11</sub> is N or C(R<sub>11</sub>),

X<sub>12</sub> is N or C(R<sub>12</sub>),

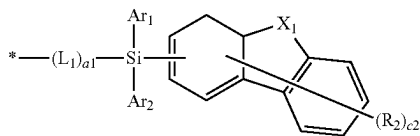
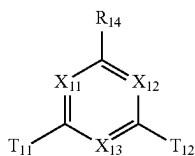
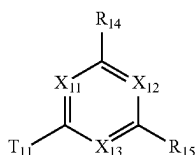
X<sub>13</sub> is N or C(R<sub>13</sub>),

at least one selected from X<sub>11</sub> to X<sub>13</sub> is N or C(CN),

R<sub>11</sub> to R<sub>15</sub> are the same as described in connection with R<sub>1</sub>, and

T<sub>11</sub> and T<sub>12</sub> are the same as described in connection with T<sub>1</sub>.

3. A heterocyclic compound represented by Formula 1(a) or 1(b):



wherein, in Formulae 1(a) and 1(b),

X<sub>11</sub> is N or C(R<sub>11</sub>),

X<sub>12</sub> is N or C(R<sub>12</sub>),

X<sub>13</sub> is N or C(R<sub>13</sub>),

T<sub>11</sub> and T<sub>12</sub> are each a group represented by Formula 2, and

wherein, in Formula 2,

X<sub>1</sub> is O or S,

L<sub>1</sub> is a single bond, a C<sub>4</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

a<sub>1</sub> is an integer from 0 to 5, and

Ar<sub>1</sub> and Ar<sub>2</sub> are each independently a Ir electron-rich C<sub>3</sub>-C<sub>60</sub> cyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, and

wherein, in Formulae 1(a), 1(b), and 2,

R<sub>2</sub> and R<sub>11</sub> to R<sub>15</sub> are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkenyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>2</sub>-C<sub>60</sub> alkynyl group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> alkoxy

172

group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> aryloxy group unsubstituted or substituted with at least one R<sub>10a</sub>, a C<sub>6</sub>-C<sub>60</sub> arylthio group unsubstituted or substituted with at least one R<sub>10a</sub>, —Si(Q<sub>1</sub>)(Q<sub>2</sub>)(Q<sub>3</sub>), —B(Q<sub>1</sub>)(Q<sub>2</sub>), —P(Q<sub>1</sub>)(Q<sub>2</sub>), or —C(=O)(Q<sub>1</sub>), wherein at least one of R<sub>11</sub> to R<sub>15</sub> is a C<sub>3</sub>-C<sub>60</sub> carboxylic group unsubstituted or substituted with at least one R<sub>10a</sub> and at least one other of R<sub>11</sub> to R<sub>15</sub> is a C<sub>1</sub>-C<sub>60</sub> heterocyclic group unsubstituted or substituted with at least one R<sub>10a</sub>,

c<sub>2</sub> in Formula 2 is an integer from 0 to 7, and

R<sub>10a</sub> is:

deuterium (—D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, or a C<sub>1</sub>-C<sub>60</sub> alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, —Si(Q<sub>11</sub>)(Q<sub>12</sub>)(Q<sub>13</sub>), —N(Q<sub>11</sub>)(Q<sub>12</sub>), —B(Q<sub>11</sub>)(Q<sub>12</sub>), —C(=O)(Q<sub>11</sub>), —S(=O)<sub>2</sub>(Q<sub>11</sub>), —P(=O)(Q<sub>12</sub>), or any combination thereof;

a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, or a C<sub>6</sub>-C<sub>60</sub> arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>2</sub>-C<sub>60</sub> alkenyl group, a C<sub>2</sub>-C<sub>60</sub> alkynyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a C<sub>3</sub>-C<sub>60</sub> carbocyclic group, a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, a C<sub>6</sub>-C<sub>60</sub> aryloxy group, a C<sub>6</sub>-C<sub>60</sub> arylthio group, a C<sub>7</sub>-C<sub>60</sub> arylalkyl group, a C<sub>2</sub>-C<sub>60</sub> heteroarylalkyl group, —Si(Q<sub>21</sub>)(Q<sub>22</sub>)(Q<sub>23</sub>), —N(Q<sub>21</sub>)(Q<sub>22</sub>), —B(Q<sub>21</sub>)(Q<sub>22</sub>), —C(=O)(Q<sub>21</sub>), —S(=O)<sub>2</sub>(Q<sub>21</sub>), —P(=O)(Q<sub>21</sub>)(Q<sub>22</sub>), or any combination thereof; or —Si(Q<sub>31</sub>)(Q<sub>32</sub>)(Q<sub>33</sub>), —N(Q<sub>31</sub>)(Q<sub>32</sub>), —B(Q<sub>31</sub>)(Q<sub>32</sub>), —C(=O)(Q<sub>31</sub>), —S(=O)<sub>2</sub>(Q<sub>31</sub>), or —P(=O)(Q<sub>31</sub>)(Q<sub>32</sub>),

wherein Q<sub>1</sub> to Q<sub>3</sub>, Q<sub>11</sub> to Q<sub>13</sub>, Q<sub>21</sub> to Q<sub>23</sub>, and Q<sub>31</sub> to Q<sub>33</sub> are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C<sub>1</sub>-C<sub>60</sub> alkyl group; a C<sub>2</sub>-C<sub>60</sub> alkenyl group; a C<sub>2</sub>-C<sub>60</sub> alkynyl group; a C<sub>1</sub>-C<sub>60</sub> alkoxy group; or a C<sub>3</sub>-C<sub>60</sub> carbocyclic group or a C<sub>1</sub>-C<sub>60</sub> heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C<sub>1</sub>-C<sub>60</sub> alkyl group, a C<sub>1</sub>-C<sub>60</sub> alkoxy group, a phenyl group, a biphenyl group, or any combination thereof,

\* in Formula 2 indicates a binding site to T<sub>11</sub> or T<sub>12</sub> in Formula 1(a) or Formula 1(b), and

wherein at least two of X<sub>11</sub> to X<sub>13</sub> are each N.

4. The heterocyclic compound of claim 2, wherein X<sub>11</sub> is C(R<sub>11</sub>),

X<sub>12</sub> is C(R<sub>12</sub>),

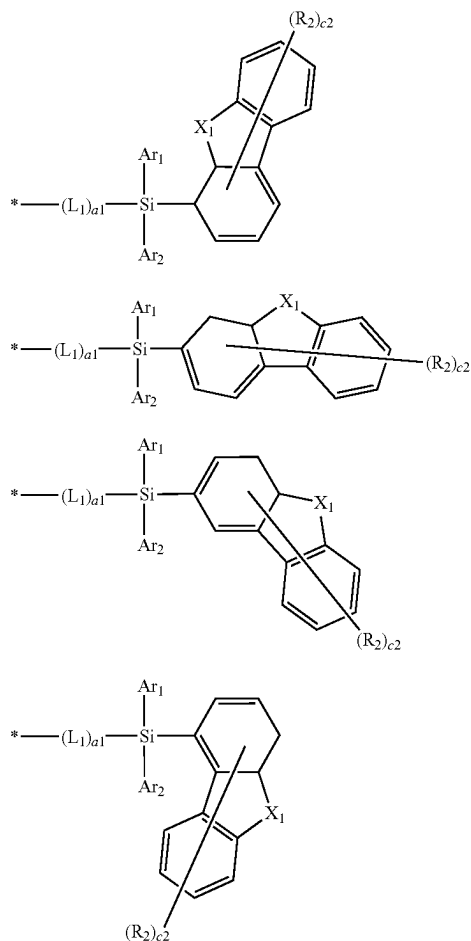
X<sub>13</sub> is C(R<sub>13</sub>), and

at least one selected from X<sub>11</sub> to X<sub>13</sub> is C (CN).

5. The heterocyclic compound of claim 1, wherein b<sub>1</sub> in Formula 1 is 1 or 2.

6. The heterocyclic compound of claim 1, wherein Formula 2 is represented by one selected from Formulae 2-1 to 2-4:

173



wherein, in Formulae 2-1 to 2-4,  
X<sub>1</sub>, L<sub>1</sub>, a<sub>1</sub>, Ar<sub>1</sub>, Ar<sub>2</sub>, R<sub>2</sub>, and c<sub>2</sub> are the same as described  
in claim 1, and

\* indicates a binding site to EW<sub>1</sub> in Formula 1.

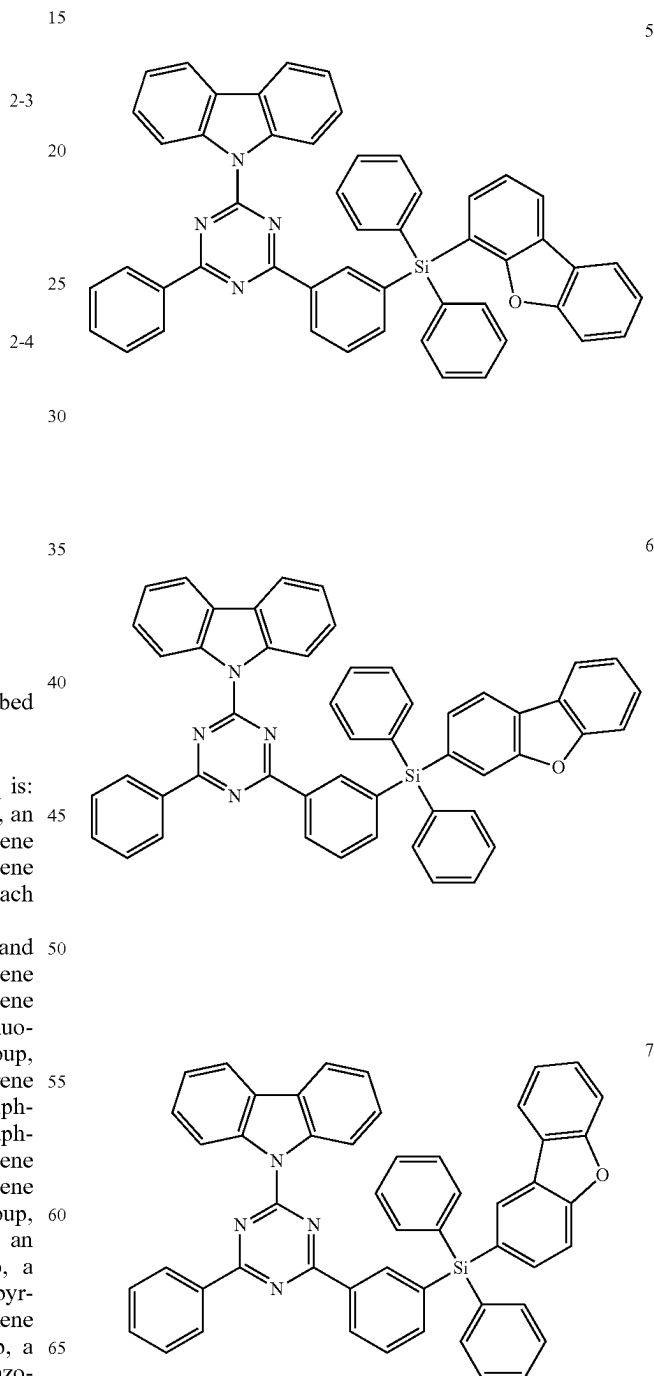
7. The heterocyclic compound of claim 1, wherein L<sub>1</sub> is:  
a single bond; or a benzene group, a naphthalene group, an  
anthracene group, a phenanthrene group, a triphenylene  
group, a pyrene group, a chrysene group, a cyclopentadiene  
group, or a 1,2,3,4-tetrahydronaphthalene group, each  
unsubstituted or substituted with at least one R<sub>10a</sub>.

8. The heterocyclic compound of claim 1, wherein Ar<sub>1</sub> and  
Ar<sub>2</sub> are each independently a benzene group, a heptalene  
group, an indene group, a naphthalene group, an azulene  
group, an indacene group, an acenaphthenyl group, a fluo-  
rene group, a spiro-bifluorene group, a benzofluorene group,  
a dibenzofluorene group, a phenalene group, a phenanthrene  
group, an anthracene group, a fluoranthene group, a triphen-  
ylene group, a pyrene group, a chrysene group, a naph-  
thalene group, a picene group, a perylene group, a pentacene  
group, a hexacene group, a pentaphene group, a rubicene  
group, a coronene group, an ovalene group, a pyrrole group,  
a furan group, a thiophene group, an isoindole group, an  
indole group, an indene group, a benzofuran group, a  
benzothiophene group, a benzosilole group, a naphtho pyr-  
role group, a naphthofuran group, a naphthothiophene  
group, a naphthosilole group, a benzocarbazole group, a  
dibenzocarbazole group, a dibenzofuran group, a dibenzo-  
thiophene group, a carbazole group, a dibenzosilole group,

174

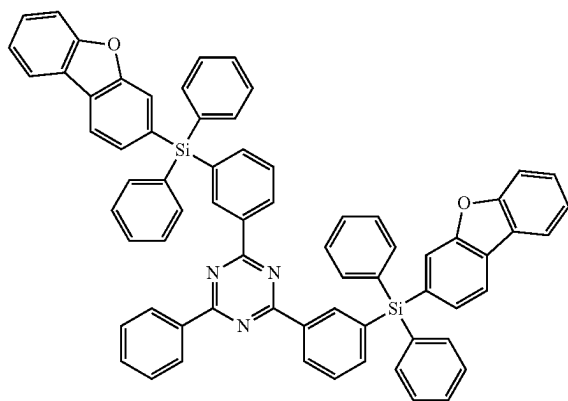
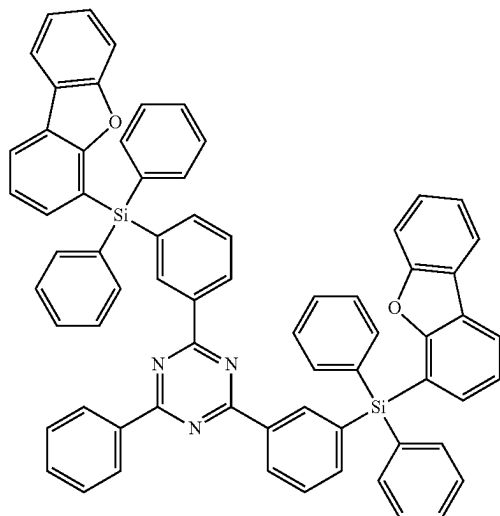
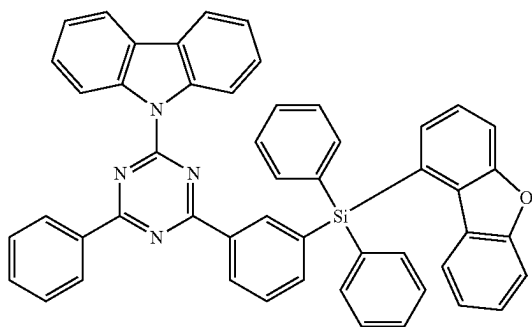
an indenocarbazole group, an indolocarbazole group, a  
benzofurocarbazole group, a benzothiencarbazole group, a  
benzosilolocarbazole group, a triindolobenzene group, a  
pyrrolophenanthrene group, a furano phenanthrene group, a  
thienophenanthrene group, a benzonaphthofuran group, a  
benzonaphthothiophene group, a (indolo) phenanthrene  
group, a (benzofurano) phenanthrene group, or a (benzoth-  
ieno) phenanthrene group, each unsubstituted or substituted  
with at least one R<sub>10a</sub>.

9. A heterocyclic compound, wherein the heterocyclic  
compound is selected from Compounds 5 to 28, 37 to 48, 53  
to 50 to 76 and 81-96:

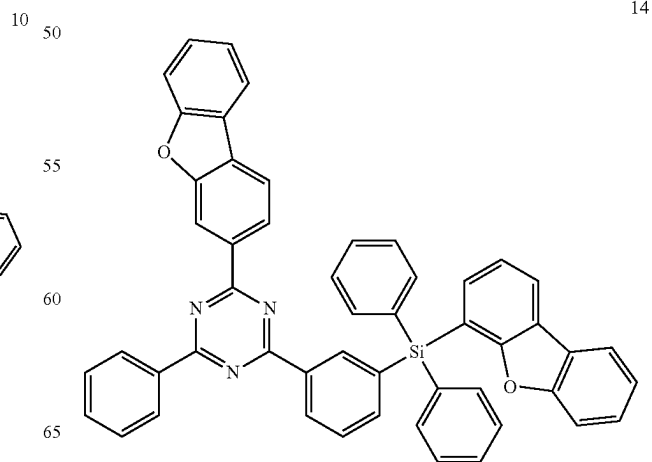
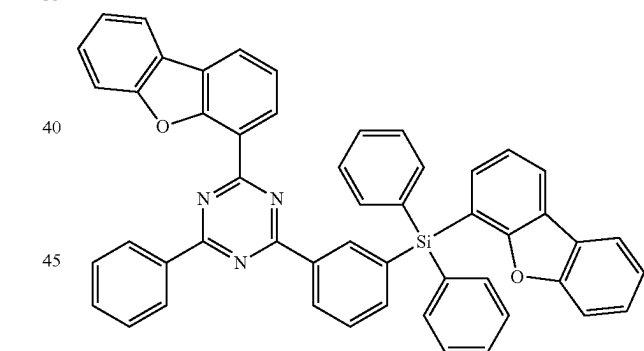
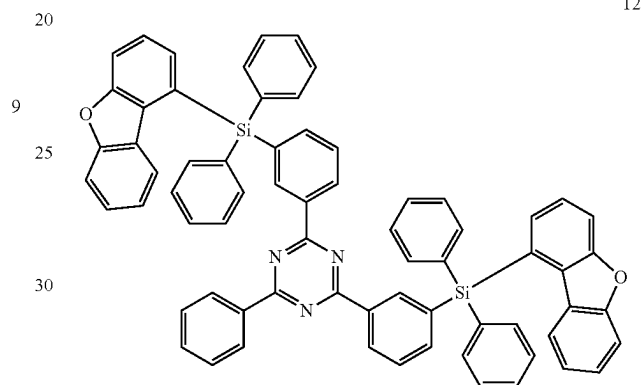
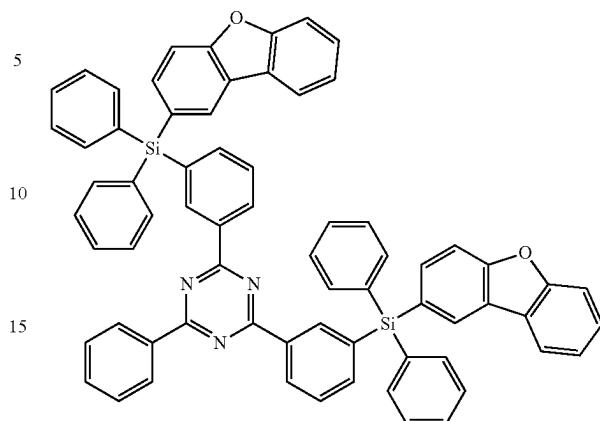


**175**

-continued

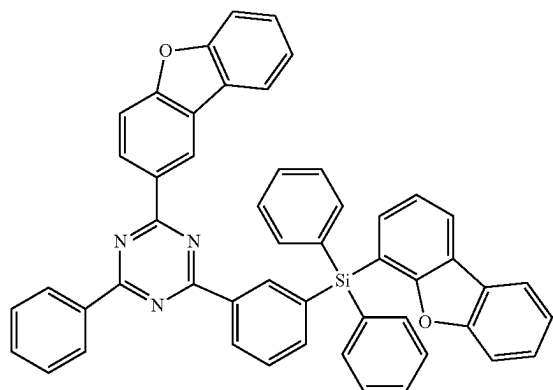
**176**

-continued



**177**

-continued



15

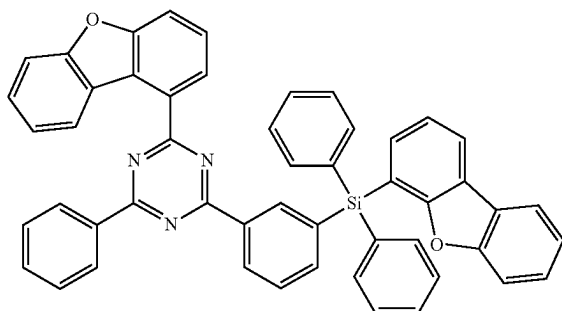
5

10

15

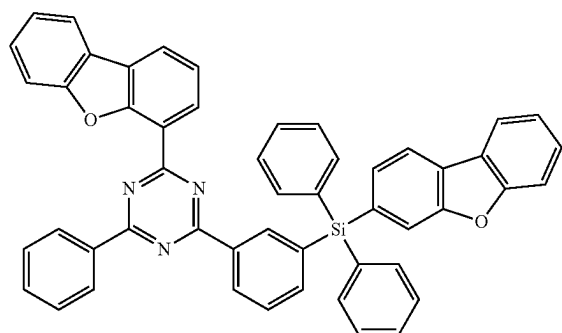
16

20



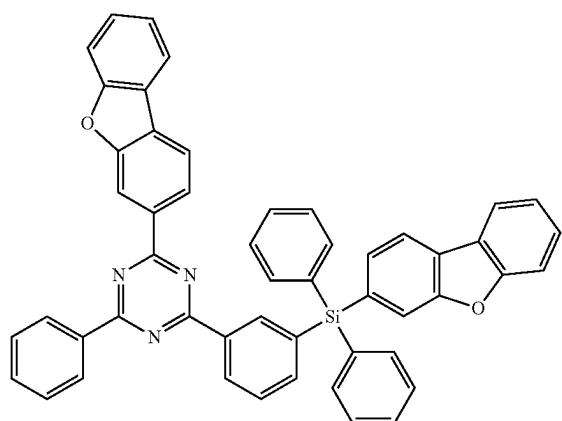
30

17



40

18



50

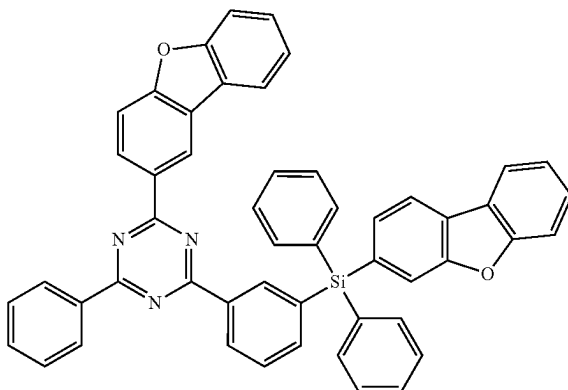
55

60

65

**178**

-continued



19

15

16

20

25

30

17

35

40

45

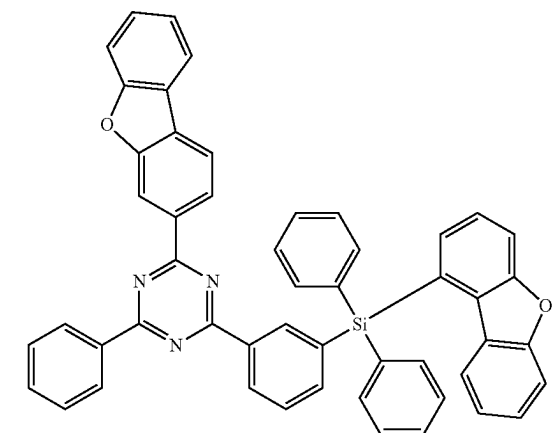
18

50

55

60

65

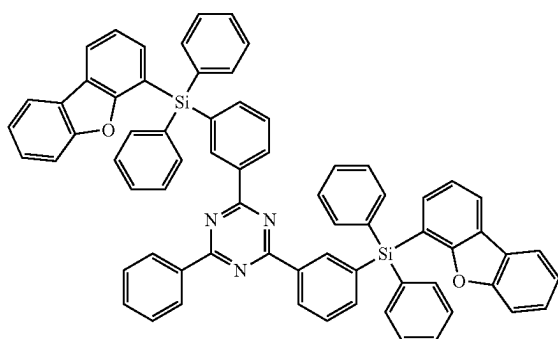
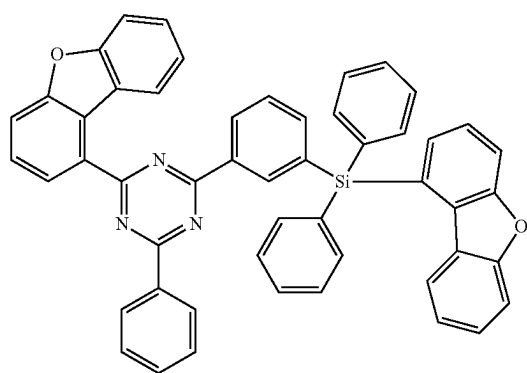
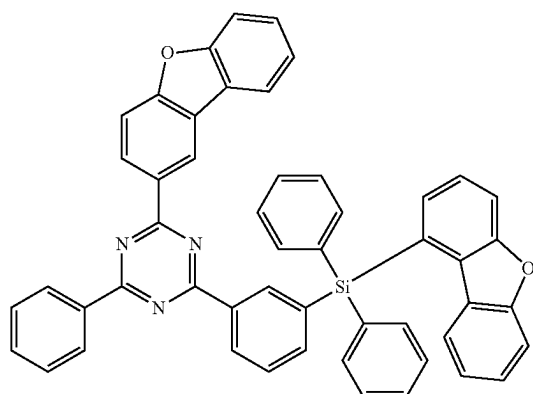


21

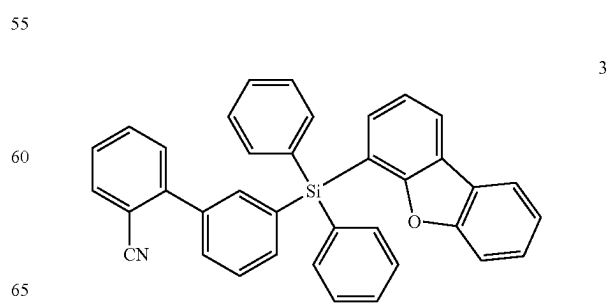
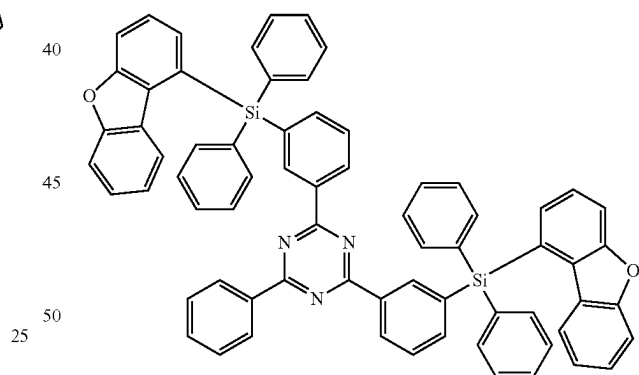
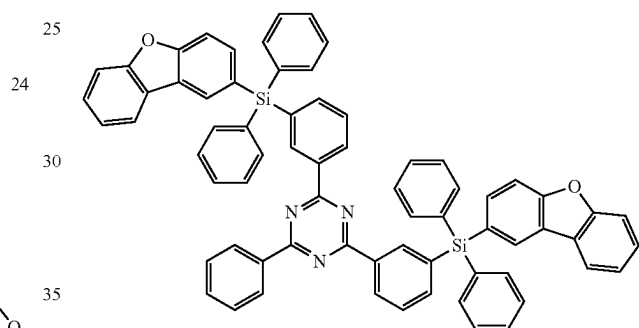
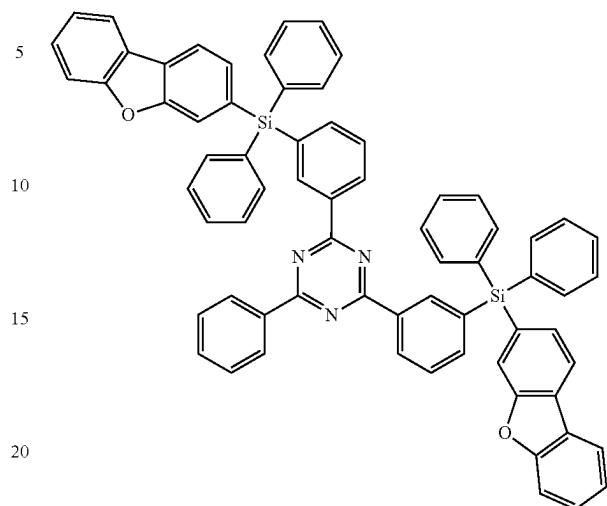
22

**179**

-continued

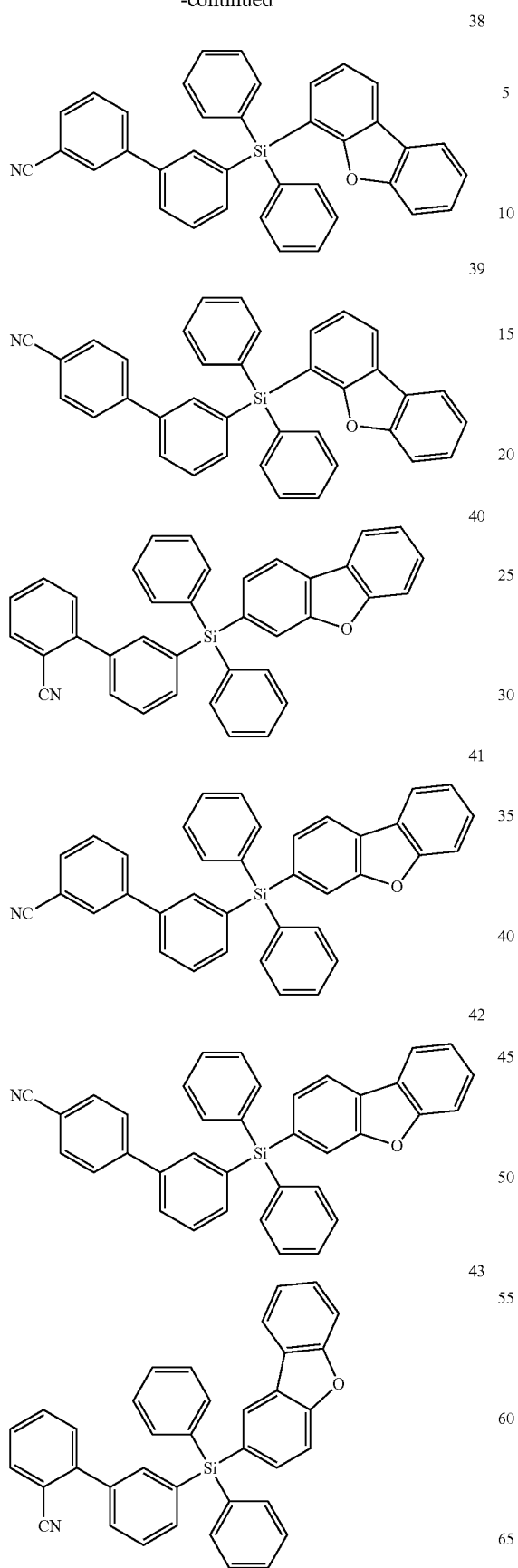
**180**

-continued

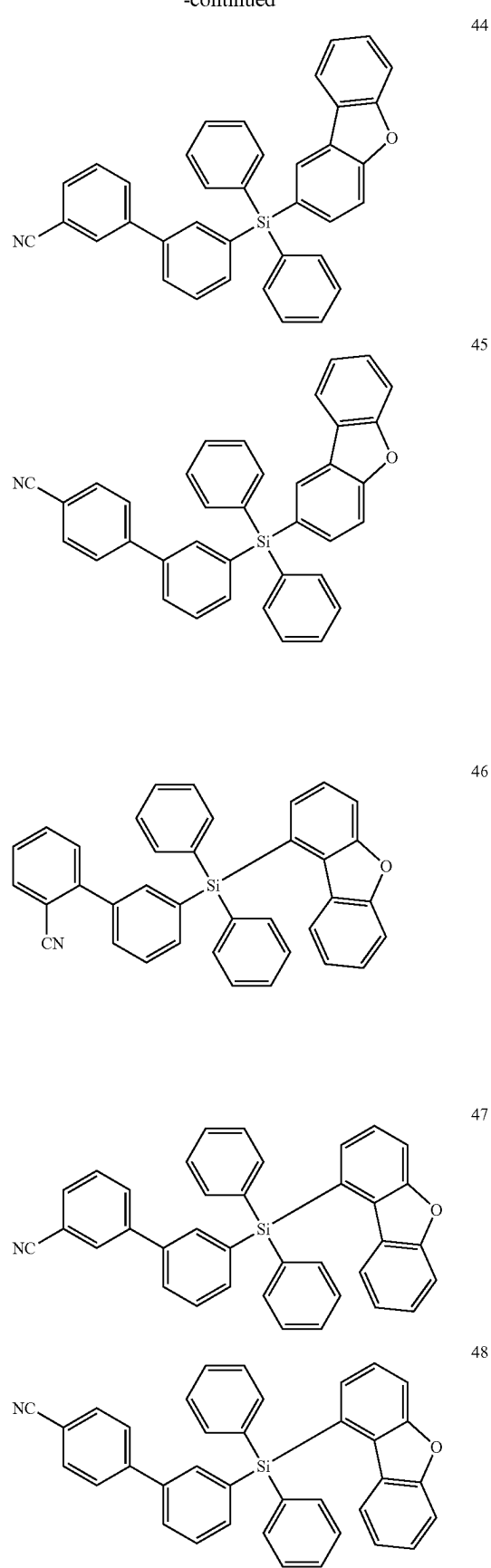


**181**

-continued

**182**

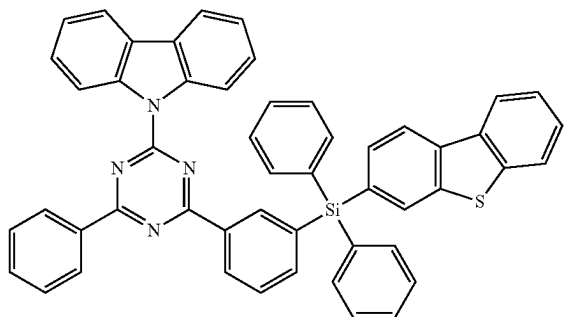
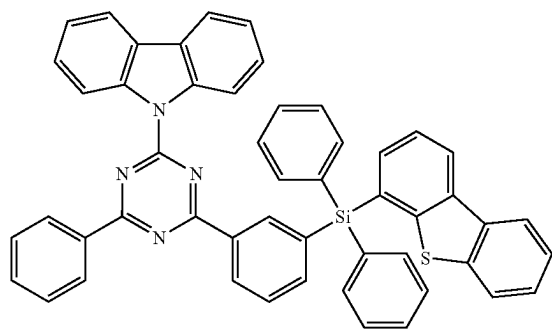
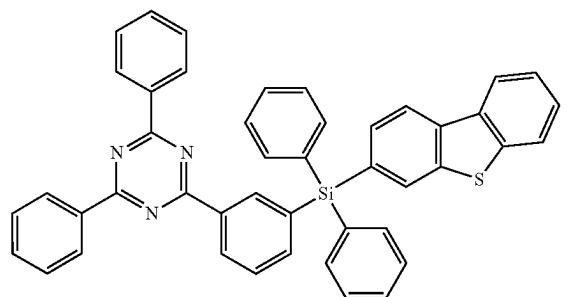
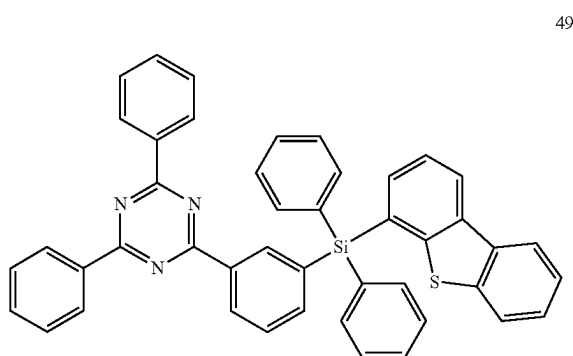
-continued



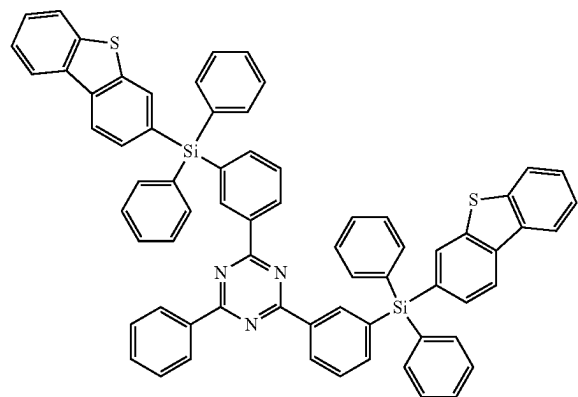
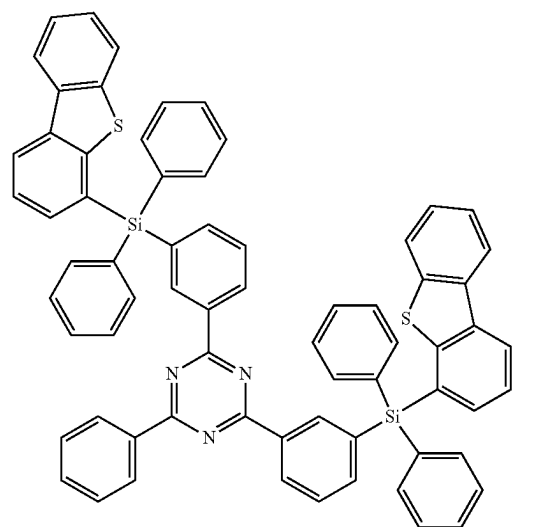
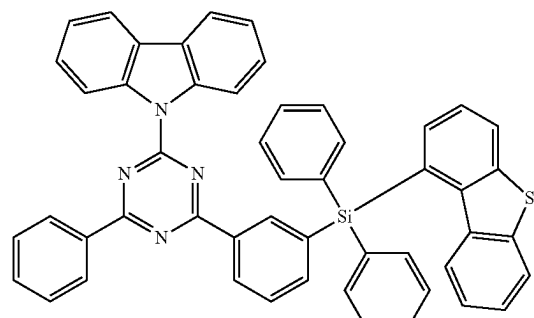
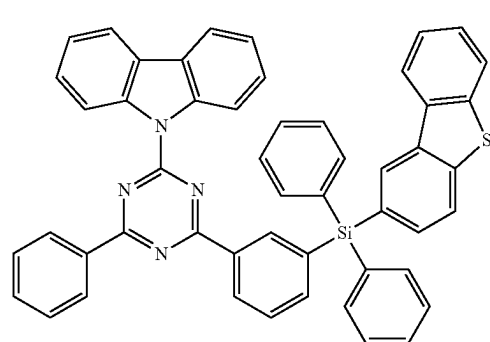


**183**

-continued

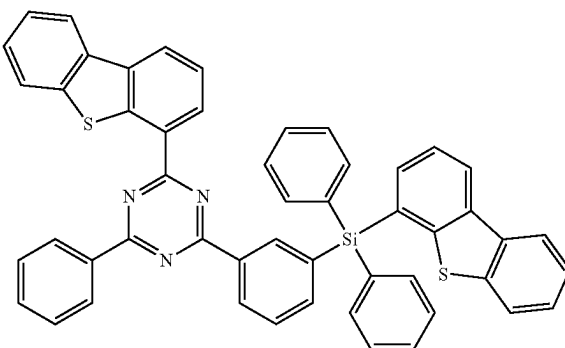
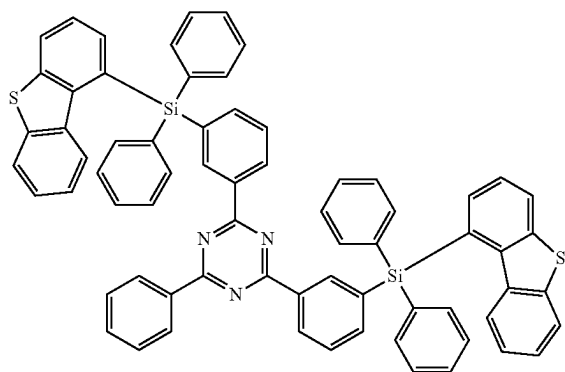
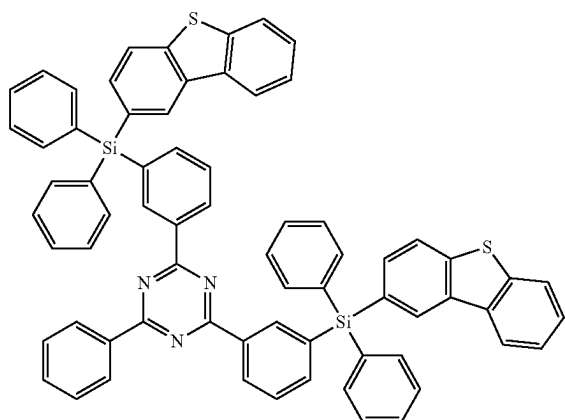
**184**

-continued

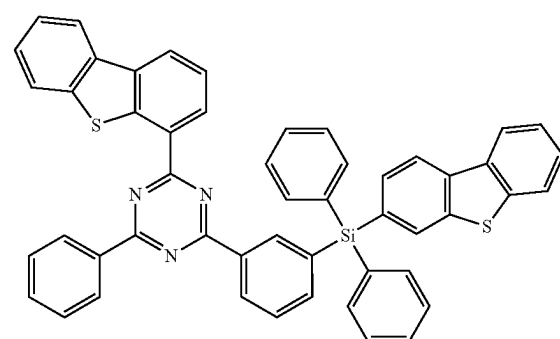
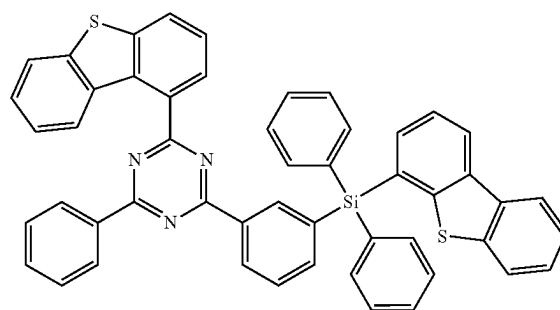
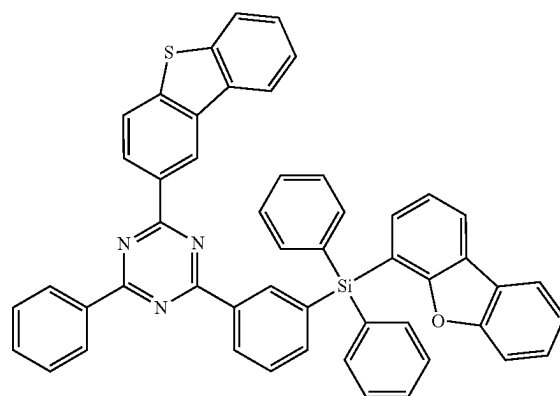
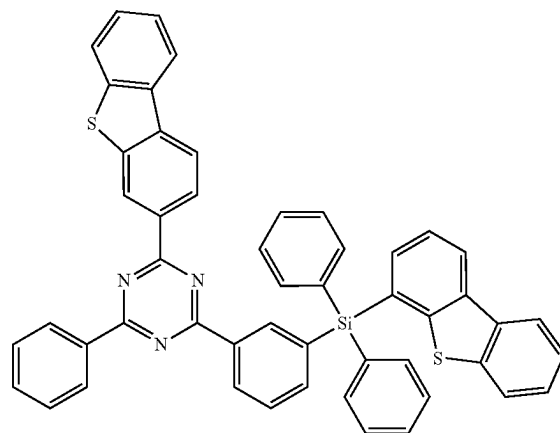


**185**

-continued

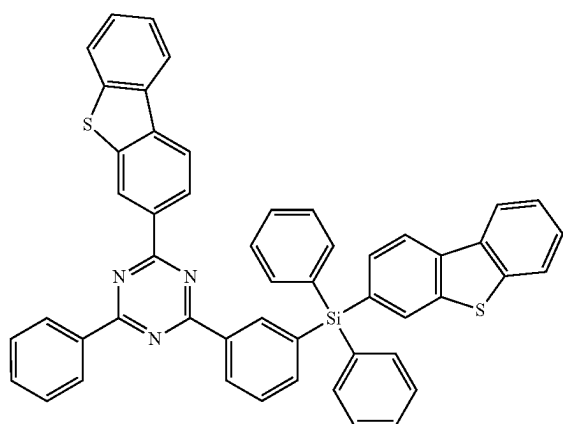
**186**

-continued



**187**

-continued



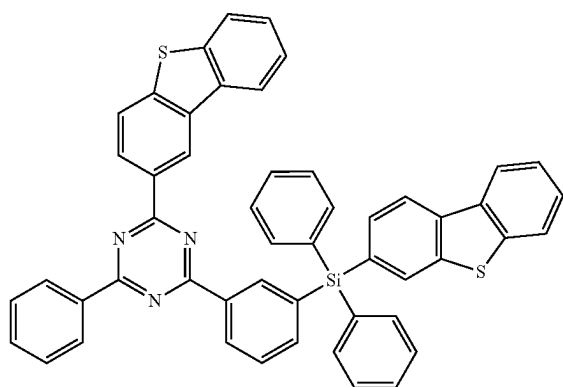
66

5

10

15

67 20

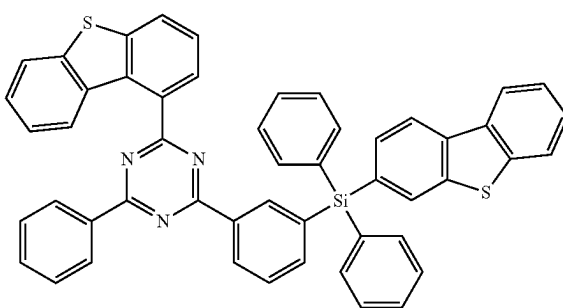


25

30

35

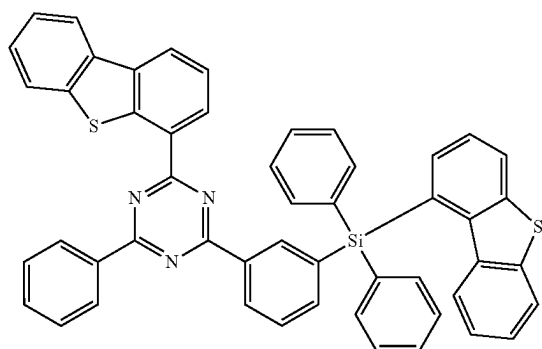
68



40

45

69 50



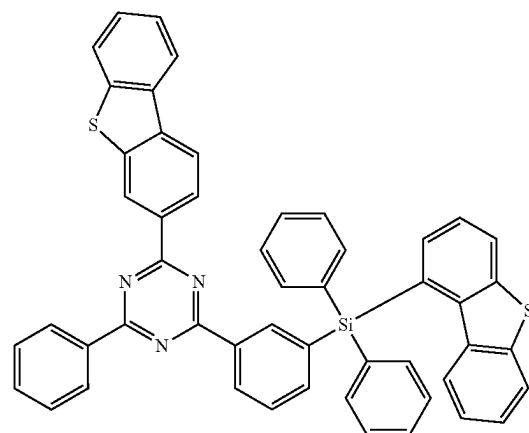
55

60

65

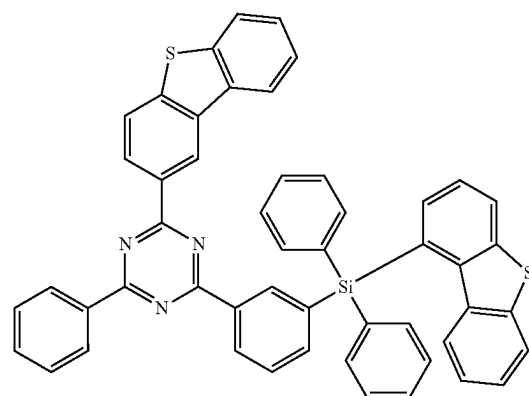
**188**

-continued

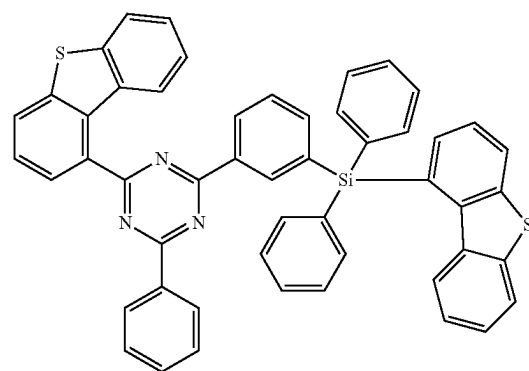


70

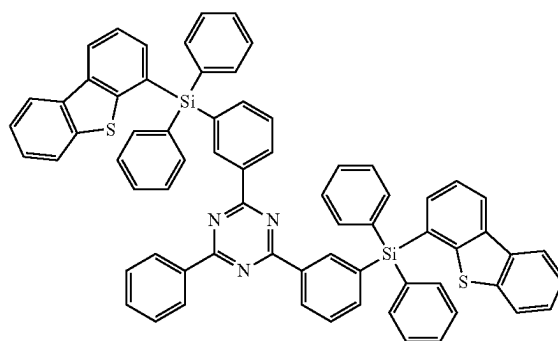
71



72

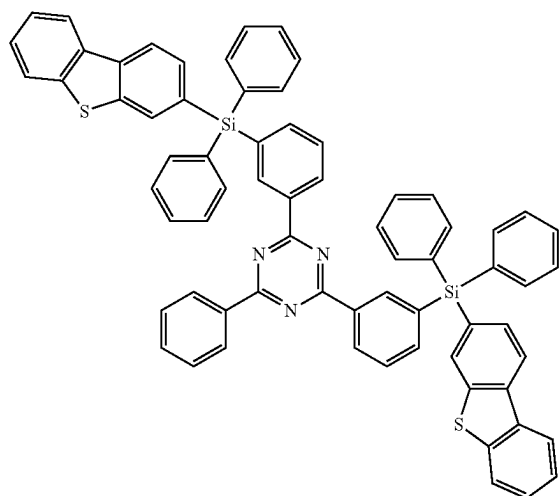


73

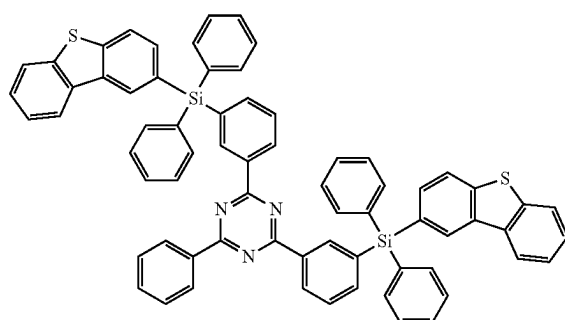


-continued

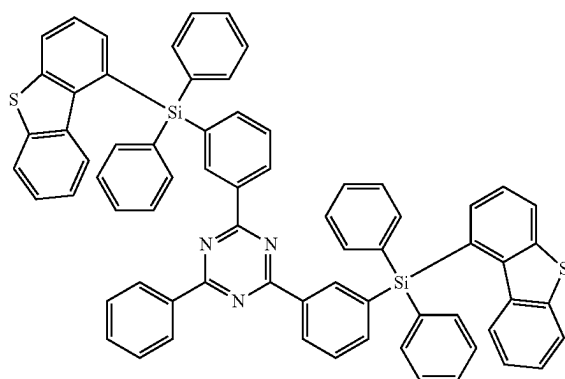
74



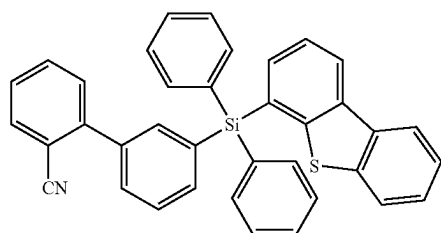
75



76



85

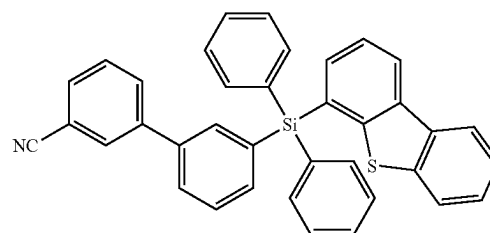


65

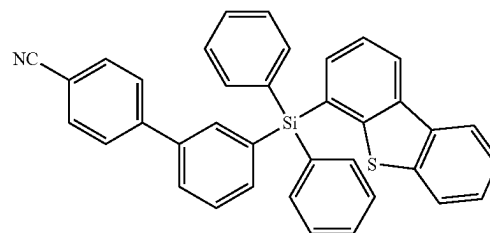
-continued

-continued

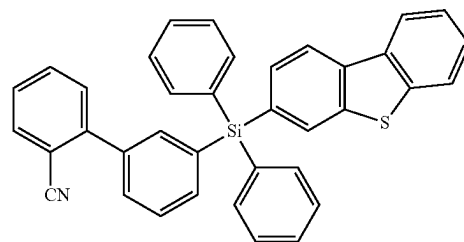
86



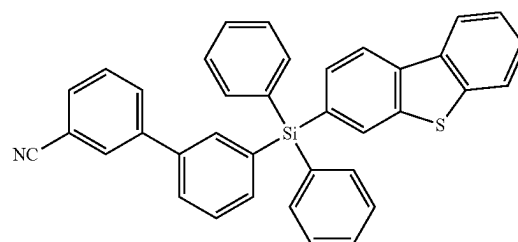
87



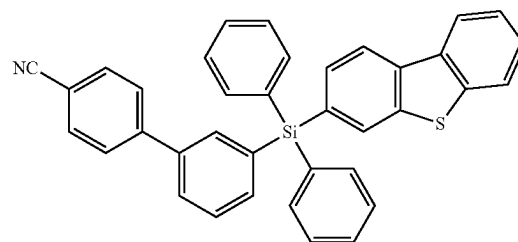
88



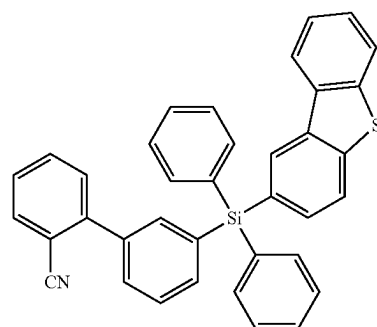
89



90

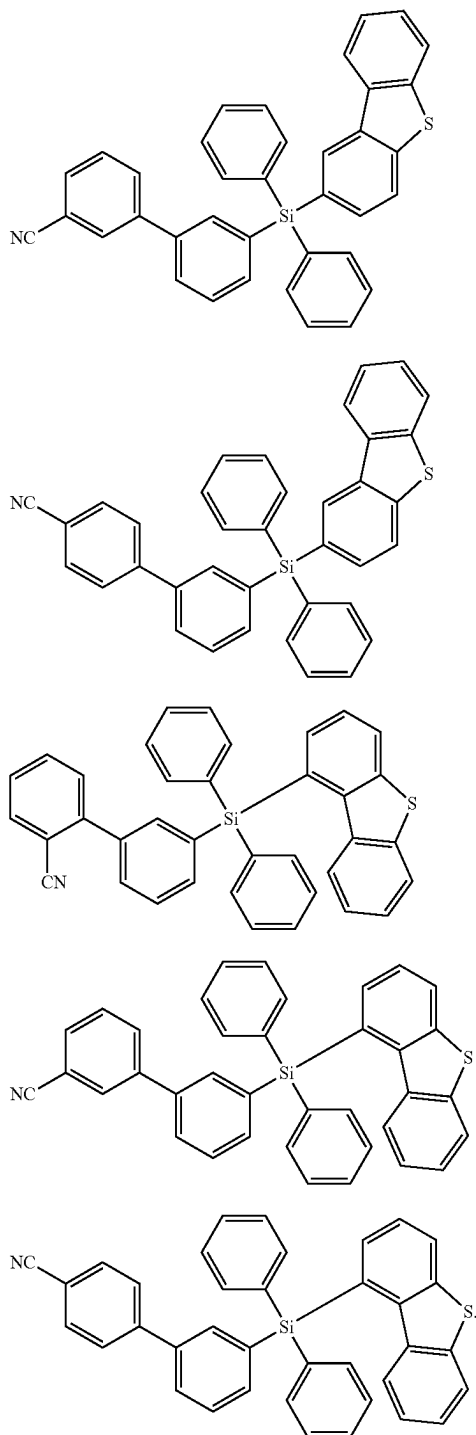


91



191

-continued



192

92 **10.** The heterocyclic compound of claim 1, wherein the heterocyclic compound emits blue light having a maximum emission wavelength of 400 nm or more and 465 nm or less, and

5 a difference between a singlet ( $S_1$ ) energy level and a triplet ( $T_1$ ) energy level of the heterocyclic compound is 0.5 eV or less.

**11.** A light-emitting device comprising:

a first electrode;

10 a second electrode facing the first electrode;

an interlayer between the first electrode and the second electrode and including an emission layer; and the heterocyclic compound represented by Formula 1 of claim 1.

93 **12.** The light-emitting device of claim 11, wherein:

the first electrode is an anode,

the second electrode is a cathode,

the interlayer further comprises a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode,

the hole transport region comprises a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and

25 the electron transport region comprises a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

94 **13.** The light-emitting device of claim 12, wherein at least one selected from the emission layer and the electron transport region comprises the heterocyclic compound.

30 **14.** The light-emitting device of claim 11, wherein the emission layer comprises a host, and the heterocyclic compound acts as the host.

35 **15.** The light-emitting device of claim 12, further comprising:

95 a first capping layer outside the first electrode and including the heterocyclic compound represented by Formula 1;

40 a second capping layer outside the second electrode and including the heterocyclic compound represented by Formula 1; or

the first capping layer and the second capping layer.

45 **16.** An electronic apparatus comprising the light-emitting device of claim 12.

96 **17.** The electronic apparatus of claim 16, further comprising a thin-film transistor, wherein the thin-film transistor comprises a source electrode and a drain electrode, and

50 the first electrode of the light-emitting device is electrically coupled to at least one selected from the source electrode and the drain electrode of the thin-film transistor.

55 **18.** The electronic apparatus of claim 16, further comprising a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof.

\* \* \* \* \*